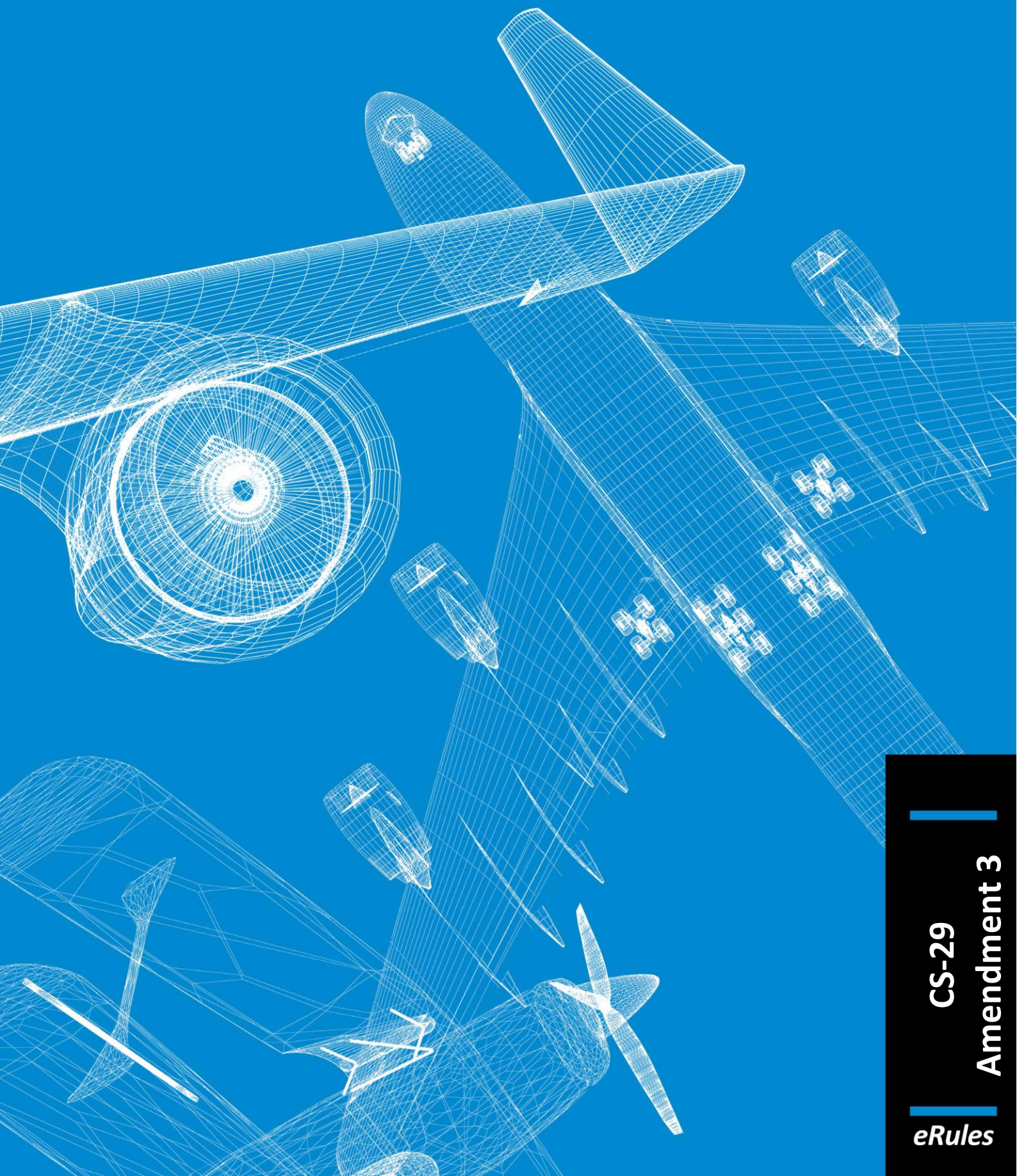




Easy Access Rules for Large Rotorcraft (CS-29) (Amendment 3)

EASA eRules: aviation rules for the 21st century

Rules and regulations are the core of the European Union civil aviation system. The aim of the **EASA eRules** project is to make them **accessible** in an efficient and reliable way to stakeholders.

EASA eRules will be a comprehensive, single system for the drafting, sharing and storing of rules. It will be the single source for all aviation safety rules applicable to European airspace users. It will offer easy (online) access to all rules and regulations as well as new and innovative applications such as rulemaking process automation, stakeholder consultation, cross-referencing, and comparison with ICAO and third countries' standards.

To achieve these ambitious objectives, the **EASA eRules** project is structured in ten modules to cover all aviation rules and innovative functionalities.

The **EASA eRules** system is developed and implemented in close cooperation with Member States and aviation industry to ensure that all its capabilities are relevant and effective.

Published June 2018¹

¹ The published date represents the date when the consolidated version of the document was generated.

DISCLAIMER

This version is issued by the European Aviation Safety Agency (EASA) in order to provide its stakeholders with an updated and easy-to-read publication. It has been prepared by putting together the certification specifications with the related acceptable means of compliance and guidance material. However, this is not an official publication and EASA accepts no liability for damage of any kind resulting from the risks inherent in the use of this document.

NOTE FROM THE EDITOR

The content of this document is arranged as follows: the certification specifications (CS) are followed by the related acceptable means of compliance (AMC) and guidance material (GM) paragraph(s).

All elements (i.e. CS, AMC and GM) are colour-coded and can be identified according to the illustration below. The EASA Executive Director (ED) decision through which the point or paragraph was introduced or last amended is indicated below the paragraph title(s) *in italics*.

Certification specification

ED decision

Acceptable means of compliance

ED decision

Guidance material

ED decision

The format of this document has been adjusted to make it user-friendly and for reference purposes. Any comments should be sent to erules@easa.europa.eu.

INCORPORATED AMENDMENTS

CS/AMC/GM (ED DECISIONS)

Incorporated ED Decision	CS/AMC Issue No, Amendment No	Applicability date
ED Decision 2003/16/RM	CS-29/ Initial issue	14/11/2003
ED Decision 2007/014/R	CS-29/ Amendment 1	30/11/2007
ED Decision 2008/010/R	CS-29/ Amendment 2	17/11/2008
ED Decision 2012/022/R	CS-29/ Amendment 3	18/12/2012

Note: To access the official versions, please click on the hyperlinks provided above.

TABLE OF CONTENTS

Disclaimer	3
Note from the editor	4
Incorporated amendments.....	5
Table of contents	6
Preamble	18
Subpart A — General	20
AMC 29 General	20
CS 29.1 Applicability	20
Subpart B — Flight	21
GENERAL	21
CS 29.21 Proof of compliance.....	21
CS 29.25 Weight limits.....	21
CS 29.27 Centre of gravity limits	22
CS 29.29 Empty weight and corresponding centre of gravity	22
CS 29.31 Removable ballast	22
CS 29.33 Main rotor speed and pitch limits	23
PERFORMANCE	24
CS 29.45 General	24
CS 29.49 Performance at minimum operating speed	24
CS 29.51 Take-off data: General.....	25
CS 29.53 Take-off: Category A.....	25
CS 29.55 Take-off Decision Point: Category A.....	25
CS 29.59 Take-off Path: Category A.....	26
CS 29.60 Elevated heliport take-off path: Category A.....	26
CS 29.61 Take-off distance: Category A	27
CS 29.62 Rejected take-off: Category A	27
CS 29.63 Take-off: Category B	27
CS 29.64 Climb: General	28
CS 29.65 Climb: All engines operating.....	28

CS 29.67 Climb: One Engine Inoperative (OEI).....	28
CS 29.71 Helicopter angle of glide: Category B	29
CS 29.75 Landing: General.....	29
CS 29.77 Landing Decision Point: Category A.....	30
CS 29.79 Landing: Category A.....	30
CS 29.81 Landing distance (ground level sites): Category A.....	30
CS 29.83 Landing: Category B.....	30
CS 29.85 Balked landing: Category A.....	31
CS 29.87 Height-velocity envelope.....	31
FLIGHT CHARACTERISTICS	32
CS 29.141 General	32
Appendix B – Airworthiness Criteria for Helicopter Instrument Flight	32
CS 29.143 Controllability and manoeuvrability	36
CS 29.151 Flight controls	37
CS 29.161 Trim control	37
CS 29.171 Stability: general	38
CS 29.173 Static longitudinal stability	38
CS 29.175 Demonstration of static longitudinal stability	38
CS 29.177 Static directional stability	39
CS 29.181 Dynamic stability: Category A rotorcraft.....	40
GROUND AND WATER HANDLING CHARACTERISTICS	41
CS 29.231 General	41
CS 29.235 Taxying condition.....	41
CS 29.239 Spray characteristics.....	41
CS 29.241 Ground resonance	41
MISCELLANEOUS FLIGHT REQUIREMENTS	42
CS 29.251 Vibration	42
Subpart C — Strength requirements	43
GENERAL	43
CS 29.301 Loads.....	43
CS 29.303 Factor of safety	43
CS 29.305 Strength and deformation.....	43
CS 29.307 Proof of structure	43
CS 29.309 Design limitations	44

FLIGHT LOADS	45
CS 29.321 General	45
CS 29.337 Limit manoeuvring load factor	45
CS 29.339 Resultant limit manoeuvring loads.....	45
CS 29.341 Gust loads	46
CS 29.351 Yawing conditions.....	46
AMC 29.351 Yaw manoeuvre conditions	46
CS 29.361 Engine torque	48
CONTROL SURFACE AND SYSTEM LOADS.....	49
CS 29.391 General	49
CS 29.395 Control system.....	49
CS 29.397 Limit pilot forces and torques	50
CS 29.399 Dual control system.....	50
CS 29.411 Ground clearance: tail rotor guard.....	50
CS 29.427 Unsymmetrical loads	50
GROUND LOADS	52
CS 29.471 General	52
CS 29.473 Ground loading conditions and assumptions.....	52
CS 29.475 Tyres and shock absorbers	52
CS 29.477 Landing gear arrangement	52
CS 29.479 Level landing conditions	52
CS 29.481 Tail-down landing conditions	53
CS 29.483 One-wheel landing conditions.....	53
CS 29.485 Lateral drift landing conditions	53
CS 29.493 Braked roll conditions.....	54
CS 29.497 Ground loading conditions: landing gear with tail wheels.....	54
CS 29.501 Ground loading conditions: landing gear with skids	56
CS 29.505 Ski landing conditions	57
CS 29.511 Ground load: unsymmetrical loads on multiple-wheel units.....	58
WATER LOADS.....	59
CS 29.519 Hull type rotorcraft: Water-based and amphibian.....	59
CS 29.521 Float landing conditions	59
MAIN COMPONENT REQUIREMENTS.....	61
CS 29.547 Main and tail rotor structure.....	61
AMC 29.547 Main rotor and tail rotor structure.....	61
CS 29.549 Fuselage and rotor pylon structures	61

CS 29.551 Auxiliary lifting surfaces.....	62
EMERGENCY LANDING CONDITIONS	63
CS 29.561 General	63
CS 29.562 Emergency landing dynamic conditions	64
CS 29.563 Structural ditching provisions.....	65
FATIGUE EVALUATION	66
CS 29.571 Fatigue Tolerance Evaluation of Metallic Structure	66
CS 29.573 Damage Tolerance and Fatigue Evaluation of Composite Rotorcraft Structures	67
 Subpart D — Design and Construction	 69
GENERAL.....	69
CS 29.601 Design	69
CS 29.602 Critical parts.....	69
CS 29.603 Materials.....	69
CS 29.605 Fabrication methods.....	69
CS 29.607 Fasteners	69
CS 29.609 Protection of structure	70
CS 29.610 Lightning and static electricity protection.....	70
CS 29.611 Inspection provisions.....	70
CS 29.613 Material strength properties and design values	71
CS 29.619 Special factors.....	71
CS 29.621 Casting factors	72
CS 29.623 Bearing factors.....	73
CS 29.625 Fitting factors.....	73
CS 29.629 Flutter and divergence	73
CS 29.631 Birdstrike	74
ROTORS	75
CS 29.653 Pressure venting and drainage of rotor blades	75
CS 29.659 Mass balance	75
CS 29.661 Rotor blade clearance.....	75
CS 29.663 Ground resonance prevention means.....	75
CONTROL SYSTEMS.....	76
CS 29.671 General	76
CS 29.672 Stability augmentation, automatic, and power-operated systems.....	76

CS 29.673 Primary flight controls	76
CS 29.674 Interconnected controls	77
CS 29.675 Stops	77
CS 29.679 Control system locks	77
CS 29.681 Limit load static tests	77
CS 29.683 Operation tests	78
CS 29.685 Control system details	78
CS 29.687 Spring devices	79
CS 29.691 Autorotation control mechanism	79
CS 29.695 Power boost and power-operated control system	79
LANDING GEAR.....	80
CS 29.723 Shock absorption tests	80
CS 29.725 Limit drop test	80
CS 29.727 Reserve energy absorption drop test	81
CS 29.729 Retracting mechanism	81
CS 29.731 Wheels	82
CS 29.733 Tyres	82
CS 29.735 Brakes	82
CS 29.737 Skis	83
FLOATS AND HULLS	84
CS 29.751 Main float buoyancy	84
CS 29.753 Main float design	84
CS 29.755 Hull buoyancy	84
CS 29.757 Hull and auxiliary float strength	84
PERSONNEL AND CARGO ACCOMMODATIONS	85
CS 29.771 Pilot compartment.....	85
CS 29.773 Pilot compartment view	85
CS 29.775 Windshields and windows	85
CS 29.777 Cockpit controls	86
CS 29.779 Motion and effect of cockpit controls	86
CS 29.783 Doors	86
CS 29.785 Seats, berths, safety belts, and harnesses.....	87
CS 29.787 Cargo and baggage compartments	89
CS 29.801 Ditching.....	89
CS 29.803 Emergency evacuation	90

Appendix D – Criteria for demonstration of emergency evacuation procedures under CS 29.803	90
CS 29.805 Flight crew emergency exits	92
CS 29.807 Passenger emergency exits	92
CS 29.809 Emergency exit arrangement	94
CS 29.811 Emergency exit marking	95
CS 29.812 Emergency lighting	96
CS 29.813 Emergency exit access	97
CS 29.815 Main aisle width	97
CS 29.831 Ventilation	98
CS 29.833 Heaters	98
FIRE PROTECTION	99
CS 29.851 Fire extinguishers	99
AMC 29.851 Fire extinguishers	99
CS 29.853 Compartment interiors	99
CS 29.855 Cargo and baggage compartments	101
CS 29.859 Combustion heater fire protection.....	102
CS 29.861 Fire protection of structure, controls, and other parts	103
CS 29.863 Flammable fluid fire protection.....	104
EXTERNAL LOADS	105
CS 29.865 External loads	105
MISCELLANEOUS	107
CS 29.871 Levelling marks	107
CS 29.873 Ballast provisions	107
Subpart E — Powerplant	108
GENERAL	108
CS 29.901 Installation	108
CS 29.903 Engines.....	108
CS 29.907 Engine vibration.....	109
CS 29.908 Cooling fans	109
ROTOR DRIVE SYSTEM	111
CS 29.917 Design	111
AMC 29.917 Rotor Drive System Design	111
CS 29.921 Rotor brake	111
CS 29.923 Rotor drive system and control mechanism tests	112

CS 29.927 Additional tests.....	115
CS 29.931 Shafting critical speed	116
CS 29.935 Shafting joints.....	117
CS 29.939 Turbine engine operating characteristics.....	117
FUEL SYSTEMS.....	118
CS 29.951 General	118
CS 29.952 Fuel system crash resistance	118
CS 29.953 Fuel system independence.....	120
CS 29.954 Fuel system lightning protection.....	120
CS 29.955 Fuel flow	121
CS 29.957 Flow between inter-connected tanks.....	121
CS 29.959 Unusable fuel supply	122
CS 29.961 Fuel system hot weather operation	122
CS 29.963 Fuel tanks: general	122
CS 29.965 Fuel tank tests.....	122
CS 29.967 Fuel tank installation	123
CS 29.969 Fuel tank expansion space.....	124
CS 29.971 Fuel tank sump	124
CS 29.973 Fuel tank filler connection.....	124
CS 29.975 Fuel tank vents and carburetor vapour vents	125
CS 29.977 Fuel tank outlet.....	125
CS 29.979 Pressure refuelling and fuelling provisions below fuel level	126
FUEL SYSTEM COMPONENTS.....	127
CS 29.991 Fuel pumps	127
CS 29.993 Fuel system lines and fittings	127
CS 29.995 Fuel valves	127
CS 29.997 Fuel strainer or filter.....	128
CS 29.999 Fuel system drains	128
CS 29.1001 Fuel jettisoning	128
OIL SYSTEM	130
CS 29.1011 Engines: General.....	130
CS 29.1013 Oil tanks.....	130
CS 29.1015 Oil tank tests.....	131
CS 29.1017 Oil lines and fittings	131
CS 29.1019 Oil strainer or filter	131

CS 29.1021 Oil system drains	132
CS 29.1023 Oil radiators	132
CS 29.1025 Oil valves.....	132
CS 29.1027 Transmissions and gearboxes: General	132
COOLING	134
CS 29.1041 General	134
CS 29.1043 Cooling tests	134
CS 29.1045 Climb cooling test procedures.....	135
CS 29.1047 Take-off cooling test procedures.....	136
CS 29.1049 Hovering cooling test procedures	137
INDUCTION SYSTEM	138
CS 29.1091 Air induction	138
CS 29.1093 Induction system icing protection	138
CS 29.1101 Carburettor air preheater design	139
CS 29.1103 Induction systems ducts and air duct systems	139
CS 29.1105 Induction system screens	140
CS 29.1107 Inter-coolers and after-coolers.....	140
CS 29.1109 Carburettor air cooling	140
EXHAUST SYSTEM.....	141
CS 29.1121 General	141
CS 29.1123 Exhaust piping	141
CS 29.1125 Exhaust heat exchangers	141
POWERPLANT CONTROLS AND ACCESSORIES	143
CS 29.1141 Powerplant controls: general	143
CS 29.1142 Auxiliary power unit controls	143
CS 29.1143 Engine controls	143
CS 29.1145 Ignition switches.....	144
CS 29.1147 Mixture controls	144
CS 29.1151 Rotor brake controls.....	144
CS 29.1157 Carburettor air temperature controls	144
CS 29.1159 Supercharger controls	144
CS 29.1163 Powerplant accessories	145
CS 29.1165 Engine ignition systems	145
POWERPLANT FIRE PROTECTION.....	146
CS 29.1181 Designated fire zones: regions included.....	146

CS 29.1183 Lines, fittings, and components.....	146
CS 29.1185 Flammable fluids	146
CS 29.1187 Drainage and ventilation of fire zones	147
CS 29.1189 Shutoff means.....	147
CS 29.1191 Firewalls.....	148
CS 29.1193 Cowling and engine compartment covering	148
CS 29.1194 Other surfaces	149
CS 29.1195 Fire extinguishing systems.....	149
CS 29.1197 Fire extinguishing agents.....	150
AMC 29.1197 Fire extinguishing agents.....	150
CS 29.1199 Extinguishing agent containers.....	151
CS 29.1201 Fire extinguishing system materials	152
CS 29.1203 Fire detector systems	152

Subpart F — Equipment 153

GENERAL..... 153

CS 29.1301 Function and installation	153
CS 29.1303 Flight and navigation instruments.....	153
CS 29.1305 Power plant instruments	154
CS 29.1307 Miscellaneous equipment	156
CS 29.1309 Equipment, systems, and installations	156

INSTRUMENTS: INSTALLATION..... 158

CS 29.1321 Arrangement and visibility	158
CS 29.1322 Warning, caution, and advisory lights	158
CS 29.1323 Airspeed indicating system.....	159
CS 29.1325 Static pressure and pressure altimeter systems	159
CS 29.1327 Magnetic direction indicator	160
CS 29.1329 Automatic pilot system.....	160
CS 29.1331 Instruments using a power supply	161
CS 29.1333 Instrument systems	161
CS 29.1335 Flight director systems	161
CS 29.1337 Powerplant instruments.....	162

ELECTRICAL SYSTEMS AND EQUIPMENT 163

CS 29.1351 General	163
CS 29.1353 Electrical equipment and installations	164
CS 29.1355 Distribution system	165

CS 29.1357 Circuit protective devices	165
CS 29.1359 Electrical system fire and smoke protection	165
CS 29.1363 Electrical system tests	166
LIGHTS	167
CS 29.1381 Instrument lights	167
CS 29.1383 Landing lights	167
CS 29.1385 Position light system installation.....	167
CS 29.1387 Position light system dihedral angles	168
CS 29.1389 Position light distribution and intensities.....	168
CS 29.1391 Minimum intensities in the horizontal plane of forward and rear position lights	169
CS 29.1393 Minimum intensities in any vertical plane of forward and rear position lights	169
CS 29.1395 Maximum intensities in overlapping beams of forward and rear position lights	169
CS 29.1397 Colour specifications	170
CS 29.1399 Riding light	170
CS 29.1401 Anti-collision light system.....	170
SAFETY EQUIPMENT	172
CS 29.1411 General	172
CS 29.1413 Safety belts: passenger warning device	172
CS 29.1415 Ditching equipment	172
CS 29.1419 Ice protection.....	173
Appendix C – Icing Certification	173
MISCELLANEOUS EQUIPMENT.....	180
CS 29.1431 Electronic equipment	180
CS 29.1433 Vacuum systems	180
CS 29.1435 Hydraulic systems.....	180
CS 29.1439 Protective breathing equipment	181
CS 29.1457 Cockpit voice recorders	181
CS 29.1459 Flight recorder	183
CS 29.1461 Equipment containing high energy rotors.....	183
CS 29.1465 Vibration Health Monitoring	184
AMC 29.1465 Vibration health monitoring	184
 Subpart G — Operating Limitations and Information	 197
GENERAL	197

CS 29.1501 General	197
OPERATING LIMITATIONS	198
CS 29.1503 Airspeed limitations: general	198
CS 29.1505 Never-exceed speed	198
CS 29.1509 Rotor speed	199
CS 29.1517 Limiting height-speed envelope	199
CS 29.1519 Weight and centre of gravity	199
CS 29.1521 Powerplant limitations	199
CS 29.1522 Auxiliary power unit limitations	202
CS 29.1523 Minimum flight crew	202
CS 29.1525 Kinds of operation	202
CS 29.1527 Maximum operating altitude	202
CS 29.1529 Instructions for Continued Airworthiness	203
Appendix A – Instructions for Continued Airworthiness	203
MARKINGS AND PLACARDS	205
CS 29.1541 General	205
CS 29.1543 Instrument markings: general	205
CS 29.1545 Airspeed indicator	205
CS 29.1547 Magnetic direction indicator	205
CS 29.1549 Powerplant instruments	206
CS 29.1551 Oil quantity indicator	206
CS 29.1553 Fuel quantity indicator	206
CS 29.1555 Control markings	206
CS 29.1557 Miscellaneous markings and placards	207
CS 29.1559 Limitations placard	207
CS 29.1561 Safety equipment	208
CS 29.1565 Tail rotor	208

ROTORCRAFT FLIGHT MANUAL	209
CS 29.1581 General	209
CS 29.1583 Operating limitations.....	209
CS 29.1585 Operating procedures.....	210
CS 29.1587 Performance information.....	210
CS 29.1589 Loading information	212
 Miscellaneous guidance	 213
MG4 Full Authority Digital Electronic Controls (FADEC)	213

PREAMBLE

ED Decision 2012/022/R

Amendment 3

The following is a list of paragraphs affected by this amendment:

Subpart C	
AMC 29.547	Created (NPA 2010-12)
CS 29.571	Created (NPA 2010-06)
CS 29.573	Created (NPA 2010-04)
Subpart D	
AMC 29.851	Created (NPA 2011-14)
Subpart E	
AMC 29.917	Created (NPA 2010-12)
CS 29.955	Editorial change
AMC 29.1197	Created (NPA 2011-14)
Subpart F	
CS 29.1401	Editorial change
CS 29.1465	Created (NPA 2010-12)
AMC 29.1465	Created (NPA 2010-12)
Subpart G	
CS-29 Appendix A A29.4	Amended (NPA 2010-04)

ED Decision 2008/010/R

Amendment 2

The following is a list of paragraphs affected by this amendment:

Subpart A	
AMC 29 General	Amended (NPA 2007-17)
Subpart B	
CS 29.143	Corrected
Subpart C	
AMC 29.351	Created (NPA 2007-17)
Subpart D	
AMC 29.602	Deleted (NPA 2007-17)
Subpart F	
CS 29.1305	Amended (NPA 2007-17)
AMC 29.1305(a)(25) and (26)	Deleted (NPA 2007-17)
Subpart G	
CS 29.1587	Amended (NPA 2007-17)
CS-29 Appendix A	Amended (NPA 2007-17)
AMC to Appendix A, A29.3(b)(2)	Deleted (NPA 2007-17)
Miscellaneous guidance	
MG4	Created (NPA 2007-17)

Amendment 1

The following is a list of paragraphs affected by this amendment:

Preamble	Preamble added
Subpart B	
CS 29.25	Amended (NPA 12/2006)
CS-29 Appendix B	Amended (NPA 12/2006)
CS 29.143	Amended (NPA 12/2006)
CS 29.173	Amended (NPA 12/2006)
CS 29.175	Amended (NPA 12/2006)
CS 29.177	Amended (NPA 12/2006)
Subpart G	
CS 29.1587	Amended (NPA 12/2006)

SUBPART A — GENERAL

AMC 29 General

ED Decision 2008/010/R

1. The AMC to CS-29 consists of FAA AC 29-2C Change 2 dated 25 April 2006 with the changes/additions given in this BOOK 2 of CS-29.
2. The primary reference for each of these AMCs is the CS-29 paragraph. Where there is an appropriate paragraph in FAA AC 29-2C Change 2 dated 25 April 2006 this is added as a secondary reference.

[Amdt. No. 29/2]

CS 29.1 Applicability

ED Decision 2003/16/RM

- (a) This Airworthiness Code is applicable to large rotorcraft.
- (b) Large rotorcraft must be certificated in accordance with either the Category A or Category B requirements. A multi-engine rotorcraft may be type certificated as both Category A and Category B with appropriate and different operating limitations for each category.
- (c) Rotorcraft with a maximum weight greater than 9072 kg (20 000 pounds) and 10 or more passenger seats must be type certificated as Category A rotorcraft.
- (d) Rotorcraft with a maximum weight greater than 9072 kg (20 000 pounds) and nine or less passenger seats may be type certificated as Category B rotorcraft provided the Category A requirements of Subparts C, D, E, and F are met.
- (e) Rotorcraft with a maximum weight of 9072 kg (20 000 pounds) or less but with 10 or more passenger seats may be type certificated as Category B rotorcraft provided the Category A requirements of [CS 29.67\(a\)\(2\)](#), [29.87](#), [29.1517](#), and of Subparts C, D, E, and F are met.
- (f) Rotorcraft with a maximum weight of 9072 kg (20 000 pounds) or less and nine or less passenger seats may be type certificated as Category B rotorcraft.

SUBPART B — FLIGHT

GENERAL

CS 29.21 Proof of compliance

ED Decision 2003/16/RM

Each requirement of this Subpart must be met at each appropriate combination of weight and centre of gravity within the range of loading conditions for which certification is requested. This must be shown:

- (a) By tests upon a rotorcraft of the type for which certification is requested, or by calculations based on, and equal in accuracy to, the results of testing; and
- (b) By systematic investigation of each required combination of weight and centre of gravity, if compliance cannot be reasonably inferred from combinations investigated.

CS 29.25 Weight limits

ED Decision 2007/014/R

- (a) *Maximum weight.* The maximum weight (the highest weight at which compliance with each applicable requirement of this CS-29 is shown) or, at the option of the applicant, the highest weight for each altitude and for each practicably separable operating condition, such as take-off, en-route operation, and landing, must be established so that it is not more than:
 - (1) The highest weight selected by the applicant;
 - (2) The design maximum weight (the highest weight at which compliance with each applicable structural loading condition of this CS-29 is shown); or
 - (3) The highest weight at which compliance with each applicable flight requirement of this CS-29 is shown.
 - (4) For Category B rotorcraft with 9 or less passenger seats, the maximum weight, altitude, and temperature at which the rotorcraft can safely operate near the ground with the maximum wind velocity determined under [CS 29.143\(c\)](#) and may include other demonstrated wind velocities and azimuths. The operating envelopes must be stated in the Limitations section of the Rotorcraft Flight Manual.
- (b) *Minimum weight.* The minimum weight (the lowest weight at which compliance with each applicable requirement of this CS-29 is shown) must be established so that it is not less than:
 - (1) The lowest weight selected by the applicant;
 - (2) The design minimum weight (the lowest weight at which compliance with each structural loading condition of this CS-29 is shown); or
 - (3) The lowest weight at which compliance with each applicable flight requirement of this CS-29 is shown.
- (c) *Total weight with jettisonable external load.* A total weight for the rotorcraft with a jettisonable external load attached that is greater than the maximum weight established under subparagraph (a) may be established for any rotorcraft-load combination if:
 - (1) The rotorcraft-load combination does not include human external cargo,

- (2) Structural component approval for external load operations under either [CS 29.865](#), or under equivalent operational standards is obtained,
- (3) The portion of the total weight that is greater than the maximum weight established under sub-paragraph (a) is made up only of the weight of all or part of the jettisonable external load,
- (4) Structural components of the rotorcraft are shown to comply with the applicable structural requirements of this CS-29 under the increased loads and stresses caused by the weight increase over that established under sub-paragraph (a), and
- (5) Operation of the rotorcraft at a total weight greater than the maximum certificated weight established under sub-paragraph (a) is limited by appropriate operating limitations under [CS 29.865\(a\) and \(d\)](#).

[Amdt. No.: 29/1]

CS 29.27 Centre of gravity limits

ED Decision 2003/16/RM

The extreme forward and aft centres of gravity and, where critical, the extreme lateral centres of gravity must be established for each weight established under [CS 29.25](#). Such an extreme may not lie beyond –

- (a) The extremes selected by the applicant;
- (b) The extremes within which the structure is proven; or
- (c) The extremes within which compliance with the applicable flight requirements is shown.

CS 29.29 Empty weight and corresponding centre of gravity

ED Decision 2003/16/RM

- (a) The empty weight and corresponding centre of gravity must be determined by weighing the rotorcraft without the crew and payload, but with:
 - (1) Fixed ballast;
 - (2) Unusable fuel; and
 - (3) Full operating fluids, including:
 - (i) Oil;
 - (ii) Hydraulic fluid; and
 - (iii) Other fluids required for normal operation of rotorcraft systems, except water intended for injection in the engines.
- (b) The condition of the rotorcraft at the time of determining empty weight must be one that is well defined and can be easily repeated, particularly with respect to the weights of fuel, oil, coolant, and installed equipment.

CS 29.31 Removable ballast

ED Decision 2003/16/RM

Removable ballast may be used in showing compliance with the flight requirements of this Subpart.

CS 29.33 Main rotor speed and pitch limits

ED Decision 2003/16/RM

- (a) *Main rotor speed limits.* A range of main rotor speeds must be established that:
- (1) With power on, provides adequate margin to accommodate the variations in rotor speed occurring in any appropriate manoeuvre, and is consistent with the kind of governor or synchroniser used; and
 - (2) With power off, allows each appropriate autorotative manoeuvre to be performed throughout the ranges of airspeed and weight for which certification is requested.
- (b) *Normal main rotor high pitch limit (power-on).* For rotorcraft, except helicopters required to have a main rotor low speed warning under sub-paragraph (e), it must be shown, with power on and without exceeding approved engine maximum limitations, that main rotor speeds substantially less than the minimum approved main rotor speed will not occur under any sustained flight condition. This must be met by:
- (1) Appropriate setting of the main rotor high pitch stop;
 - (2) Inherent rotorcraft characteristics that make unsafe low main rotor speeds unlikely; or
 - (3) Adequate means to warn the pilot of unsafe main rotor speeds.
- (c) *Normal main rotor low pitch limit (power-off).* It must be shown, with power off, that:
- (1) The normal main rotor low pitch limit provides sufficient rotor speed, in any autorotative condition, under the most critical combinations of weight and airspeed; and
 - (2) It is possible to prevent overspeeding of the rotor without exceptional piloting skill.
- (d) *Emergency high pitch.* If the main rotor high pitch stop is set to meet sub-paragraph (b)(1), and if that stop cannot be exceeded inadvertently, additional pitch may be made available for emergency use.
- (e) *Main rotor low speed warning for helicopters.* For each single engine helicopter, and each multi-engine helicopter that does not have an approved device that automatically increases power on the operating engines when one engine fails, there must be a main rotor low speed warning which meets the following requirements:
- (1) The warning must be furnished to the pilot in all flight conditions, including power-on and power-off flight, when the speed of a main rotor approaches a value that can jeopardise safe flight.
 - (2) The warning may be furnished either through the inherent aerodynamic qualities of the helicopter or by a device.
 - (3) The warning must be clear and distinct under all conditions, and must be clearly distinguishable from all other warnings. A visual device that requires the attention of the crew within the cockpit is not acceptable by itself.
 - (4) If a warning device is used, the device must automatically deactivate and reset when the low-speed condition is corrected. If the device has an audible warning, it must also be equipped with a means for the pilot to manually silence the audible warning before the low-speed condition is corrected.

PERFORMANCE

CS 29.45 General

ED Decision 2003/16/RM

- (a) The performance prescribed in this subpart must be determined:
 - (1) With normal piloting skill; and
 - (2) Without exceptionally favourable conditions.
- (b) Compliance with the performance requirements of this subpart must be shown:
 - (1) For still air at sea-level with a standard atmosphere; and
 - (2) For the approved range of atmospheric variables.
- (c) The available power must correspond to engine power, not exceeding the approved power, less:
 - (1) Installation losses; and
 - (2) The power absorbed by the accessories and services at the values for which certification is requested and approved.
- (d) For reciprocating engine-powered rotorcraft, the performance, as affected by engine power, must be based on a relative humidity of 80% in a standard atmosphere.
- (e) For turbine engine-powered rotorcraft, the performance, as affected by engine power, must be based on a relative humidity of:
 - (1) 80%, at and below standard temperature; and
 - (2) 34%, at and above standard temperature plus 28°C (50°F).

Between these two temperatures, the relative humidity must vary linearly.
- (f) For turbine-engine-powered rotorcraft, a means must be provided to permit the pilot to determine prior to take-off that each engine is capable of developing the power necessary to achieve the applicable rotorcraft performance prescribed in this subpart.

CS 29.49 Performance at minimum operating speed

ED Decision 2003/16/RM

- (a) For each Category A helicopter, the hovering performance must be determined over the ranges of weight, altitude and temperature for which take-off data are scheduled:
 - (1) With not more than take-off power;
 - (2) With the landing gear extended; and
 - (3) At a height consistent with the procedure used in establishing the take-off, climbout and rejected take-off paths.
- (b) For each Category B helicopter, the hovering performance must be determined over the ranges of weight, altitude and temperature for which certification is requested, with:
 - (1) Take-off power;
 - (2) The landing gear extended; and

- (3) The helicopter in ground effect at a height consistent with normal take-off procedures.
- (c) For each helicopter, the out-of ground- effect hovering performance must be determined over the ranges of weight, altitude and temperature for which certification is requested, with take-off power.
- (d) For rotorcraft other than helicopters, the steady rate of climb at the minimum operating speed must be determined over the ranges of weight, altitude and temperature for which certification is requested, with:
 - (1) Take-off power; and
 - (2) The landing gear extended.

CS 29.51 Take-off data: General

ED Decision 2003/16/RM

- (a) The take-off data required by [CS 29.53](#), [29.55](#), [29.59](#), [29.60](#), [29.61](#), [29.62](#), [29.63](#) and [29.67](#) must be determined:
 - (1) At each weight, altitude, and temperature selected by the applicant; and
 - (2) With the operating engines within approved operating limitations.
- (b) Take-off data must:
 - (1) Be determined on a smooth, dry, hard surface; and
 - (2) Be corrected to assume a level take-off surface.
- (c) No take-off made to determine the data required by this paragraph may require exceptional piloting skill or alertness, or exceptionally favourable conditions.

CS 29.53 Take-off: Category A

ED Decision 2003/16/RM

The take-off performance must be determined and scheduled so that, if one engine fails at any time after the start of take-off, the rotorcraft can:

- (a) Return to and stop safely on, the take-off area; or
- (b) Continue the take-off and climb-out, and attain a configuration and airspeed allowing compliance with [CS 29.67\(a\)\(2\)](#).

CS 29.55 Take-off Decision Point: Category A

ED Decision 2003/16/RM

- (a) The take-off decision point (TDP) is the first point from which a continued take-off capability is assured under [CS 29.59](#) and is the last point in the take-off path from which a rejected take-off is assured within the distance determined under [CS 29.62](#).
- (b) The TDP must be established in relation to the take-off path using no more than two parameters, such as airspeed and height, to designate the TDP.
- (c) Determination of the TDP must include the pilot recognition time interval following failure of the critical engine.

CS 29.59 Take-off Path: Category A

ED Decision 2003/16/RM

- (a) The take-off path extends from the point of commencement of the take-off procedure to a point at which the rotorcraft is 305 m (1000 ft) above the take-off surface and compliance with [CS 29.67\(a\)\(2\)](#) is shown. In addition:
- (1) The take-off path must remain clear of the height-velocity envelope established in accordance with [CS 29.87](#);
 - (2) The rotorcraft must be flown to the engine failure point at which point the critical engine must be made inoperative and remain inoperative for the rest of the take-off;
 - (3) After the critical engine is made inoperative, the rotorcraft must continue to the TDP, and then attain V_{TOSS} .
 - (4) Only primary controls may be used while attaining V_{TOSS} and while establishing a positive rate of climb. Secondary controls that are located on the primary controls may be used after a positive rate of climb and V_{TOSS} are established but in no case less than 3 seconds after the critical engine is made inoperative; and
 - (5) After attaining V_{TOSS} and a positive rate of climb, the landing gear may be retracted.
- (b) During the take-off path determination made in accordance with sub-paragraph (a) and after attaining V_{TOSS} and a positive rate of climb, the climb must be continued at a speed as close as practicable to, but not less than, V_{TOSS} until the rotorcraft is 61 m (200 ft) above the take-off surface. During this interval, the climb performance must meet or exceed that required by [CS 29.67\(a\)\(1\)](#).
- (c) During the continued take-off the rotorcraft shall not descend below 4.6 m (15 ft) above the take-off surface when the TDP is above 4.6 m (15 ft).
- (d) From 61 m (200 ft) above the take-off surface, the rotorcraft take-off path must be level or positive until a height 305 m (1000 ft) above the take-off surface is attained with not less than the rate of climb required by [CS 29.67\(a\)\(2\)](#). Any secondary or auxiliary control may be used after attaining 61 m (200 ft) above the take-off surface.
- (e) Take-off distance will be determined in accordance with [CS 29.61](#).

CS 29.60 Elevated heliport take-off path: Category A

ED Decision 2003/16/RM

- (a) The elevated heliport take-off path extends from the point of commencement of the take-off procedure to a point in the take-off path at which the rotorcraft is 305 m (1 000 ft) above the take-off surface and compliance with [CS 29.67\(a\)\(2\)](#) is shown. In addition:
- (1) The requirements of [CS 29.59\(a\)](#) must be met;
 - (2) While attaining V_{TOSS} and a positive rate of climb, the rotorcraft may descend below the level of the take-off surface if, in so doing and when clearing the elevated heliport edge, every part of the rotorcraft clears all obstacles by at least 4.6 m (15 ft);
 - (3) The vertical magnitude of any descent below the take-off surface must be determined; and
 - (4) After attaining V_{TOSS} and a positive rate of climb, the landing gear may be retracted.

- (b) The scheduled take-off weight must be such that the climb requirements of [CS 29.67\(a\)\(1\)](#) and [CS 29.67\(a\)\(2\)](#) will be met.
- (c) Take-off distance will be determined in accordance with [CS 29.61](#).

CS 29.61 Take-off distance: Category A

ED Decision 2003/16/RM

- (a) The normal take-off distance is the horizontal distance along the take-off path from the start of the take-off to the point at which the rotorcraft attains and remains at least 11 m (35 ft) above the take-off surface, attains and maintains a speed of at least V_{TOSS} ; and establishes a positive rate of climb, assuming the critical engine failure occurs at the engine failure point prior to the TDP.
- (b) For elevated heliports, the take-off distance is the horizontal distance along the take-off path from the start of the take-off to the point at which the rotorcraft attains and maintains a speed of at least V_{TOSS} and establishes a positive rate of climb, assuming the critical engine failure occurs at the engine failure point prior to the TDP.

CS 29.62 Rejected take-off: Category A

ED Decision 2003/16/RM

The rejected take-off distance and procedures for each condition where take-off is approved will be established with:

- (a) The take-off path requirements of [CS 29.59](#) and [29.60](#) being used up to the TDP where the critical engine failure is recognised, and the rotorcraft landed and brought to a stop on the take-off surface;
- (b) The remaining engines operating within approved limits;
- (c) The landing gear remaining extended throughout the entire rejected take-off; and
- (d) The use of only the primary controls until the rotorcraft is on the ground. Secondary controls located on the primary control may not be used until the rotorcraft is on the ground. Means other than wheel brakes may be used to stop the rotorcraft if the means are safe and reliable and consistent results can be expected under normal operating conditions.

CS 29.63 Take-off: Category B

ED Decision 2003/16/RM

The horizontal distance required to take-off and climb over a 15 m (50-foot) obstacle must be established with the most unfavourable centre of gravity. The take-off may be begun in any manner if –

- (a) The take-off surface is defined;
- (b) Adequate safeguards are maintained to ensure proper centre of gravity and control positions; and
- (c) A landing can be made safely at any point along the flight path if an engine fails.

CS 29.64 Climb: General

ED Decision 2003/16/RM

Compliance with the requirements of [CS 29.65](#) and [29.67](#) must be shown at each weight, altitude and temperature within the operational limits established for the rotorcraft and with the most unfavourable centre of gravity for each configuration. Cowl flaps, or other means of controlling the engine-cooling air supply, will be in the position that provides adequate cooling at the temperatures and altitudes for which certification is requested.

CS 29.65 Climb: All engines operating

ED Decision 2003/16/RM

- (a) The steady rate of climb must be determined:
 - (1) With maximum continuous power;
 - (2) With the landing gear retracted; and
 - (3) At V_Y for standard sea-level conditions and at speeds selected by the applicant for other conditions.
- (b) For each Category B rotorcraft except helicopters, the rate of climb determined under subparagraph (a) must provide a steady climb gradient of at least 1:6 under standard sea-level conditions.

CS 29.67 Climb: One Engine Inoperative (OEI)

ED Decision 2003/16/RM

- (a) For Category A rotorcraft, in the critical take-off configuration existing along the take-off path, the following apply:
 - (1) The steady rate of climb without ground effect, 61 m (200 ft) above the take-off surface, must be at least 30 m (100 ft) per minute, for each weight, altitude, and temperature for which take-off data are to be scheduled with:
 - (i) The critical engine inoperative and the remaining engines within approved operating limitations, except that for rotorcraft for which the use of 30-second/2-minute OEI power is requested, only the 2-minute OEI power may be used in showing compliance with this paragraph;
 - (ii) The landing gear extended; and
 - (iii) The take-off safety speed selected by the applicant.
 - (2) The steady rate of climb without ground effect, 305 m (1 000 ft) above the take-off surface, must be at least 46 m (150 ft) per minute, for each weight, altitude, and temperature for which take-off data are to be scheduled with:
 - (i) The critical engine inoperative and the remaining engines at maximum continuous power including continuous OEI power, if approved, or at 30-minute OEI power for rotorcraft for which certification for use of 30-minute OEI power is requested;
 - (ii) The landing gear retracted; and
 - (iii) The speed selected by the applicant.

- (3) The steady rate of climb (or descent), in feet per minute, at each altitude and temperature at which the rotorcraft is expected to operate and at each weight within the range of weights for which certification is requested, must be determined with:
 - (i) The critical engine inoperative and the remaining engines at maximum continuous power including continuous OEI power, if approved, and at 30-minute OEI power for rotorcraft for which certification for the use of 30-minute OEI power is requested;
 - (ii) The landing gear retracted; and
 - (iii) The speed selected by the applicant.
- (b) For multi-engine Category B rotorcraft meeting the Category A engine isolation requirements, the steady rate of climb (or descent) must be determined at the speed for best rate of climb (or minimum rate of descent) at each altitude, temperature, and weight at which the rotorcraft is expected to operate, with the critical engine inoperative and the remaining engines at maximum continuous power including continuous OEI power, if approved, and at 30-minute OEI power for rotorcraft for which certification for the use of 30-minute OEI power is requested.

CS 29.71 Helicopter angle of glide: Category B

ED Decision 2003/16/RM

For each Category B helicopter, except multi-engine helicopters meeting the requirements of [CS 29.67\(b\)](#) and the powerplant installation requirements of Category A, the steady angle of glide must be determined in autorotation:

- (a) At the forward speed for minimum rate of descent as selected by the applicant;
- (b) At the forward speed for best glide angle;
- (c) At maximum weight; and
- (d) At the rotor speed or speeds selected by the applicant.

CS 29.75 Landing: General

ED Decision 2003/16/RM

- (a) For each rotorcraft:
 - (1) The corrected landing data must be determined for a smooth, dry, hard and level surface;
 - (2) The approach and landing must not require exceptional piloting skill or exceptionally favourable conditions; and,
 - (3) The landing must be made without excessive vertical acceleration or tendency to bounce, nose over, ground loop, porpoise, or water loop.
- (b) The landing data required by [CS 29.77](#), [29.79](#), [29.81](#), [29.83](#) and [29.85](#) must be determined:
 - (1) At each weight, altitude and temperature for which landing data are approved;
 - (2) With each operating engine within approved operating limitations; and
 - (3) With the most unfavourable centre of gravity.

CS 29.77 Landing Decision Point: Category A

ED Decision 2003/16/RM

- (a) The landing decision point (LDP) is the last point in the approach and landing path from which a balked landing can be accomplished in accordance with [CS 29.85](#).
- (b) Determination of the LDP must include the pilot recognition time interval following failure of the critical engine.

CS 29.79 Landing: Category A

ED Decision 2003/16/RM

- (a) For Category A rotorcraft:
 - (1) The landing performance must be determined and scheduled so that if the critical engine fails at any point in the approach path, the rotorcraft can either land and stop safely or climb out and attain a rotorcraft configuration and speed allowing compliance with the climb requirement of [CS 29.67\(a\)\(2\)](#);
 - (2) The approach and landing paths must be established with the critical engine inoperative so that the transition between each stage can be made smoothly and safely;
 - (3) The approach and landing speeds must be selected for the rotorcraft and must be appropriate to the type of rotorcraft; and
 - (4) The approach and landing path must be established to avoid the critical areas of the height-velocity envelope determined in accordance with [CS 29.87](#).
- (b) It must be possible to make a safe landing on a prepared landing surface after complete power failure occurring during normal cruise.

CS 29.81 Landing distance (ground level sites): Category A

ED Decision 2003/16/RM

The horizontal distance required to land and come to a complete stop (or to a speed of approximately 5.6 km/h (3 knots) for water landings) from a point 15 m (50 ft) above the landing surface must be determined from the approach and landing paths established in accordance with [CS 29.79](#).

CS 29.83 Landing: Category B

ED Decision 2003/16/RM

- (a) For each Category B rotorcraft, the horizontal distance required to land and come to a complete stop (or to a speed of approximately 5.6 km/h (3 knots) for water landings) from a point 15 m (50 ft) above the landing surface must be determined with:
 - (1) Speeds appropriate to the type of rotorcraft and chosen by the applicant to avoid the critical areas of the height-velocity envelope established under [CS 29.87](#); and
 - (2) The approach and landing made with power on and within approved limits.
- (b) Each multi-engine Category B rotorcraft that meets the powerplant installation requirements for Category A must meet the requirements of:
 - (1) [CS 29.79](#) and [29.81](#); or
 - (2) Sub-paragraph (a).

- (c) It must be possible to make a safe landing on a prepared landing surface if complete power failure occurs during normal cruise.

CS 29.85 Balked landing: Category A

ED Decision 2003/16/RM

For Category A rotorcraft, the balked landing path with the critical engine inoperative must be established so that:

- (a) The transition from each stage of the manoeuvre to the next stage can be made smoothly and safely;
- (b) From the LDP on the approach path selected by the applicant, a safe climbout can be made at speeds allowing compliance with the climb requirements of [CS 29.67\(a\)\(1\) and \(2\)](#); and
- (c) The rotorcraft does not descend below 4.6 m (15 ft) above the landing surface. For elevated heliport operations, descent may be below the level of the landing surface provided the deck edge clearance of [CS 29.60](#) is maintained and the descent (loss of height) below the landing surface is determined.

CS 29.87 Height-velocity envelope

ED Decision 2003/16/RM

- (a) If there is any combination of height and forward velocity (including hover) under which a safe landing cannot be made after failure of the critical engine and with the remaining engines (where applicable) operating within approved limits, a height-velocity envelope must be established for:
 - (1) All combinations of pressure altitude and ambient temperature for which take-off and landing are approved; and
 - (2) Weight, from the maximum weight (at sea-level) to the highest weight approved for take-off and landing at each altitude. For helicopters, this weight need not exceed the highest weight allowing hovering out of ground effect at each altitude.
- (b) For single engine or multi-engine rotorcraft that do not meet the Category A engine isolation requirements, the height-velocity envelope for complete power failure must be established.

FLIGHT CHARACTERISTICS

CS 29.141 General

ED Decision 2003/16/RM

The rotorcraft must:

- (a) Except as specifically required in the applicable paragraph, meet the flight characteristics requirements of this Subpart:
 - (1) At the approved operating altitudes and temperatures;
 - (2) Under any critical loading condition within the range of weights and centres of gravity for which certification is requested; and
 - (3) For power-on operations, under any condition of speed, power, and rotor rpm for which certification is requested; and
 - (4) For power-off operations, under any condition of speed, and rotor rpm for which certification is requested that is attainable with the controls rigged in accordance with the approved rigging instructions and tolerances;
- (b) Be able to maintain any required flight condition and make a smooth transition from any flight condition to any other flight condition without exceptional piloting skill, alertness, or strength, and without danger of exceeding the limit load factor under any operating condition probable for the type, including:
 - (1) Sudden failure of one engine, for multi-engine rotorcraft meeting Category A engine isolation requirements;
 - (2) Sudden, complete power failure, for other rotorcraft; and
 - (3) Sudden, complete control system failures specified in [CS 29.695](#); and
- (c) Have any additional characteristics required for night or instrument operation, if certification for those kinds of operation is requested. Requirements for helicopter instrument flight are contained in [appendix B](#).

Appendix B – Airworthiness Criteria for Helicopter Instrument Flight

ED Decision 2007/014/R

- I. *General.* A large helicopter may not be type certificated for operation under the instrument flight rules (IFR) unless it meets the design and installation requirements contained in this appendix.
- II. *Definitions*
 - (a) V_{YI} means instrument climb speed, utilised instead of V_Y for compliance with the climb requirements for instrument flight.
 - (b) V_{NEI} means instrument flight never-exceed speed, utilised instead of V_{NE} for compliance with maximum limit speed requirements for instrument flight.
 - (c) V_{MINI} means instrument flight minimum speed, utilised in complying with minimum limit speed requirements for instrument flight.
- III. *Trim.* It must be possible to trim the cyclic, collective, and directional control forces to zero at all approved IFR airspeeds, power settings, and configurations appropriate to the type.

- (a) *General.* The helicopter must possess positive static longitudinal control force stability at critical combinations of weight and centre of gravity at the conditions specified in sub-paragraphs IV (b) to (f) of this appendix. The stick force must vary with speed so that any substantial speed change results in a stick force clearly perceptible to the pilot. The airspeed must return to within 10% of the trim speed when the control force is slowly released for each trim condition specified in sub-paragraphs IV (b) to (f) of this appendix.
- (b) *Climb.* Stability must be shown in climb throughout the speed range 37 km/h (20 knots) either side of trim with:
 - (1) The helicopter trimmed at V_{YI} ;
 - (2) Landing gear retracted (if retractable); and
 - (3) Power required for limit climb rate (at least 5.1 m/s (1000 fpm)) at V_{YI} or maximum continuous power, whichever is less.
- (c) *Cruise.* Stability must be shown throughout the speed range from 0.7 to 1.1 V_H or V_{NEI} , whichever is lower, not to exceed ± 37 km/h (± 20 knots) from trim with:
 - (1) The helicopter trimmed and power adjusted for level flight at 0.9 V_H or 0.9 V_{NEI} , whichever is lower; and
 - (2) Landing gear retracted (if retractable).
- (d) *Slow cruise.* Stability must be shown throughout the speed range from 0.9 V_{MINI} to 1.3 V_{MINI} or 37 km/h (20 knots) above trim speed, whichever is greater, with:
 - (1) The helicopter trimmed and power adjusted for level flight at 1.1 V_{MINI} ; and
 - (2) Landing gear retracted (if retractable).
- (e) *Descent.* Stability must be shown throughout the speed range 37 km/h (20 knots) either side of trim with:
 - (1) The helicopter trimmed at 0.8 V_H or 0.8 V_{NEI} (or 0.8 V_{LE} for the landing gear extended case), whichever is lower;
 - (2) Power required for 5.1 m/s (1000 fpm) descent at trim speed; and
 - (3) Landing gear extended and retracted, if applicable.
- (f) *Approach.* Stability must be shown throughout the speed range from 0.7 times the minimum recommended approach speed to 37 km/h (20 knots) above the maximum recommended approach speed with:
 - (1) The helicopter trimmed at the recommended approach speed or speeds;
 - (2) Landing gear extended and retracted, if applicable; and
 - (3) Power required to maintain a 3° glide path and power required to maintain the steepest approach gradient for which approval is requested.

V. *Static lateral-directional stability*

- (a) Static directional stability must be positive throughout the approved ranges of airspeed, power, and vertical speed. In straight and steady sideslips up to $\pm 10^\circ$ from trim, directional control position must increase without discontinuity with the angle of sideslip, except for a small range of sideslip angles around trim. At greater angles up to the maximum sideslip angle appropriate to the type, increased directional control position

must produce increased angle of sideslip. It must be possible to maintain balanced flight without exceptional pilot skill or alertness.

- (b) During sideslips up to $\pm 10^\circ$ from trim throughout the approved ranges of airspeed, power, and vertical speed there must be no negative dihedral stability perceptible to the pilot through lateral control motion or force. Longitudinal cyclic movement with sideslip must not be excessive.

VI. *Dynamic stability*

- (a) Any oscillation having a period of less than 5 seconds must damp to $\frac{1}{2}$ amplitude in not more than one cycle.
- (b) Any oscillation having a period of 5 seconds or more but less than 10 seconds must damp to $\frac{1}{2}$ amplitude in not more than two cycles.
- (c) Any oscillation having a period of 10 seconds or more but less than 20 seconds must be damped.
- (d) Any oscillation having a period of 20 seconds or more may not achieve double amplitude in less than 20 seconds.
- (e) Any aperiodic response may not achieve double amplitude in less than 9 seconds.

VII. *Stability augmentation system (SAS)*

- (a) If a SAS is used, the reliability of the SAS must be related to the effects of its failure. Any SAS failure condition that would prevent continued safe flight and landing must be extremely improbable. It must be shown that, for any failure condition of the SAS that is not shown to be extremely improbable:
 - (1) The helicopter is safely controllable when the failure or malfunction occurs at any speed or altitude within the approved IFR operating limitations; and
 - (2) The overall flight characteristics of the helicopter allow for prolonged instrument flight without undue pilot effort. Additional unrelated probable failures affecting the control system must be considered. In addition:
 - (i) The controllability and manoeuvrability requirements in Subpart B of CS-29 must be met throughout a practical flight envelope;
 - (ii) The flight control, trim, and dynamic stability characteristics must not be impaired below a level needed to allow continued safe flight and landing;
 - (iii) For Category A helicopters, the dynamic stability requirements of Subpart B of CS-29 must also be met throughout a practical flight envelope; and
 - (iv) The static longitudinal and static directional stability requirements of Subpart B of CS-29 must be met throughout a practical flight envelope.
- (b) The SAS must be designed so that it cannot create a hazardous deviation in flight path or produce hazardous loads on the helicopter during normal operation or in the event of malfunction or failure, assuming corrective action begins within an appropriate period of time. Where multiple systems are installed, subsequent malfunction conditions must be considered in sequence unless their occurrence is shown to be improbable.

VIII. *Equipment, systems, and installation.* The basic equipment and installation must comply with Subpart F of CS-29 with the following exceptions and additions:

- (a) *Flight and navigation instruments*

- (1) A magnetic gyro-stabilised direction indicator instead of the gyroscopic direction indicator required by [CS 29.1303\(h\)](#); and
 - (2) A standby attitude indicator which meets the requirements of [CS 29.1303\(g\)\(1\) to \(7\)](#), instead of a rate-of-turn indicator required by [CS 29.1303\(g\)](#). If standby batteries are provided, they may be charged from the aircraft electrical system if adequate isolation is incorporated. The system must be designed so that the standby batteries may not be used for engine starting.
- (b) *Miscellaneous requirements*
- (1) Instrument systems and other systems essential for IFR flight that could be adversely affected by icing must be provided with adequate ice protection whether or not the rotorcraft is certificated for operation in icing conditions.
 - (2) There must be means in the generating system to automatically de-energise and disconnect from the main bus any power source developing hazardous overvoltage.
 - (3) Each required flight instrument using a power supply (electric, vacuum etc.) must have a visual means integral with the instrument to indicate the adequacy of the power being supplied.
 - (4) When multiple systems performing like functions are required, each system must be grouped, routed, and spaced so that physical separation between systems is provided to ensure that a single malfunction will not adversely affect more than one system.
 - (5) For systems that operate the required flight instruments at each pilot's station:
 - (i) Only the required flight instruments for the first pilot may be connected to that operating system;
 - (ii) Additional instruments, systems, or equipment may not be connected to an operating system for a second pilot unless provisions are made to ensure the continued normal functioning of the required instruments in the event of any malfunction of the additional instruments, systems, or equipment which is not shown to be extremely improbable;
 - (iii) The equipment, systems, and installations must be designed so that one display of the information essential to the safety of flight which is provided by the instruments will remain available to a pilot, without additional crew member action, after any single failure or combination of failures that is not shown to be extremely improbable; and
 - (iv) For single-pilot configurations, instruments which require a static source must be provided with a means of selecting an alternate source and that source must be calibrated.
 - (6) In determining compliance with the requirements of [CS 29.1351\(d\)\(2\)](#), the supply of electrical power to all systems necessary for flight under IFR must be included in the evaluation.
- (c) *Thunderstorm lights*. In addition to the instrument lights required by [CS 29.1381\(a\)](#), thunderstorm lights which provide high intensity white flood lighting to the basic flight instruments must be provided. The thunderstorm lights must be installed to meet the requirements of [CS 29.1381\(b\)](#).

- IX. *Rotorcraft flight manual.* A rotorcraft flight manual or rotorcraft flight manual IFR Supplement must be provided and must contain –
- (a) *Limitations.* The approved IFR flight envelope, the IFR flightcrew composition, the revised kinds of operation, and the steepest IFR precision approach gradient for which the helicopter is approved;
 - (b) *Procedures.* Required information for proper operation of IFR systems and the recommended procedures in the event of stability augmentation or electrical system failures; and
 - (c) *Performance.* If V_{YI} differs from V_Y , climb performance at V_{YI} and with maximum continuous power throughout the ranges of weight, altitude, and temperature for which approval is requested.

[Amdt. No.: 29/1]

CS 29.143 Controllability and manoeuvrability

ED Decision 2008/010/R

- (a) The rotorcraft must be safely controllable and manoeuvrable:
 - (1) During steady flight; and
 - (2) During any manoeuvre appropriate to the type, including:
 - (i) Take-off,
 - (ii) Climb;
 - (iii) Level flight;
 - (iv) Turning flight;
 - (v) Autorotation; and
 - (vi) Landing (power on and power off).
- (b) The margin of cyclic control must allow satisfactory roll and pitch control a V_{NE} with:
 - (1) Critical weight;
 - (2) Critical centre of gravity;
 - (3) Critical rotor rpm; and
 - (4) Power off (except for helicopters demonstrating compliance with sub-paragraph (f) and power on).
- (c) Wind velocities from zero to at least 31 km/h (17 knots), from all azimuths, must be established in which the rotorcraft can be operated without loss of control on or near the ground in any manoeuvre appropriate to the type (such as crosswind take-offs, sideward flight, and rearward flight), with:
 - (1) Critical weight;
 - (2) Critical centre of gravity;
 - (3) Critical rotor rpm; and
 - (4) Altitude from standard sea-level conditions to the maximum take-off and landing altitude capability of the rotorcraft.

- (d) Wind velocities from zero to at least 31 km/h (17 knots), from all azimuths, must be established in which the rotorcraft can be operated without loss of control out-of-ground effect, with:
 - (1) Weight selected by the applicant;
 - (2) Critical center of gravity;
 - (3) Rotor rpm selected by the applicant; and
 - (4) Altitude, from standard sea-level conditions to the maximum take-off and landing altitude capability of the rotorcraft.
- (e) The rotorcraft, after failure of one engine, in the case of multi-engine rotorcraft that meet Category A engine isolation requirements, or complete power failure in the case of other rotorcraft, must be controllable over the range of speeds and altitudes for which certification is requested when such power failure occurs with maximum continuous power and critical weight. No corrective action time delay for any condition following power failure may be less than:
 - (1) For the cruise condition, one second, or normal pilot reaction time (whichever is greater); and
 - (2) For any other condition, normal pilot reaction time.
- (f) For helicopters for which a V_{NE} (power-off) is established under [CS 29.1505\(c\)](#), compliance must be demonstrated with the following requirements with critical weight, critical centre of gravity, and critical rotor rpm:
 - (1) The helicopter must be safely slowed to V_{NE} (power-off), without exceptional pilot skill after the last operating engine is made inoperative at power-on V_{NE} .
 - (2) At a speed of $1.1 V_{NE}$ (power-off), the margin of cyclic control must allow satisfactory roll and pitch control with power off.

[Amdt. No.: 29/1]

[Amdt. No. 29/2]

CS 29.151 Flight controls

ED Decision 2003/16/RM

- (a) Longitudinal, lateral, directional, and collective controls may not exhibit excessive breakout force, friction, or preload.
- (b) Control system forces and free play may not inhibit a smooth, direct rotorcraft response to control system input.

CS 29.161 Trim control

ED Decision 2003/16/RM

The trim control:

- (a) Must trim any steady longitudinal, lateral, and collective control forces to zero in level flight at any appropriate speed; and
- (b) May not introduce any undesirable discontinuities in control force gradients.

CS 29.171 Stability: general

ED Decision 2003/16/RM

The rotorcraft must be able to be flown, without undue pilot fatigue or strain, in any normal manoeuvre for a period of time as long as that expected in normal operation. At least three landings and take-offs must be made during this demonstration.

CS 29.173 Static longitudinal stability

ED Decision 2007/014/R

- (a) The longitudinal control must be designed so that a rearward movement of the control is necessary to obtain an airspeed less than the trim speed, and a forward movement of the control is necessary to obtain an airspeed more than the trim speed.
- (b) Throughout the full range of altitude for which certification is requested, with the throttle and collective pitch held constant during the manoeuvres specified in [CS 29.175\(a\) through \(d\)](#), the slope of the control position versus airspeed curve must be positive. However, in limited flight conditions or modes of operation determined by the Agency to be acceptable, the slope of the control position versus airspeed curve may be neutral or negative if the rotorcraft possesses flight characteristics that allow the pilot to maintain airspeed within ± 9 km/h (± 5 knots) of the desired trim airspeed without exceptional piloting skill or alertness.

[Amdt. No.: 29/1]

CS 29.175 Demonstration of static longitudinal stability

ED Decision 2007/014/R

- (a) *Climb.* Static longitudinal stability must be shown in the climb condition at speeds from $V_Y - 19$ km/h (10 knots) to $V_Y + 19$ km/h (10 knots), with:
 - (1) Critical weight;
 - (2) Critical centre of gravity;
 - (3) Maximum continuous power;
 - (4) The landing gear retracted; and
 - (5) The rotorcraft trimmed at V_Y .
- (b) *Cruise.* Static longitudinal stability must be shown in the cruise condition at speeds from $0.8 V_{NE} - 19$ km/h (10 knots) to $0.8 V_{NE} + 19$ km/h (10 knots) or, if V_H is less than $0.8 V_{NE}$, from $0.8 V_{NE} - 19$ km/h (10 knots) to $0.8 V_{NE} + 19$ km/h (10 knots), with:
 - (1) Critical weight;
 - (2) Critical centre of gravity;
 - (3) Power for level flight at $0.8 V_{NE}$ or V_H , whichever is less;
 - (4) The landing gear retracted; and
 - (5) The rotorcraft trimmed at $0.8 V_{NE}$ or V_H , whichever is less.
- (c) V_{NE} . Static longitudinal stability must be shown at speeds from $V_{NE} - 37$ km/h (20 knots) to V_{NE} with:
 - (1) Critical weight;
 - (2) Critical center of gravity;

- (3) Power required for level flight at $V_{NE} - 19$ km/h (10 knots) or maximum continuous power, whichever is less;
 - (4) The landing gear retracted; and
 - (5) The rotorcraft trimmed at $V_{NE} - 19$ km/h (10 knots).
- (d) *Autorotation*. Static longitudinal stability must be shown in autorotation at:
- (1) Airspeeds from the minimum rate of descent airspeed $- 19$ km/h (10 knots) to the minimum rate of descent airspeed $+ 19$ km/h (10 knots), with:
 - (i) Critical weight;
 - (ii) Critical center of gravity;
 - (iii) The landing gear extended; and
 - (iv) The rotorcraft trimmed at the minimum rate of descent airspeed.
 - (2) Airspeeds from the best angle-of-glide airspeed $- 19$ km/h (10 knots) to the best angle-of-glide airspeed $+ 19$ km/h (10 knots), with:
 - (i) Critical weight;
 - (ii) Critical center of gravity;
 - (iii) The landing gear retracted; and
 - (iv) The rotorcraft trimmed at the best angle-of-glide airspeed.

[Amdt. No.: 29/1]

CS 29.177 Static directional stability

ED Decision 2007/014/R

- (a) The directional controls must operate in such a manner that the sense and direction of motion of the rotorcraft following control displacement are in the direction of the pedal motion with throttle and collective controls held constant at the trim conditions specified in [CS 29.175\(a\), \(b\), \(c\) and \(d\)](#). Sideslip angles must increase with steadily increasing directional control deflection for sideslip angles up to the lesser of:
 - (1) ± 25 degrees from trim at a speed of 28 km/h (15 knots) less than the speed for minimum rate of descent varying linearly to ± 10 degrees from trim at V_{NE} ;
 - (2) The steady state sideslip angles established by [CS 29.351](#);
 - (3) A sideslip angle selected by the applicant which corresponds to a sideforce of at least 0.1g; or,
 - (4) The sideslip angle attained by maximum directional control input.
- (b) Sufficient cues must accompany the sideslip to alert the pilot when approaching sideslip limits.
- (c) During the manoeuvre specified in sub-paragraph (a) of this paragraph, the sideslip angle versus directional control position curve may have a negative slope within a small range of angles around trim, provided the desired heading can be maintained without exceptional piloting skill or alertness.

[Amdt. No.: 29/1]

CS 29.181 Dynamic stability: Category A rotorcraft

ED Decision 2003/16/RM

Any short period oscillation occurring at any speed from V_Y to V_{NE} must be positively damped with the primary flight controls free and in a fixed position.

GROUND AND WATER HANDLING CHARACTERISTICS

CS 29.231 General

ED Decision 2003/16/RM

The rotorcraft must have satisfactory ground and water handling characteristics, including freedom from uncontrollable tendencies in any condition expected in operation.

CS 29.235 Taxying condition

ED Decision 2003/16/RM

The rotorcraft must be designed to withstand the loads that would occur when the rotorcraft is taxied over the roughest ground that may reasonably be expected in normal operation.

CS 29.239 Spray characteristics

ED Decision 2003/16/RM

If certification for water operation is requested, no spray characteristics during taxying, take-off, or landing may obscure the vision of the pilot or damage the rotors, propellers, or other parts of the rotorcraft.

CS 29.241 Ground resonance

ED Decision 2003/16/RM

The rotorcraft may have no dangerous tendency to oscillate on the ground with the rotor turning.

MISCELLANEOUS FLIGHT REQUIREMENTS

CS 29.251 Vibration

ED Decision 2003/16/RM

Each part of the rotorcraft must be free from excessive vibration under each appropriate speed and power condition.

SUBPART C — STRENGTH REQUIREMENTS

GENERAL

CS 29.301 Loads

ED Decision 2003/16/RM

- (a) Strength requirements are specified in terms of limit loads (the maximum loads to be expected in service) and ultimate loads (limit loads multiplied by prescribed factors of safety). Unless otherwise provided, prescribed loads are limit loads.
- (b) Unless otherwise provided, the specified air, ground, and water loads must be placed in equilibrium with inertia forces, considering each item of mass in the rotorcraft. These loads must be distributed to closely approximate or conservatively represent actual conditions.
- (c) If deflections under load would significantly change the distribution of external or internal loads, this redistribution must be taken into account.

CS 29.303 Factor of safety

ED Decision 2003/16/RM

Unless otherwise provided, a factor of safety of 1.5 must be used. This factor applies to external and inertia loads unless its application to the resulting internal stresses is more conservative.

CS 29.305 Strength and deformation

ED Decision 2003/16/RM

- (a) The structure must be able to support limit loads without detrimental or permanent deformation. At any load up to limit loads, the deformation may not interfere with safe operation.
- (b) The structure must be able to support ultimate loads without failure. This must be shown by:
 - (1) Applying ultimate loads to the structure in a static test for at least 3 seconds; or
 - (2) Dynamic tests simulating actual load application.

CS 29.307 Proof of structure

ED Decision 2003/16/RM

- (a) Compliance with the strength and deformation requirements of this Subpart must be shown for each critical loading condition accounting for the environment to which the structure will be exposed in operation. Structural analysis (static or fatigue) may be used only if the structure conforms to those for which experience has shown this method to be reliable. In other cases, substantiating load tests must be made.
- (b) Proof of compliance with the strength requirements of this Subpart must include:
 - (1) Dynamic and endurance tests of rotors, rotor drives, and rotor controls;
 - (2) Limit load tests of the control system, including control surfaces;
 - (3) Operation tests of the control system;
 - (4) Flight stress measurement tests;

- (5) Landing gear drop tests; and
- (6) Any additional tests required for new or unusual design features.

CS 29.309 Design limitations

ED Decision 2003/16/RM

The following values and limitations must be established to show compliance with the structural requirements of this Subpart:

- (a) The design maximum and design minimum weights.
- (b) The main rotor rpm ranges, power on and power off.
- (c) The maximum forward speeds for each main rotor rpm within the ranges determined under sub-paragraph (b).
- (d) The maximum rearward and sideward flight speeds.
- (e) The centre of gravity limits corresponding to the limitations determined under sub-paragraphs (b), (c) and (d).
- (f) The rotational speed ratios between each powerplant and each connected rotating component.
- (g) The positive and negative limit manoeuvring load factors.

FLIGHT LOADS

CS 29.321 General

ED Decision 2003/16/RM

- (a) The flight load factor must be assumed to act normal to the longitudinal axis of the rotorcraft, and to be equal in magnitude and opposite in direction to the rotorcraft inertia load factor at the centre of gravity.
- (b) Compliance with the flight load requirements of this Subpart must be shown:
 - (1) At each weight from the design minimum weight to the design maximum weight; and
 - (2) With any practical distribution of disposable load within the operating limitations in the rotorcraft flight manual.

CS 29.337 Limit manoeuvring load factor

ED Decision 2003/16/RM

The rotorcraft must be designed for –

- (a) A limit manoeuvring load factor ranging from a positive limit of 3.5 to a negative limit of -1.0; or
- (b) Any positive limit manoeuvring load factor not less than 2.0 and any negative limit manoeuvring load factor of not less than -0.5 for which:
 - (1) The probability of being exceeded is shown by analysis and flight tests to be extremely remote; and
 - (2) The selected values are appropriate to each weight condition between the design maximum and design minimum weights.

CS 29.339 Resultant limit manoeuvring loads

ED Decision 2003/16/RM

The loads resulting from the application of limit manoeuvring load factors are assumed to act at the centre of each rotor hub and at each auxiliary lifting surface, and to act in directions and with distributions of load among the rotors and auxiliary lifting surfaces, so as to represent each critical manoeuvring condition, including power-on and power-off flight with the maximum design rotor tip speed ratio. The rotor tip speed ratio is the ratio of the rotorcraft flight velocity component in the plane of the rotor disc to the rotational tip speed of the rotor blades and is expressed as follows:

$$\mu = \frac{V \cos \alpha}{\Omega R}$$

where:

- V = The airspeed along the flight path (m/s (fps));
- α = The angle between the projection, in the plane of symmetry, of the axis of no feathering and a line perpendicular to the flight path (radians, positive when axis is pointing aft);
- Ω = The angular velocity of rotor (radians per second); and
- R = The rotor radius (m (ft)).

CS 29.341 Gust loads

ED Decision 2003/16/RM

Each rotorcraft must be designed to withstand, at each critical airspeed including hovering, the loads resulting from vertical and horizontal gusts of 9.1 metres per second (30 ft/s).

CS 29.351 Yawing conditions

ED Decision 2003/16/RM

- (a) Each rotorcraft must be designed for the loads resulting from the manoeuvres specified in sub-paragraphs (b) and (c), with:
 - (1) Unbalanced aerodynamic moments about the centre of gravity which the aircraft reacts to in a rational or conservative manner considering the principal masses furnishing the reacting inertia forces; and
 - (2) Maximum main rotor speed.
- (b) To produce the load required in sub-paragraph (a), in unaccelerated flight with zero yaw, at forward speeds from zero up to $0.6 V_{NE}$.
 - (1) Displace the cockpit directional control suddenly to the maximum deflection limited by the control stops or by the maximum pilot force specified in [CS 29.397\(a\)](#);
 - (2) Attain a resulting sideslip angle or 90° , whichever is less; and
 - (3) Return the directional control suddenly to neutral.
- (c) To produce the load required in sub-paragraph (a), in unaccelerated flight with zero yaw, at forward speeds from $0.6 V_{NE}$ up to V_{NE} or V_H , whichever is less:
 - (1) Displace the cockpit directional control suddenly to the maximum deflection limited by the control stops or by the maximum pilot force specified in [CS 29.397\(a\)](#);
 - (2) Attain a resulting sideslip angle or 15° , whichever is less, at the lesser speed of V_{NE} or V_H ;
 - (3) Vary the sideslip angles of sub-paragraphs (b)(2) and (c)(2) directly with speed; and
 - (4) Return the directional control suddenly to neutral.

AMC 29.351 Yaw manoeuvre conditions

ED Decision 2008/010/R

1. Introduction

This AMC provides further guidance and acceptable means of compliance to supplement FAA AC 29-2C Change 2 (AC 29.351b. § 29.351 (Amendment 29-40) YAWING CONDITIONS), to meet the Agency's interpretation of [CS 29.351](#). As such it should be used in conjunction with the FAA AC but take precedence over it, where stipulated, in the showing of compliance.

Specifically, this AMC addresses two areas where the FAA AC has been deemed by the Agency as being unclear or at variance to the Agency's interpretation. These areas are as follows:

a. Aerodynamic Loads

The certification specification [CS 29.351](#) provides a minimum safety standard for the design of rotorcraft structural components that are subjected in flight to critical loads combinations of anti-torque system thrust (e.g. tail rotor), inertia and aerodynamics. A typical example of these structural components is the tailboom.

However, compliance with this standard according to FAA AC 29-2c Change 2 may not necessarily be adequate for the design of rotorcraft structural components that are principally subjected in flight to significant aerodynamic loads (e.g. vertical empennage, fins, cowlings and doors).

For these components and their supporting structure, suitable design criteria should be developed by the Applicant and agreed with the Agency.

In lieu of acceptable design criteria developed by the applicant, a suitable combination of sideslip angle and airspeed for the design of rotorcraft components subjected to aerodynamic loads may be obtained from a simulation of the yaw manoeuvre of [CS 29.351](#), starting from the initial directional control input specified in [CS 29.351\(b\)\(1\) and \(c\)\(1\)](#), until the rotorcraft reaches the maximum overswing sideslip angle resulting from its motion around the yaw axis.

b. Interaction of System and Structure

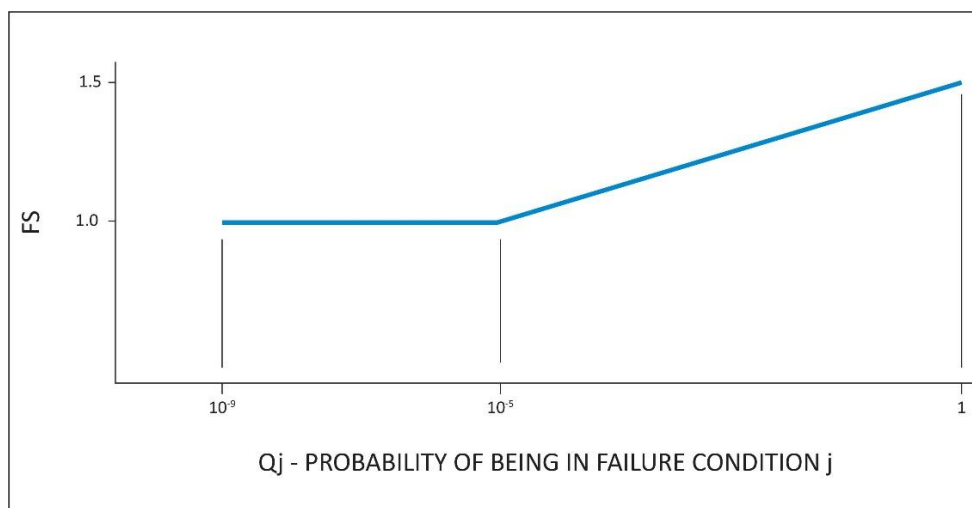
Maximum displacement of the directional control, except as limited by pilot effort ([CS 29.397\(a\)](#)), is required for the conditions cited in the certification specification. In the load evaluation credit may be taken for consideration of the effects of control system limiting devices.

However, the probability of failure or malfunction of these system(s) should also be considered and if it is shown not to be extremely improbable then further load conditions with the system in the failed state should be evaluated. This evaluation may include Flight Manual Limitations, if failure of the system is reliably indicated to the crew.

A yaw limiting device is a typical example of a system whose failed condition should be investigated in the assessment of the loads requested by [CS 29.351](#).

An acceptable methodology to investigate the effects of all system failures not shown to be extremely improbable on the loading conditions of [CS 29.351](#) is as follows:

- (i) With the system in the failed state and considering any appropriate reconfiguration and flight limitations, it should be shown that the rotorcraft structure can withstand without failure the loading conditions of [CS 29.351](#), when the manoeuvre is performed in accordance with the provisions of the this AMC.
- (ii) The factor of safety to apply to the above specified loading conditions to comply with [CS 29.305](#) is defined in the figure below.



$$Q_j = (T_j)(P_j)$$

where:

T_j = Average flight time spent with a failed limiting system j (in hours)

P_j = Probability of occurrence of failure of control limiting system j (per hour)

Note: If P_j is greater than 1×10^{-3} per flight hour then a 1.5 factor of safety should be applied to all limit load conditions evaluated for the system failure under consideration.

[Amdt. No. 29/2]

CS 29.361 Engine torque

ED Decision 2003/16/RM

The limit engine torque may not be less than the following:

- (a) For turbine engines, the highest of:
 - (1) The mean torque for maximum continuous power multiplied by 1.25;
 - (2) The torque required by [CS 29.923](#);
 - (3) The torque required by [CS 29.927](#); or
 - (4) The torque imposed by sudden engine stoppage due to malfunction or structural failure (such as compressor jamming).
- (b) For reciprocating engines, the mean torque for maximum continuous power multiplied by:
 - (1) 1.33, for engines with five or more cylinders; and
 - (2) Two, three, and four, for engines with four, three, and two cylinders, respectively.

CONTROL SURFACE AND SYSTEM LOADS

CS 29.391 General

ED Decision 2003/16/RM

Each auxiliary rotor, each fixed or movable stabilising or control surface, and each system operating any flight control must meet the requirements of [CS 29.395](#) to [29.427](#).

CS 29.395 Control system

ED Decision 2003/16/RM

- (a) The reaction to the loads prescribed in [CS 29.397](#) must be provided by:
 - (1) The control stops only;
 - (2) The control locks only;
 - (3) The irreversible mechanism only (with the mechanism locked and with the control surface in the critical positions for the effective parts of the system within its limit of motion);
 - (4) The attachment of the control system to the rotor blade pitch control horn only (with the control in the critical positions for the affected parts of the system within the limits of its motion); and
 - (5) The attachment of the control system to the control surface horn (with the control in the critical positions for the affected parts of the system within the limits of its motion).
- (b) Each primary control system, including its supporting structure, must be designed as follows:
 - (1) The system must withstand loads resulting from the limit pilot forces prescribed in [CS 29.397](#);
 - (2) Notwithstanding sub-paragraph (b)(3), when power-operated actuator controls or power boost controls are used, the system must also withstand the loads resulting from the limit pilot forces prescribed in [CS 29.397](#) in conjunction with the forces output of each normally energised power device, including any single power boost or actuator system failure;
 - (3) If the system design or the normal operating loads are such that a part of the system cannot react to the limit pilot forces prescribed in [CS 29.397](#), that part of the system must be designed to withstand the maximum loads that can be obtained in normal operation. The minimum design loads must, in any case, provide a rugged system for service use, including consideration of fatigue, jamming, ground gusts, control inertia and friction loads. In the absence of a rational analysis, the design loads resulting from 0.60 of the specified limit pilot forces are acceptable minimum design loads; and
 - (4) If operational loads may be exceeded through jamming, ground gusts, control inertia, or friction, the system must withstand the limit pilot forces specified in [CS 29.397](#), without yielding.

CS 29.397 Limit pilot forces and torques

ED Decision 2003/16/RM

- (a) Except as provided in sub-paragraph (b), the limit pilot forces are as follows:
 - (1) For foot controls, 578 N (130 lbs).
 - (2) For stick controls, 445 N (100 lbs) fore and aft, and 298 N (67 lbs) laterally.
- (b) For flap, tab, stabiliser, rotor brake and landing gear operating controls, the following apply:
 - (1) Crank, wheel, and lever controls, $(25.4 + R) \times 2.919$ N, where R = radius in millimetres ($\left[\frac{1+R}{3}\right] \times 50$ lbs, where R = radius in inches), but not less than 222 N (50 lbs) nor more than 445 N (100 lbs) for hand-operated controls or 578 N (130 lbs) for foot-operated controls, applied at any angle within 20° of the plane of motion of the control.
 - (2) Twist controls, $356 \times R$ Newton-millimetres, where R = radius in millimetres ($80 \times R$ inch-pounds where R = radius in inches).

CS 29.399 Dual control system

ED Decision 2003/16/RM

Each dual primary flight control system must be able to withstand the loads that result when pilot forces not less than 0.75 times those obtained under [CS 29.395](#) are applied:

- (a) In opposition; and
- (b) In the same direction.

CS 29.411 Ground clearance: tail rotor guard

ED Decision 2003/16/RM

- (a) It must be impossible for the tail rotor to contact the landing surface during a normal landing.
- (b) If a tail rotor guard is required to show compliance with sub-paragraph (a):
 - (1) Suitable design loads must be established for the guard; and
 - (2) The guard and its supporting structure must be designed to withstand those loads.

CS 29.427 Unsymmetrical loads

ED Decision 2003/16/RM

- (a) Horizontal tail surfaces and their supporting structure must be designed for unsymmetrical loads arising from yawing and rotor wake effects in combination with the prescribed flight conditions.
- (b) To meet the design criteria of sub-paragraph (a), in the absence of more rational data, both of the following must be met:
 - (1) 100% of the maximum loading from the symmetrical flight conditions acts on the surface on one side of the plane of symmetry, and no loading acts on the other side.
 - (2) 50% of the maximum loading from the symmetrical flight conditions acts on the surface on each side of the plane of symmetry, in opposite directions.

-
- (c) For empennage arrangements where the horizontal tail surfaces are supported by the vertical tail surfaces, the vertical tail surfaces and supporting structure must be designed for the combined vertical and horizontal surface loads resulting from each prescribed flight condition, considered separately. The flight conditions must be selected so that the maximum design loads are obtained on each surface. In the absence of more rational data, the unsymmetrical horizontal tail surface loading distributions described in this paragraph must be assumed.

GROUND LOADS

CS 29.471 General

ED Decision 2003/16/RM

- (a) *Loads and equilibrium.* For limit ground loads:
- (1) The limit ground loads obtained in the landing conditions in this CS-29 must be considered to be external loads that would occur in the rotorcraft structure if it were acting as a rigid body; and
 - (2) In each specified landing condition, the external loads must be placed in equilibrium with linear and angular inertia loads in a rational or conservative manner.
- (b) *Critical centres of gravity.* The critical centres of gravity within the range for which certification is requested must be selected so that the maximum design loads are obtained in each landing gear element.

CS 29.473 Ground loading conditions and assumptions

ED Decision 2003/16/RM

- (a) For specified landing conditions, a design maximum weight must be used that is not less than the maximum weight. A rotor lift may be assumed to act through the centre of gravity throughout the landing impact. This lift may not exceed two-thirds of the design maximum weight.
- (b) Unless otherwise prescribed, for each specified landing condition, the rotorcraft must be designed for a limit load factor of not less than the limit inertia load factor substantiated under [CS 29.725](#).
- (c) Triggering or actuating devices for additional or supplementary energy absorption may not fail under loads established in the tests prescribed in [CS 29.725](#) and [29.727](#), but the factor of safety prescribed in [CS 29.303](#) need not be used.

CS 29.475 Tyres and shock absorbers

ED Decision 2003/16/RM

Unless otherwise prescribed, for each specified landing condition, the tyres must be assumed to be in their static position and the shock absorbers to be in their most critical position.

CS 29.477 Landing gear arrangement

ED Decision 2003/16/RM

Paragraphs [CS 29.235](#), [29.479](#) to [29.485](#), and [29.493](#) apply to landing gear with two wheels aft, and one or more wheels forward, of the centre of gravity.

CS 29.479 Level landing conditions

ED Decision 2003/16/RM

- (a) *Attitudes.* Under each of the loading conditions prescribed in sub-paragraph (b), the rotorcraft is assumed to be in each of the following level landing attitudes:
- (1) An attitude in which each wheel contacts the ground simultaneously.

- (2) An attitude in which the aft wheels contact the ground with the forward wheels just clear of the ground.
- (b) *Loading conditions.* The rotorcraft must be designed for the following landing loading conditions:
 - (1) Vertical loads applied under [CS 29.471](#).
 - (2) The loads resulting from a combination of the loads applied under sub-paragraph (b)(1) with drag loads at each wheel of not less than 25% of the vertical load at that wheel.
 - (3) The vertical load at the instant of peak drag load combined with a drag component simulating the forces required to accelerate the wheel rolling assembly up to the specified ground speed, with:
 - (i) The ground speed for determination of the spin-up loads being at least 75% of the optimum forward flight speed for minimum rate of descent in autorotation; and
 - (ii) The loading conditions of sub-paragraph (b) applied to the landing gear and its attaching structure only.
 - (4) If there are two wheels forward, a distribution of the loads applied to those wheels under sub-paragraphs (b)(1) and (2) in a ratio of 40:60.
- (c) *Pitching moments.* Pitching moments are assumed to be resisted by:
 - (1) In the case of the attitude in sub-paragraph (a)(1), the forward landing gear; and
 - (2) In the case of the attitude in sub-paragraph (a)(2), the angular inertia forces.

CS 29.481 Tail-down landing conditions

ED Decision 2003/16/RM

- (a) The rotorcraft is assumed to be in the maximum nose-up attitude allowing ground clearance by each part of the rotorcraft.
- (b) In this attitude, ground loads are assumed to act perpendicular to the ground.

CS 29.483 One-wheel landing conditions

ED Decision 2003/16/RM

For the one-wheel landing condition, the rotorcraft is assumed to be in the level attitude and to contact the ground on one aft wheel. In this attitude:

- (a) The vertical load must be the same as that obtained on that side under [CS 29.479\(b\)\(1\)](#); and
- (b) The unbalanced external loads must be reacted by rotorcraft inertia.

CS 29.485 Lateral drift landing conditions

ED Decision 2003/16/RM

- (a) The rotorcraft is assumed to be in the level landing attitude, with:
 - (1) Side loads combined with one-half of the maximum ground reactions obtained in the level landing conditions of [CS 29.479\(b\)\(1\)](#); and
 - (2) The loads obtained under sub-paragraph (a)(1) applied:
 - (i) At the ground contact point; or

- (ii) For full-swivelling gear, at the centre of the axle.
- (b) The rotorcraft must be designed to withstand, at ground contact:
 - (1) When only the aft wheels contact the ground, side loads of 0.8 times the vertical reaction acting inward on one side and 0.6 times the vertical reaction acting outward on the other side, all combined with the vertical loads specified in sub-paragraph (a); and
 - (2) When the wheels contact the ground simultaneously:
 - (i) For the aft wheels, the side loads specified in sub-paragraph (b)(1); and
 - (ii) For the forward wheels, a side load of 0.8 times the vertical reaction combined with the vertical load specified in sub-paragraph (a).

CS 29.493 Braked roll conditions

ED Decision 2003/16/RM

Under braked roll conditions with the shock absorbers in their static positions:

- (a) The limit vertical load must be based on a load factor of at least –
 - (1) 1.33, for the attitude specified in [CS 29.479\(a\)\(1\)](#); and
 - (2) 1.0, for the attitude specified in [CS 29.479\(a\)\(2\)](#); and
- (b) The structure must be designed to withstand, at the ground contact point of each wheel with brakes, a drag load of at least the lesser of:
 - (1) The vertical load multiplied by a coefficient of friction of 0.8; and
 - (2) The maximum value based on limiting brake torque.

CS 29.497 Ground loading conditions: landing gear with tail wheels

ED Decision 2003/16/RM

- (a) *General.* Rotorcraft with landing gear with two wheels forward and one wheel aft of the centre of gravity must be designed for loading conditions as prescribed in this paragraph..
- (b) *Level landing attitude with only the forward wheels contacting the ground.* In this attitude:
 - (1) The vertical loads must be applied under [CS 29.471](#) to [CS 29.475](#);
 - (2) The vertical load at each axle must be combined with a drag load at that axle of not less than 25% of that vertical load; and
 - (3) Unbalanced pitching moments are assumed to be resisted by angular inertia forces.
- (c) *Level landing attitude with all wheels contacting the ground simultaneously.* In this attitude, the rotorcraft must be designed for landing loading conditions as prescribed in sub-paragraph (b).
- (d) *Maximum nose-up attitude with only the rear wheel contacting the ground.* The attitude for this condition must be the maximum nose-up attitude expected in normal operation, including autorotative landings. In this attitude:
 - (1) The appropriate ground loads specified in sub-paragraphs (b)(1) and (2) must be determined and applied, using a rational method to account for the moment arm between the rear wheel ground reaction and the rotorcraft centre of gravity; or
 - (2) The probability of landing with initial contact on the rear wheel must be shown to be extremely remote.

-
- (e) *Level landing attitude with only one forward wheel contacting the ground.* In this attitude, the rotorcraft must be designed for ground loads as specified in sub-paragraphs (b)(1) and (3).
- (f) *Side loads in the level landing attitude.* In the attitudes specified in sub-paragraphs (b) and (c), the following apply:
- (1) The side loads must be combined at each wheel with one-half of the maximum vertical ground reactions obtained for that wheel under sub-paragraphs (b) and (c). In this condition, the side loads must be:
 - (i) For the forward wheels, 0.8 times the vertical reaction (on one side) acting inward and 0.6 times the vertical reaction (on the other side) acting outward; and
 - (ii) For the rear wheel, 0.8 times the vertical reaction.
 - (2) The loads specified in sub-paragraph (f)(1) must be applied:
 - (i) At the ground contact point with the wheel in the trailing position (for non-full swivelling landing gear or for full swivelling landing gear with a lock, steering device, or shimmy damper to keep the wheel in the trailing position); or
 - (ii) At the centre of the axle (for full swivelling landing gear without a lock, steering device, or shimmy damper).
- (g) *Braked roll conditions in the level landing attitude.* In the attitudes specified in sub-paragraphs (b) and (c), and with the shock absorbers in their static positions, the rotorcraft must be designed for braked roll loads as follows:
- (1) The limit vertical load must be based on a limit vertical load factor of not less than:
 - (i) 1.0, for the attitude specified in sub-paragraph (b); and
 - (ii) 1.33, for the attitude specified in sub-paragraph (c).
 - (2) For each wheel with brakes, a drag load must be applied, at the ground contact point, of not less than the lesser of:
 - (i) 0.8 times the vertical load; and
 - (ii) The maximum based on limiting brake torque.
- (h) *Rear wheel turning loads in the static ground attitude.* In the static ground attitude, and with the shock absorbers and tyres in their static positions, the rotorcraft must be designed for rear wheel turning loads as follows:
- (1) A vertical ground reaction equal to the static load on the rear wheel must be combined with an equal side load.
 - (2) The load specified in sub-paragraph (h)(1) must be applied to the rear landing gear:
 - (i) Through the axle, if there is a swivel (the rear wheel being assumed to be swivelled 90°, to the longitudinal axis of the rotorcraft); or
 - (ii) At the ground contact point if there is a lock, steering device or shimmy damper (the rear wheel being assumed to be in the trailing position).
- (i) *Taxying condition.* The rotorcraft and its landing gear must be designed for the loads that would occur when the rotorcraft is taxied over the roughest ground that may reasonably be expected in normal operation.

CS 29.501 Ground loading conditions: landing gear with skids

ED Decision 2003/16/RM

- (a) *General.* Rotorcraft with landing gear with skids must be designed for the loading conditions specified in this paragraph. In showing compliance with this paragraph, the following apply:
- (1) The design maximum weight, centre of gravity, and load factor must be determined under [CS 29.471](#) to [29.475](#).
 - (2) Structural yielding of elastic spring members under limit loads is acceptable.
 - (3) Design ultimate loads for elastic spring members need not exceed those obtained in a drop test of the gear with:
 - (i) A drop height of 1.5 times that specified in [CS 29.725](#); and
 - (ii) An assumed rotor lift of not more than 1.5 times that used in the limit drop tests prescribed in [CS 29.725](#).
 - (4) Compliance with sub-paragraphs (b) to (e) must be shown with:
 - (i) The gear in its most critically deflected position for the landing condition being considered; and
 - (ii) The ground reactions rationally distributed along the bottom of the skid tube.
- (b) *Vertical reactions in the level landing attitude.* In the level attitude, and with the rotorcraft contacting the ground along the bottom of both skids, the vertical reactions must be applied as prescribed in sub-paragraph (a).
- (c) *Drag reactions in the level landing attitude.* In the level attitude, and with the rotorcraft contacting the ground along the bottom of both skids, the following apply:
- (1) The vertical reactions must be combined with horizontal drag reactions of 50% of the vertical reaction applied at the ground.
 - (2) The resultant ground loads must equal the vertical load specified in sub-paragraph (b).
- (d) *Sideloads in the level landing attitude.* In the level attitude, and with the rotorcraft contacting the ground along the bottom of both skids, the following apply:
- (1) The vertical ground reaction must be:
 - (i) Equal to the vertical loads obtained in the condition specified in sub-paragraph (b); and
 - (ii) Divided equally among the skids.
 - (2) The vertical ground reactions must be combined with a horizontal sideload of 25% of their value.
 - (3) The total sideload must be applied equally between skids and along the length of the skids.
 - (4) The unbalanced moments are assumed to be resisted by angular inertia.
 - (5) The skid gear must be investigated for:
 - (i) Inward acting sideloads; and
 - (ii) Outward acting sideloads.

- (e) *One-skid landing loads in the level attitude.* In the level attitude, and with the rotorcraft contacting the ground along the bottom of one skid only, the following apply:
- (1) The vertical load on the ground contact side must be the same as that obtained on that side in the condition specified in sub-paragraph (b).
 - (2) The unbalanced moments are assumed to be resisted by angular inertia.
- (f) *Special conditions.* In addition to the specified in sub-paragraphs (b) and (c), the rotorcraft must be designed for the following ground reactions:
- (1) A ground reaction load acting up and aft at an angle of 45°, to the longitudinal axis of the rotorcraft. This load must be:
 - (i) Equal to 1.33 times the maximum weight;
 - (ii) Distributed symmetrically among the skids;
 - (iii) Concentrated at the forward end of the straight part of the skid tube; and
 - (iv) Applied only to the forward end of the skid tube and its attachment to the rotorcraft.
 - (2) With the rotorcraft in the level landing attitude, a vertical ground reaction load equal to one-half of the vertical load determined under sub-paragraph (b). This load must be:
 - (i) Applied only to the skid tube and its attachment to the rotorcraft; and
 - (ii) Distributed equally over 33.3% of the length between the skid tube attachments and centrally located midway between the skid tube attachments.

CS 29.505 Ski landing conditions

ED Decision 2003/16/RM

If certification for ski operation is requested, the rotorcraft, with skis, must be designed to withstand the following loading conditions (where P is the maximum static weight on each ski with the rotorcraft at design maximum weight, and n is the limit load factor determined under [CS 29.473\(b\)](#)):

- (a) Up-load conditions in which:
- (1) A vertical load of Pn and a horizontal load of $Pn/4$ are simultaneously applied at the pedestal bearings; and
 - (2) A vertical load of $1.33 P$ is applied at the pedestal bearings.
- (b) A side load condition in which a side load of $0.35 Pn$ is applied at the pedestal bearings in a horizontal plane perpendicular to the centreline of the rotorcraft.
- (c) A torque-load condition in which a torque load of $1.33 P$ (in foot-pounds) is applied to the ski about the vertical axis through the centreline of the pedestal bearings.

CS 29.511 Ground load: unsymmetrical loads on multiple-wheel units

ED Decision 2003/16/RM

- (a) In dual-wheel gear units, 60% of the total ground reaction for the gear unit must be applied to one wheel and 40% to the other.
- (b) To provide for the case of one deflated tyre, 60% of the specified load for the gear unit must be applied to either wheel, except that the vertical ground reaction may not be less than the full static value.
- (c) In determining the total load on a gear unit, the transverse shift in the load centroid, due to unsymmetrical load distribution on the wheels, may be neglected.

WATER LOADS

CS 29.519 Hull type rotorcraft: Water-based and amphibian

ED Decision 2003/16/RM

- (a) *General.* For hull type rotorcraft, the structure must be designed to withstand the water loading set forth in sub-paragraphs (b), (c), and (d) considering the most severe wave heights and profiles for which approval is desired. The loads for the landing conditions of sub-paragraphs (b) and (c) must be developed and distributed along and among the hull and auxiliary floats, if used, in a rational and conservative manner, assuming a rotor lift not exceeding two-thirds of the rotorcraft weight to act throughout the landing impact.
- (b) *Vertical landing conditions.* The rotorcraft must initially contact the most critical wave surface at zero forward speed in likely pitch and roll attitudes which result in critical design loadings. The vertical descent velocity may not be less than 1.98 metres per second (6.5 ft/s) relative to the mean water surface.
- (c) *Forward speed landing conditions.* The rotorcraft must contact the most critical wave at forward velocities from zero up to 56 km/h (30 knots) in likely pitch, roll, and yaw attitudes and with a vertical descent velocity of not less than 1.98 metres per second (6.5 ft/s) relative to the mean water surface. A maximum forward velocity of less than 56 km/h (30 knots) may be used in design if it can be demonstrated that the forward velocity selected would not be exceeded in a normal one-engine-out landing.
- (d) *Auxiliary float immersion condition.* In addition to the loads from the landing conditions, the auxiliary float, and its support and attaching structure in the hull, must be designed for the load developed by a fully immersed float unless it can be shown that full immersion of the float is unlikely, in which case the highest likely float buoyancy load must be applied that considers loading of the float immersed to create restoring moments compensating for upsetting moments caused by side wind, asymmetrical rotorcraft loading, water wave action and rotorcraft inertia.

CS 29.521 Float landing conditions

ED Decision 2003/16/RM

If certification for float operation (including float amphibian operation) is requested, the rotorcraft, with floats, must be designed to withstand the following loading conditions (where the limit load factor is determined under [CS 29.473\(b\)](#) or assumed to be equal to that determined for wheel landing gear):

- (a) Up-load conditions in which:
 - (1) A load is applied so that, with the rotorcraft in the static level attitude, the resultant water reaction passes vertically through the centre of gravity; and
 - (2) The vertical load prescribed in sub-paragraph (a)(1) is applied simultaneously with an aft component of 0.25 times the vertical component.

- (b) A side load condition in which:
- (1) A vertical load of 0.75 times the total vertical load specified in sub-paragraph (a)(1) is divided equally among the floats; and
 - (2) For each float, the load share determined under sub-paragraph (b)(1), combined with a total side load of 0.25 times the total vertical load specified in sub-paragraph (b)(1), is applied to that float only.

MAIN COMPONENT REQUIREMENTS

CS 29.547 Main and tail rotor structure

ED Decision 2003/16/RM

- (a) A rotor is an assembly of rotating components, which includes the rotor hub, blades, blade dampers, the pitch control mechanisms, and all other parts that rotate with the assembly.
- (b) Each rotor assembly must be designed as prescribed in this paragraph and must function safely for the critical flight load and operating conditions. A design assessment must be performed, including a detailed failure analysis to identify all failures that will prevent continued safe flight or safe landing, and must identify the means to minimise the likelihood of their occurrence.
- (c) The rotor structure must be designed to withstand the following loads prescribed in [CS 29.337](#) to [29.341](#), and [CS 29.351](#):
 - (1) Critical flight loads.
 - (2) Limit loads occurring under normal conditions of autorotation.
- (d) The rotor structure must be designed to withstand loads simulating:
 - (1) For the rotor blades, hubs and flapping hinges, the impact force of each blade against its stop during ground operation; and
 - (2) Any other critical condition expected in normal operation.
- (e) The rotor structure must be designed to withstand the limit torque at any rotational speed, including zero. In addition:
 - (1) The limit torque need not be greater than the torque defined by a torque limiting device (where provided), and may not be less than the greater of:
 - (i) The maximum torque likely to be transmitted to the rotor structure, in either direction, by the rotor drive or by sudden application of the rotor brake; and
 - (ii) For the main rotor, the limit engine torque specified in [CS 29.361](#).
 - (2) The limit torque must be equally and rationally distributed to the rotor blades.

AMC 29.547 Main rotor and tail rotor structure

ED Decision 2012/022/R

Where Vibration Health Monitoring is used as a compensating provision to meet [CS 29.547\(b\)](#), the design and performance of the vibration health monitoring system should be approved by requesting compliance with [CS 29.1465\(a\)](#).

[Amdt 29/3]

CS 29.549 Fuselage and rotor pylon structures

ED Decision 2003/16/RM

- (a) Each fuselage and rotor pylon structure must be designed to withstand:
 - (1) The critical loads prescribed in [CS 29.337](#) to [29.341](#), and [CS 29.351](#);

- (2) The applicable ground loads prescribed in [CS 29.235](#), [29.471](#) to [29.485](#), [CS 29.493](#), [29.497](#), [29.505](#), and [29.521](#); and
- (3) The loads prescribed in [CS 29.547\(d\)\(1\)](#) and [\(e\)\(1\)\(i\)](#).
- (b) Auxiliary rotor thrust, the torque reaction of each rotor drive system, and the balancing air and inertia loads occurring under accelerated flight conditions, must be considered.
- (c) Each engine mount and adjacent fuselage structure must be designed to withstand the loads occurring under accelerated flight and landing conditions, including engine torque.
- (d) Reserved.
- (e) If approval for the use of 2½-minute OEI power is requested, each engine mount and adjacent structure must be designed to withstand the loads resulting from a limit torque equal to 1.25 times the mean torque for 2½-minute power OEI combined with 1g flight loads.

CS 29.551 Auxiliary lifting surfaces

ED Decision 2003/16/RM

Each auxiliary lifting surface must be designed to withstand:

- (a) The critical flight loads in [CS 29.337](#) to [29.341](#), and [CS 29.351](#);
- (b) The applicable ground loads in [CS 29.235](#), [29.471](#) to [29.485](#), [CS 29.493](#), [29.505](#), and [29.521](#); and
- (c) Any other critical condition expected in normal operation.

EMERGENCY LANDING CONDITIONS

CS 29.561 General

ED Decision 2003/16/RM

- (a) The rotorcraft, although it may be damaged in emergency landing conditions on land or water, must be designed as prescribed in this paragraph to protect the occupants under those conditions.
- (b) The structure must be designed to give each occupant every reasonable chance of escaping serious injury in a crash landing when:
 - (1) Proper use is made of seats, belts, and other safety design provisions;
 - (2) The wheels are retracted (where applicable); and
 - (3) Each occupant and each item of mass inside the cabin that could injure an occupant is restrained when subjected to the following ultimate inertial load factors relative to the surrounding structure:
 - (i) Upward – 4 g
 - (ii) Forward – 16 g
 - (iii) Sideward – 8 g
 - (iv) Downward – 20 g, after the intended displacement of the seat device
 - (v) Rearward – 1.5 g.
- (c) The supporting structure must be designed to restrain under any ultimate inertial load factor up to those specified in this paragraph, any item of mass above and/or behind the crew and passenger compartment that could injure an occupant if it came loose in an emergency landing. Items of mass to be considered include, but are not limited to, rotors, transmission and engines. The items of mass must be restrained for the following ultimate inertial load factors:
 - (1) Upward – 1.5 g
 - (2) Forward – 12 g
 - (3) Sideward – 6 g
 - (4) Downward – 12 g
 - (5) Rearward – 1.5 g.
- (d) Any fuselage structure in the area of internal fuel tanks below the passenger floor level must be designed to resist the following ultimate inertia factors and loads, and to protect the fuel tanks from rupture, if rupture is likely when those loads are applied to that area:
 - (1) Upward – 1.5 g
 - (2) Forward – 4.0 g
 - (3) Sideward – 2.0 g
 - (4) Downward – 4.0 g

CS 29.562 Emergency landing dynamic conditions

ED Decision 2003/16/RM

- (a) The rotorcraft, although it may be damaged in a crash landing, must be designed to reasonably protect each occupant when:
 - (1) The occupant properly uses the seats, safety belts, and shoulder harnesses provided in the design; and
 - (2) The occupant is exposed to loads equivalent to those resulting from the conditions prescribed in this paragraph.
- (b) Each seat type design or other seating device approved for crew or passenger occupancy during take-off and landing must successfully complete dynamic tests or be demonstrated by rational analysis based on dynamic tests of a similar type seat in accordance with the following criteria. The tests must be conducted with an occupant simulated by a 77 kg (170-pound) anthropomorphic test dummy (ATD), sitting in the normal upright position.
 - (1) A change in downward velocity of not less than 9.1 metres per second (30 ft/s) when the seat or other seating device is oriented in its nominal position with respect to the rotorcraft's reference system, the rotorcraft's longitudinal axis is canted upward 60°, with respect to the impact velocity vector, and the rotorcraft's lateral axis is perpendicular to a vertical plane containing the impact velocity vector and the rotorcraft's longitudinal axis. Peak floor deceleration must occur in not more than 0.031 seconds after impact and must reach a minimum of 30 g.
 - (2) A change in forward velocity of not less than 12.8 metres per second (42 ft/s) when the seat or other seating device is oriented in its nominal position with respect to the rotorcraft's reference system, the rotorcraft's longitudinal axis is yawed 10°, either right or left of the impact velocity vector (whichever would cause the greatest load on the shoulder harness), the rotorcraft's lateral axis is contained in a horizontal plane containing the impact velocity vector, and the rotorcraft's vertical axis is perpendicular to a horizontal plane containing the impact velocity vector. Peak floor deceleration must occur in not more than 0.071 seconds after impact and must reach a minimum of 18.4 g.
 - (3) Where floor rails or floor or sidewall attachment devices are used to attach the seating devices to the airframe structure for the conditions of this paragraph, the rails or devices must be misaligned with respect to each other by at least 10° vertically (i.e. pitch out of parallel) and by at least a 10° lateral roll, with the directions optional, to account for possible floor warp.
- (c) Compliance with the following must be shown:
 - (1) The seating device system must remain intact although it may experience separation intended as part of its design.
 - (2) The attachment between the seating device and the airframe structure must remain intact, although the structure may have exceeded its limit load.
 - (3) The ATD's shoulder harness strap or straps must remain on or in the immediate vicinity of the ATD's shoulder during the impact.
 - (4) The safety belt must remain on the ATD's pelvis during the impact.

- (5) The ATD's head either does not contact any portion of the crew or passenger compartment, or if contact is made, the head impact does not exceed a head injury criteria (HIC) of 1000 as determined by this equation.

$$HIC = (t_2 - t_1) \left[\frac{1}{(t_2 - t_1)} \int_{t_1}^{t_2} a(t) dt \right]^{2.5}$$

Where – a(t) is the resultant acceleration at the centre of gravity of the head form expressed as a multiple of g (the acceleration of gravity) and t₂–t₁ is the time duration, in seconds, of major head impact, not to exceed 0.05 seconds.

- (6) Loads in individual shoulder harness straps must not exceed 7784 N (1750 lbs). If dual straps are used for retaining the upper torso, the total harness strap loads must not exceed 8896 N (2000 lbs).
- (7) The maximum compressive load measured between the pelvis and the lumbar column of the ATD must not exceed 6674 N (1500 lbs).
- (d) An alternate approach that achieves an equivalent or greater level of occupant protection, as required by this paragraph, must be substantiated on a rational basis.

CS 29.563 Structural ditching provisions

ED Decision 2003/16/RM

If certification with ditching provisions is requested, structural strength for ditching must meet the requirements of this paragraph and [CS 29.801\(e\)](#).

- (a) *Forward speed landing conditions.* The rotorcraft must initially contact the most critical wave for reasonably probable water conditions at forward velocities from zero up to 56 km/h (30 knots) in likely pitch, roll, and yaw attitudes. The rotorcraft limit vertical descent velocity may not be less than 1.5 metres per second (5 ft/s) relative to the mean water surface. Rotor lift may be used to act through the centre of gravity throughout the landing impact. This lift may not exceed two-thirds of the design maximum weight. A maximum forward velocity of less than 30 knots may be used in design if it can be demonstrated that the forward velocity selected would not be exceeded in a normal one-engine-out touchdown.
- (b) *Auxiliary or emergency float conditions*
- (1) *Floats fixed or deployed before initial water contact.* In addition to the landing loads in sub-paragraph (a), each auxiliary or emergency float, or its support and attaching structure in the airframe or fuselage, must be designed for the load developed by a fully immersed float unless it can be shown that full and pitch angles determined from compliance with [CS 29.801\(d\)](#) may be used, if significant, to determine the extent of immersion of each float. If the floats are deployed in flight, appropriate air loads derived from the flight limitations with the floats deployed shall be used in substantiation of the floats and their attachment to the rotorcraft. For this purpose, the design airspeed for limit load is the float deployed airspeed operating limit multiplied by 1.11.
- (2) *Floats deployed after initial water contact.* Each float must be designed for full or partial immersion prescribed in sub-paragraph (b)(1). In addition, each float must be designed for combined vertical and drag loads using a relative limit speed of 37 km/h (20 knots) between the rotorcraft and the water. The vertical load may not be less than the highest likely buoyancy load determined under paragraph (b)(1).

FATIGUE EVALUATION

CS 29.571 Fatigue Tolerance Evaluation of Metallic Structure

ED Decision 2012/022/R

- (a) A fatigue tolerance evaluation of each Principal Structural Element (PSE) must be performed, and appropriate inspections and retirement time or approved equivalent means must be established to avoid Catastrophic Failure during the operational life of the rotorcraft.
- (b) Reserved
- (c) Reserved
- (d) Each PSE must be identified. Structure to be considered must include the rotors, rotor drive systems between the engines and rotor hubs, controls, fuselage, fixed and movable control surfaces, engine and transmission mountings, landing gear, and their related primary attachments.
- (e) Each fatigue tolerance evaluation must include:
 - (1) In-flight measurements to determine the fatigue loads or stresses for the PSEs identified in sub-paragraph (d) in all critical conditions throughout the range of design limitations required in [CS 29.309](#) (including altitude effects), except that manoeuvring load factors need not exceed the maximum values expected in operations.
 - (2) The loading spectra as severe as those expected in operations based on loads or stresses determined under sub-paragraph (e)(1), including external load operations, if applicable, and other high frequency power-cycle operations.
 - (3) Take-off, landing, and taxi loads when evaluating the landing gear (including skis and floats) and other affected PSEs.
 - (4) For each PSE identified in sub-paragraph (d), a threat assessment, which includes a determination of the probable locations, types, and sizes of damage taking into account fatigue, environmental effects, intrinsic and discrete flaws, or accidental damage that may occur during manufacture or operation.
 - (5) A determination of the fatigue tolerance characteristics for the PSE with the damage identified in sub-paragraph (e)(4) that supports the inspection and retirement times, or other approved equivalent means.
 - (6) Analyses supported by test evidence and, if available, service experience.
- (f) A residual strength determination is required that substantiates the maximum damage size assumed in the fatigue tolerance evaluation. In determining inspection intervals based on damage growth, the residual strength evaluation must show that the remaining structure, after damage growth, is able to withstand design limit loads without failure.
- (g) The effect of damage on stiffness, dynamic behaviour, loads and functional performance must be considered.
- (h) The inspection and retirement times or approved equivalent means established under this paragraph must be included in the Airworthiness Limitation Section of the Instructions for Continued Airworthiness required by [CS 29.1529](#) and paragraph A29.4 of [Appendix A](#).
- (i) If inspections for any of the damage types identified in sub-paragraph (e)(4) cannot be established within the limitations of geometry, inspectability, or good design practice, then

supplemental procedures, in conjunction with the PSE retirement time, must be established to minimize the risk of occurrence of these types of damage that could result in a catastrophic failure during the operational life of the rotorcraft.

[Amdt 29/3]

CS 29.573 Damage Tolerance and Fatigue Evaluation of Composite Rotorcraft Structures

ED Decision 2012/022/R

- (a) Composite rotorcraft structure must be evaluated under the damage tolerance requirements of sub-paragraph (d) unless the applicant establishes that a damage tolerance evaluation is impractical within the limits of geometry, inspectability, and good design practice. In such a case, the composite rotorcraft structure must undergo a fatigue evaluation in accordance with sub-paragraph (e).
- (b) Reserved
- (c) Reserved
- (d) Damage Tolerance Evaluation:
 - (1) Damage tolerance evaluations of composite structures must show that Catastrophic Failure due to static and fatigue loads is avoided throughout the operational life or prescribed inspection intervals of the rotorcraft.
 - (2) The damage tolerance evaluation must include PSEs of the airframe, main and tail rotor drive systems, main and tail rotor blades and hubs, rotor controls, fixed and movable control surfaces, engine and transmission mountings, landing gear, and any other detail design points or parts whose failure or detachment could prevent continued safe flight and landing.
 - (3) Each damage tolerance evaluation must include:
 - (i) The identification of the structure being evaluated;
 - (ii) A determination of the structural loads or stresses for all critical conditions throughout the range of limits in [CS 29.309](#) (including altitude effects), supported by in-flight and ground measurements, except that manoeuvring load factors need not exceed the maximum values expected in service;
 - (iii) The loading spectra as severe as those expected in service based on loads or stresses determined under sub-paragraph (d)(3)(ii), including external load operations, if applicable, and other operations including high torque events;
 - (iv) A Threat Assessment for all structure being evaluated that specifies the locations, types, and sizes of damage, considering fatigue, environmental effects, intrinsic and discrete flaws, and impact or other accidental damage (including the discrete source of the accidental damage) that may occur during manufacture or operation;
 - (v) An assessment of the residual strength and fatigue characteristics of all structure being evaluated that supports the replacement times and inspection intervals established under sub-paragraph (d)(4); and
 - (vi) allowances for the detrimental effects of material, fabrication techniques, and process variability.

-
- (4) Replacement times, inspections, or other procedures must be established to require the repair or replacement of damaged parts to prevent Catastrophic Failure. These replacement times, inspections, or other procedures must be included in the Airworthiness Limitations Section of the Instructions for Continued Airworthiness required by [CS 29.1529](#).
- (i) Replacement times must be determined by tests, or by analysis supported by tests to show that throughout its life the structure is able to withstand the repeated loads of variable magnitude expected in-service. In establishing these replacement times, the following items must be considered:
- (A) Damage identified in the Threat Assessment required by sub-paragraph (d)(3)(iv);
- (B) Maximum acceptable manufacturing defects and in-service damage (i.e., those that do not lower the residual strength below ultimate design loads and those that can be repaired to restore ultimate strength); and
- (C) Ultimate load strength capability after applying repeated loads.
- (ii) Inspection intervals must be established to reveal any damage identified in the Threat Assessment required by sub-paragraph (d)(3)(iv) that may occur from fatigue or other in-service causes before such damage has grown to the extent that the component cannot sustain the required residual strength capability. In establishing these inspection intervals, the following items must be considered:
- (A) The growth rate, including no-growth, of the damage under the repeated loads expected in-service determined by tests or analysis supported by tests; and
- (B) The required residual strength for the assumed damage established after considering the damage type, inspection interval, detectability of damage, and the techniques adopted for damage detection. The minimum required residual strength is limit load.
- (5) The effects of damage on stiffness, dynamic behaviour, loads and functional performance must be taken into account when substantiating the maximum assumed damage size and inspection interval.
- (e) Fatigue Evaluation:
- If an applicant establishes that the damage tolerance evaluation described in sub-paragraph (d) is impractical within the limits of geometry, inspectability, or good design practice, the applicant must do a fatigue evaluation of the particular composite rotorcraft structure and:
- (1) Identify structure considered in the fatigue evaluation;
- (2) Identify the types of damage considered in the fatigue evaluation;
- (3) Establish supplemental procedures to minimise the risk of Catastrophic Failure associated with damage identified in sub-paragraph (e)(2); and
- (4) Include these supplemental procedures in the Airworthiness Limitations section of the Instructions for Continued Airworthiness required by [CS 29.1529](#).

[Amdt 29/3]

SUBPART D — DESIGN AND CONSTRUCTION

GENERAL

CS 29.601 Design

ED Decision 2003/16/RM

- (a) The rotorcraft may have no design features or details that experience has shown to be hazardous or unreliable.
- (b) The suitability of each questionable design detail and part must be established by tests.

CS 29.602 Critical parts

ED Decision 2003/16/RM

- (a) Critical part - A critical part is a part, the failure of which could have a catastrophic effect upon the rotorcraft, and for which critical characteristics have been identified which must be controlled to ensure the required level of integrity.
- (b) If the type design includes critical parts, a critical parts list shall be established. Procedures shall be established to define the critical design characteristics, identify processes that affect those characteristics, and identify the design change and process change controls necessary for showing compliance with the quality assurance requirements of Part-21.

CS 29.603 Materials

ED Decision 2003/16/RM

The suitability and durability of materials used for parts, the failure of which could adversely affect safety, must –

- (a) Be established on the basis of experience or tests;
- (b) Meet approved specifications that ensure their having the strength and other properties assumed in the design data; and
- (c) Take into account the effects of environmental conditions, such as temperature and humidity, expected in service.

CS 29.605 Fabrication methods

ED Decision 2003/16/RM

- (a) The methods of fabrication used must produce consistently sound structures. If a fabrication process (such as gluing, spot welding, or heat- treating) requires close control to reach this objective, the process must be performed according to an approved process specification.
- (b) Each new aircraft fabrication method must be substantiated by a test program.

CS 29.607 Fasteners

ED Decision 2003/16/RM

- (a) Each removable bolt, screw, nut, pin or other fastener whose loss could jeopardise the safe operation of the rotorcraft must incorporate two separate locking devices. The fastener and its

locking devices may not be adversely affected by the environmental conditions associated with the particular installation.

- (b) No self-locking nut may be used on any bolt subject to rotation in operation unless a non-friction locking device is used in addition to the self-locking device.

CS 29.609 Protection of structure

ED Decision 2003/16/RM

Each part of the structure must:

- (a) Be suitably protected against deterioration or loss of strength in service due to any cause, including:
 - (1) Weathering;
 - (2) Corrosion; and
 - (3) Abrasion; and
- (b) Have provisions for ventilation and drainage where necessary to prevent the accumulation of corrosive, flammable, or noxious fluids.

CS 29.610 Lightning and static electricity protection

ED Decision 2003/16/RM

- (a) The rotorcraft structure must be protected against catastrophic effects from lightning.
- (b) For metallic components, compliance with sub-paragraph (a) may be shown by:
 - (1) Electrically bonding the components properly to the airframe; or
 - (2) Designing the components so that a strike will not endanger the rotorcraft.
- (c) For non-metallic components, compliance with sub-paragraph (a) may be shown by:
 - (1) Designing the components to minimise the effect of a strike; or
 - (2) Incorporating acceptable means of diverting the resulting electrical current to not endanger the rotorcraft.
- (d) The electrical bonding and protection against lightning and static electricity must:
 - (1) Minimise the accumulation of electrostatic charge;
 - (2) Minimise the risk of electrical shock to crew, passengers, and servicing and maintenance personnel using normal precautions;
 - (3) Provide an electrical return path, under both normal and fault conditions, on rotorcraft having grounded electrical systems; and
 - (4) Reduce to an acceptable level the effects of lightning and static electricity on the functioning of essential electrical and electronic equipment.

CS 29.611 Inspection provisions

ED Decision 2003/16/RM

There must be means to allow close examination of each part that requires:

- (a) Recurring inspection;

- (b) Adjustment for proper alignment and functioning; or
- (c) Lubrication.

CS 29.613 Material strength properties and design values

ED Decision 2003/16/RM

- (a) Material strength properties must be based on enough tests of material meeting specifications to establish design values on a statistical basis.
- (b) Design values must be chosen to minimise the probability of structural failure due to material variability. Except as provided in subparagraphs (d) and (e), compliance with this paragraph must be shown by selecting design values that assure material strength with the following probability:
 - (1) Where applied loads are eventually distributed through a single member within an assembly, the failure of which would result in loss of structural integrity of the component, 99% probability with 95% confidence; and
 - (2) For redundant structures, those in which the failure of individual elements would result in applied loads being safely distributed to other load-carrying members, 90% probability with 95% confidence.
- (c) The strength, detail design, and fabrication of the structure must minimise the probability of disastrous fatigue failure, particularly at points of stress concentration.
- (d) Material specifications must be those contained in documents accepted by the Agency.
- (e) Other design values may be used if a selection of the material is made in which a specimen of each individual item is tested before use and it is determined that the actual strength properties of that particular item will equal or exceed those used in design.

CS 29.619 Special factors

ED Decision 2003/16/RM

- (a) The special factors prescribed in [CS 29.621](#) to [29.625](#) apply to each part of the structure whose strength is:
 - (1) Uncertain;
 - (2) Likely to deteriorate in service before normal replacement; or
 - (3) Subject to appreciable variability due to:
 - (i) Uncertainties in manufacturing processes; or
 - (ii) Uncertainties in inspection methods.
- (b) For each part of the rotorcraft to which [CS 29.621](#) to [29.625](#) apply, the factor of safety prescribed in [CS 29.303](#) must be multiplied by a special factor equal to:
 - (1) The applicable special factors prescribed in [CS 29.621](#) to [29.625](#); or
 - (2) Any other factor great enough to ensure that the probability of the part being under strength because of the uncertainties specified in sub-paragraph (a) is extremely remote.

CS 29.621 Casting factors

ED Decision 2003/16/RM

- (a) *General.* The factors, tests, and inspections specified in sub-paragraphs (b) and (c) must be applied in addition to those necessary to establish foundry quality control. The inspections must meet approved specifications. Subparagraphs (c) and (d) apply to structural castings except castings that are pressure tested as parts of hydraulic or other fluid systems and do not support structural loads.
- (b) *Bearing stressed and surfaces.* The casting factors specified in sub-paragraphs (c) and (d):
 - (1) Need not exceed 1.25 with respect to bearing stresses regardless of the method of inspection used; and
 - (2) Need not be used with respect to the bearing surfaces of a part whose bearing factor is larger than the applicable casting factor.
- (c) *Critical castings.* For each casting whose failure would preclude continued safe flight and landing of the rotorcraft or result in serious injury to any occupant, the following apply:
 - (1) Each critical casting must:
 - (i) Have a casting factor of not less than 1.25; and
 - (ii) Receive 100% inspection by visual, radiographic, and magnetic particle (for ferromagnetic materials) or penetrant (for non ferromagnetic materials) inspection methods or approved equivalent inspection methods.
 - (2) For each critical casting with a casting factor less than 1.50, three sample castings must be static tested and shown to meet:
 - (i) The strength requirements of [CS 29.305](#) at an ultimate load corresponding to a casting factor of 1.25; and
 - (ii) The deformation requirements of [CS 29.305](#) at a load of 1.15 times the limit load.
- (d) *Non critical castings.* For each casting other than those specified in sub-paragraph (c), the following apply:
 - (1) Except as provided in sub-paragraphs (d)(2) and (3), the casting factors and corresponding inspections must meet the following table:

Casting factor	Inspection
2.0 or greater	100% visual.
Less than 2.0 greater than 1.5	100% visual, and magnetic particle (ferromagnetic materials), penetrant (non ferro-magnetic materials), or approved equivalent inspection methods.
1.25 through 1.50.....	100% visual, and magnetic particle (ferromagnetic materials), penetrant (non ferro-magnetic materials), and radiographic or approved equivalent inspection methods.

- (2) The percentage of castings inspected by non visual methods may be reduced below that specified in sub-paragraph (d)(1) when an approved quality control procedure is established.
- (3) For castings procured to a specification that guarantees the mechanical properties of the material in the casting and provides for demonstration of these properties by test of coupons cut from the castings on a sampling basis:

- (i) A casting factor of 1.0 may be used; and
- (ii) The castings must be inspected as provided in sub-paragraph (d)(1) for casting factors of '1.25 to 1.50' and tested under sub-paragraph (c)(2).

CS 29.623 Bearing factors

ED Decision 2003/16/RM

- (a) Except as provided in sub-paragraph (b), each part that has clearance (free fit), and that is subject to pounding or vibration, must have a bearing factor large enough to provide for the effects of normal relative motion.
- (b) No bearing factor need be used on a part for which any larger special factor is prescribed.

CS 29.625 Fitting factors

ED Decision 2003/16/RM

For each fitting (part or terminal used to join one structural member to another) the following apply:

- (a) For each fitting whose strength is not proven by limit and ultimate load tests in which actual stress conditions are simulated in the fitting and surrounding structures, a fitting factor of at least 1.15 must be applied to each part of:
 - (1) The fitting;
 - (2) The means of attachment; and
 - (3) The bearing on the joined members.
- (b) No fitting factor need be used:
 - (1) For joints made under approved practices and based on comprehensive test data (such as continuous joints in metal plating, welded joints, and scarf joints in wood); and
 - (2) With respect to any bearing surface for which a larger special factor is used.
- (c) For each integral fitting, the part must be treated as a fitting up to the point at which the section properties become typical of the member.
- (d) Each seat, berth, litter, safety belt, and harness attachment to the structure must be shown by analysis, tests, or both, to be able to withstand the inertia forces prescribed in [CS 29.561\(b\)\(3\)](#) multiplied by a fitting factor of 1.33.

CS 29.629 Flutter and divergence

ED Decision 2003/16/RM

Each aerodynamic surface of the rotorcraft must be free from flutter and divergence under each appropriate speed and power condition.

CS 29.631 Birdstrike

ED Decision 2003/16/RM

The rotorcraft must be designed to assure capability of continued safe flight and landing (for Category A) or safe landing (for Category B) after impact with a 1 kg bird, when the velocity of the rotorcraft (relative to the bird along the flight path of the rotorcraft) is equal to V_{NE} or V_H (whichever is the lesser) at altitudes up to 2438 m (8 000 ft). Compliance must be shown by tests, or by analysis based on tests carried out on sufficiently representative structures of similar design.

ROTORS

CS 29.653 Pressure venting and drainage of rotor blades

ED Decision 2003/16/RM

- (a) For each rotor blade:
 - (1) There must be means for venting the internal pressure of the blade;
 - (2) Drainage holes must be provided for the blade; and
 - (3) The blade must be designed to prevent water from becoming trapped in it.
- (b) Sub-paragraphs (a)(1) and (2) do not apply to sealed rotor blades capable of withstanding the maximum pressure differentials expected in service.

CS 29.659 Mass balance

ED Decision 2003/16/RM

- (a) The rotor and blades must be mass balanced as necessary to:
 - (1) Prevent excessive vibration; and
 - (2) Prevent flutter at any speed up to the maximum forward speed.
- (b) The structural integrity of the mass balance installation must be substantiated.

CS 29.661 Rotor blade clearance

ED Decision 2003/16/RM

There must be enough clearance between the rotor blades and other parts of the structure to prevent the blades from striking any part of the structure during any operating condition.

CS 29.663 Ground resonance prevention means

ED Decision 2003/16/RM

- (a) The reliability of the means for preventing ground resonance must be shown either by analysis and tests, or reliable service experience, or by showing through analysis or tests that malfunction or failure of a single means will not cause ground resonance.
- (b) The probable range of variations, during service, of the damping action of the ground resonance prevention means must be established and must be investigated during the test required by [CS 29.241](#).

CONTROL SYSTEMS

CS 29.671 General

ED Decision 2003/16/RM

- (a) Each control and control system must operate with the ease, smoothness, and positiveness appropriate to its function.
- (b) Each element of each flight control system must be designed, or distinctively and permanently marked, to minimise the probability of any incorrect assembly that could result in the malfunction of the system.
- (c) A means must be provided to allow full control movement of all primary flight controls prior to flight, or a means must be provided that will allow the pilot to determine that full control authority is available prior to flight.

CS 29.672 Stability augmentation, automatic, and power-operated systems

ED Decision 2003/16/RM

If the functioning of stability augmentation or other automatic or power-operated system is necessary to show compliance with flight characteristics requirements of CS-29, the system must comply with [CS 29.671](#) and the following:

- (a) A warning which is clearly distinguishable to the pilot under expected flight conditions without requiring the pilot's attention must be provided for any failure in the stability augmentation system or in any other automatic or power-operated system which could result in an unsafe condition if the pilot is unaware of the failure. Warning systems must not activate the control systems.
- (b) The design of the stability augmentation system or of any other automatic or power-operated system must allow initial counteraction of failures without requiring exceptional pilot skill or strength, by overriding the failure by moving the flight controls in the normal sense, and by deactivating the failed system.
- (c) It must be shown that after any single failure of the stability augmentation system or any other automatic or power-operated system:
 - (1) The rotorcraft is safely controllable when the failure or malfunction occurs at any speed or altitude within the approved operating limitations;
 - (2) The controllability and manoeuvrability requirements of CS-29 are met within a practical operational flight envelope (for example, speed, altitude, normal acceleration, and rotorcraft configurations) which is described in the rotorcraft flight manual; and
 - (3) The trim and stability characteristics are not impaired below a level needed to allow continued safe flight and landing.

CS 29.673 Primary flight controls

ED Decision 2003/16/RM

Primary flight controls are those used by the pilot for immediate control of pitch, roll, yaw, and vertical motion of the rotorcraft.

CS 29.674 Interconnected controls

ED Decision 2003/16/RM

Each primary flight control system must provide for safe flight and landing and operate independently after a malfunction, failure, or jam of any auxiliary interconnected control.

CS 29.675 Stops

ED Decision 2003/16/RM

- (a) Each control system must have stops that positively limit the range of motion of the pilot's controls.
- (b) Each stop must be located in the system so that the range of travel of its control is not appreciably affected by:
 - (1) Wear;
 - (2) Slackness; or
 - (3) Take-up adjustments.
- (c) Each stop must be able to withstand the loads corresponding to the design conditions for the system.
- (d) For each main rotor blade:
 - (1) Stops that are appropriate to the blade design must be provided to limit travel of the blade about its hinge points; and
 - (2) There must be means to keep the blade from hitting the droop stops during any operation other than starting and stopping the rotor.

CS 29.679 Control system locks

ED Decision 2003/16/RM

If there is a device to lock the control system with the rotorcraft on the ground or water, there must be means to:

- (a) Automatically disengage the lock when the pilot operates the controls in a normal manner, or limit the operation of the rotorcraft so as to give unmistakable warning to the pilot before take-off, and
- (b) Prevent the lock from engaging in flight.

CS 29.681 Limit load static tests

ED Decision 2003/16/RM

- (a) Compliance with the limit load requirements of this Code must be shown by tests in which:
 - (1) The direction of the test loads produces the most severe loading in the control system; and
 - (2) Each fitting, pulley, and bracket used in attaching the system to the main structure is included.
- (b) Compliance must be shown (by analyses or individual load tests) with the special factor requirements for control system joints subject to angular motion.

CS 29.683 Operation tests

ED Decision 2003/16/RM

It must be shown by operation tests that, when the controls are operated from the pilot compartment with the control system loaded to correspond with loads specified for the system, the system is free from:

- (a) Jamming;
- (b) Excessive friction; and
- (c) Excessive deflection.

CS 29.685 Control system details

ED Decision 2003/16/RM

- (a) Each detail of each control system must be designed to prevent jamming, chafing, and interference from cargo, passengers, loose objects, or the freezing of moisture.
- (b) There must be means in the cockpit to prevent the entry of foreign objects into places where they would jam the system.
- (c) There must be means to prevent the slapping of cables or tubes against other parts.
- (d) Cable systems must be designed as follows:
 - (1) Cables, cable fittings, turnbuckles, splices, and pulleys must be of an acceptable kind.
 - (2) The design of cable systems must prevent any hazardous change in cable tension throughout the range of travel under any operating conditions and temperature variations.
 - (3) No cable smaller than 3.2 mm (1/8 inch) diameter may be used in any primary control system.
 - (4) Pulley kinds and sizes must correspond to the cables with which they are used.
 - (5) Pulleys must have close fitting guards to prevent the cables from being displaced or fouled.
 - (6) Pulleys must lie close enough to the plane passing through the cable to prevent the cable from rubbing against the pulley flange.
 - (7) No fairlead may cause a change in cable direction of more than 3°.
 - (8) No clevis pin subject to load or motion and retained only by cotter pins may be used in the control system.
 - (9) Turnbuckles attached to parts having angular motion must be installed to prevent binding throughout the range of travel.
 - (10) There must be means for visual inspection at each fairlead, pulley, terminal, and turnbuckle.
- (e) Control system joints subject to angular motion must incorporate the following special factors with respect to the ultimate bearing strength of the softest material used as a bearing:
 - (1) 3.33 for push-pull systems other than ball and roller bearing systems.
 - (2) 2.0 for cable systems.

- (f) For control system joints, the manufacturer's static, non-Brinell rating of ball and roller bearings may not be exceeded.

CS 29.687 Spring devices

ED Decision 2003/16/RM

- (a) Each control system spring device whose failure could cause flutter or other unsafe characteristics must be reliable.
- (b) Compliance with sub-paragraph (a) must be shown by tests simulating service conditions.

CS 29.691 Autorotation control mechanism

ED Decision 2003/16/RM

Each main rotor blade pitch control mechanism must allow rapid entry into autorotation after power failure.

CS 29.695 Power boost and power-operated control system

ED Decision 2003/16/RM

- (a) If a power boost or power-operated control system is used, an alternate system must be immediately available that allows continued safe flight and landing in the event of –
- (1) Any single failure in the power portion of the system; or
 - (2) The failure of all engines.
- (b) Each alternate system may be a duplicate power portion or a manually operated mechanical system. The power portion includes the power source (such as hydraulic pumps), and such items as valves, lines, and actuators.
- (c) The failure of mechanical parts (such as piston rods and links), and the jamming of power cylinders, must be considered unless they are extremely improbable.

LANDING GEAR

CS 29.723 Shock absorption tests

ED Decision 2003/16/RM

The landing inertia load factor and the reserve energy absorption capacity of the landing gear must be substantiated by the tests prescribed in [CS 29.725](#) and [29.727](#), respectively. These tests must be conducted on the complete rotorcraft or on units consisting of wheel, tyre, and shock absorber in their proper relation.

CS 29.725 Limit drop test

ED Decision 2003/16/RM

The limit drop test must be conducted as follows:

- (a) The drop height must be at least 20 cm (8 inches).
- (b) If considered, the rotor lift specified in [CS 29.473\(a\)](#) must be introduced into the drop test by appropriate energy absorbing devices or by the use of an effective mass.
- (c) Each landing gear unit must be tested in the attitude simulating the landing condition that is most critical from the standpoint of the energy to be absorbed by it.
- (d) When an effective mass is used in showing compliance with sub-paragraph (b), the following formulae may be used instead of more rational computations:

$$W_e = W \left(\frac{h + (1 - L)d}{h + d} \right); \text{ and}$$
$$n = n_j \frac{W_e}{W} + L$$

where:

W_e = the effective weight to be used in the drop test (N (lb)).

W = W_M for main gear units (N (lb)), equal to the static reaction on the particular unit with the rotorcraft in the most critical attitude. A rational method may be used in computing a main gear static reaction, taking into consideration the moment arm between the main wheel reaction and the rotorcraft centre of gravity.

W = W_N for nose gear units (N (lb)), equal to the vertical component of the static reaction that would exist at the nose wheel, assuming that the mass of the rotorcraft acts at the centre of gravity and exerts a force of 1.0 g downward and 0.25 g forward.

W = W_T for tailwheel units (N (lb)) equal to whichever of the following is critical:

- (1) The static weight on the tailwheel with the rotorcraft resting on all wheels; or
- (2) The vertical component of the ground reaction that would occur at the tailwheel assuming that the mass of the rotorcraft acts at the centre of gravity and exerts a force of 1 g downward with the rotorcraft in the maximum nose-up attitude considered in the nose-up landing conditions.

h = specified free drop height (m (inches)).

L = ratio of assumed rotor lift to the rotorcraft weight.

- d = deflection under impact of the tyre (at the proper inflation pressure) plus the vertical component of the axle travel (m (inches)) relative to the drop mass.
- n = limit inertia load factor.
- n_j = the load factor developed, during impact, on the mass used in the drop test (i.e., the acceleration dv/dt in g recorded in the drop test plus 1.0).

CS 29.727 Reserve energy absorption drop test

ED Decision 2003/16/RM

The reserve energy absorption drop test must be conducted as follows:

- (a) The drop height must be 1.5 times that specified in [CS 29.725\(a\)](#).
- (b) Rotor lift, where considered in a manner similar to that prescribed in [CS 29.725\(b\)](#), may not exceed 1.5 times the lift allowed under that paragraph.
- (c) The landing gear must withstand this test without collapsing. Collapse of the landing gear occurs when a member of the nose, tail, or main gear will not support the rotorcraft in the proper attitude or allows the rotorcraft structure, other than landing gear and external accessories, to impact the landing surface.

CS 29.729 Retracting mechanism

ED Decision 2003/16/RM

For rotorcraft with retractable landing gear, the following apply:

- (a) *Loads.* The landing gear, retracting mechanism, wheel well doors, and supporting structure must be designed for:
 - (1) The loads occurring in any manoeuvring condition with the gear retracted;
 - (2) The combined friction, inertia, and air loads occurring during retraction and extension at any airspeed up to the design maximum landing gear operating speed; and
 - (3) The flight loads, including those in yawed flight, occurring with the gear extended at any airspeed up to the design maximum landing gear extended speed.
- (b) *Landing gear lock.* A positive means must be provided to keep the gear extended.
- (c) *Emergency operation.* When other than manual power is used to operate the gear, emergency means must be provided for extending the gear in the event of:
 - (1) Any reasonably probable failure in the normal retraction system; or
 - (2) The failure of any single source of hydraulic, electric, or equivalent energy.
- (d) *Operation tests.* The proper functioning of the retracting mechanism must be shown by operation tests.
- (e) *Position indicator.* There must be means to indicate to the pilot when the gear is secured in the extreme positions.
- (f) *Control.* The location and operation of the retraction control must meet the requirements of [CS 29.777](#) and [29.779](#).
- (g) *Landing gear warning.* An aural or equally effective landing gear warning device must be provided that functions continuously when the rotorcraft is in a normal landing mode and the landing gear is not fully extended and locked. A manual shutoff capability must be provided for

the warning device and the warning system must automatically reset when the rotorcraft is no longer in the landing mode.

CS 29.731 Wheels

ED Decision 2003/16/RM

- (a) Each landing gear wheel must be approved.
- (b) The maximum static load rating of each wheel may not be less than the corresponding static ground reaction with:
 - (1) Maximum weight; and
 - (2) Critical centre of gravity.
- (c) The maximum limit load rating of each wheel must equal or exceed the maximum radial limit load determined under the applicable ground load requirements of CS-29.

CS 29.733 Tyres

ED Decision 2003/16/RM

Each landing gear wheel must have a tyre:

- (a) That is a proper fit on the rim of the wheel; and
- (b) Of a rating that is not exceeded under:
 - (1) The design maximum weight;
 - (2) A load on each main wheel tyre equal to the static ground reaction corresponding to the critical centre of gravity; and
 - (3) A load on nose wheel tyres to be compared with the dynamic rating established for those tyres equal to the reaction obtained at the nose wheel, assuming that the mass of the rotorcraft acts as the most critical centre of gravity and exerts a force of 1.0 g downward and 0.25 g forward, the reactions being distributed to the nose and main wheels according to the principles of statics with the drag reaction at the ground applied only at wheels with brakes.
- (c) Each tyre installed on a retractable landing gear system must, at the maximum size of the tyre type expected in service, have a clearance to surrounding structure and systems that is adequate to prevent contact between the tyre and any part of the structure or systems.

CS 29.735 Brakes

ED Decision 2003/16/RM

For rotorcraft with wheel-type landing gear, a braking device must be installed that is:

- (a) Controllable by the pilot;
- (b) Usable during power-off landings; and
- (c) Adequate to:
 - (1) Counteract any normal unbalanced torque when starting or stopping the rotor; and
 - (2) Hold the rotorcraft parked on a 10° slope on a dry, smooth pavement.

CS 29.737 Skis

ED Decision 2003/16/RM

- (a) The maximum limit load rating of each ski must equal or exceed the maximum limit load determined under the applicable ground load requirements of CS-29.
- (b) There must be a stabilising means to maintain the ski in an appropriate position during flight. This means must have enough strength to withstand the maximum aerodynamic and inertia loads on the ski.

FLOATS AND HULLS

CS 29.751 Main float buoyancy

ED Decision 2003/16/RM

- (a) For main floats, the buoyancy necessary to support the maximum weight of the rotorcraft in fresh water must be exceeded by:
 - (1) 50%, for single floats; and
 - (2) 60%, for multiple floats.
- (b) Each main float must have enough watertight compartments so that, with any single main float compartment flooded, the main floats will provide a margin of positive stability great enough to minimise the probability of capsizing.

CS 29.753 Main float design

ED Decision 2003/16/RM

- (a) *Bag floats.* Each bag float must be designed to withstand:
 - (1) The maximum pressure differential that might be developed at the maximum altitude for which certification with the float is requested; and
 - (2) The vertical loads prescribed in [CS 29.521\(a\)](#), distributed along the length of the bag over three-quarters of its projected area.
- (b) *Rigid floats.* Each rigid float must be able to withstand the vertical, horizontal, and side loads prescribed in [CS 29.521](#). An appropriate load distribution under critical conditions must be used.

CS 29.755 Hull buoyancy

ED Decision 2003/16/RM

Water-based and amphibian rotorcraft. The hull and auxiliary floats, if used, must have enough watertight compartments so that, with any single compartment of the hull or auxiliary floats flooded, the buoyancy of the hull and auxiliary floats, and wheel tyres if used, provides a margin of positive water stability great enough to minimise the probability of capsizing the rotorcraft for the worst combination of wave heights and surface winds for which approval is desired.

CS 29.757 Hull and auxiliary float strength

ED Decision 2003/16/RM

The hull, and auxiliary floats if used, must withstand the water loads prescribed by [CS 29.519](#) with a rational and conservative distribution of local and distributed water pressures over the hull and float bottom.

PERSONNEL AND CARGO ACCOMMODATIONS

CS 29.771 Pilot compartment

ED Decision 2003/16/RM

For each pilot compartment:

- (a) The compartment and its equipment must allow each pilot to perform his duties without unreasonable concentration or fatigue;
- (b) If there is provision for a second pilot, the rotorcraft must be controllable with equal safety from either pilot position. Flight and powerplant controls must be designed to prevent confusion or inadvertent operation when the rotorcraft is piloted from either position;
- (c) The vibration and noise characteristics of cockpit appurtenances may not interfere with safe operation;
- (d) Inflight leakage of rain or snow that could distract the crew or harm the structure must be prevented.

CS 29.773 Pilot compartment view

ED Decision 2003/16/RM

- (a) *Non precipitation conditions.* For non precipitation conditions, the following apply:
 - (1) Each pilot compartment must be arranged to give the pilots a sufficiently extensive, clear, and undistorted view for safe operation.
 - (2) Each pilot compartment must be free of glare and reflection that could interfere with the pilot's view. If certification for night operation is requested, this must be shown by night flight tests.
- (b) *Precipitation conditions.* For precipitation conditions, the following apply:
 - (1) Each pilot must have a sufficiently extensive view for safe operation:
 - (i) In heavy rain at forward speeds up to V_H ; and
 - (ii) In the most severe icing condition for which certification is requested.
 - (2) The first pilot must have a window that:
 - (i) Is openable under the conditions prescribed in sub-paragraph (b)(1); and
 - (ii) Provides the view prescribed in that paragraph.

CS 29.775 Windshields and windows

ED Decision 2003/16/RM

Windshields and windows must be made of material that will not break into dangerous fragments.

CS 29.777 Cockpit controls

ED Decision 2003/16/RM

Cockpit controls must be:

- (a) Located to provide convenient operation and to prevent confusion and inadvertent operation; and
- (b) Located and arranged with respect to the pilot's seats so that there is full and unrestricted movement of each control without interference from the cockpit structure or the pilot's clothing when pilots from 1.57 m (5ft 2ins) to 1.8 m (6ft) in height are seated.

CS 29.779 Motion and effect of cockpit controls

ED Decision 2003/16/RM

Cockpit controls must be designed so that they operate in accordance with the following movements and actuation:

- (a) Flight controls, including the collective pitch control, must operate with a sense of motion which corresponds to the effect on the rotorcraft.
- (b) Twist-grip engine power controls must be designed so that, for left-hand operation, the motion of the pilot's hand is clockwise to increase power when the hand is viewed from the edge containing the index finger. Other engine power controls, excluding the collective control, must operate with a forward motion to increase power.
- (c) Normal landing gear controls must operate downward to extend the landing gear.

CS 29.783 Doors

ED Decision 2003/16/RM

- (a) Each closed cabin must have at least one adequate and easily accessible external door.
- (b) Each external door must be located, and appropriate operating procedures must be established, to ensure that persons using the door will not be endangered by the rotors, propellers, engine intakes, and exhausts when the operating procedures are used.
- (c) There must be means for locking crew and external passenger doors and for preventing their opening in flight inadvertently or as a result of mechanical failure. It must be possible to open external doors from inside and outside the cabin with the rotorcraft on the ground even though persons may be crowded against the door on the inside of the rotorcraft. The means of opening must be simple and obvious and so arranged and marked that it can be readily located and operated.
- (d) There must be reasonable provisions to prevent the jamming of any external door in a minor crash as a result of fuselage deformation under the following ultimate inertial forces except for cargo or service doors not suitable for use as an exit in an emergency:
 - (1) Upward – 1.5 g
 - (2) Forward – 4.0 g
 - (3) Sideward – 2.0 g
 - (4) Downward – 4.0 g

- (e) There must be means for direct visual inspection of the locking mechanism by crew members to determine whether the external doors (including passenger, crew, service, and cargo doors) are fully locked. There must be visual means to signal to appropriate crew members when normally used external doors are closed and fully locked.
- (f) For outward opening external doors usable for entrance or egress, there must be an auxiliary safety latching device to prevent the door from opening when the primary latching mechanism fails. If the door does not meet the requirements of sub-paragraph (c) with this device in place, suitable operating procedures must be established to prevent the use of the device during take-off and landing.
- (g) If an integral stair is installed in a passenger entry door that is qualified as a passenger emergency exit, the stair must be designed so that under the following conditions the effectiveness of passenger emergency egress will not be impaired:
 - (1) The door, integral stair, and operating mechanism have been subjected to the inertial forces specified in sub-paragraph (d), acting separately relative to the surrounding structure.
 - (2) The rotorcraft is in the normal ground attitude and in each of the attitudes corresponding to collapse of one or more legs, or primary members, as applicable, of the landing gear.
- (h) Non jettisonable doors used as ditching emergency exits must have means to enable them to be secured in the open position and remain secure for emergency egress in sea state conditions prescribed for ditching.

CS 29.785 Seats, berths, safety belts, and harnesses

ED Decision 2003/16/RM

- (a) Each seat, safety belt, harness, and adjacent part of the rotorcraft at each station designated for occupancy during take-off and landing must be free of potentially injurious objects, sharp edges, protuberances, and hard surfaces and must be designed so that a person making proper use of these facilities will not suffer serious injury in an emergency landing as a result of the inertial factors specified in [CS 29.561\(b\)](#) and dynamic conditions specified in [CS 29.562](#).
- (b) Each occupant must be protected from serious head injury by a safety belt plus a shoulder harness that will prevent the head from contacting any injurious object except as provided for in [CS 29.562\(c\)\(5\)](#). A shoulder harness (upper torso restraint), in combination with the safety belt, constitutes a torso restraint system as described in ETSO-C114.
- (c) Each occupant's seat must have a combined safety belt and shoulder harness with a single-point release. Each pilot's combined safety belt and shoulder harness must allow each pilot when seated with safety belt and shoulder harness fastened to perform all functions necessary for flight operations. There must be a means to secure belts and harnesses, when not in use, to prevent interference with the operation of the rotorcraft and with rapid egress in an emergency.
- (d) If seat backs do not have a firm handhold, there must be hand grips or rails along each aisle to let the occupants steady themselves while using the aisle in moderately rough air.
- (e) Each projecting object that would injure persons seated or moving about in the rotorcraft in normal flight must be padded.
- (f) Each seat and its supporting structure must be designed for an occupant weight of at least 77 kg (170 pounds) considering the maximum load factors, inertial forces, and reactions between the

occupant, seat, and safety belt or harness corresponding with the applicable flight and ground load conditions, including the emergency landing conditions of [CS 29.561\(b\)](#). In addition:

- (1) Each pilot seat must be designed for the reactions resulting from the application of the pilot forces prescribed in [CS 29.397](#); and
 - (2) The inertial forces prescribed in [CS 29.561\(b\)](#) must be multiplied by a factor of 1.33 in determining the strength of the attachment of:
 - (i) Each seat to the structure; and
 - (ii) Each safety belt or harness to the seat or structure.
- (g) When the safety belt and shoulder harness are combined, the rated strength of the safety belt and shoulder harness may not be less than that corresponding to the inertial forces specified in [CS 29.561\(b\)](#), considering the occupant weight of at least 77 kg (170 pounds), considering the dimensional characteristics of the restraint system installation, and using a distribution of at least a 60% load to the safety belt and at least a 40% load to the shoulder harness. If the safety belt is capable of being used without the shoulder harness, the inertial forces specified must be met by the safety belt alone.
- (h) When a headrest is used, the headrest and its supporting structure must be designed to resist the inertia forces specified in [CS 29.561](#), with a 1.33 fitting factor and a head weight of at least 5.9 kg (13 pounds).
- (i) Each seating device system includes the device such as the seat, the cushions, the occupant restraint system, and attachment devices.
- (j) Each seating device system may use design features such as crushing or separation of certain parts of the seat in the design to reduce occupant loads for the emergency landing dynamic conditions of [CS 29.562](#); otherwise, the system must remain intact and must not interfere with rapid evacuation of the rotorcraft.
- (k) For the purposes of this paragraph, a litter is defined as a device designed to carry a non ambulatory person, primarily in a recumbent position, into and on the rotorcraft. Each berth or litter must be designed to withstand the load reaction of an occupant weight of at least 77 kg (170 pounds) when the occupant is subjected to the forward inertial factors specified in [CS 29.561\(b\)](#). A berth or litter installed within 15° or less of the longitudinal axis of the rotorcraft must be provided with a padded end- board, cloth diaphragm, or equivalent means that can withstand the forward load reaction. A berth or litter oriented greater than 15° with the longitudinal axis of the rotorcraft must be equipped with appropriate restraints, such as straps or safety belts, to withstand the forward reaction. In addition:
- (1) The berth or litter must have a restraint system and must not have corners or other protuberances likely to cause serious injury to a person occupying it during emergency landing conditions; and
 - (2) The berth or litter attachment and the occupant restraint system attachments to the structure must be designed to withstand the critical loads resulting from flight and ground load conditions and from the conditions prescribed in [CS 29.561\(b\)](#). The fitting factor required by [CS 29.625\(d\)](#) shall be applied.

CS 29.787 Cargo and baggage compartments

ED Decision 2003/16/RM

- (a) Each cargo and baggage compartment must be designed for its placarded maximum weight of contents and for the critical load distributions at the appropriate maximum load factors corresponding to the specified flight and ground load conditions, except the emergency landing conditions of [CS 29.561](#).
- (b) There must be means to prevent the contents of any compartment from becoming a hazard by shifting under the loads specified in subparagraph (a).
- (c) Under the emergency landing conditions of [CS 29.561](#), cargo and baggage compartments must:
 - (1) Be positioned so that if the contents break loose they are unlikely to cause injury to the occupants or restrict any of the escape facilities provided for use after an emergency landing; or
 - (2) Have sufficient strength to withstand the conditions specified in [CS 29.561](#), including the means of restraint and their attachments required by sub-paragraph (b). Sufficient strength must be provided for the maximum authorised weight of cargo and baggage at the critical loading distribution.
- (d) If cargo compartment lamps are installed, each lamp must be installed so as to prevent contact between lamp bulb and cargo.

CS 29.801 Ditching

ED Decision 2003/16/RM

- (a) If certification with ditching provisions is requested, the rotorcraft must meet the requirements of this paragraph and [CS 29.807\(d\)](#), [29.1411](#) and [29.1415](#).
- (b) Each practicable design measure, compatible with the general characteristics of the rotorcraft, must be taken to minimise the probability that in an emergency landing on water, the behaviour of the rotorcraft would cause immediate injury to the occupants or would make it impossible for them to escape.
- (c) The probable behaviour of the rotorcraft in a water landing must be investigated by model tests or by comparison with rotorcraft of similar configuration for which the ditching characteristics are known. Scoops, flaps, projections, and any other factors likely to affect the hydrodynamic characteristics of the rotorcraft must be considered.
- (d) It must be shown that, under reasonably probable water conditions, the flotation time and trim of the rotorcraft will allow the occupants to leave the rotorcraft and enter the life rafts required by [CS 29.1415](#). If compliance with this provision is shown by buoyancy and trim computations, appropriate allowances must be made for probable structural damage and leakage. If the rotorcraft has fuel tanks (with fuel jettisoning provisions) that can reasonably be expected to withstand a ditching without leakage, the jettisonable volume of fuel may be considered as buoyancy volume.
- (e) Unless the effects of the collapse of external doors and windows are accounted for in the investigation of the probable behaviour of the rotorcraft in a water landing (as prescribed in sub-paragraphs (c) and (d)), the external doors and windows must be designed to withstand the probable maximum local pressures.

CS 29.803 Emergency evacuation

ED Decision 2003/16/RM

- (a) Each crew and passenger area must have means for rapid evacuation in a crash landing, with the landing gear:
 - (1) extended; and
 - (2) retracted;considering the possibility of fire.
- (b) Passenger entrance, crew, and service doors may be considered as emergency exits if they meet the requirements of this paragraph and of [CS 29.805](#) to [29.815](#).
- (c) Reserved.
- (d) Except as provided in sub-paragraph (e), the following categories of rotorcraft must be tested in accordance with the requirements of [Appendix D](#) to demonstrate that the maximum seating capacity, including the crew-members required by the operating rules, can be evacuated from the rotorcraft to the ground within 90 seconds:
 - (1) Rotorcraft with a seating capacity of more than 44 passengers.
 - (2) Rotorcraft with all of the following:
 - (i) Ten or more passengers per passenger exit as determined under [CS 29.807\(b\)](#).
 - (ii) No main aisle, as described in [CS 29.815](#), for each row of passenger seats.
 - (iii) Access to each passenger exit for each passenger by virtue of design features of seats, such as folding or break- over seat backs or folding seats.
- (e) A combination of analysis and tests may be used to show that the rotorcraft is capable of being evacuated within 90 seconds under the conditions specified in [CS 29.803\(d\)](#) if the Agency finds that the combination of analysis and tests will provide data, with respect to the emergency evacuation capability of the rotorcraft, equivalent to that which would be obtained by actual demonstration.

Appendix D – Criteria for demonstration of emergency evacuation procedures under CS 29.803

ED Decision 2003/16/RM

- (a) The demonstration must be conducted either during the dark of the night or during daylight with the dark of night simulated. If the demonstration is conducted indoors during daylight hours, it must be conducted inside a darkened hangar having doors and windows covered. In addition, the doors and windows of the rotorcraft must be covered if the hangar illumination exceeds that of a moonless night. Illumination on the floor or ground may be used, but it must be kept low and shielded against shining into the rotorcraft's windows or doors.
- (b) The rotorcraft must be in a normal attitude with landing gear extended.
- (c) Safety equipment such as mats or inverted liferafts may be placed on the floor or ground to protect participants. No other equipment that is not part of the rotorcraft's emergency evacuation equipment may be used to aid the participants in reaching the ground.
- (d) Except as provided in paragraph (a), only the rotorcraft's emergency lighting system may provide illumination.

- (e) All emergency equipment required for the planned operation of the rotorcraft must be installed.
- (f) Each external door and exit and each internal door or curtain must be in the take-off configuration.
- (g) Each crewmember must be seated in the normally assigned seat for take-off and must remain in that seat until receiving the signal for commencement of the demonstration. For compliance with this paragraph, each crewmember must be:
 - (1) A member of a regularly scheduled line crew; or
 - (2) A person having knowledge of the operation of exits and emergency equipment.
- (h) A representative passenger load of persons in normal health must be used as follows:
 - (1) At least 25% must be over 50 years of age, with at least 40% of these being females.
 - (2) The remaining 75% or less, must be 50 years of age or younger, with at least 30% of these being females.
 - (3) Three life-size dolls, not included as part of the total passenger load, must be carried by passengers to simulate live infants 2 years old or younger, except for a total passenger load of fewer than 44 but more than 19, one doll must be carried. A doll is not required for a 19 or fewer passenger load.
 - (4) Crewmembers, mechanics, and training personnel who maintain or operate the rotorcraft in the normal course of their duties may not be used as passengers.
- (i) No passenger may be assigned a specific seat except as the Agency may require. Except as required by paragraph (g), no employee of the applicant may be seated next to an emergency exit, except as allowed by the Agency.
- (j) Seat belts and shoulder harnesses (as required) must be fastened.
- (k) Before the start of the demonstration, approximately one-half of the total average amount of carry-on baggage, blankets, pillows and other similar articles must be distributed at several locations in the aisles and emergency exit access ways to create minor obstructions.
- (l) No prior indication may be given to any crewmember or passenger of the particular exits to be used in the demonstration.
- (m) There must not be any practising, rehearsing or description of the demonstration for the participants nor may any participant have taken part in this type of demonstration within the preceding 6 months.
- (n) A pre-take-off passenger briefing may be given. The passengers may also be advised to follow directions of crewmembers, but not be instructed on the procedures to be followed in the demonstration.
- (o) If safety equipment, as allowed by paragraph (c), is provided, either all passenger and cockpit windows must be blacked out or all emergency exits must have safety equipment to prevent disclosure of the available emergency exits.
- (p) Not more than 50% of the emergency exits in the sides of the fuselage of a rotorcraft that meet all of the requirements applicable to the required emergency exits for that rotorcraft may be used for demonstration. Exits that are not to be used for the demonstration must have the exit handle deactivated or must be indicated by red lights, red tape, or other acceptable means placed outside the exits to indicate fire or other reasons why they are unusable. The exits to be

used must be representative of all the emergency exits on the rotorcraft and must be designated subject to approval by the Agency. If installed, at least one floor level exit (Type I; [CS 29.807\(a\)\(1\)](#)) must be used as required by [CS 29.807\(c\)](#).

- (q) All evacuees must leave the rotorcraft by a means provided as part of the rotorcraft's equipment.
- (r) Approved procedures must be fully utilised during the demonstration.
- (s) The evacuation time period is completed when the last occupant has evacuated the rotorcraft and is on the ground.

CS 29.805 Flight crew emergency exits

ED Decision 2003/16/RM

- (a) For rotorcraft with passenger emergency exits that are not convenient to the flight crew, there must be flight crew emergency exits, on both sides of the rotorcraft or as a top hatch, in the flight crew area.
- (b) Each flight crew emergency exit must be of sufficient size and must be located so as to allow rapid evacuation of the flight crew. This must be shown by test.
- (c) Each exit must not be obstructed by water or flotation devices after a ditching. This must be shown by test, demonstration, or analysis.

CS 29.807 Passenger emergency exits

ED Decision 2003/16/RM

- (a) *Type.* For the purpose of this CS-29, the types of passenger emergency exit are as follows:
 - (1) *Type I.* This type must have a rectangular opening of not less than 0.61 m wide by 1.22 m (24 inches wide by 48 inches) high, with corner radii not greater than one-third the width of the exit, in the passenger area in the side of the fuselage at floor level and as far away as practicable from areas that might become potential fire hazards in a crash.
 - (2) *Type II.* This type is the same as Type I, except that the opening must be at least 0.51 m wide by 1.12 m (20 inches wide by 44 inches) high.
 - (3) *Type III.* This type is the same as Type I, except that:
 - (i) The opening must be at least 0.51 m wide by 0.91 m (20 inches wide by 36 inches) high; and
 - (ii) The exits need not be at floor level.
 - (4) *Type IV.* This type must have a rectangular opening of not less than 0.48 m wide by 0.66 m (19 inches wide by 26 inches) high, with corner radii not greater than one-third the width of the exit, in the side of the fuselage with a step-up inside the rotorcraft of not more than 0.74 m (29 inches).

Openings with dimensions larger than those specified in this paragraph may be used, regardless of shape, if the base of the opening has a flat surface of not less than the specified width.

- (b) *Passenger emergency exits: side-of fuselage.* Emergency exits must be accessible to the passengers and, except as provided in sub-paragraph (d), must be provided in accordance with the following table:

Passenger seating capacity	Emergency exits for each side of the fuselage			
	(Type I)	(Type II)	(Type III)	(Type IV)
1 to 10				1
11 to 19			1 or	2
20 to 39		1		1
40 to 59	1			1
60 to 79	1		1 or	2

- (c) *Passenger emergency exits; other than side of-fuselage.* In addition to the requirements of subparagraph (b):
- (1) There must be enough openings in the top, bottom, or ends of the fuselage to allow evacuation with the rotorcraft on its side; or
 - (2) The probability of the rotorcraft coming to rest on its side in a crash landing must be extremely remote.
- (d) *Ditching emergency exits for passengers.* If certification with ditching provisions is requested, ditching emergency exits must be provided in accordance with the following requirements and must be proven by test, demonstration, or analysis unless the emergency exits required by subparagraph (b) already meet these requirements:
- (1) For rotorcraft that have a passenger seating configuration, excluding pilots seats, of nine seats or less, one exit above the waterline in each side of the rotorcraft, meeting at least the dimensions of a Type IV exit.
 - (2) For rotorcraft that have a passenger seating configuration, excluding pilots seats, of 10 seats or more, one exit above the waterline in a side of the rotorcraft meeting at least the dimensions of a Type III exit, for each unit (or part of a unit) of 35 passenger seats, but no less than two such exits in the passenger cabin, with one on each side of the rotorcraft. However, where it has been shown through analysis, ditching demonstrations, or any other tests found necessary by the Agency, that the evacuation capability of the rotorcraft during ditching is improved by the use of larger exits, or by other means, the passenger seat to exit ratio may be increased.
 - (3) Flotation devices, whether stowed or deployed, may not interfere with or obstruct the exits.
- (e) *Ramp exits.* One Type I exit only, or one Type II exit only, that is required in the side of the fuselage under sub-paragraph (b), may be installed instead in the ramp of floor ramp rotorcraft if:
- (1) Its installation in the side of the fuselage is impractical; and
 - (2) Its installation in the ramp meets [CS 29.813](#).
- (f) *Tests.* The proper functioning of each emergency exit must be shown by test.

CS 29.809 Emergency exit arrangement

ED Decision 2003/16/RM

- (a) Each emergency exit must consist of a movable door or hatch in the external walls of the fuselage and must provide an unobstructed opening to the outside.
- (b) Each emergency exit must be openable from the inside and from the outside.
- (c) The means of opening each emergency exit must be simple and obvious and may not require exceptional effort.
- (d) There must be means for locking each emergency exit and for preventing opening in flight inadvertently or as a result of mechanical failure.
- (e) There must be means to minimise the probability of the jamming of any emergency exit in a minor crash landing as a result of fuselage deformation under the ultimate inertial forces in [CS 29.783\(d\)](#).
- (f) Except as provided in sub-paragraph (h), each land-based rotorcraft emergency exit must have an approved slide as stated in sub-paragraph (g), or its equivalent, to assist occupants in descending to the ground from each floor level exit and an approved rope, or its equivalent, for all other exits, if the exit threshold is more than 1.8 m (6 ft) above the ground:
 - (1) With the rotorcraft on the ground and with the landing gear extended;
 - (2) With one or more legs or part of the landing gear collapsed, broken, or not extended; and
 - (3) With the rotorcraft resting on its side, if required by [CS 29.803\(d\)](#).
- (g) The slide for each passenger emergency exit must be a self-supporting slide or equivalent, and must be designed to meet the following requirements:
 - (1) It must be automatically deployed, and deployment must begin during the interval between the time the exit opening means is actuated from inside the rotorcraft and the time the exit is fully opened. However, each passenger emergency exit which is also a passenger entrance door or a service door must be provided with means to prevent deployment of the slide when the exit is opened from either the inside or the outside under non-emergency conditions for normal use.
 - (2) It must be automatically erected within 10 seconds after deployment is begun.
 - (3) It must be of such length after full deployment that the lower end is self-supporting on the ground and provides safe evacuation of occupants to the ground after collapse of one or more legs or part of the landing gear.
 - (4) It must have the capability, in 12.9 m/s (25-knot) winds directed from the most critical angle, to deploy and, with the assistance of only one person, to remain usable after full deployment to evacuate occupants safely to the ground.
 - (5) Each slide installation must be qualified by five consecutive deployment and inflation tests conducted (per exit) without failure, and at least three tests of each such five-test series must be conducted using a single representative sample of the device. The sample devices must be deployed and inflated by the system's primary means after being subjected to the inertia forces specified in [CS 29.561\(b\)](#). If any part of the system fails or does not function properly during the required tests, the cause of the failure or malfunction must be corrected by positive means and after that, the full series of five consecutive deployment and inflation tests must be conducted without failure.

- (h) For rotorcraft having 30 or fewer passenger seats and having an exit threshold more than 1.8 m (6 ft) above the ground, a rope or other assist means may be used in place of the slide specified in sub-paragraph (f), provided an evacuation demonstration is accomplished as prescribed in [CS 29.803\(d\) or \(e\)](#).
- (i) If a rope, with its attachment, is used for compliance with sub-paragraph (f), (g) or (h), it must -
 - (1) Withstand a 182 kg (400-pound) static load; and
 - (2) Attach to the fuselage structure at or above the top of the emergency exit opening, or at another approved location if the stowed rope would reduce the pilot's view in flight.

CS 29.811 Emergency exit marking

ED Decision 2003/16/RM

- (a) Each passenger emergency exit, its means of access, and its means of opening must be conspicuously marked for the guidance of occupants using the exits in daylight or in the dark. Such markings must be designed to remain visible for rotorcraft equipped for overwater flights if the rotorcraft is capsized and the cabin is submerged.
- (b) The identity and location of each passenger emergency exit must be recognisable from a distance equal to the width of the cabin.
- (c) The location of each passenger emergency exit must be indicated by a sign visible to occupants approaching along the main passenger aisle. There must be a locating sign:
 - (1) Next to or above the aisle near each floor emergency exit, except that one sign may serve two exits if both exits can be seen readily from that sign; and
 - (2) On each bulkhead or divider that prevents fore and aft vision along the passenger cabin, to indicate emergency exits beyond and obscured by it, except that if this is not possible the sign may be placed at another appropriate location.
- (d) Each passenger emergency exit marking and each locating sign must have white letters 25 mm (1 inch) high on a red background 51 mm (2 inches) high, be self or electrically illuminated, and have a minimum luminescence (brightness) of at least 0.51 candela/m² (160 microlamberts). The colours may be reversed if this will increase the emergency illumination of the passenger compartment.
- (e) The location of each passenger emergency exit operating handle and instructions for opening must be shown:
 - (1) For each emergency exit, by a marking on or near the exit that is readable from a distance of 0.76 m (30 inches); and
 - (2) For each Type I or Type II emergency exit with a locking mechanism released by rotary motion of the handle, by:
 - (i) A red arrow, with a shaft at least 19 mm (¾ inch) wide and a head twice the width of the shaft, extending along at least 70° of arc at a radius approximately equal to three-fourths of the handle length; and
 - (ii) The word 'open' in red letters 25 mm (1 inch) high, placed horizontally near the head of the arrow.
- (f) Each emergency exit, and its means of opening, must be marked on the outside of the rotorcraft. In addition, the following apply:

- (1) There must be a 51 mm (2-inch) coloured band outlining each passenger emergency exit, except small rotorcraft with a maximum weight of 5 670 kg (12 500 pounds) or less may have a 51 mm (2-inch) coloured band outlining each exit release lever or device of passenger emergency exits which are normally used doors.
 - (2) Each outside marking, including the band, must have colour contrast to be readily distinguishable from the surrounding fuselage surface. The contrast must be such that, if the reflectance of the darker colour is 15% or less, the reflectance of the lighter colour must be at least 45%. 'Reflectance' is the ratio of the luminous flux reflected by a body to the luminous flux it receives. When the reflectance of the darker colour is greater than 15%, at least a 30% difference between its reflectance and the reflectance of the lighter colour must be provided.
- (g) Exits marked as such, though in excess of the required number of exits, must meet the requirements for emergency exits of the particular type. Emergency exits need only be marked with the word 'Exit'.

CS 29.812 Emergency lighting

ED Decision 2003/16/RM

For transport Category A rotorcraft, the following apply:

- (a) A source of light with its power supply independent of the main lighting system must be installed to:
 - (1) Illuminate each passenger emergency exit marking and locating sign; and
 - (2) Provide enough general lighting in the passenger cabin so that the average illumination, when measured at 1.02 m (40-inch) intervals at seat armrest height on the centre line of the main passenger aisle, is at least 0.5 lux (0.05 foot-candle).
- (b) Exterior emergency lighting must be provided at each emergency exit. The illumination may not be less than 0.5 lux (0.05 foot-candle) (measured normal to the direction of incident light) for minimum width on the ground surface, with landing gear extended, equal to the width of the emergency exit where an evacuee is likely to make first contact with the ground outside the cabin. The exterior emergency lighting may be provided by either interior or exterior sources with light intensity measurements made with the emergency exits open.
- (c) Each light required by sub-paragraph (a) or (b) must be operable manually from the cockpit station and from a point in the passenger compartment that is readily accessible. The cockpit control device must have an 'on', 'off', and 'armed' position so that when turned on at the cockpit or passenger compartment station or when armed at the cockpit station, the emergency lights will either illuminate or remain illuminated upon interruption of the rotorcraft's normal electric power.
- (d) Any means required to assist the occupants in descending to the ground must be illuminated so that the erected assist means is visible from the rotorcraft.
 - (1) The assist means must be provided with an illumination of not less than 0.3 lux (0.03 foot-candle) (measured normal to the direction of the incident light) at the ground end of the erected assist means where an evacuee using the established escape route would normally make first contact with the ground, with the rotorcraft in each of the attitudes corresponding to the collapse of one or more legs of the landing gear.

- (2) If the emergency lighting subsystem illuminating the assist means is independent of the rotorcraft's main emergency lighting system, it:
 - (i) Must automatically be activated when the assist means is erected;
 - (ii) Must provide the illumination required by sub-paragraph (d)(1); and
 - (iii) May not be adversely affected by stowage.
- (e) The energy supply to each emergency lighting unit must provide the required level of illumination for at least 10 minutes at the critical ambient conditions after an emergency landing.
- (f) If storage batteries are used as the energy supply for the emergency lighting system, they may be recharged from the rotorcraft's main electrical power system provided the charging circuit is designed to preclude inadvertent battery discharge into charging circuit faults.

CS 29.813 Emergency exit access

ED Decision 2003/16/RM

- (a) Each passageway between passenger compartments, and each passageway leading to Type I and Type II emergency exits, must be:
 - (1) Unobstructed; and
 - (2) At least 0.51 m (20 inches) wide.
- (b) For each emergency exit covered by [CS 29.809\(f\)](#), there must be enough space adjacent to that exit to allow a crew member to assist in the evacuation of passengers without reducing the unobstructed width of the passageway below that required for that exit.
- (c) There must be access from each aisle to each Type III and Type IV exit; and
 - (1) For rotorcraft that have a passenger seating configuration, excluding pilot seats, of 20 or more, the projected opening of the exit provided must not be obstructed by seats, berths, or other protrusions (including seatbacks in any position) for a distance from that exit of not less than the width of the narrowest passenger seat installed on the rotorcraft;
 - (2) For rotorcraft that have a passenger seating configuration, excluding pilot seats, of 19 or less, there may be minor obstructions in the region described in sub-paragraph (1), if there are compensating factors to maintain the effectiveness of the exit.

CS 29.815 Main aisle width

ED Decision 2003/16/RM

The main passenger aisle width between seats must equal or exceed the values in the following table:

	Minimum main passenger aisle width	
	Less than 0.64 m (25 in) from floor m (in)	0.64 m (25 in) and more from floor m (in)
10 or less	0.30 (12)*	0.38 (15)
11 to 19	0.30 (12)	0.51 (20)
20 or more	0.38 (15)	0.51 (20)

* A narrower width not less than 0.23 m (9 inches) may be approved when substantiated by tests found necessary by the Agency.

CS 29.831 Ventilation

ED Decision 2003/16/RM

- (a) Each passenger and crew compartment must be ventilated, and each crew compartment must have enough fresh air (but not less than 0.3 m³ (10 cu ft) per minute per crew member) to let crew members perform their duties without undue discomfort or fatigue.
- (b) Crew and passenger compartment air must be free from harmful or hazardous concentrations of gases or vapours.
- (c) The concentration of carbon monoxide may not exceed one part in 20 000 parts of air during forward flight. If the concentration exceeds this value under other conditions, there must be suitable operating restrictions.
- (d) There must be means to ensure compliance with sub-paragraphs (b) and (c) under any reasonably probable failure of any ventilating, heating, or other system or equipment.

CS 29.833 Heaters

ED Decision 2003/16/RM

Each combustion heater must be approved.

FIRE PROTECTION

CS 29.851 Fire extinguishers

ED Decision 2003/16/RM

- (a) *Hand fire extinguishers.* For hand fire extinguishers the following apply:
- (1) Each hand fire extinguisher must be approved.
 - (2) The kinds and quantities of each extinguishing agent used must be appropriate to the kinds of fires likely to occur where that agent is used.
 - (3) Each extinguisher for use in a personnel compartment must be designed to minimise the hazard of toxic gas concentrations.
- (b) *Built-in fire extinguishers.* If a built-in fire extinguishing system is required:
- (1) The capacity of each system, in relation to the volume of the compartment where used and the ventilation rate, must be adequate for any fire likely to occur in that compartment.
 - (2) Each system must be installed so that:
 - (i) No extinguishing agent likely to enter personnel compartments will be present in a quantity that is hazardous to the occupants; and
 - (ii) No discharge of the extinguisher can cause structural damage.

AMC 29.851 Fire extinguishers

ED Decision 2012/022/R

Based on EU legislation¹, in new installations of hand fire extinguishers for which the certification application is submitted after 31 December 2014, Halon 1211, 1301 and Halon 2402 are unacceptable extinguishing agents.

The guidance regarding hand fire extinguishers in FAA Advisory Circular AC 20-42D is considered acceptable by the Agency. See [AMC 29.1197](#) for more information on Halon alternatives.

[Amdt 29/3]

CS 29.853 Compartment interiors

ED Decision 2003/16/RM

For each compartment to be used by the crew or passengers:

- (a) The materials (including finishes or decorative surfaces applied to the materials) must meet the following test criteria as applicable:
- (1) Interior ceiling panels, interior wall panels, partitions, galley structure, large cabinet walls, structural flooring, and materials used in the construction of stowage compartments (other than underseat stowage compartments and compartments for stowing small items such as magazines and maps) must be self-extinguishing when tested vertically in accordance with the applicable portions of Appendix F of CS-25, or other approved

¹ Commission Regulation (EU) No 744/2010 of 18 August 2010 amending Regulation (EC) No 1005/2009 of the European Parliament and of the Council on substances that deplete the ozone layer, with regard to the critical uses of halon (OJ L 218, 19.8.2010, p. 2).

equivalent methods. The average burn length may not exceed 0.15 m (6 in) and the average flame time after removal of the flame source may not exceed 15 seconds. Drippings from the test specimen may not continue to flame for more than an average of 3 seconds after falling.

- (2) Floor covering, textiles (including draperies and upholstery), seat cushions, padding, decorative and non-decorative coated fabrics, leather, trays and galley furnishings, electrical conduit, thermal and acoustical insulation and insulation covering, air ducting, joint and edge covering, cargo compartment liners, insulation blankets, cargo covers, and transparencies, moulded and thermoformed parts, air ducting joints, and trim strips (decorative and chafing) that are constructed of materials not covered in sub-paragraph (a)(3), must be self-extinguishing when tested vertically in accordance with the applicable portion of Appendix F of CS-25, or other approved equivalent methods. The average burn length may not exceed 0.20 m (8 in) and the average flame time after removal of the flame source may not exceed 15 seconds. Drippings from the test specimen may not continue to flame for more than an average of 5 seconds after falling.
 - (3) Acrylic windows and signs, parts constructed in whole or in part of elastomeric materials, edge lighted instrument assemblies consisting of two or more instruments in a common housing, seat belts, shoulder harnesses, and cargo and baggage tiedown equipment, including containers, bins, pallets, etc., used in passenger or crew compartments, may not have an average burn rate greater than 64 mm (2.5 in) per minute when tested horizontally in accordance with the applicable portions of Appendix F of CS-25, or other approved equivalent methods.
 - (4) Except for electrical wire and cable insulation, and for small parts (such as knobs, handles, rollers, fasteners, clips, grommets, rub strips, pulleys, and small electrical parts) that the Agency finds would not contribute significantly to the propagation of a fire, materials in items not specified in sub-paragraphs (a)(1), (a)(2), or (a)(3) may not have a burn rate greater than 0.10 m (4 in) per minute when tested horizontally in accordance with the applicable portions of Appendix F of CS-25, or other approved equivalent methods.
- (b) In addition to meeting the requirements of sub-paragraph (a)(2), seat cushions, except those on flight-crew member seats, must meet the test requirements of Part II of Appendix F of CS-25, or equivalent.
- (c) If smoking is to be prohibited, there must be a placard so stating, and if smoking is to be allowed:
- (1) There must be an adequate number of self-contained, removable ashtrays; and
 - (2) Where the crew compartment is separated from the passenger compartment, there must be at least one illuminated sign (using either letters or symbols) notifying all passengers when smoking is prohibited. Signs which notify when smoking is prohibited must:
 - (i) When illuminated, be legible to each passenger seated in the passenger cabin under all probable lighting conditions; and
 - (ii) Be so constructed that the crew can turn the illumination on and off.
- (d) Each receptacle for towels, paper, or waste must be at least fire-resistant and must have means for containing possible fires;
- (e) There must be a hand fire extinguisher for the flight-crew members; and

- (f) At least the following number of hand fire extinguishers must be conveniently located in passenger compartments:

Passenger capacity	Fire extinguishers
7 to 30	1
31 to 60	2
61 or more	3

CS 29.855 Cargo and baggage compartments

ED Decision 2003/16/RM

- (a) Each cargo and baggage compartment must be constructed of, or lined with, materials in accordance with the following:
- (1) For accessible and inaccessible compartments not occupied by passengers or crew, the material must be at least fire-resistant.
 - (2) Materials must meet the requirements in [CS 29.853\(a\)\(1\), \(a\)\(2\), and \(a\)\(3\)](#) for cargo or baggage compartments in which:
 - (i) The presence of a compartment fire would be easily discovered by a crew member while at the crew member's station;
 - (ii) Each part of the compartment is easily accessible in flight;
 - (iii) The compartment has a volume of 5.6 m³ (200 cu ft) or less; and
 - (iv) Notwithstanding [CS 29.1439\(a\)](#), protective breathing equipment is not required.
- (b) No compartment may contain any controls, wiring, lines, equipment, or accessories whose damage or failure would affect safe operation, unless those items are protected so that:
- (1) They cannot be damaged by the movement of cargo in the compartment; and
 - (2) Their breakage or failure will not create a fire hazard.
- (c) The design and sealing of inaccessible compartments must be adequate to contain compartment fires until a landing and safe evacuation can be made.
- (d) Each cargo and baggage compartment that is not sealed so as to contain cargo compartment fires completely without endangering the safety of a rotorcraft or its occupants must be designed, or must have a device, to ensure detection of fires or smoke by a crew member while at his station and to prevent the accumulation of harmful quantities of smoke, flame, extinguishing agents, and other noxious gases in any crew or passenger compartment. This must be shown in flight.
- (e) For rotorcraft used for the carriage of cargo only, the cabin area may be considered a cargo compartment and, in addition to sub-paragraphs (a) to (d), the following apply:
- (1) There must be means to shut off the ventilating airflow to or within the compartment. Controls for this purpose must be accessible to the flight crew in the crew compartment.
 - (2) Required crew emergency exits must be accessible under all cargo loading conditions.
 - (3) Sources of heat within each compartment must be shielded and insulated to prevent igniting the cargo.

CS 29.859 Combustion heater fire protection

ED Decision 2003/16/RM

- (a) *Combustion heater fire zones.* The following combustion heater fire zones must be protected against fire under the applicable provisions of [CS 29.1181](#) to [29.1191](#), and [CS 29.1195](#) to [29.1203](#):
- (1) The region surrounding any heater, if that region contains any flammable fluid system components (including the heater fuel system), that could:
 - (i) Be damaged by heater malfunctioning; or
 - (ii) Allow flammable fluids or vapours to reach the heater in case of leakage.
 - (2) Each part of any ventilating air passage that:
 - (i) Surrounds the combustion chamber; and
 - (ii) Would not contain (without damage to other rotorcraft components) any fire that may occur within the passage.
- (b) *Ventilating air ducts.* Each ventilating air duct passing through any fire zone must be fireproof. In addition –
- (1) Unless isolation is provided by fireproof valves or by equally effective means, the ventilating air duct downstream of each heater must be fireproof for a distance great enough to ensure that any fire originating in the heater can be contained in the duct; and
 - (2) Each part of any ventilating duct passing through any region having a flammable fluid system must be so constructed or isolated from that system that the malfunctioning of any component of that system cannot introduce flammable fluids or vapours into the ventilating airstream.
- (c) *Combustion air ducts.* Each combustion air duct must be fireproof for a distance great enough to prevent damage from backfiring or reverse flame propagation. In addition:
- (1) No combustion air duct may communicate with the ventilating airstream unless flames from backfires or reverse burning cannot enter the ventilating airstream under any operating condition, including reverse flow or malfunction of the heater or its associated components; and
 - (2) No combustion air duct may restrict the prompt relief of any backfire that, if so restricted, could cause heater failure.
- (d) *Heater controls; general.* There must be means to prevent the hazardous accumulation of water or ice on or in any heater control component, control system tubing, or safety control.
- (e) *Heater safety controls.* For each combustion heater, safety control means must be provided as follows:
- (1) Means independent of the components provided for the normal continuous control of air temperature, airflow, and fuel flow must be provided, for each heater, to automatically shut off the ignition and fuel supply of that heater at a point remote from that heater when any of the following occurs:
 - (i) The heat exchanger temperature exceeds safe limits.
 - (ii) The ventilating air temperature exceeds safe limits.
 - (iii) The combustion airflow becomes inadequate for safe operation.

- (iv) The ventilating airflow becomes inadequate for safe operation.
 - (2) The means of complying with sub-paragraph (e)(1) for any individual heater must:
 - (i) Be independent of components serving any other heater whose heat output is essential for safe operation; and
 - (ii) Keep the heater off until restarted by the crew.
 - (3) There must be means to warn the crew when any heater whose heat output is essential for safe operation has been shut off by the automatic means prescribed in sub-paragraph (e)(1).
- (f) *Air intakes.* Each combustion and ventilating air intake must be where no flammable fluids or vapours can enter the heater system under any operating condition:
- (1) During normal operation; or
 - (2) As a result of the malfunction of any other component.
- (g) *Heater exhaust.* Each heater exhaust system must meet the requirements of [CS 29.1121](#) and [29.1123](#). In addition:
- (1) Each exhaust shroud must be sealed so that no flammable fluids or hazardous quantities of vapours can reach the exhaust systems through joints; and
 - (2) No exhaust system may restrict the prompt relief of any backfire that, if so restricted, could cause heater failure.
- (h) *Heater fuel systems.* Each heater fuel system must meet the powerplant fuel system requirements affecting safe heater operation. Each heater fuel system component in the ventilating airstream must be protected by shrouds so that no leakage from those components can enter the ventilating airstream.
- (i) *Drains.* There must be means for safe drainage of any fuel that might accumulate in the combustion chamber or the heat exchanger. In addition –
- (1) Each part of any drain that operates at high temperatures must be protected in the same manner as heater exhausts; and
 - (2) Each drain must be protected against hazardous ice accumulation under any operating condition.

CS 29.861 Fire protection of structure, controls, and other parts

ED Decision 2003/16/RM

Each part of the structure, controls, and the rotor mechanism, and other parts essential to controlled landing and (for Category A) flight that would be affected by powerplant fires must be isolated under [CS 29.1191](#), or must be:

- (a) For Category A rotorcraft, fire-proof; and
- (b) For Category B rotorcraft, fire-proof or protected so that they can perform their essential functions for at least 5 minutes under any foreseeable powerplant fire conditions.

CS 29.863 Flammable fluid fire protection

ED Decision 2003/16/RM

- (a) In each area where flammable fluids or vapours might escape by leakage of a fluid system, there must be means to minimise the probability of ignition of the fluids and vapours, and the resultant hazards if ignition does occur.
- (b) Compliance with sub-paragraph (a) must be shown by analysis or tests, and the following factors must be considered:
 - (1) Possible sources and paths of fluid leakage, and means of detecting leakage.
 - (2) Flammability characteristics of fluids, including effects of any combustible or absorbing materials.
 - (3) Possible ignition sources, including electrical faults, overheating of equipment, and malfunctioning of protective devices.
 - (4) Means available for controlling or extinguishing a fire, such as stopping flow of fluids, shutting down equipment, fireproof containment, or use of extinguishing agents.
 - (5) Ability of rotorcraft components that are critical to safety of flight to withstand fire and heat.
- (c) If action by the flight crew is required to prevent or counteract a fluid fire (e.g. equipment shutdown or actuation of a fire extinguisher), quick acting means must be provided to alert the crew.
- (d) Each area where flammable fluids or vapours might escape by leakage of a fluid system must be identified and defined.

EXTERNAL LOADS

CS 29.865 External loads

ED Decision 2003/16/RM

- (a) It must be shown by analysis, test, or both, that the rotorcraft external load attaching means for rotorcraft-load combinations to be used for non-human external cargo applications can withstand a limit static load equal to 2.5, or some lower load factor approved under [CS 29.337](#) through [29.341](#), multiplied by the maximum external load for which authorisation is requested. It must be shown by analysis, test, or both that the rotorcraft external load attaching means and corresponding personnel-carrying device system for rotorcraft-load combinations to be used for human external cargo applications can withstand a limit static load equal to 3.5 or some lower load factor, not less than 2.5, approved under [CS 29.337](#) through [29.341](#), multiplied by the maximum external load for which authorisation is requested. The load for any rotorcraft-load combination class, for any external cargo type, must be applied in the vertical direction. For jettisonable rotorcraft-load combinations, for any applicable external cargo type, the load must also be applied in any direction making the maximum angle with the vertical that can be achieved in service but not less than 30°. However, the 30° angle may be reduced to a lesser angle if:
- (1) An operating limitation is established limiting external load operations to such angles for which compliance with this paragraph has been shown; or
 - (2) It is shown that the lesser angle cannot be exceeded in service.
- (b) The external load attaching means, for jettisonable rotorcraft-load combinations, must include a quick-release system to enable the pilot to release the external load quickly during flight. The quick-release system must consist of a primary quick-release subsystem and a backup quick-release subsystem that are isolated from one another. The quick-release system, and the means by which it is controlled, must comply with the following:
- (1) A control for the primary quick-release subsystem must be installed either on one of the pilot's primary controls or in an equivalently accessible location and must be designed and located so that it may be operated by either the pilot or a crew member without hazardously limiting the ability to control the rotorcraft during an emergency situation.
 - (2) A control for the backup quick-release subsystem, readily accessible to either the pilot or another crew member, must be provided.
 - (3) Both the primary and backup quick-release subsystems must:
 - (i) Be reliable, durable, and function properly with all external loads up to and including the maximum external limit load for which authorisation is requested.
 - (ii) Be protected against electromagnetic interference (EMI) from external and internal sources and against lightning to prevent inadvertent load release.
 - (A) The minimum level of protection required for jettisonable rotorcraft-load combinations used for non human external cargo is a radio frequency field strength of 20 volts per metre.
 - (B) The minimum level of protection required for jettisonable rotorcraft-load combinations used for human external cargo is a radio frequency field strength of 200 volts per metre.

- (iii) Be protected against any failure that could be induced by a failure mode of any other electrical or mechanical rotorcraft system.
- (c) For rotorcraft-load combinations to be used for human external cargo applications, the rotorcraft must:
 - (1) For jettisonable external loads, have a quick-release system that meets the requirements of sub-paragraph (b) and that:
 - (i) Provides a dual actuation device for the primary quick-release subsystem, and
 - (ii) Provides a separate dual actuation device for the backup quick-release subsystem.
 - (2) Have a reliable, approved personnel-carrying device system that has the structural capability and personnel safety features essential for external occupant safety,
 - (3) Have placards and markings at all appropriate locations that clearly state the essential system operating instructions and, for the personnel carrying device system, ingress and egress instructions,
 - (4) Have equipment to allow direct intercommunication among required crew members and external occupants,
 - (5) Have the appropriate limitations and procedures incorporated in the flight manual for conducting human external cargo operations, and
 - (6) For human external cargo applications requiring use of Category A rotorcraft, have one-engine-inoperative hover performance data and procedures in the flight manual for the weights, altitudes, and temperatures for which external load approval is requested.
- (d) The critically configured jettisonable external loads must be shown by a combination of analysis, ground tests, and flight tests to be both transportable and releasable throughout the approved operational envelope without hazard to the rotorcraft during normal flight conditions. In addition, these external loads must be shown to be releasable without hazard to the rotorcraft during emergency flight conditions.
- (e) A placard or marking must be installed next to the external-load attaching means clearly stating any operational limitations and the maximum authorised external load as demonstrated under [CS 29.25](#) and this paragraph.
- (f) The fatigue evaluation of [CS 29.571](#) does not apply to rotorcraft-load combinations to be used for non-human external cargo except for the failure of critical structural elements that would result in a hazard to the rotorcraft. For rotorcraft-load combinations to be used for human external cargo, the fatigue evaluation of [CS 29.571](#) applies to the entire quick-release and personnel-carrying device structural systems and their attachments.

MISCELLANEOUS

CS 29.871 Levelling marks

ED Decision 2003/16/RM

There must be reference marks for levelling the rotorcraft on the ground.

CS 29.873 Ballast provisions

ED Decision 2003/16/RM

Ballast provisions must be designed and constructed to prevent inadvertent shifting of ballast in flight.

SUBPART E — POWERPLANT

GENERAL

CS 29.901 Installation

ED Decision 2003/16/RM

- (a) For the purpose of this Code, the powerplant installation includes each part of the rotorcraft (other than the main and auxiliary rotor structures) that:
 - (1) Is necessary for propulsion;
 - (2) Affects the control of the major propulsive units; or
 - (3) Affects the safety of the major propulsive units between normal inspections or overhauls.
- (b) For each powerplant installation:
 - (1) The installation must comply with:
 - (i) The installation instructions provided under CS-E; and
 - (ii) The applicable provisions of this Subpart.
 - (2) Each component of the installation must be constructed, arranged, and installed to ensure its continued safe operation between normal inspections or overhauls for the range of temperature and altitude for which approval is requested.
 - (3) Accessibility must be provided to allow any inspection and maintenance necessary for continued airworthiness.
 - (4) Electrical interconnections must be provided to prevent differences of potential between major components of the installation and the rest of the rotorcraft.
 - (5) Axial and radial expansion of turbine engines may not affect the safety of the installation; and
 - (6) Design precautions must be taken to minimise the possibility of incorrect assembly of components and equipment essential to safe operation of the rotorcraft, except where operation with the incorrect assembly can be shown to be extremely improbable.
- (c) For each powerplant and auxiliary power unit installation, it must be established that no single failure or malfunction or probable combination of failures will jeopardise the safe operation of the rotorcraft except that the failure of structural elements need not be considered if the probability of any such failure is extremely remote.
- (d) Each auxiliary power unit installation must meet the applicable provisions of this Subpart.

CS 29.903 Engines

ED Decision 2003/16/RM

- (a) *(Reserved)*
- (b) *Category A; engine isolation.* For each Category A rotorcraft, the powerplants must be arranged and isolated from each other to allow operation, in at least one configuration, so that the failure or malfunction of any engine, or the failure of any system that can affect any engine, will not –
 - (1) Prevent the continued safe operation of the remaining engines; or

- (2) Require immediate action, other than normal pilot action with primary flight controls, by any crew member to maintain safe operation.
- (c) *Category A; control of engine rotation.* For each Category A rotorcraft, there must be a means for stopping the rotation of any engine individually in flight, except that, for turbine engine installations, the means for stopping the engine need be provided only where necessary for safety. In addition –
 - (1) Each component of the engine stopping system that is located on the engine side of the firewall, and that might be exposed to fire, must be at least fire resistant; or
 - (2) Duplicate means must be available for stopping the engine and the controls must be where all are not likely to be damaged at the same time in case of fire.
- (d) *Turbine engine installation.* For turbine engine installations,
 - (1) Design precautions must be taken to minimise the hazards to the rotorcraft in the event of an engine rotor failure; and,
 - (2) The powerplant systems associated with engine control devices, systems, and instrumentation must be designed to give reasonable assurance that those engine operating limitations that adversely affect engine rotor structural integrity will not be exceeded in service.
- (e) *Restart capability:*
 - (1) A means to restart any engine in flight must be provided.
 - (2) Except for the in-flight shutdown of all engines, engine restart capability must be demonstrated throughout a flight envelope for the rotorcraft.
 - (3) Following the in-flight shutdown of all engines, in-flight engine restart capability must be provided.

CS 29.907 Engine vibration

ED Decision 2003/16/RM

- (a) Each engine must be installed to prevent the harmful vibration of any part of the engine or rotorcraft.
- (b) The addition of the rotor and the rotor drive system to the engine may not subject the principal rotating parts of the engine to excessive vibration stresses. This must be shown by a vibration investigation.

CS 29.908 Cooling fans

ED Decision 2003/16/RM

For cooling fans that are a part of a powerplant installation the following apply:

- (a) *Category A.* For cooling fans installed in Category A rotorcraft, it must be shown that a fan blade failure will not prevent continued safe flight either because of damage caused by the failed blade or loss of cooling air.
- (b) *Category B.* For cooling fans installed in Category B rotorcraft, there must be means to protect the rotorcraft and allow a safe landing if a fan blade fails. It must be shown that :
 - (1) The fan blade would be contained in the case of a failure;
 - (2) Each fan is located so that a fan blade failure will not jeopardise safety; or

-
- (3) Each fan blade can withstand an ultimate load of 1.5 times the centrifugal force expected in service, limited by either:
- (i) The highest rotational speeds achievable under uncontrolled conditions; or
 - (ii) An overspeed limiting device.
- (c) *Fatigue evaluation.* Unless a fatigue evaluation under [CS 29.571](#) is conducted, it must be shown that cooling fan blades are not operating at resonant conditions within the operating limits of the rotorcraft.

ROTOR DRIVE SYSTEM

CS 29.917 Design

ED Decision 2003/16/RM

- (a) *General.* The rotor drive system includes any part necessary to transmit power from the engines to the rotor hubs. This includes gearboxes, shafting, universal joints, couplings, rotor brake assemblies, clutches, supporting bearings for shafting, any attendant accessory pads or drives, and any cooling fans that are a part of, attached to, or mounted on the rotor drive system.
- (b) *Design assessment.* A design assessment must be performed to ensure that the rotor drive system functions safely over the full range of conditions for which certification is sought. The design assessment must include a detailed failure analysis to identify all failures that will prevent continued safe flight or safe landing, and must identify the means to minimise the likelihood of their occurrence.
- (c) *Arrangement.* Rotor drive systems must be arranged as follows:
 - (1) Each rotor drive system of multi-engine rotorcraft must be arranged so that each rotor necessary for operation and control will continue to be driven by the remaining engines if any engine fails.
 - (2) For single-engine rotorcraft, each rotor drive system must be so arranged that each rotor necessary for control in autorotation will continue to be driven by the main rotors after disengagement of the engine from the main and auxiliary rotors.
 - (3) Each rotor drive system must incorporate a unit for each engine to automatically disengage that engine from the main and auxiliary rotors if that engine fails.
 - (4) If a torque limiting device is used in the rotor drive system, it must be located so as to allow continued control of the rotorcraft when the device is operating.
 - (5) If the rotors must be phased for intermeshing, each system must provide constant and positive phase relationship under any operating condition.
 - (6) If a rotor dephasing device is incorporated, there must be means to keep the rotors locked in proper phase before operation.

AMC 29.917 Rotor Drive System Design

ED Decision 2012/022/R

Where Vibration Health Monitoring is used as a compensating provision to meet [CS 29.917\(b\)](#), the design and performance of the vibration health monitoring system should be approved by requesting compliance with [CS 29.1465\(a\)](#).

[Amdt 29/3]

CS 29.921 Rotor brake

ED Decision 2003/16/RM

If there is a means to control the rotation of the rotor drive system independently of the engine, any limitations on the use of that means must be specified, and the control for that means must be guarded to prevent inadvertent operation.

CS 29.923 Rotor drive system and control mechanism tests

ED Decision 2003/16/RM

- (a) *Endurance tests, general.* Each rotor drive system and rotor control mechanism must be tested, as prescribed in sub-paragraphs (b) to (n) and (p), for at least 200 hours plus the time required to meet the requirements of sub-paragraphs (b)(2), (b)(3) and (k). These tests must be conducted as follows:
- (1) Ten-hour test cycles must be used, except that the test cycle must be extended to include the OEI test of sub-paragraphs (b)(2) and (k), if OEI ratings are requested.
 - (2) The tests must be conducted on the rotorcraft.
 - (3) The test torque and rotational speed must be:
 - (i) Determined by the powerplant limitations; and
 - (ii) Absorbed by the rotors to be approved for the rotorcraft.
- (b) *Endurance tests, take-off run.* The take-off run must be conducted as follows:
- (1) Except as prescribed in sub-paragraphs (b)(2) and (b)(3), the take-off torque run must consist of 1 hour of alternate runs of 5 minutes at take-off torque and the maximum speed for use with take-off torque, and 5 minutes at as low an engine idle speed as practicable. The engine must be declutched from the rotor drive system, and the rotor brake, if furnished and so intended, must be applied during the first minute of the idle run. During the remaining 4 minutes of the idle run, the clutch must be engaged so that the engine drives the rotors at the minimum practical rpm. The engine and the rotor drive system must be accelerated at the maximum rate. When declutching the engine, it must be decelerated rapidly enough to allow the operation of the overrunning clutch.
 - (2) For helicopters for which the use of a 2½-minute OEI rating is requested, the take-off run must be conducted as prescribed in subparagraph (b)(1), except for the third and sixth runs for which the take-off torque and the maximum speed for use with take-off torque are prescribed in that paragraph. For these runs, the following apply:
 - (i) Each run must consist of at least one period of 2½ minutes with take-off torque and the maximum speed for use with take-off torque on all engines.
 - (ii) Each run must consist of at least one period, for each engine in sequence, during which that engine simulates a power failure and the remaining engines are run at the 2½-minute OEI torque and the maximum speed for use with 2½-minute OEI torque for 2½ minutes.
 - (3) For multi-engine, turbine-powered rotorcraft for which the use of 30-second/2-minute OEI power is requested, the take-off run must be conducted as prescribed in subparagraph (b)(1) except for the following:
 - (i) Immediately following any one 5-minute power-on run required by subparagraph (b)(1), simulate a failure, for each power source in turn, and apply the maximum torque and the maximum speed for use with the 30-second OEI power to the remaining affected drive system power inputs for not less than 30 seconds. Each application of 30-second OEI power must be followed by two applications of the maximum torque and the maximum speed for use with the 2 minute OEI power for not less than 2 minutes each; the second application must follow a period at stabilised continuous or 30-minute OEI power (whichever is requested by the applicant.) At least one run sequence must be conducted from a simulated 'flight

- idle' condition. When conducted on a bench test, the test sequence must be conducted following stabilisation at take-off power.
- (ii) For the purpose of this paragraph, an affected power input includes all parts of the rotor drive system which can be adversely affected by the application of higher or asymmetric torque and speed prescribed by the test.
 - (iii) This test may be conducted on a representative bench test facility when engine limitations either preclude repeated use of this power or would result in premature engine removals during the test. The loads, the vibration frequency, and the methods of application to the affected rotor drive system components must be representative of rotorcraft conditions. Test components must be those used to show compliance with the remainder of this paragraph.
- (c) *Endurance tests, maximum continuous run.* Three hours of continuous operation at maximum continuous torque and the maximum speed for use with maximum continuous torque must be conducted as follows:
- (1) The main rotor controls must be operated at a minimum of 15 times each hour through the main rotor pitch positions of maximum vertical thrust, maximum forward thrust component, maximum aft thrust component, maximum left thrust component, and maximum right thrust component, except that the control movements need not produce loads or blade flapping motion exceeding the maximum loads of motions encountered in flight.
 - (2) The directional controls must be operated at a minimum of 15 times each hour through the control extremes of maximum right turning torque, neutral torque as required by the power applied to the main rotor, and maximum left turning torque.
 - (3) Each maximum control position must be held for at least 10 seconds, and the rate of change of control position must be at least as rapid as that for normal operation.
- (d) *Endurance tests: 90% of maximum continuous run.* One hour of continuous operation at 90% of maximum continuous torque and the maximum speed for use with 90% of maximum continuous torque must be conducted.
- (e) *Endurance tests; 80% of maximum continuous run.* One hour of continuous operation at 80% of maximum continuous torque and the minimum speed for use with 80% of maximum continuous torque must be conducted.
- (f) *Endurance tests; 60% of maximum continuous run.* Two hours or, for helicopters for which the use of either 30-minute OEI power or continuous OEI power is requested, 1 hour of continuous operation at 60% of maximum continuous torque and the minimum speed for use with 60% of maximum continuous torque must be conducted.
- (g) *Endurance tests: engine malfunctioning run.* It must be determined whether malfunctioning of components, such as the engine fuel or ignition systems, or whether unequal engine power can cause dynamic conditions detrimental to the drive system. If so, a suitable number of hours of operation must be accomplished under those conditions, 1 hour of which must be included in each cycle, and the remaining hours of which must be accomplished at the end of the 20 cycles. If no detrimental condition results, an additional hour of operation in compliance with sub-paragraph (b) must be conducted in accordance with the run schedule of sub-paragraph (b)(1) without consideration of sub-paragraph (b)(2).

- (h) *Endurance tests; overspeed run.* One hour of continuous operation must be conducted at maximum continuous torque and the maximum power-on overspeed expected in service, assuming that speed and torque limiting devices, if any, function properly.
- (i) *Endurance tests: rotor control positions.* When the rotor controls are not being cycled during the endurance tests, the rotor must be operated, using the procedures prescribed in subparagraph (c), to produce each of the maximum thrust positions for the following percentages of test time (except that the control positions need not produce loads or blade flapping motion exceeding the maximum loads or motions encountered in flight):
 - (1) For full vertical thrust, 20%.
 - (2) For the forward thrust component, 50%
 - (3) For the right thrust component, 10%.
 - (4) For the left thrust component, 10%.
 - (5) For the aft thrust component, 10%.
- (j) *Endurance tests, clutch and brake engagements.* A total of at least 400 clutch and brake engagements, including the engagements of sub-paragraph (b), must be made during the take-off torque runs and, if necessary, at each change of torque and speed throughout the test. In each clutch engagement, the shaft on the driven side of the clutch must be accelerated from rest. The clutch engagements must be accomplished at the speed and by the method prescribed by the applicant. During deceleration after each clutch engagement, the engines must be stopped rapidly enough to allow the engines to be automatically disengaged from the rotors and rotor drives. If a rotor brake is installed for stopping the rotor, the clutch, during brake engagements, must be disengaged above 40% of maximum continuous rotor speed and the rotors allowed to decelerate to 40% of maximum continuous rotor speed, at which time the rotor brake must be applied. If the clutch design does not allow stopping the rotors with the engine running, or if no clutch is provided, the engine must be stopped before each application of the rotor brake, and then immediately be started after the rotors stop.
- (k) *Endurance tests, OEI power run.*
 - (1) For rotorcraft for which the use of 30-minute OEI power is requested, a run at 30-minute OEI torque and the maximum speed for use with 30-minute OEI torque must be conducted as follows. For each engine, in sequence, that engine must be inoperative and the remaining engines must be run for a 30-minute period.
 - (2) For rotorcraft for which the use of continuous OEI power is requested, a run at continuous OEI torque and the maximum speed for use with continuous OEI torque must be conducted as follows. For each engine, in sequence, that engine must be inoperative and the remaining engines must be run for 1 hour.
 - (3) The number of periods prescribed in sub-paragraph (k)(1) or (k)(2) may not be less than the number of engines, nor may it be less than two.
- (l) Reserved.
- (m) Any components that are affected by manoeuvring and gust loads must be investigated for the same flight conditions as are the main rotors, and their service lives must be determined by fatigue tests or by other acceptable methods. In addition, a level of safety equal to that of the main rotors must be provided for:
 - (1) Each component in the rotor drive system whose failure would cause an uncontrolled landing;

- (2) Each component essential to the phasing of rotors on multi-rotor rotorcraft, or that furnishes a driving link for the essential control of rotors in autorotation; and
- (3) Each component common to two or more engines on multi-engine rotorcraft.
- (n) *Special tests.* Each rotor drive system designed to operate at two or more gear ratios must be subjected to special testing for durations necessary to substantiate the safety of the rotor drive system.
- (o) Each part tested as prescribed in this paragraph must be in a serviceable condition at the end of the tests. No intervening disassembly which might affect test results may be conducted.
- (p) *Endurance tests; operating lubricants.* To be approved for use in rotor drive and control systems, lubricants must meet the specifications of lubricants used during the tests prescribed by this paragraph. Additional or alternate lubricants may be qualified by equivalent testing or by comparative analysis of lubricant specifications and rotor drive and control system characteristics. In addition:
 - (1) At least three 10-hour cycles required by this paragraph must be conducted with transmission and gearbox lubricant temperatures, at the location prescribed for measurement, not lower than the maximum operating temperature for which approval is requested;
 - (2) For pressure lubricated systems, at least three 10-hour cycles required by this paragraph must be conducted with the lubricant pressure, at the location prescribed for measurement, not higher than the minimum operating pressure for which approval is requested; and
 - (3) The test conditions of sub-paragraphs (p)(1) and (p)(2) must be applied simultaneously and must be extended to include operation at any one-engine-inoperative rating for which approval is requested.

CS 29.927 Additional tests

ED Decision 2003/16/RM

- (a) Any additional dynamic, endurance, and operational tests, and vibratory investigations necessary to determine that the rotor drive mechanism is safe, must be performed.
- (b) If turbine engine torque output to the transmission can exceed the highest engine or transmission torque limit, and that output is not directly controlled by the pilot under normal operating conditions (such as where the primary engine power control is accomplished through the flight control), the following test must be made:
 - (1) Under conditions associated with all engines operating, make 200 applications, for 10 seconds each, of torque that is at least equal to the lesser of:
 - (i) The maximum torque used in meeting [CS 29.923](#) plus 10%; or
 - (ii) The maximum torque attainable under probable operating conditions, assuming that torque limiting devices, if any, function properly.
 - (2) For multi-engine rotorcraft under conditions associated with each engine, in turn, becoming inoperative, apply to the remaining transmission torque inputs the maximum torque attainable under probable operating conditions, assuming that torque limiting devices, if any, function properly. Each transmission input must be tested at this maximum torque for at least 15 minutes.

- (c) *Lubrication system failure.* For lubrication systems required for proper operation of rotor drive systems, the following apply:
- (1) Category A. Unless such failures are extremely remote, it must be shown by test that any failure which results in loss of lubricant in any normal use lubrication system will not prevent continued safe operation, although not necessarily without damage, at a torque and rotational speed prescribed by the applicant for continued flight, for at least 30 minutes after perception by the flight crew of the lubrication system failure or loss of lubricant.
 - (2) The requirements of Category A apply except that the rotor drive system need only be capable of operating under autorotative conditions for at least 15 minutes.
- (d) *Overspeed test.* The rotor drive system must be subjected to 50 overspeed runs, each 30 ± 3 seconds in duration, at not less than either the higher of the rotational speed to be expected from an engine control device failure or 105% of the maximum rotational speed, including transients, to be expected in service. If speed and torque limiting devices are installed, are independent of the normal engine control, and are shown to be reliable, their rotational speed limits need not be exceeded. These runs must be conducted as follows:
- (1) Overspeed runs must be alternated with stabilising runs of from 1 to 5 minutes duration each at 60 to 80% of maximum continuous speed.
 - (2) Acceleration and deceleration must be accomplished in a period not longer than 10 seconds (except where maximum engine acceleration rate will require more than 10 seconds), and the time for changing speeds may not be deducted from the specified time for the overspeed runs.
 - (3) Overspeed runs must be made with the rotors in the flattest pitch for smooth operation.
- (e) The tests prescribed in sub-paragraphs (b) and (d) must be conducted on the rotorcraft and the torque must be absorbed by the rotors to be installed, except that other ground or flight test facilities with other appropriate methods of torque absorption may be used if the conditions of support and vibration closely simulate the conditions that would exist during a test on the rotorcraft.
- (f) Each test prescribed by this paragraph must be conducted without intervening disassembly and, except for the lubrication system failure test required by sub-paragraph (c), each part tested must be in a serviceable condition at the conclusion of the test.

CS 29.931 Shafting critical speed

ED Decision 2003/16/RM

- (a) The critical speeds of any shafting must be determined by demonstration except that analytical methods may be used if reliable methods of analysis are available for the particular design.
- (b) If any critical speed lies within, or close to, the operating ranges for idling, power-on, and autorotative conditions, the stresses occurring at that speed must be within safe limits. This must be shown by tests.
- (c) If analytical methods are used and show that no critical speed lies within the permissible operating ranges, the margins between the calculated critical speeds and the limits of the allowable operating ranges must be adequate to allow for possible variations between the computed and actual values.

CS 29.935 Shafting joints

ED Decision 2003/16/RM

Each universal joint, slip joint, and other shafting joints whose lubrication is necessary for operation must have provision for lubrication.

CS 29.939 Turbine engine operating characteristics

ED Decision 2003/16/RM

- (a) Turbine engine operating characteristics must be investigated in flight to determine that no adverse characteristics (such as stall, surge, or flameout) are present, to a hazardous degree, during normal and emergency operation within the range of operating limitations of the rotorcraft and of the engine.
- (b) The turbine engine air inlet system may not, as a result of airflow distortion during normal operation, cause vibration harmful to the engine.
- (c) For governor-controlled engines, it must be shown that there exists no hazardous torsional instability of the drive system associated with critical combinations of power, rotational speed, and control displacement.

FUEL SYSTEMS

CS 29.951 General

ED Decision 2003/16/RM

- (a) Each fuel system must be constructed and arranged to ensure a flow of fuel at a rate and pressure established for proper engine and auxiliary power unit functioning under any likely operating conditions, including the manoeuvres for which certification is requested and during which the engine or auxiliary power unit is permitted to be in operation.
- (b) Each fuel system must be arranged so that:
 - (1) No engine or fuel pump can draw fuel from more than one tank at a time; or
 - (2) There are means to prevent introducing air into the system.
- (c) Each fuel system for a turbine engine must be capable of sustained operation throughout its flow and pressure range with fuel initially saturated with water at 27°C (80°F) and having 0.20 cm³ of free water per litre (0.75 cc per US-gallon) added and cooled to the most critical condition for icing likely to be encountered in operation.

CS 29.952 Fuel system crash resistance

ED Decision 2003/16/RM

Unless other means acceptable to the Agency are employed to minimise the hazard of fuel fires to occupants following an otherwise survivable impact (crash landing), the fuel systems must incorporate the design features of this paragraph. These systems must be shown to be capable of sustaining the static and dynamic deceleration loads of this paragraph, considered as ultimate loads acting alone, measured at the system component's centre of gravity without structural damage to the system components, fuel tanks, or their attachments that would leak fuel to an ignition source.

- (a) *Drop test requirements.* Each tank, or the most critical tank, must be drop-tested as follows:
 - (1) The drop height must be at least 15.2m (50 ft).
 - (2) The drop impact surface must be non deforming.
 - (3) The tanks must be filled with water to 80% of the normal, full capacity.
 - (4) The tank must be enclosed in a surrounding structure representative of the installation unless it can be established that the surrounding structure is free of projections or other design features likely to contribute to rupture of the tank.
 - (5) The tank must drop freely and impact in a horizontal position $\pm 10^\circ$.
 - (6) After the drop test, there must be no leakage.
- (b) *Fuel tank load factors.* Except for fuel tanks located so that tank rupture with fuel release to either significant ignition sources, such as engines, heaters, and auxiliary power units, or occupants is extremely remote, each fuel tank must be designed and installed to retain its contents under the following ultimate inertial load factors, acting alone.
 - (1) For fuel tanks in the cabin –
 - (i) Upward – 4 g.
 - (ii) Forward – 16 g.
 - (iii) Sideward – 8 g.

- (iv) Downward – 20 g.
 - (2) For fuel tanks located above or behind the crew or passenger compartment that, if loosened, could injure an occupant in an emergency landing –
 - (i) Upward – 1.5 g.
 - (ii) Forward – 8 g.
 - (iii) Sideward – 2 g.
 - (iv) Downward – 4 g.
 - (3) For fuel tanks in other areas –
 - (i) Upward – 1.5 g.
 - (ii) Forward – 4 g.
 - (iii) Sideward – 2 g.
 - (iv) Downward – 4 g.
- (c) *Fuel line self-sealing breakaway couplings.* Self-sealing breakaway couplings must be installed unless hazardous relative motion of fuel system components to each other or to local rotorcraft structure is demonstrated to be extremely improbable or unless other means are provided. The couplings or equivalent devices must be installed at all fuel tank- to-fuel line connections, tank-to-tank interconnects, and at other points in the fuel system where local structural deformation could lead to release of fuel.
- (1) The design and construction of self- sealing breakaway couplings must incorporate the following design features:
 - (i) The load necessary to separate a breakaway coupling must be between 25 and 50% of the minimum ultimate failure load (ultimate strength) of the weakest component in the fluid-carrying line. The separation load must in no case be less than 1334 N (300 pounds), regardless of the size of the fluid line.
 - (ii) A breakaway coupling must separate whenever its ultimate load (as defined in subparagraph (c)(1)(i)) is applied in the failure modes most likely to occur.
 - (iii) All breakaway coupling must incorporate design provisions to visually ascertain that the coupling is locked together (leak-free) and is open during normal installation and service.
 - (iv) All breakaway couplings must incorporate design provisions to prevent uncoupling or unintended closing due to operational shocks, vibrations, or accelerations.
 - (v) No breakaway coupling design may allow the release of fuel once the coupling has performed its intended function.
 - (2) All individual breakaway couplings, coupling fuel feed systems, or equivalent means must be designed, tested, installed, and maintained so inadvertent fuel shutoff in flight is improbable in accordance with [CS 29.955\(a\)](#) and must comply with the fatigue evaluation requirements of [CS 29.571](#) without leaking.
 - (3) Alternate, equivalent means to the use of breakaway couplings must not create a survivable impact-induced load on the fuel line to which it is installed greater than 25 to 50% of the ultimate load (strength) of the weakest component in the line and must comply with the fatigue requirements of [CS 29.571](#) without leaking.

- (d) *Frangible or deformable structural attachments.* Unless hazardous relative motion of fuel tanks and fuel system components to local rotorcraft structure is demonstrated to be extremely improbable in an otherwise survivable impact, frangible or locally deformable attachments of fuel tanks and fuel system components to local rotorcraft structure must be used. The attachment of fuel tanks and fuel system components to local rotorcraft structure, whether frangible or locally deformable, must be designed such that its separation or relative local deformation will occur without rupture or local tearout of the fuel tank or fuel system component that will cause fuel leakage. The ultimate strength of frangible or deformable attachments must be as follows:
- (1) The load required to separate a frangible attachment from its support structure, or deform a locally deformable attachment relative to its support structure, must be between 25 and 50% of the minimum ultimate load (ultimate strength) of the weakest component in the attached system. In no case may the load be less than 1334 N (300 pounds).
 - (2) A frangible or locally deformable attachment must separate or locally deform as intended whenever its ultimate load (as defined in sub-paragraph (d)(1)) is applied in the modes most likely to occur.
 - (3) All frangible or locally deformable attachments must comply with the fatigue requirements of [CS 29.571](#).
- (e) *Separation of fuel and ignition sources.* To provide maximum crash resistance, fuel must be located as far as practicable from all occupiable areas and from all potential ignition sources.
- (f) *Other basic mechanical design criteria.* Fuel tanks, fuel lines, electrical wires and electrical devices must be designed, constructed, and installed, as far as practicable, to be crash resistant.
- (g) *Rigid or semi-rigid fuel tanks.* Rigid or semi-rigid fuel tank or bladder walls must be impact and tear resistant.

CS 29.953 Fuel system independence

ED Decision 2003/16/RM

- (a) For Category A rotorcraft:
- (1) The fuel system must meet the requirements of [CS 29.903\(b\)](#); and
 - (2) Unless other provisions are made to meet sub-paragraph (a)(1), the fuel system must allow fuel to be supplied to each engine through a system independent of those parts of each system supplying fuel to other engines.
- (b) Each fuel system for a multi-engine Category B rotorcraft must meet the requirements of sub-paragraph (a)(2). However, separate fuel tanks need not be provided for each engine.

CS 29.954 Fuel system lightning protection

ED Decision 2003/16/RM

The fuel system must be designed and arranged to prevent the ignition of fuel vapour within the system by:

- (a) Direct lightning strikes to areas having a high probability of stroke attachment;
- (b) Swept lightning strokes to areas where swept strokes are highly probable; and
- (c) Corona and streamer at fuel vent outlets.

CS 29.955 Fuel flow

ED Decision 2012/022/R

- (a) *General.* The fuel system for each engine must provide the engine with at least 100% of the fuel required under all operating and manoeuvring conditions to be approved for the rotorcraft, including, as applicable, the fuel required to operate the engines under the test conditions required by [CS 29.927](#). Unless equivalent methods are used, compliance must be shown by test during which the following provisions are met, except that combinations of conditions which are shown to be improbable need not be considered.
- (1) The fuel pressure, corrected for accelerations (load factors), must be within the limits specified by the engine type certificate data sheet.
 - (2) The fuel level in the tank may not exceed that established as the unusable fuel supply for that tank under [CS 29.959](#), plus that necessary to conduct the test.
 - (3) The fuel head between the tank and the engine must be critical with respect to rotorcraft flight attitudes.
 - (4) The fuel flow transmitter, if installed, and the critical fuel pump (for pump-fed systems) must be installed to produce (by actual or simulated failure) the critical restriction to fuel flow to be expected from component failure.
 - (5) Critical values of engine rotational speed, electrical power, or other sources of fuel pump motive power must be applied.
 - (6) Critical values of fuel properties which adversely affect fuel flow are applied during demonstrations of fuel flow capability.
 - (7) The fuel filter required by [CS 29.997](#) is blocked to the degree necessary to simulate the accumulation of fuel contamination required to activate the indicator required by [CS 29.1305\(a\)\(18\)](#).
- (b) *Fuel transfer system.* If normal operation of the fuel system requires fuel to be transferred to another tank, the transfer must occur automatically via a system which has been shown to maintain the fuel level in the receiving tank within acceptable limits during flight or surface operation of the rotorcraft.
- (c) *Multiple fuel tanks.* If an engine can be supplied with fuel from more than one tank, the fuel system, in addition to having appropriate manual switching capability, must be designed to prevent interruption of fuel flow to the engine, without attention by the flight crew, when any tank supplying fuel to that engine is depleted of usable fuel during normal operation and any other tank that normally supplies fuel to that engine alone contains usable fuel.

[Amdt 29/3]

CS 29.957 Flow between inter-connected tanks

ED Decision 2003/16/RM

- (a) Where tank outlets are interconnected and allow fuel to flow between them due to gravity or flight accelerations, it must be impossible for fuel to flow between tanks in quantities great enough to cause overflow from the tank vent in any sustained flight condition.
- (b) If fuel can be pumped from one tank to another in flight:
- (1) The design of the vents and the fuel transfer system must prevent structural damage to tanks from overfilling; and

- (2) There must be means to warn the crew before overflow through the vents occurs.

CS 29.959 Unusable fuel supply

ED Decision 2003/16/RM

The unusable fuel supply for each tank must be established as not less than the quantity at which the first evidence of malfunction occurs under the most adverse fuel feed condition occurring under any intended operations and flight manoeuvres involving that tank.

CS 29.961 Fuel system hot weather operation

ED Decision 2003/16/RM

Each suction lift fuel system and other fuel systems conducive to vapour formation must be shown to operate satisfactorily (within certification limits) when using fuel at the most critical temperature for vapour formation under critical operating conditions including, if applicable, the engine operating conditions defined by [CS 29.927\(b\)\(1\) and \(b\)\(2\)](#).

CS 29.963 Fuel tanks: general

ED Decision 2003/16/RM

- (a) Each fuel tank must be able to withstand, without failure, the vibration, inertia, fluid, and structural loads to which it may be subjected in operation.
- (b) Each flexible fuel tank bladder or liner must be approved or shown to be suitable for the particular application and must be puncture resistant. Puncture resistance must be shown by meeting the ETSO-C80, paragraph 16.0, requirements using a minimum puncture force of 1646 N (370 pounds).
- (c) Each integral fuel tank must have facilities for inspection and repair of its interior.
- (d) The maximum exposed surface temperature of all components in the fuel tank must be less by a safe margin than the lowest expected auto-ignition temperature of the fuel or fuel vapour in the tank. Compliance with this requirement must be shown under all operating conditions and under all normal or malfunction conditions of all components inside the tank.
- (e) Each fuel tank installed in personnel compartments must be isolated by fume-proof and fuel-proof enclosures that are drained and vented to the exterior of the rotorcraft. The design and construction of the enclosures must provide necessary protection for the tank, must be crash resistant during a survivable impact in accordance with [CS 29.952](#), and must be adequate to withstand loads and abrasions to be expected in personnel compartments.

CS 29.965 Fuel tank tests

ED Decision 2003/16/RM

- (a) Each fuel tank must be able to withstand the applicable pressure tests in this paragraph without failure or leakage. If practicable, test pressures may be applied in a manner simulating the pressure distribution in service.
- (b) Each conventional metal tank, each non-metallic tank with walls that are not supported by the rotorcraft structure, and each integral tank must be subjected to a pressure of 24 kPa (3.5 psi) unless the pressure developed during maximum limit acceleration or emergency deceleration with a full tank exceeds this value, in which case a hydrostatic head, or equivalent test, must be

applied to duplicate the acceleration loads as far as possible. However, the pressure need not exceed 24 kPa (3.5 psi) on surfaces not exposed to the acceleration loading.

- (c) Each non-metallic tank with walls supported by the rotorcraft structure must be subjected to the following tests:
 - (1) A pressure test of at least 14 kPa (2.0 psi). This test may be conducted on the tank alone in conjunction with the test specified in subparagraph (c)(2).
 - (2) A pressure test, with the tank mounted in the rotorcraft structure, equal to the load developed by the reaction of the contents, with the tank full, during maximum limit acceleration or emergency deceleration. However, the pressure need not exceed 14 kPa (2.0 psi) on surfaces not exposed to the acceleration loading.
- (d) Each tank with large unsupported or unstiffened flat areas, or with other features whose failure or deformation could cause leakage, must be subjected to the following test or its equivalent:
 - (1) Each complete tank assembly and its supports must be vibration tested while mounted to simulate the actual installation.
 - (2) The tank assembly must be vibrated for 25 hours while two-thirds full of any suitable fluid. The amplitude of vibration may not be less than 0.8 mm (one thirty-second of an inch), unless otherwise substantiated.
 - (3) The test frequency of vibration must be as follows:
 - (i) If no frequency of vibration resulting from any rpm within the normal operating range of engine or rotor system speeds is critical, the test frequency of vibration, in number of cycles per minute, must, unless a frequency based on a more rational analysis is used, be the number obtained by averaging the maximum and minimum power-on engine speeds (rpm) for reciprocating engine powered rotorcraft or 2000 cpm for turbine engine powered rotorcraft.
 - (ii) If only one frequency of vibration resulting from any rpm within the normal operating range of engine or rotor system speeds is critical, that frequency of vibration must be the test frequency.
 - (iii) If more than one frequency of vibration resulting from any rpm within the normal operating range of engine or rotor system speeds is critical, the most critical of these frequencies must be the test frequency.
 - (4) Under sub-paragraph (d)(3)(ii) and (iii), the time of test must be adjusted to accomplish the same number of vibration cycles as would be accomplished in 25 hours at the frequency specified in sub-paragraph (d)(3)(i).
 - (5) During the test the tank assembly must be rocked at the rate of 16 to 20 complete cycles per minute through an angle of 15° on both sides of the horizontal (30° total), about the most critical axis, for 25 hours. If motion about more than one axis is likely to be critical, the tank must be rocked about each critical axis for 12½ hours.

CS 29.967 Fuel tank installation

ED Decision 2003/16/RM

- (a) Each fuel tank must be supported so that tank loads are not concentrated on unsupported tank surfaces. In addition:
 - (1) There must be pads, if necessary, to prevent chafing between each tank and its supports;

- (2) The padding must be non-absorbent or treated to prevent the absorption of fuel;
- (3) If flexible tank liners are used, they must be supported so that they are not required to withstand fluid loads; and
- (4) Each interior surface of tank compartments must be smooth and free of projections that could cause wear of the liner, unless:
 - (i) There are means for protection of the liner at those points; or
 - (ii) The construction of the liner itself provides such protection.
- (b) Any spaces adjacent to tank surfaces must be adequately ventilated to avoid accumulation of fuel or fumes in those spaces due to minor leakage. If the tank is in a sealed compartment, ventilation may be limited to drain holes that prevent clogging and that prevent excessive pressure resulting from altitude changes. If flexible tank liners are installed, the venting arrangement for the spaces between the liner and its container must maintain the proper relationship to tank vent pressures for any expected flight condition.
- (c) The location of each tank must meet the requirements of [CS 29.1185\(b\) and \(c\)](#).
- (d) No rotorcraft skin immediately adjacent to a major air outlet from the engine compartment may act as the wall of an integral tank.

CS 29.969 Fuel tank expansion space

ED Decision 2003/16/RM

Each fuel tank or each group of fuel tanks with interconnected vent systems must have an expansion space of not less than 2% of the combined tank capacity. It must be impossible to fill the fuel tank expansion space inadvertently with the rotorcraft in the normal ground attitude.

CS 29.971 Fuel tank sump

ED Decision 2003/16/RM

- (a) Each fuel tank must have a sump with a capacity of not less than the greater of:
 - (1) 0.10% of the tank capacity; or
 - (2) 0.24 litres (0.05 Imperial gallon/one sixteenth US gallon).
- (b) The capacity prescribed in sub-paragraph (a) must be effective with the rotorcraft in any normal attitude, and must be located so that the sump contents cannot escape through the tank outlet opening.
- (c) Each fuel tank must allow drainage of hazardous quantities of water from each part of the tank to the sump with the rotorcraft in any ground attitude to be expected in service.
- (d) Each fuel tank sump must have a drain that allows complete drainage of the sump on the ground.

CS 29.973 Fuel tank filler connection

ED Decision 2003/16/RM

- (a) Each fuel tank filler connection must prevent the entrance of fuel into any part of the rotorcraft other than the tank itself during normal operations and must be crash resistant during a survivable impact in accordance with [CS 29.952\(c\)](#). In addition:
 - (1) Each filler must be marked as prescribed in [CS 29.1557\(c\)\(1\)](#);

- (2) Each recessed filler connection that can retain any appreciable quantity of fuel must have a drain that discharges clear of the entire rotorcraft; and
 - (3) Each filler cap must provide a fuel- tight seal under the fluid pressure expected in normal operation and in a survivable impact.
- (b) Each filler cap or filler cap cover must warn when the cap is not fully locked or seated on the filler connection.

CS 29.975 Fuel tank vents and carburetor vapour vents

ED Decision 2003/16/RM

- (a) *Fuel tank vents.* Each fuel tank must be vented from the top part of the expansion space so that venting is effective under normal flight conditions. In addition:
- (1) The vents must be arranged to avoid stoppage by dirt or ice formation;
 - (2) The vent arrangement must prevent siphoning of fuel during normal operation;
 - (3) The venting capacity and vent pressure levels must maintain acceptable differences of pressure between the interior and exterior of the tank, during:
 - (i) Normal flight operation;
 - (ii) Maximum rate of ascent and descent; and
 - (iii) Refuelling and defuelling (where applicable);
 - (4) Airspaces of tanks with interconnected outlets must be interconnected;
 - (5) There may be no point in any vent line where moisture can accumulate with the rotorcraft in the ground attitude or the level flight attitude, unless drainage is provided;
 - (6) No vent or drainage provision may end at any point:
 - (i) Where the discharge of fuel from the vent outlet would constitute a fire hazard; or
 - (ii) From which fumes could enter personnel compartments; and
 - (7) The venting system must be designed to minimise spillage of fuel through the vents to an ignition source in the event of a rollover during landing, ground operations, or a survivable impact.
- (b) *Carburettor vapour vents.* Each carburettor with vapour elimination connections must have a vent line to lead vapours back to one of the fuel tanks. In addition –
- (1) Each vent system must have means to avoid stoppage by ice; and
 - (2) If there is more than one fuel tank, and it is necessary to use the tanks in a definite sequence, each vapour vent return line must lead back to the fuel tank used for take-off and landing.

CS 29.977 Fuel tank outlet

ED Decision 2003/16/RM

- (a) There must be a fuel strainer for the fuel tank outlet or for the booster pump. This strainer must:
- (1) For reciprocating engine powered rotorcraft, have 3 to 6 meshes per cm (8 to 16 meshes per inch); and

- (2) For turbine engine powered rotorcraft, prevent the passage of any object that could restrict fuel flow or damage any fuel system component.
- (b) The clear area of each fuel tank outlet strainer must be at least five times the area of the outlet line.
- (c) The diameter of each strainer must be at least that of the fuel tank outlet.
- (d) Each finger strainer must be accessible for inspection and cleaning.

CS 29.979 Pressure refuelling and fuelling provisions below fuel level

ED Decision 2003/16/RM

- (a) Each fuelling connection below the fuel level in each tank must have means to prevent the escape of hazardous quantities of fuel from that tank in case of malfunction of the fuel entry valve.
- (b) For systems intended for pressure refuelling, a means in addition to the normal means for limiting the tank content must be installed to prevent damage to the tank in case of failure of the normal means.
- (c) The rotorcraft pressure fuelling system (not fuel tanks and fuel tank vents) must withstand an ultimate load that is 2.0 times the load arising from the maximum pressure, including surge, that is likely to occur during fuelling. The maximum surge pressure must be established with any combination of tank valves being either intentionally or inadvertently closed.
- (d) The rotorcraft defuelling system (not including fuel tanks and fuel tank vents) must withstand an ultimate load that is 2.0 times the load arising from the maximum permissible defuelling pressure (positive or negative) at the rotorcraft fuelling connection.

FUEL SYSTEM COMPONENTS

CS 29.991 Fuel pumps

ED Decision 2003/16/RM

- (a) Compliance with [CS 29.955](#) must not be jeopardised by failure of:
 - (1) Any one pump except pumps that are approved and installed as parts of a type certificated engine; or
 - (2) Any component required for pump operation except the engine served by that pump.
- (b) The following fuel pump installation requirements apply:
 - (1) When necessary to maintain the proper fuel pressure:
 - (i) A connection must be provided to transmit the carburettor air intake static pressure to the proper fuel pump relief valve connection; and
 - (ii) The gauge balance lines must be independently connected to the carburettor inlet pressure to avoid incorrect fuel pressure readings.
 - (2) The installation of fuel pumps having seals or diaphragms that may leak must have means for draining leaking fuel.
 - (3) Each drain line must discharge where it will not create a fire hazard.

CS 29.993 Fuel system lines and fittings

ED Decision 2003/16/RM

- (a) Each fuel line must be installed and supported to prevent excessive vibration and to withstand loads due to fuel pressure, valve actuation, and accelerated flight conditions.
- (b) Each fuel line connected to components of the rotorcraft between which relative motion could exist must have provisions for flexibility.
- (c) Each flexible connection in fuel lines that may be under pressure or subjected to axial loading must use flexible hose assemblies.
- (d) Flexible hose must be approved.
- (e) No flexible hose that might be adversely affected by high temperatures may be used where excessive temperatures will exist during operation or after engine shutdown.

CS 29.995 Fuel valves

ED Decision 2003/16/RM

In addition to meeting the requirements of [CS 29.1189](#), each fuel valve must:

- (a) Reserved.
- (b) Be supported so that no loads resulting from their operation or from accelerated flight conditions are transmitted to the lines attached to the valve.

CS 29.997 Fuel strainer or filter

ED Decision 2003/16/RM

There must be a fuel strainer or filter between the fuel tank outlet and the inlet of the first fuel system component which is susceptible to fuel contamination, including but not limited to the fuel metering device or an engine positive displacement pump, whichever is nearer the fuel tank outlet. This fuel strainer or filter must:

- (a) Be accessible for draining and cleaning and must incorporate a screen or element which is easily removable;
- (b) Have a sediment trap and drain, except that it need not have a drain if the strainer or filter is easily removable for drain purposes;
- (c) Be mounted so that its weight is not supported by the connecting lines or by the inlet or outlet connections of the strainer or filter itself, unless adequate strength margins under all loading conditions are provided in the lines and connections; and
- (d) Provide a means to remove from the fuel any contaminant which would jeopardise the flow of fuel through rotorcraft or engine fuel system components required for proper rotorcraft or engine fuel system operation.

CS 29.999 Fuel system drains

ED Decision 2003/16/RM

- (a) There must be at least one accessible drain at the lowest point in each fuel system to completely drain the system with the rotorcraft in any ground attitude to be expected in service.
- (b) Each drain required by sub-paragraph (a) including the drains prescribed in [CS 29.971](#) must:
 - (1) Discharge clear of all parts of the rotorcraft;
 - (2) Have manual or automatic means to ensure positive closure in the off position; and
 - (3) Have a drain valve:
 - (i) That is readily accessible and which can be easily opened and closed; and
 - (ii) That is either located or protected to prevent fuel spillage in the event of a landing with landing gear retracted.

CS 29.1001 Fuel jettisoning

ED Decision 2003/16/RM

If a fuel jettisoning system is installed, the following apply:

- (a) Fuel jettisoning must be safe during all flight regimes for which jettisoning is to be authorised.
- (b) In showing compliance with sub-paragraph (a), it must be shown that:
 - (1) The fuel jettisoning system and its operation are free from fire hazard;
 - (2) No hazard results from fuel or fuel vapours which impinge on any part of the rotorcraft during fuel jettisoning; and
 - (3) Controllability of the rotorcraft remains satisfactory throughout the fuel jettisoning operation.

- (c) Means must be provided to automatically prevent jettisoning fuel below the level required for an all-engine climb at maximum continuous power from sea-level to 1524 m (5000 ft) altitude and cruise thereafter for 30 minutes at maximum range engine power.
- (d) The controls for any fuel jettisoning system must be designed to allow flight personnel (minimum crew) to safely interrupt fuel jettisoning during any part of the jettisoning operation.
- (e) The fuel jettisoning system must be designed to comply with the powerplant installation requirements of [CS 29.901\(c\)](#).
- (f) An auxiliary fuel jettisoning system which meets the requirements of sub-paragraphs (a), (b), (d) and (e) may be installed to jettison additional fuel provided it has separate and independent controls.

OIL SYSTEM

CS 29.1011 Engines: General

ED Decision 2003/16/RM

- (a) Each engine must have an independent oil system that can supply it with an appropriate quantity of oil at a temperature not above that safe for continuous operation.
- (b) The usable oil capacity of each system may not be less than the product of the endurance of the rotorcraft under critical operating conditions and the maximum allowable oil consumption of the engine under the same conditions, plus a suitable margin to ensure adequate circulation and cooling. Instead of a rational analysis of endurance and consumption, a usable oil capacity of 3.8 litres (0.83 Imperial gallon/1 US gallon) for each 151 litres (33.3 Imperial gallons/40 US gallons) of usable fuel may be used for reciprocating engine installations.
- (c) Oil-fuel ratios lower than those prescribed in sub-paragraph (b) may be used if they are substantiated by data on the oil consumption of the engine.
- (d) The ability of the engine oil cooling provisions to maintain the oil temperature at or below the maximum established value must be shown under the applicable requirements of [CS 29.1041](#) to [29.1049](#).

CS 29.1013 Oil tanks

ED Decision 2003/16/RM

- (a) *Installation.* Each oil tank installation must meet the requirements of [CS 29.967](#).
- (b) *Expansion space.* Oil tank expansion space must be provided so that –
 - (1) Each oil tank used with a reciprocating engine has an expansion space of not less than the greater of 10% of the tank capacity or 1.9 litres (0.42 Imperial gallon/0.5 US gallon), and each oil tank used with a turbine engine has an expansion space of not less than 10% of the tank capacity;
 - (2) Each reserve oil tank not directly connected to any engine has an expansion space of not less than 2% of the tank capacity; and
 - (3) It is impossible to fill the expansion space inadvertently with the rotorcraft in the normal ground attitude.
- (c) *Filler connections.* Each recessed oil tank filler connection that can retain any appreciable quantity of oil must have a drain that discharges clear of the entire rotorcraft. In addition –
 - (1) Each oil tank filler cap must provide an oil-tight seal under the pressure expected in operation;
 - (2) For Category A rotorcraft, each oil tank filler cap or filler cap cover must incorporate features that provide a warning when caps are not fully locked or seated on the filler connection; and
 - (3) Each oil filler must be marked under [CS 29.1557\(c\)\(2\)](#).
- (d) *Vent.* Oil tanks must be vented as follows:
 - (1) Each oil tank must be vented from the top part of the expansion space so that venting is effective under all normal flight conditions.

- (2) Oil tank vents must be arranged so that condensed water vapour that might freeze and obstruct the line cannot accumulate at any point.
- (e) *Outlet.* There must be means to prevent entrance into the tank itself, or into the tank outlet, of any object that might obstruct the flow of oil through the system. No oil tank outlet may be enclosed by a screen or guard that would reduce the flow of oil below a safe value at any operating temperature. There must be a shutoff valve at the outlet of each oil tank used with a turbine engine unless the external portion of the oil system (including oil tank supports) is fireproof.
- (f) *Flexible liners.* Each flexible oil tank liner must be approved or shown to be suitable for the particular installation.

CS 29.1015 Oil tank tests

ED Decision 2003/16/RM

Each oil tank must be designed and installed so that –

- (a) It can withstand, without failure, any vibration, inertia, and fluid loads to which it may be subjected in operation; and
- (b) It meets the requirements of [CS 29.965](#), except that instead of the pressure specified in [CS 29.965\(b\)](#) –
 - (1) For pressurised tanks used with a turbine engine, the test pressure may not be less than 34 kPa (5 psi) plus the maximum operating pressure of the tank; and
 - (2) For all other tanks, the test pressure may not be less than 34 kPa (5 psi).

CS 29.1017 Oil lines and fittings

ED Decision 2003/16/RM

- (a) Each oil line must meet the requirements of [CS 29.993](#).
- (b) Breather lines must be arranged so that –
 - (1) Condensed water vapour that might freeze and obstruct the line cannot accumulate at any point;
 - (2) The breather discharge will not constitute a fire hazard if foaming occurs, or cause emitted oil to strike the pilot's windshield; and
 - (3) The breather does not discharge into the engine air induction system.

CS 29.1019 Oil strainer or filter

ED Decision 2003/16/RM

- (a) Each turbine engine installation must incorporate an oil strainer or filter through which all of the engine oil flows and which meets the following requirements:
 - (1) Each oil strainer or filter that has a bypass must be constructed and installed so that oil will flow at the normal rate through the rest of the system with the strainer or filter completely blocked.
 - (2) The oil strainer or filter must have the capacity (with respect to operating limitations established for the engine) to ensure that engine oil system functioning is not impaired

when the oil is contaminated to a degree (with respect to particle size and density) that is greater than that established for the engine under CS-E.

- (3) The oil strainer or filter, unless it is installed at an oil tank outlet, must incorporate a means to indicate contamination before it reaches the capacity established in accordance with subparagraph (a)(2).
- (4) The bypass of a strainer or filter must be constructed and installed so that the release of collected contaminants is minimised by appropriate location of the bypass to ensure that collected contaminants are not in the bypass flow path.
- (5) An oil strainer or filter that has no bypass, except one that is installed at an oil tank outlet, must have a means to connect it to the warning system required in [CS 29.1305\(a\)\(18\)](#).
- (b) Each oil strainer or filter in a powerplant installation using reciprocating engines must be constructed and installed so that oil will flow at the normal rate through the rest of the system with the strainer or filter element completely blocked.

CS 29.1021 Oil system drains

ED Decision 2003/16/RM

A drain (or drains) must be provided to allow safe drainage of the oil system. Each drain must –

- (a) Be accessible; and
- (b) Have manual or automatic means for positive locking in the closed position.

CS 29.1023 Oil radiators

ED Decision 2003/16/RM

- (a) Each oil radiator must be able to withstand any vibration, inertia, and oil pressure loads to which it would be subjected in operation.
- (b) Each oil radiator air duct must be located, or equipped, so that, in case of fire, and with the airflow as it would be with and without the engine operating, flames cannot directly strike the radiator.

CS 29.1025 Oil valves

ED Decision 2003/16/RM

- (a) Each oil shutoff must meet the requirements of [CS 29.1189](#).
- (b) The closing of oil shutoffs may not prevent autorotation.
- (c) Each oil valve must have positive stops or suitable index provisions in the 'on' and 'off' positions and must be supported so that no loads resulting from its operation or from accelerated flight conditions are transmitted to the lines attached to the valve.

CS 29.1027 Transmissions and gearboxes: General

ED Decision 2003/16/RM

- (a) The oil system for components of the rotor drive system that require continuous lubrication must be sufficiently independent of the lubrication systems of the engine(s) to ensure:
 - (1) Operation with any engine inoperative; and
 - (2) Safe autorotation.

-
- (b) Pressure lubrication systems for transmissions and gearboxes must comply with the requirements of [CS 29.1013](#), sub-paragraphs (c), (d) and (f) only, [CS 29.1015](#), [29.1017](#), [29.1021](#), [29.1023](#) and [29.1337\(d\)](#). In addition, the system must have:
- (1) An oil strainer or filter through which all the lubricant flows, and must:
 - (i) Be designed to remove from the lubricant any contaminant which may damage transmission and drive system components or impede the flow of lubricant to a hazardous degree; and
 - (ii) Be equipped with a bypass constructed and installed so that:
 - (A) The lubricant will flow at the normal rate through the rest of the system with the strainer or filter completely blocked; and
 - (B) The release of collected contaminants is minimised by appropriate location of the bypass to ensure that collected contaminants are not in the bypass flow path;
 - (iii) Be equipped with a means to indicate collection of contaminants on the filter or strainer at or before opening of the bypass;
 - (2) For each lubricant tank or sump outlet supplying lubrication to rotor drive systems and rotor drive system components, a screen to prevent entrance into the lubrication system of any object that might obstruct the flow of lubricant from the outlet to the filter required by sub-paragraph (b)(1). The requirements of sub-paragraph (b)(1) do not apply to screens installed at lubricant tank or sump outlets.
- (c) Splash type lubrication systems for rotor drive system gearboxes must comply with [CS 29.1021](#) and [29.1337\(d\)](#).

COOLING

CS 29.1041 General

ED Decision 2003/16/RM

- (a) The powerplant and auxiliary power unit cooling provisions must be able to maintain the temperatures of powerplant components, engine fluids, and auxiliary power unit components and fluids within the temperature limits established for these components and fluids, under ground, water, and flight operating conditions for which certification is requested, and after normal engine or auxiliary power shut-down, or both.
- (b) There must be cooling provisions to maintain the fluid temperatures in any power transmission within safe values under any critical surface (ground or water) and flight operating conditions.
- (c) Except for ground-use-only auxiliary power units, compliance with sub-paragraphs (a) and (b) must be shown by flight tests in which the temperatures of selected powerplant component and auxiliary power unit component, engine, and transmission fluids are obtained under the conditions prescribed in those paragraphs.

CS 29.1043 Cooling tests

ED Decision 2003/16/RM

- (a) *General.* For the tests prescribed in [CS 29.1041\(c\)](#), the following apply:
 - (1) If the tests are conducted under conditions deviating from the maximum ambient atmospheric temperature specified in sub-paragraph (b), the recorded powerplant temperatures must be corrected under sub-paragraphs (c) and (d), unless a more rational correction method is applicable.
 - (2) No corrected temperature determined under sub-paragraph (a)(1) may exceed established limits.
 - (3) The fuel used during the cooling tests must be of the minimum grade approved for the engines, and the mixture settings must be those used in normal operation.
 - (4) The test procedures must be as prescribed in [CS 29.1045](#) to [29.1049](#).
 - (5) For the purposes of the cooling tests, a temperature is 'stabilised' when its rate of change is less than 1°C (2°F) per minute.
- (b) *Maximum ambient atmospheric pressure.* A maximum ambient atmospheric temperature corresponding to sea-level conditions of at least 38°C (100°F) must be established. The assumed temperature lapse rate is 2.0°C (3.6°F) per thousand feet of altitude above sea-level until a temperature of -56.5°C (-69.7°F) is reached, above which altitude the temperature is considered constant at -56.5°C (-69.7°F). However, for winterisation installations, the applicant may select a maximum ambient atmospheric temperature corresponding to sea-level conditions of less than 38°C (100°F).
- (c) *Correction factor (except cylinder barrels).* Unless a more rational correction applies, temperatures of engine fluids and powerplant components (except cylinder barrels) for which temperature limits are established, must be corrected by adding to them the difference between the maximum ambient atmospheric temperature and the temperature of the ambient air at the time of the first occurrence of the maximum component or fluid temperature recorded during the cooling test.

- (d) *Correction factor for cylinder barrel temperatures.* Cylinder barrel temperatures must be corrected by adding to them 0.7 times the difference between the maximum ambient atmospheric temperature and the temperature of the ambient air at the time of the first occurrence of the maximum cylinder barrel temperature recorded during the cooling test.

CS 29.1045 Climb cooling test procedures

ED Decision 2003/16/RM

- (a) Climb cooling tests must be conducted under this paragraph for:
- (1) Category A rotorcraft; and
 - (2) Multi-engine Category B rotorcraft for which certification is requested under the Category A powerplant installation requirements, and under the requirements of [CS 29.861\(a\)](#) at the steady rate of climb or descent established under [CS 29.67\(b\)](#).
- (b) The climb or descent cooling tests must be conducted with the engine inoperative that produces the most adverse cooling conditions for the remaining engines and powerplant components.
- (c) Each operating engine must:
- (1) For helicopters for which the use of 30-minute OEI power is requested, be at 30-minute OEI power for 30 minutes, and then at maximum continuous power (or at full throttle, when above the critical altitude);
 - (2) For helicopters for which the use of continuous OEI power is requested, be at continuous OEI power (or at full throttle when above the critical altitude); and
 - (3) For other rotorcraft, be at maximum continuous power (or at full throttle when above the critical altitude).
- (d) After temperatures have stabilised in flight, the climb must be:
- (1) Begun from an altitude not greater than the lower of:
 - (i) 305 m (1000 ft) below the engine critical altitude; and
 - (ii) 305 m (1000 ft) below the maximum altitude at which the rate of climb is 0.76 m/s (150 fpm); and
 - (2) Continued for at least 5 minutes after the occurrence of the highest temperature recorded, or until the rotorcraft reaches the maximum altitude for which certification is requested.
- (e) For Category B rotorcraft without a positive rate of climb, the descent must begin at the all-engine-critical altitude and end at the higher of:
- (1) The maximum altitude at which level flight can be maintained with one engine operative; and
 - (2) Sea-level.
- (f) The climb or descent must be conducted at an airspeed representing a normal operational practice for the configuration being tested. However, if the cooling provisions are sensitive to rotorcraft speed, the most critical airspeed must be used, but need not exceed the speeds established under [CS 29.67\(a\)\(2\)](#) or [29.67\(b\)](#). The climb cooling test may be conducted in conjunction with the take-off cooling test of [CS 29.1047](#).

CS 29.1047 Take-off cooling test procedures

ED Decision 2003/16/RM

- (a) *Category A.* For each Category A rotorcraft, cooling must be shown during take-off and subsequent climb as follows:
- (1) Each temperature must be stabilised while hovering in ground effect with:
 - (i) The power necessary for hovering;
 - (ii) The appropriate cowl flap and shutter settings; and
 - (iii) The maximum weight.
 - (2) After the temperatures have stabilised, a climb must be started at the lowest practicable altitude and must be conducted with one engine inoperative.
 - (3) The operating engines must be at the greatest power for which approval is sought (or at full throttle when above the critical altitude) for the same period as this power is used in determining the take-off climbout path under [CS 29.59](#).
 - (4) At the end of the time interval prescribed in sub-paragraph (b)(3), the power must be changed to that used in meeting [CS 29.67\(a\)\(2\)](#) and the climb must be continued for:
 - (i) 30 minutes, if 30-minute OEI power is used; or
 - (ii) At least 5 minutes after the occurrence of the highest temperature recorded, if continuous OEI power or maximum continuous power is used.
 - (5) The speeds must be those used in determining the take-off flight path under [CS 29.59](#).
- (b) *Category B.* For each Category B rotorcraft, cooling must be shown during take-off and subsequent climb as follows:
- (1) Each temperature must be stabilised while hovering in ground effect with:
 - (i) The power necessary for hovering;
 - (ii) The appropriate cowl flap and shutter settings; and
 - (iii) The maximum weight.
 - (2) After the temperatures have stabilised, a climb must be started at the lowest practicable altitude with take-off power.
 - (3) Take-off power must be used for the same time interval as take-off power is used in determining the take-off flight path under [CS 29.63](#).
 - (4) At the end of the time interval prescribed in sub-paragraph (a)(3), the power must be reduced to maximum continuous power and the climb must be continued for at least 5 minutes after the occurrence of the highest temperature recorded.
 - (5) The cooling test must be conducted at an airspeed corresponding to normal operating practice for the configuration being tested. However, if the cooling provisions are sensitive to rotorcraft speed, the most critical airspeed must be used, but need not exceed the speed for best rate of climb with maximum continuous power.

CS 29.1049 Hovering cooling test procedures

ED Decision 2003/16/RM

The hovering cooling provisions must be shown –

- (a) At maximum weight or at the greatest weight at which the rotorcraft can hover (if less), at sea-level, with the power required to hover but not more than maximum continuous power, in the ground effect in still air, until at least 5 minutes after the occurrence of the highest temperature recorded; and
- (b) With maximum continuous power, maximum weight, and at the altitude resulting in zero rate of climb for this configuration, until at least 5 minutes after the occurrence of the highest temperature recorded.

INDUCTION SYSTEM

CS 29.1091 Air induction

ED Decision 2003/16/RM

- (a) The air induction system for each engine and auxiliary power unit must supply the air required by that engine and auxiliary power unit under the operating conditions for which certification is requested.
- (b) Each engine and auxiliary power unit air induction system must provide air for proper fuel metering and mixture distribution with the induction system valves in any position.
- (c) No air intake may open within the engine accessory section or within other areas of any powerplant compartment where emergence of backfire flame would constitute a fire hazard.
- (d) Each reciprocating engine must have an alternate air source.
- (e) Each alternate air intake must be located to prevent the entrance of rain, ice, or other foreign matter.
- (f) For turbine engine powered rotorcraft and rotorcraft incorporating auxiliary power units:
 - (1) There must be means to prevent hazardous quantities of fuel leakage or overflow from drains, vents, or other components of flammable fluid systems from entering the engine or auxiliary power unit intake system; and
 - (2) The air inlet ducts must be located or protected so as to minimise the ingestion of foreign matter during take-off, landing, and taxiing.

CS 29.1093 Induction system icing protection

ED Decision 2003/16/RM

- (a) *Reciprocating engines.* Each reciprocating engine air induction system must have means to prevent and eliminate icing. Unless this is done by other means, it must be shown that, in air free of visible moisture at a temperature of -1°C (30°F) and with the engines at 60% of maximum continuous power –
 - (1) Each rotorcraft with sea-level engines using conventional venturi carburetors has a preheater that can provide a heat rise of 50°C (90°F);
 - (2) Each rotorcraft with sea-level engines using carburetors tending to prevent icing has a preheater that can provide a heat rise of 39°C (70°F);
 - (3) Each rotorcraft with altitude engines using conventional venturi carburetors has a preheater that can provide a heat rise of 67°C (120°F); and
 - (4) Each rotorcraft with altitude engines using carburetors tending to prevent icing has a preheater that can provide a heat rise of 56°C (100°F).
- (b) *Turbine engines:*
 - (1) It must be shown that each turbine engine and its air inlet system can operate throughout the flight power range of the engine (including idling):
 - (i) Without accumulating ice on engine or inlet system components that would adversely affect engine operation or cause a serious loss of power under the icing conditions specified in [Appendix C](#); and

- (ii) In snow, both falling and blowing, without adverse effect on engine operation, within the limitations established for the rotorcraft.
- (2) Each turbine engine must idle for 30 minutes on the ground, with the air bleed available for engine icing protection at its critical condition, without adverse effect, in an atmosphere that is at a temperature between -9°C and -1°C (between 15°F and 30°F) and has a liquid water content not less than 0.3 grams per cubic meter in the form of drops having a mean effective diameter not less than 20 microns, followed by momentary operation at take-off power or thrust. During the 30 minutes of idle operation, the engine may be run up periodically to a moderate power or thrust setting in a manner acceptable to the Agency.
- (c) *Supercharged reciprocating engines.* For each engine having a supercharger to pressurise the air before it enters the carburettor, the heat rise in the air caused by that supercharging at any altitude may be utilised in determining compliance with subparagraph (a) if the heat rise utilised is that which will be available, automatically, for the applicable altitude and operation condition because of supercharging.

CS 29.1101 Carburettor air preheater design

ED Decision 2003/16/RM

Each carburettor air preheater must be designed and constructed to:

- (a) Ensure ventilation of the preheater when the engine is operated in cold air;
- (b) Allow inspection of the exhaust manifold parts that it surrounds; and
- (c) Allow inspection of critical parts of the preheater itself.

CS 29.1103 Induction systems ducts and air duct systems

ED Decision 2003/16/RM

- (a) Each induction system duct upstream of the first stage of the engine supercharger and of the auxiliary power unit compressor must have a drain to prevent the hazardous accumulation of fuel and moisture in the ground attitude. No drain may discharge where it might cause a fire hazard.
- (b) Each duct must be strong enough to prevent induction system failure from normal backfire conditions.
- (c) Each duct connected to components between which relative motion could exist must have means for flexibility.
- (d) Each duct within any fire zone for which a fire-extinguishing system is required must be at least:
 - (1) Fireproof, if it passes through any firewall; or
 - (2) Fire resistant, for other ducts, except that ducts for auxiliary power units must be fireproof within the auxiliary power unit fire zone.
- (e) Each auxiliary power unit induction system duct must be fireproof for a sufficient distance upstream of the auxiliary power unit compartment to prevent hot gas reverse flow from burning through auxiliary power unit ducts and entering any other compartment or area of the rotorcraft in which a hazard would be created resulting from the entry of hot gases. The materials used to form the remainder of the induction system duct and plenum chamber of the auxiliary power unit must be capable of resisting the maximum heat conditions likely to occur.

- (f) Each auxiliary power unit induction system duct must be constructed of materials that will not absorb or trap hazardous quantities of flammable fluids that could be ignited in the event of a surge or reverse flow condition.

CS 29.1105 Induction system screens

ED Decision 2003/16/RM

If induction system screens are used:

- (a) Each screen must be upstream of the carburettor;
- (b) No screen may be in any part of the induction system that is the only passage through which air can reach the engine, unless it can be deiced by heated air;
- (c) No screen may be deiced by alcohol alone; and
- (d) It must be impossible for fuel to strike any screen.

CS 29.1107 Inter-coolers and after-coolers

ED Decision 2003/16/RM

Each inter-cooler and after-cooler must be able to withstand the vibration, inertia, and air pressure loads to which it would be subjected in operation.

CS 29.1109 Carburettor air cooling

ED Decision 2003/16/RM

It must be shown under [CS 29.1043](#) that each installation using two-stage superchargers has means to maintain the air temperature, at the carburettor inlet, at or below the maximum established value.

EXHAUST SYSTEM

CS 29.1121 General

ED Decision 2003/16/RM

For powerplant and auxiliary power unit installations the following apply:

- (a) Each exhaust system must ensure safe disposal of exhaust gases without fire hazard or carbon monoxide contamination in any personnel compartment.
- (b) Each exhaust system part with a surface hot enough to ignite flammable fluids or vapours must be located or shielded so that leakage from any system carrying flammable fluids or vapours will not result in a fire caused by impingement of the fluids or vapours on any part of the exhaust system including shields for the exhaust system.
- (c) Each component upon which hot exhaust gases could impinge, or that could be subjected to high temperatures from exhaust system parts, must be fireproof. Each exhaust system component must be separated by a fireproof shield from adjacent parts of the rotorcraft that are outside the engine and auxiliary power unit compartments.
- (d) No exhaust gases may discharge so as to cause a fire hazard with respect to any flammable fluid vent or drain.
- (e) No exhaust gases may discharge where they will cause a glare seriously affecting pilot vision at night.
- (f) Each exhaust system component must be ventilated to prevent points of excessively high temperature.
- (g) Each exhaust shroud must be ventilated or insulated to avoid, during normal operation, a temperature high enough to ignite any flammable fluids or vapours outside the shroud.
- (h) If significant traps exist, each turbine engine exhaust system must have drains discharging clear of the rotorcraft, in any normal ground and flight attitudes, to prevent fuel accumulation after the failure of an attempted engine start.

CS 29.1123 Exhaust piping

ED Decision 2003/16/RM

- (a) Exhaust piping must be heat and corrosion resistant, and must have provisions to prevent failure due to expansion by operating temperatures.
- (b) Exhaust piping must be supported to withstand any vibration and inertia loads to which it would be subjected in operation.
- (c) Exhaust piping connected to components between which relative motion could exist must have provisions for flexibility.

CS 29.1125 Exhaust heat exchangers

ED Decision 2003/16/RM

For reciprocating engine powered rotorcraft the following apply:

- (a) Each exhaust heat exchanger must be constructed and installed to withstand the vibration, inertia, and other loads to which it would be subjected in operation. In addition:

-
- (1) Each exchanger must be suitable for continued operation at high temperatures and resistant to corrosion from exhaust gases;
 - (2) There must be means for inspecting the critical parts of each exchanger;
 - (3) Each exchanger must have cooling provisions wherever it is subject to contact with exhaust gases; and
 - (4) No exhaust heat exchanger or muff may have stagnant areas or liquid traps that would increase the probability of ignition of flammable fluids or vapours that might be present in case of the failure or malfunction of components carrying flammable fluids.
- (b) If an exhaust heat exchanger is used for heating ventilating air used by personnel –
- (1) There must be a secondary heat exchanger between the primary exhaust gas heat exchanger and the ventilating air system; or
 - (2) Other means must be used to prevent harmful contamination of the ventilating air.

POWERPLANT CONTROLS AND ACCESSORIES

CS 29.1141 Powerplant controls: general

ED Decision 2003/16/RM

- (a) Powerplant controls must be located and arranged under [CS 29.777](#) and marked under [CS 29.1555](#).
- (b) Each control must be located so that it cannot be inadvertently operated by persons entering, leaving or moving normally in the cockpit.
- (c) Each flexible powerplant control must be approved.
- (d) Each control must be able to maintain any set position without:
 - (1) Constant attention; or
 - (2) Tendency to creep due to control loads or vibration.
- (e) Each control must be able to withstand operating loads without excessive deflection.
- (f) Controls of powerplant valves required for safety must have:
 - (1) For manual valves, positive stops or in the case of fuel valves suitable index provisions, in the open and closed position; and
 - (2) For power-assisted valves, a means to indicate to the flight crew when the valve:
 - (i) Is in the fully open or fully closed position; or
 - (ii) Is moving between the fully open and fully closed position.

CS 29.1142 Auxiliary power unit controls

ED Decision 2003/16/RM

Means must be provided on the flight deck for starting, stopping, and emergency shutdown of each installed auxiliary power unit.

CS 29.1143 Engine controls

ED Decision 2003/16/RM

- (a) There must be a separate power control for each engine.
- (b) Power controls must be arranged to allow ready synchronisation of all engines by:
 - (1) Separate control of each engine; and
 - (2) Simultaneous control of all engines.
- (c) Each power control must provide a positive and immediately responsive means of controlling its engine.
- (d) Each fluid injection control other than fuel system control must be in the corresponding power control. However, the injection system pump may have a separate control.
- (e) If a power control incorporates a fuel shutoff feature, the control must have a means to prevent the inadvertent movement of the control into the shutoff position. The means must –
 - (1) Have a positive lock or stop at the idle position; and

- (2) Require a separate and distinct operation to place the control in the shutoff position.
- (f) For rotorcraft to be certificated for a 30-second OEI power rating, a means must be provided to automatically activate and control the 30-second OEI power and prevent any engine from exceeding the installed engine limits associated with the 30-second OEI power rating approved for the rotorcraft.

CS 29.1145 Ignition switches

ED Decision 2003/16/RM

- (a) Ignition switches must control each ignition circuit on each engine.
- (b) There must be means to quickly shut off all ignition by the grouping of switches or by a master ignition control.
- (c) Each group of ignition switches, except ignition switches for turbine engines for which continuous ignition is not required, and each master ignition control, must have a means to prevent its inadvertent operation.

CS 29.1147 Mixture controls

ED Decision 2003/16/RM

- (a) If there are mixture controls, each engine must have a separate control, and the controls must be arranged to allow:
 - (1) Separate control of each engine; and
 - (2) Simultaneous control of all engines.
- (b) Each intermediate position of the mixture controls that corresponds to a normal operating setting must be identifiable by feel and sight.

CS 29.1151 Rotor brake controls

ED Decision 2003/16/RM

- (a) It must be impossible to apply the rotor brake inadvertently in flight.
- (b) There must be means to warn the crew if the rotor brake has not been completely released before take-off.

CS 29.1157 Carburettor air temperature controls

ED Decision 2003/16/RM

There must be a separate carburettor air temperature control for each engine.

CS 29.1159 Supercharger controls

ED Decision 2003/16/RM

Each supercharger control must be accessible to:

- (a) The pilots; or
- (b) (If there is a separate flight engineer station with a control panel) the flight engineer.

CS 29.1163 Powerplant accessories

ED Decision 2003/16/RM

- (a) Each engine-mounted accessory must:
 - (1) Be approved for mounting on the engine involved;
 - (2) Use the provisions on the engine for mounting; and
 - (3) Be sealed in such a way as to prevent contamination of the engine oil system and accessory system.
- (b) Electrical equipment subject to arcing or sparking must be installed, to minimise the probability of igniting flammable fluids or vapours.
- (c) If continued rotation of an engine-driven cabin supercharger or any remote accessory driven by the engine will be a hazard if they malfunction, there must be means to prevent their hazardous rotation without interfering with the continued operation of the engine.
- (d) Unless other means are provided, torque limiting means must be provided for accessory drives located on any component of the transmission and rotor drive system to prevent damage to these components from excessive accessory load.

CS 29.1165 Engine ignition systems

ED Decision 2003/16/RM

- (a) Each battery ignition system must be supplemented with a generator that is automatically available as an alternate source of electrical energy to allow continued engine operation if any battery becomes depleted.
- (b) The capacity of batteries and generators must be large enough to meet the simultaneous demands of the engine ignition system and the greatest demands of any electrical system components that draw from the same source.
- (c) The design of the engine ignition system must account for:
 - (1) The condition of an inoperative generator;
 - (2) The condition of a completely depleted battery with the generator running at its normal operating speed; and
 - (3) The condition of a completely depleted battery with the generator operating at idling speed, if there is only one battery.
- (d) Magneto ground wiring (for separate ignition circuits) that lies on the engine side of any firewall must be installed, located, or protected, to minimise the probability of the simultaneous failure of two or more wires as a result of mechanical damage, electrical fault or other cause.
- (e) No ground wire for any engine may be routed through a fire zone of another engine unless each part of that wire within that zone is fireproof.
- (f) Each ignition system must be independent of any electrical circuit that is not used for assisting, controlling, or analysing the operation of that system.
- (g) There must be means to warn appropriate crew members if the malfunctioning of any part of the electrical system is causing the continuous discharge of any battery necessary for engine ignition.

POWERPLANT FIRE PROTECTION

CS 29.1181 Designated fire zones: regions included

ED Decision 2003/16/RM

- (a) Designated fire zones are:
- (1) The engine power section of reciprocating engines;
 - (2) The engine accessory section of reciprocating engines;
 - (3) Any complete powerplant compartment in which there is no isolation between the engine power section and the engine accessory section, for reciprocating engines;
 - (4) Any auxiliary power unit compartment;
 - (5) Any fuel-burning heater and other combustion equipment installation described in [CS 29.859](#);
 - (6) The compressor and accessory sections of turbine engines; and
 - (7) The combustor, turbine, and tailpipe sections of turbine engine installations except sections that do not contain lines and components carrying flammable fluids or gases and are isolated from the designated fire zone prescribed in sub-paragraph (a)(6) by a firewall that meets [CS 29.1191](#).
- (b) Each designated fire zone must meet the requirements of [CS 29.1183](#) to [29.1203](#).

CS 29.1183 Lines, fittings, and components

ED Decision 2003/16/RM

- (a) Except as provided in sub-paragraph (b), each line, fitting, and other component carrying flammable fluid in any area subject to engine fire conditions and each component which conveys or contains flammable fluid in a designated fire zone must be fire resistant, except that flammable fluid tanks and supports in a designated fire zone must be fireproof or be enclosed by a fireproof shield unless damage by fire to any non-fireproof part will not cause leakage or spillage of flammable fluid. Components must be shielded or located so as to safeguard against the ignition of leaking flammable fluid. An integral oil sump of less than 24 litres (5.2 Imperial gallons/25 US-quart) capacity on a reciprocating engine need not be fireproof nor be enclosed by a fireproof shield.
- (b) Sub-paragraph (a) does not apply to:
- (1) Lines, fittings, and components which are already approved as part of a type certificated engine; and
 - (2) Vent and drain lines, and their fittings, whose failure will not result in or add to, a fire hazard.

CS 29.1185 Flammable fluids

ED Decision 2003/16/RM

- (a) No tank or reservoir that is part of a system containing flammable fluids or gases may be in a designated fire zone unless the fluid contained, the design of the system, the materials used in the tank and its supports, the shutoff means, and the connections, lines, and controls provide a

degree of safety equal to that which would exist if the tank or reservoir were outside such a zone.

- (b) Each fuel tank must be isolated from the engines by a firewall or shroud.
- (c) There must be at least 13 mm ($\frac{1}{2}$ inch) of clear airspace between each tank or reservoir and each firewall or shroud isolating a designated fire zone, unless equivalent means are used to prevent heat transfer from the fire zone to the flammable fluid.
- (d) Absorbent material close to flammable fluid system components that might leak must be covered or treated to prevent the absorption of hazardous quantities of fluids.

CS 29.1187 Drainage and ventilation of fire zones

ED Decision 2003/16/RM

- (a) There must be complete drainage of each part of each designated fire zone to minimise the hazards resulting from failure or malfunction of any component containing flammable fluids. The drainage means must be:
 - (1) Effective under conditions expected to prevail when drainage is needed; and
 - (2) Arranged so that no discharged fluid will cause an additional fire hazard.
- (b) Each designated fire zone must be ventilated to prevent the accumulation of flammable vapours.
- (c) No ventilation opening may be where it would allow the entry of flammable fluids, vapours, or flame from other zones.
- (d) Ventilation means must be arranged so that no discharged vapours will cause an additional fire hazard.
- (e) For Category A rotorcraft there must be means to allow the crew to shut off the sources of forced ventilation in any fire zone (other than the engine power section of the powerplant compartment) unless the amount of extinguishing agent and the rate of discharge are based on the maximum airflow through that zone.

CS 29.1189 Shutoff means

ED Decision 2003/16/RM

- (a) There must be means to shut off or otherwise prevent hazardous quantities of fuel, oil, de-icing fluid, and other flammable fluids from flowing into, within, or through any designated fire zone, except that this means need not be provided:
 - (1) For lines, fittings, and components forming an integral part of an engine;
 - (2) For oil systems for turbine engine installations in which all components of the oil system, including oil tanks, are fireproof or located in areas not subject to engine fire conditions; or
 - (3) For engine oil systems in Category B rotorcraft using reciprocating engines of less than 8195 cm³ (500 cubic inches) displacement.
- (b) The closing of any fuel shutoff valve for any engine may not make fuel unavailable to the remaining engines.

- (c) For Category A rotorcraft no hazardous quantity of flammable fluid may drain into any designated fire zone after shutoff has been accomplished, nor may the closing of any fuel shutoff valve for an engine make fuel unavailable to the remaining engines.
- (d) The operation of any shutoff may not interfere with the later emergency operation of any other equipment, such as the means for declutching the engine from the rotor drive.
- (e) Each shutoff valve and its control must be designed, located, and protected to function properly under any condition likely to result from fire in a designated fire zone.
- (f) Except for ground-use-only auxiliary power unit installations, there must be means to prevent inadvertent operation of each shutoff and to make it possible to re-open it in flight after it has been closed.

CS 29.1191 Firewalls

ED Decision 2003/16/RM

- (a) Each engine, including the combustor, turbine, and tailpipe sections of turbine engine installations, must be isolated by a firewall, shroud, or equivalent means, from personnel compartments, structures, controls, rotor mechanisms, and other parts that are:
 - (1) Essential to controlled flight and landing; and
 - (2) Not protected under [CS 29.861](#).
- (b) Each auxiliary power unit, combustion heater, and other combustion equipment to be used in flight, must be isolated from the rest of the rotorcraft by firewalls, shrouds, or equivalent means.
- (c) Each firewall or shroud must be constructed so that no hazardous quantity of air, fluid, or flame can pass from any engine compartment to other parts of the rotorcraft.
- (d) Each opening in the firewall or shroud must be sealed with close-fitting fireproof grommets, bushings, or firewall fittings.
- (e) Each firewall and shroud must be fireproof and protected against corrosion.
- (f) In meeting this paragraph, account must be taken of the probable path of a fire as affected by the airflow in normal flight and in autorotation.

CS 29.1193 Cowling and engine compartment covering

ED Decision 2003/16/RM

- (a) Each cowling and engine compartment covering must be constructed and supported so that it can resist the vibration, inertia and air loads to which it may be subjected in operation.
- (b) Cowling must meet the drainage and ventilation requirements of [CS 29.1187](#).
- (c) On rotorcraft with a diaphragm isolating the engine power section from the engine accessory section, each part of the accessory section cowling subject to flame in case of fire in the engine power section of the powerplant must:
 - (1) Be fireproof; and
 - (2) Meet the requirements of [CS 29.1191](#).
- (d) Each part of the cowling or engine compartment covering subject to high temperatures due to its nearness to exhaust system parts or exhaust gas impingement must be fireproof.

- (e) Each rotorcraft must:
 - (1) Be designed and constructed so that no fire originating in any fire zone can enter, either through openings or by burning through external skin, any other zone or region where it would create additional hazards;
 - (2) Meet the requirements of sub-paragraph (e)(1) with the landing gear retracted (if applicable); and
 - (3) Have fireproof skin in areas subject to flame if a fire starts in or burns out of any designated fire zone.
- (f) A means of retention for each openable or readily removable panel, cowling, or engine or rotor drive system covering must be provided to preclude hazardous damage to rotors or critical control components in the event of:
 - (1) Structural or mechanical failure of the normal retention means, unless such failure is extremely improbable; or
 - (2) Fire in a fire zone, if such fire could adversely affect the normal means of retention.

CS 29.1194 Other surfaces

ED Decision 2003/16/RM

All surfaces aft of, and near, engine compartments and designated fire zones, other than tail surfaces not subject to heat, flames, or sparks emanating from a designated fire zone or engine compartment, must be at least fire resistant.

CS 29.1195 Fire extinguishing systems

ED Decision 2003/16/RM

- (a) Each turbine engine powered rotorcraft and Category A reciprocating engine powered rotorcraft, and each Category B reciprocating engine powered rotorcraft with engines of more than 24 581 cm³ (1500 cubic inches) must have a fire extinguishing system for the designated fire zones. The fire extinguishing system for a powerplant must be able to simultaneously protect all zones of the powerplant compartment for which protection is provided.
- (b) For multi-engine powered rotorcraft, the fire extinguishing system, the quantity of extinguishing agent, and the rate of discharge must:
 - (1) For each auxiliary power unit and combustion equipment, provide at least one adequate discharge; and
 - (2) For each other designated fire zone, provide two adequate discharges.
- (c) For single engine rotorcraft, the quantity of extinguishing agent and the rate of discharge must provide at least one adequate discharge for the engine compartment.
- (d) It must be shown by either actual or simulated flight tests that under critical airflow conditions in flight the discharge of the extinguishing agent in each designated fire zone will provide an agent concentration capable of extinguishing fires in that zone and of minimising the probability of re-ignition.

CS 29.1197 Fire extinguishing agents

ED Decision 2003/16/RM

- (a) Fire extinguishing agents must:
- (1) Be capable of extinguishing flames emanating from any burning of fluids or other combustible materials in the area protected by the fire extinguishing system; and
 - (2) Have thermal stability over the temperature range likely to be experienced in the compartment in which they are stored.
- (b) If any toxic extinguishing agent is used, it must be shown by test that entry of harmful concentrations of fluid or fluid vapours into any personnel compartment (due to leakage during normal operation of the rotorcraft, or discharge on the ground or in flight) is prevented, even though a defect may exist in the extinguishing system.

AMC 29.1197 Fire extinguishing agents

ED Decision 2012/022/R

1. This AMC addresses alternatives to Halon and provides further guidance and acceptable means of compliance to supplement FAA AC 29-2C AC 29.1197. As such it should be used in conjunction with the FAA AC and take precedence over it in the showing of compliance.
2. The Montreal Protocol, in existence since 1987, is an international agreement to phase out production and use of ozone-depleting substances, including halogenated hydrocarbons also known as Halon. A European regulation¹ governing substances that deplete the ozone layer was published in 2000 containing initial provisions for Halon phase-out, but also exemptions for critical uses of Halon, including fire extinguishing in aviation.
3. 'Cut-off' dates (i.e. Halon no longer acceptable in new applications for type certification) and 'end' dates (i.e. Halon no longer acceptable for use in rotorcraft) have been subsequently established by a new regulation in 2010², as presented in Table 3.1 below:

Table 3.1: 'Cut-off' and 'end' dates

Rotorcraft compartment	Type of extinguisher	Type of halon	Dates	
			Cut-off	End
Lavatory waste receptacles	Built-in	1301 1211 2402	31 December 2011	31 December 2020
Cabins and crew compartments	Hand (portable)	1211 2402	31 December 2014	31 December 2025
Propulsion systems and Auxiliary Power Units	Built-in	1301 1211 2402	31 December 2014	31 December 2040
Normally unoccupied cargo compartments	Built-in	1301 1211 2402	31 December 2018	31 December 2040

¹ Regulation (EC) No 2037/2000 of the European Parliament and of the Council of 29 June 2000 on substances that deplete the ozone layer.

² Commission Regulation (EU) No 744/2010 of 18 August 2010 amending Regulation (EC) No 1005/2009 of the European Parliament and of the Council on substances that deplete the ozone layer, with regard to the critical uses of halon (OJ L 218, 19.8.2010, p. 2).

4. In the course of Halon replacement, novel agent types such as fluorine ketone liquids and aerosols are being developed. In contrast to the gaseous agents, e.g. Halon 1301, which disperse more or less easily inside a given volume when released, liquid and powder-type substances require the evaluation of precise spray vectors and more complex piping configurations inside the compartment in order to achieve the concentration-over-time certification limits as required to act as an effective fire agent.

5. Hand fire extinguishers and agents

Historically, Halon 1211 has been the most widespread agent in hand (portable) fire extinguishers to be used in rotorcraft compartments and cabins. Minimum Performance Standards (MPS) for the agents are laid down in Appendix A to Report DOT/FAA/AR-01/37 of August 2002, while acceptable criteria to select the fire extinguishers containing said agents are laid down in the FAA Advisory Circular AC 20-42D. Three agent alternatives to Halon are presently known to meet the MPS: HFC-227ea, HFC-236fa and HFC Blend B. However, these agents are significantly heavier and occupy a greater volume than Halon 1211. This may indirectly (i.e. additional weight of the fire extinguisher and additional weight of the structures supporting it), increase CO₂ emissions. Furthermore some of these agents have also been identified for having a global warming potential much higher than Halon. Therefore, further research is underway to develop additional alternatives to Halon 1211 for hand fire extinguishers.

Should an applicant wish to propose, even before the end of 2014, any alternative agent for hand fire extinguishers meeting the mentioned MPS, the Agency will initiate a Certification Review Item addressing the use of such an alternate fire extinguishing agent.

6. Fire protection of propulsion systems and APU

Historically, Halon 1301 has been the most widespread agent used in engine or APU compartments to protect against Class B fires (i.e. fuel or other flammable fluids). The MPS for agents to be used in these compartments are particularly demanding, because of the presence of fuel and other volatile fluids in close proximity to high temperature surfaces. Various alternatives are being developed (e.g. FK-5-1-12), while the FAA is aiming at issuing a report containing the MPS.

Should an applicant wish to propose, even before the end of 2014, any alternative agent for Class B fire extinction in engine or APU compartments, even in the absence of a published MPS, the Agency will initiate a Certification Review Item addressing the use of such an alternate fire extinguishing agent.

[Amdt 29/3]

CS 29.1199 Extinguishing agent containers

ED Decision 2003/16/RM

- (a) Each extinguishing agent container must have a pressure relief to prevent bursting of the container by excessive internal pressures.
- (b) The discharge end of each discharge line from a pressure relief connection must be located so that discharge of the fire extinguishing agent would not damage the rotorcraft. The line must also be located or protected to prevent clogging caused by ice or other foreign matter.
- (c) There must be a means for each fire extinguishing agent container to indicate that the container has discharged or that the charging pressure is below the established minimum necessary for proper functioning.

- (d) The temperature of each container must be maintained, under intended operating conditions, to prevent the pressure in the container from:
 - (1) Falling below that necessary to provide an adequate rate of discharge; or
 - (2) Rising high enough to cause premature discharge.

CS 29.1201 Fire extinguishing system materials

ED Decision 2003/16/RM

- (a) No materials in any fire extinguishing system may react chemically with any extinguishing agent so as to create a hazard.
- (b) Each system component in an engine compartment must be fireproof.

CS 29.1203 Fire detector systems

ED Decision 2003/16/RM

- (a) For each turbine engine powered rotorcraft and Category A reciprocating engine powered rotorcraft, and for each Category B reciprocating engine powered rotorcraft with engines of more than 14 748 cm³ (900 cubic inches) displacement there must be approved, quick-acting fire detectors in designated fire zones and in the combustor, turbine, and tailpipe sections of turbine installations (whether or not such sections are designated fire zones) in numbers and locations ensuring prompt detection of fire in those zones.
- (b) Each fire detector must be constructed and installed to withstand any vibration, inertia and other loads to which it would be subjected in operation.
- (c) No fire detector may be affected by any oil, water, other fluids, or fumes that might be present.
- (d) There must be means to allow crew members to check, in flight, the functioning of each fire detector system electrical circuit.
- (e) The wiring and other components of each fire detector system in an engine compartment must be at least fire resistant.
- (f) No fire detector system component for any fire zone may pass through another fire zone, unless –
 - (1) It is protected against the possibility of false warnings resulting from fires in zones through which it passes; or
 - (2) The zones involved are simultaneously protected by the same detector and extinguishing systems.

SUBPART F — EQUIPMENT

GENERAL

CS 29.1301 Function and installation

ED Decision 2003/16/RM

Each item of installed equipment must:

- (a) Be of a kind and design appropriate to its intended function;
- (b) Be labelled as to its identification, function, or operating limitations, or any applicable combination of these factors;
- (c) Be installed according to limitations specified for that equipment; and
- (d) Function properly when installed.

CS 29.1303 Flight and navigation instruments

ED Decision 2003/16/RM

The following are required flight and navigational instruments:

- (a) An airspeed indicator. For Category A rotorcraft with V_{NE} less than a speed at which unmistakable pilot cues provide overspeed warning, a maximum allowable airspeed indicator must be provided. If maximum allowable airspeed varies with weight, altitude, temperature, or rpm, the indicator must show that variation.
- (b) A sensitive altimeter.
- (c) A magnetic direction indicator.
- (d) A clock displaying hours, minutes, and seconds with a sweep-second pointer or digital presentation.
- (e) A free-air temperature indicator.
- (f) A non-tumbling gyroscopic bank and pitch indicator.
- (g) A gyroscopic rate-of-turn indicator combined with an integral slip-skid indicator (turn-and-bank indicator) except that only a slip- skid indicator is required on rotorcraft with a third attitude instrument system that:
 - (1) Is usable through flight attitudes of $\pm 80^\circ$ of pitch and $\pm 120^\circ$ of roll;
 - (2) Is powered from a source independent of the electrical generating system;
 - (3) Continues reliable operation for a minimum of 30 minutes after total failure of the electrical generating system;
 - (4) Operates independently of any other attitude indicating system;
 - (5) Is operative without selection after total failure of the electrical generating system;
 - (6) Is located on the instrument panel in a position acceptable to the Agency that will make it plainly visible to and usable by any pilot at his station; and
 - (7) Is appropriately lighted during all phases of operation.
- (h) A gyroscopic direction indicator.

- (i) A rate-of-climb (vertical speed) indicator.
- (j) For Category A rotorcraft, a speed warning device when V_{NE} is less than the speed at which unmistakable overspeed warning is provided by other pilot cues. The speed warning device must give effective aural warning (differing distinctly from aural warnings used for other purposes) to the pilots whenever the indicated speed exceeds V_{NE} plus 5.6 km/h (3 knots) and must operate satisfactorily throughout the approved range of altitudes and temperatures.

CS 29.1305 Power plant instruments

ED Decision 2008/010/R

The following are required power plant instruments:

- (a) For each rotorcraft:
 - (1) A carburettor air temperature indicator for each reciprocating engine;
 - (2) A cylinder head temperature indicator for each air-cooled reciprocating engine, and a coolant temperature indicator for each liquid-cooled reciprocating engine;
 - (3) A fuel quantity indicator for each fuel tank;
 - (4) A low fuel warning device for each fuel tank which feeds an engine. This device must:
 - (i) Provide a warning to the crew when approximately 10 minutes of usable fuel remains in the tank; and
 - (ii) Be independent of the normal fuel quantity indicating system.
 - (5) A manifold pressure indicator, for each reciprocating engine of the altitude type;
 - (6) An oil pressure indicator for each pressure-lubricated gearbox;
 - (7) An oil pressure warning device for each pressure-lubricated gearbox to indicate when the oil pressure falls below a safe value;
 - (8) An oil quantity indicator for each oil tank and each rotor drive gearbox, if lubricant is self-contained;
 - (9) An oil temperature indicator for each engine;
 - (10) An oil temperature warning device to indicate unsafe oil temperatures in each main rotor drive gearbox, including gearboxes necessary for rotor phasing;
 - (11) A gas temperature indicator for each turbine engine;
 - (12) A gas producer rotor tachometer for each turbine engine;
 - (13) A tachometer for each engine that, if combined with the applicable instrument required by sub-paragraph (a)(14), indicates rotor rpm during autorotation;
 - (14) At least one tachometer to indicate, as applicable:
 - (i) The rpm of the single main rotor;
 - (ii) The common rpm of any main rotors whose speeds cannot vary appreciably with respect to each other; and
 - (iii) The rpm of each main rotor whose speed can vary appreciably with respect to that of another main rotor;
 - (15) A free power turbine tachometer for each turbine engine;

-
- (16) A means, for each turbine engine, to indicate power for that engine;
 - (17) For each turbine engine, an indicator to indicate the functioning of the power plant ice protection system;
 - (18) An indicator for the fuel filter required by [CS 29.997](#) to indicate the occurrence of contamination of the filter to the degree established in compliance with [CS 29.955](#);
 - (19) For each turbine engine, a warning means for the oil strainer or filter required by [CS 29.1019](#), if it has no bypass, to warn the pilot of the occurrence of contamination of the strainer or filter before it reaches the capacity established in accordance with [CS 29.1019\(a\)\(2\)](#);
 - (20) An indicator to indicate the functioning of any selectable or controllable heater used to prevent ice clogging of fuel system components;
 - (21) An individual fuel pressure indicator for each engine, unless the fuel system which supplies that engine does not employ any pumps, filters, or other components subject to degradation or failure which may adversely affect fuel pressure at the engine;
 - (22) A means to indicate to the flight crew the failure of any fuel pump installed to show compliance with [CS 29.955](#);
 - (23) Warning or caution devices to signal to the flight crew when ferromagnetic particles are detected by the chip detector required by [CS 29.1337\(e\)](#); and
 - (24) For auxiliary power units, an individual indicator, warning or caution device, or other means to advise the flight crew that limits are being exceeded, if exceeding these limits can be hazardous, for:
 - (i) Gas temperature;
 - (ii) Oil pressure; and
 - (iii) Rotor speed.
 - (25) For rotorcraft for which a 30-second/2-minute OEI power rating is requested, a means must be provided to alert the pilot when the engine is at the 30-second and 2-minute OEI power levels, when the event begins, and when the time interval expires.
 - (26) For each turbine engine utilising 30-second/2-minute OEI power, a device or system must be provided for use by ground personnel which:
 - (i) Automatically records each usage and duration of power at the 30-second and 2-minute OEI levels;
 - (ii) Permits retrieval of the recorded data;
 - (iii) Can be reset only by ground maintenance personnel; and
 - (iv) Has a means to verify proper operation of the system or device.
- (b) For Category A rotorcraft:
- (1) An individual oil pressure indicator for each engine, and either an independent warning device for each engine or a master warning device for the engines with means for isolating the individual warning circuit from the master warning device;
 - (2) An independent fuel pressure warning device for each engine or a master warning device for all engines with provision for isolating the individual warning device from the master warning device; and

- (3) Fire warning indicators.
- (c) For Category B rotorcraft:
 - (1) An individual oil pressure indicator for each engine; and
 - (2) Fire warning indicators, when fire detection is required.

[Amdt. No. 29/2]

CS 29.1307 Miscellaneous equipment

ED Decision 2003/16/RM

The following is required miscellaneous equipment:

- (a) An approved seat for each occupant.
- (b) A master switch arrangement for electrical circuits other than ignition.
- (c) Hand fire extinguishers.
- (d) A windshield wiper or equivalent device for each pilot station.
- (e) A two-way radio communication system.

CS 29.1309 Equipment, systems, and installations

ED Decision 2003/16/RM

- (a) The equipment, systems, and installations whose functioning is required by this CS-29 must be designed and installed to ensure that they perform their intended functions under any foreseeable operating condition.
- (b) The rotorcraft systems and associated components, considered separately and in relation to other systems, must be designed so that –
 - (1) For Category B rotorcraft, the equipment, systems, and installations must be designed to prevent hazards to the rotorcraft if they malfunction or fail; or
 - (2) For Category A rotorcraft:
 - (i) The occurrence of any failure condition which would prevent the continued safe flight and landing of the rotorcraft is extremely improbable; and
 - (ii) The occurrence of any other failure conditions which would reduce the capability of the rotorcraft or the ability of the crew to cope with adverse operating conditions is improbable.
- (c) Warning information must be provided to alert the crew to unsafe system operating conditions and to enable them to take appropriate corrective action. Systems, controls, and associated monitoring and warning means must be designed to minimise crew errors which could create additional hazards.
- (d) Compliance with the requirements of sub-paragraph (b)(2) must be shown by analysis and, where necessary, by appropriate ground, flight, or simulator tests. The analysis must consider:
 - (1) Possible modes of failure, including malfunctions and damage from external sources;
 - (2) The probability of multiple failures and undetected failures;
 - (3) The resulting effects on the rotorcraft and occupants, considering the stage of flight and operating conditions; and

- (4) The crew warning cues, corrective action required, and the capability of detecting faults.
- (e) For Category A rotorcraft, each installation whose functioning is required by this CS-29 and which requires a power supply is an 'essential load' on the power supply. The power sources and the system must be able to supply the following power loads in probable operating combinations and for probable durations:
 - (1) Loads connected to the system with the system functioning normally.
 - (2) Essential loads, after failure of any one prime mover, power converter, or energy storage device.
 - (3) Essential loads, after failure of:
 - (i) Any one engine, on rotorcraft with two engines; and
 - (ii) Any two engines, on rotorcraft with three or more engines.
- (f) In determining compliance with sub-paragraphs (e)(2) and (3), the power loads may be assumed to be reduced under a monitoring procedure consistent with safety in the kinds of operations authorised. Loads not required for controlled flight need not be considered for the two-engine-inoperative condition on rotorcraft with three or more engines.
- (g) In showing compliance with sub-paragraphs (a) and (b) with regard to the electrical system and to equipment design and installation, critical environmental conditions must be considered. For electrical generation, distribution and utilisation equipment required by or used in complying with this CS-29, except equipment covered by European Technical Standard Orders containing environmental test procedures, the ability to provide continuous, safe service under foreseeable environmental conditions may be shown by environmental tests, design analysis, or reference to previous comparable service experience on other aircraft.
- (h) In showing compliance with sub-paragraphs (a) and (b), the effects of lightning strikes on the rotorcraft must be considered.

INSTRUMENTS: INSTALLATION

CS 29.1321 Arrangement and visibility

ED Decision 2003/16/RM

- (a) Each flight, navigation, and powerplant instrument for use by any pilot must be easily visible to him from his station with the minimum practicable deviation from his normal position and line of vision when he is looking forward along the flight path.
- (b) Each instrument necessary for safe operation, including the airspeed indicator, gyroscopic direction indicator, gyroscopic bank-and-pitch indicator, slip-skid indicator, altimeter, rate-of-climb indicator, rotor tachometers, and the indicator most representative of engine power, must be grouped and centred as nearly as practicable about the vertical plane of the pilot's forward vision. In addition, for rotorcraft approved for IFR flight:
 - (1) The instrument that most effectively indicates attitude must be on the panel in the top centre position;
 - (2) The instrument that most effectively indicates direction of flight must be adjacent to and directly below the attitude instrument;
 - (3) The instrument that most effectively indicates airspeed must be adjacent to and to the left of the attitude instrument; and
 - (4) The instrument that most effectively indicates altitude or is most frequently utilised in control of altitude must be adjacent to and to the right of the attitude instrument.
- (c) Other required powerplant instruments must be closely grouped on the instrument panel.
- (d) Identical powerplant instruments for the engines must be located so as to prevent any confusion as to which engine each instrument relates.
- (e) Each powerplant instrument vital to safe operation must be plainly visible to appropriate crew members.
- (f) Instrument panel vibration may not damage, or impair the readability or accuracy of, any instrument.
- (g) If a visual indicator is provided to indicate malfunction of an instrument, it must be effective under all probable cockpit lighting conditions.

CS 29.1322 Warning, caution, and advisory lights

ED Decision 2003/16/RM

If warning, caution or advisory lights are installed in the cockpit they must, unless otherwise approved by the Agency, be:

- (a) Red, for warning lights (lights indicating a hazard which may require immediate corrective action);
- (b) Amber, for caution lights (lights indicating the possible need for future corrective action);
- (c) Green, for safe operation lights; and
- (d) Any other colour, including white, for lights not described in sub-paragraphs (a) to (c), provided the colour differs sufficiently from the colours prescribed in sub-paragraphs (a) to (c) to avoid possible confusion.

CS 29.1323 Airspeed indicating system

ED Decision 2003/16/RM

For each airspeed indicating system, the following apply:

- (a) Each airspeed indicating instrument must be calibrated to indicate true airspeed (at sea-level with a standard atmosphere) with a minimum practicable instrument calibration error when the corresponding pitot and static pressures are applied.
- (b) Each system must be calibrated to determine system error excluding airspeed instrument error. This calibration must be determined:
 - (1) In level flight at speeds of 37 km/h (20 knots) and greater, and over an appropriate range of speeds for flight conditions of climb and autorotation; and
 - (2) During take-off, with repeatable and readable indications that ensure:
 - (i) Consistent realisation of the field lengths specified in the Rotorcraft Flight Manual; and
 - (ii) Avoidance of the critical areas of the height-velocity envelope as established under [CS 29.87](#).
- (c) For Category A rotorcraft:
 - (1) The indication must allow consistent definition of the take-off decision point; and
 - (2) The system error, excluding the airspeed instrument calibration error, may not exceed –
 - (i) 3% or 9.3 km/h (5 knots), whichever is greater, in level flight at speeds above 80% of take-off safety speed; and
 - (ii) 19 km/h (10 knots) in climb at speeds from 19 km/h (10 knots) below take-off safety speed to 19 km/h (10 knots) above V_Y .
- (d) For Category B rotorcraft, the system error, excluding the airspeed instrument calibration error, may not exceed 3% or 9.3 km/h (5 knots), whichever is greater, in level flight at speeds above 80% of the climbout speed attained at 15 m (50 ft) when complying with [CS 29.63](#).
- (e) Each system must be arranged, so far as practicable, to prevent malfunction or serious error due to the entry of moisture, dirt, or other substances.
- (f) Each system must have a heated pitot tube or an equivalent means of preventing malfunction due to icing.

CS 29.1325 Static pressure and pressure altimeter systems

ED Decision 2003/16/RM

- (a) Each instrument with static air case connections must be vented to the outside atmosphere through an appropriate piping system.
- (b) Each vent must be located where its orifices are least affected by airflow variation, moisture, or other foreign matter.
- (c) Each static pressure port must be designed and located in such manner that the correlation between air pressure in the static pressure system and true ambient atmospheric static pressure is not altered when the rotorcraft encounters icing conditions. An anti-icing means or an alternate source of static pressure may be used in showing compliance with this requirement.

If the reading of the altimeter, when on the alternate static pressure system, differs from the reading of the altimeter when on the primary static system by more than 15 m (50 ft), a correction card must be provided for the alternate static system.

- (d) Except for the vent into the atmosphere, each system must be airtight.
- (e) Each pressure altimeter must be approved and calibrated to indicate pressure altitude in a standard atmosphere with a minimum practicable calibration error when the corresponding static pressures are applied.
- (f) Each system must be designed and installed so that an error in indicated pressure altitude, at sea-level, with a standard atmosphere, excluding instrument calibration error, does not result in an error of more than ± 9 m (± 30 ft) per 185 km/h (100 knots) speed. However, the error need not be less than ± 9 m (± 30 ft).
- (g) Except as provided in sub-paragraph (h) if the static pressure system incorporates both a primary and an alternate static pressure source, the means for selecting one or the other source must be designed so that:
 - (1) When either source is selected, the other is blocked off; and
 - (2) Both sources cannot be blocked off simultaneously.
- (h) For unpressurised rotorcraft, sub-paragraph (g)(1) does not apply if it can be demonstrated that the static pressure system calibration, when either static pressure source is selected, is not changed by the other static pressure source being open or blocked.

CS 29.1327 Magnetic direction indicator

ED Decision 2003/16/RM

- (a) Each magnetic direction indicator must be installed so that its accuracy is not excessively affected by the rotorcraft's vibration or magnetic fields.
- (b) The compensated installation may not have a deviation, in level flight, greater than 10° on any heading.

CS 29.1329 Automatic pilot system

ED Decision 2003/16/RM

- (a) Each automatic pilot system must be designed so that the automatic pilot can:
 - (1) Be sufficiently overpowered by one pilot to allow control of the rotorcraft; and
 - (2) Be readily and positively disengaged by each pilot to prevent it from interfering with the control of the rotorcraft.
- (b) Unless there is automatic synchronisation, each system must have a means to readily indicate to the pilot the alignment of the actuating device in relation to the control system it operates.
- (c) Each manually operated control for the system's operation must be readily accessible to the pilots.
- (d) The system must be designed and adjusted so that, within the range of adjustment available to the pilot, it cannot produce hazardous loads on the rotorcraft, or create hazardous deviations in the flight path, under any flight condition appropriate to its use, either during normal operation or in the event of a malfunction, assuming that corrective action begins within a reasonable period of time.

- (e) If the automatic pilot integrates signals from auxiliary controls or furnishes signals for operation of other equipment, there must be positive interlocks and sequencing of engagement to prevent improper operation.
- (f) If the automatic pilot system can be coupled to airborne navigation equipment, means must be provided to indicate to the pilots the current mode of operation. Selector switch position is not acceptable as a means of indication.

CS 29.1331 Instruments using a power supply

ED Decision 2003/16/RM

For Category A rotorcraft:

- (a) Each required flight instrument using a power supply must have –
 - (1) Two independent sources of power;
 - (2) A means of selecting either power source; and
 - (3) A visual means integral with each instrument to indicate when the power adequate to sustain proper instrument performance is not being supplied. The power must be measured at or near the point where it enters the instrument. For electrical instruments, the power is considered to be adequate when the voltage is within approved limits; and
- (b) The installation and power supply system must be such that failure of any flight instrument connected to one source, or of the energy supply from one source, or a fault in any part of the power distribution system does not interfere with the proper supply of energy from any other source.

CS 29.1333 Instrument systems

ED Decision 2003/16/RM

For systems that operate the required flight instruments which are located at each pilot's station, the following apply:

- (a) Only the required flight instruments for the first pilot may be connected to that operating system.
- (b) The equipment, systems, and installations must be designed so that one display of the information essential to the safety of flight which is provided by the flight instruments remains available to a pilot, without additional crew member action, after any single failure or combination of failures that are not shown to be extremely improbable.
- (c) Additional instruments, systems, or equipment may not be connected to the operating system for a second pilot unless provisions are made to ensure the continued normal functioning of the required flight instruments in the event of any malfunction of the additional instruments, systems, or equipment which is not shown to be extremely improbable.

CS 29.1335 Flight director systems

ED Decision 2003/16/RM

If a flight director system is installed, means must be provided to indicate to the flight crew its current mode of operation. Selector switch position is not acceptable as a means of indication.

CS 29.1337 Powerplant instruments

ED Decision 2003/16/RM

- (a) *Instruments and instrument lines*
- (1) Each powerplant and auxiliary power unit instrument line must meet the requirements of [CS 29.993](#) and [29.1183](#).
 - (2) Each line carrying flammable fluids under pressure must:
 - (i) Have restricting orifices or other safety devices at the source of pressure to prevent the escape of excessive fluid if the line fails; and
 - (ii) Be installed and located so that the escape of fluids would not create a hazard.
 - (3) Each power plant and auxiliary power unit instrument that utilises flammable fluids must be installed and located so that the escape of fluid would not create a hazard.
- (b) *Fuel quantity indicator.* There must be means to indicate to the flight-crew members the quantity, in US-gallons or equivalent units, of usable fuel in each tank during flight. In addition:
- (1) Each fuel quantity indicator must be calibrated to read 'zero' during level flight when the quantity of fuel remaining in the tank is equal to the unusable fuel supply determined under [CS 29.959](#);
 - (2) When two or more tanks are closely interconnected by a gravity feed system and vented, and when it is impossible to feed from each tank separately, at least one fuel quantity indicator must be installed;
 - (3) Tanks with interconnected outlets and airspaces may be treated as one tank and need not have separate indicators; and
 - (4) Each exposed sight gauge used as a fuel quantity indicator must be protected against damage.
- (c) *Fuel flowmeter system.* If a fuel flowmeter system is installed, each metering component must have a means for bypassing the fuel supply if malfunction of that component severely restricts fuel flow.
- (d) *Oil quantity indicator.* There must be a stick gauge or equivalent means to indicate the quantity of oil:
- (1) In each tank; and
 - (2) In each transmission gearbox.
- (e) Rotor drive system transmissions and gearboxes utilising ferromagnetic materials must be equipped with chip detectors designed to indicate the presence of ferromagnetic particles resulting from damage or excessive wear within the transmission or gearbox. Each chip detector must:
- (1) Be designed to provide a signal to the indicator required by [CS 29.1305\(a\)\(23\)](#); and
 - (2) Be provided with a means to allow crew members to check, in flight, the function of each detector electrical circuit and signal.

ELECTRICAL SYSTEMS AND EQUIPMENT

CS 29.1351 General

ED Decision 2003/16/RM

- (a) *Electrical system capacity.* The required generating capacity and the number and kind of power sources must:
- (1) Be determined by an electrical load analysis; and
 - (2) Meet the requirements of [CS 29.1309](#).
- (b) *Generating system.* The generating system includes electrical power sources, main power busses, transmission cables, and associated control, regulation, and protective devices. It must be designed so that:
- (1) Power sources function properly when independent and when connected in combination;
 - (2) No failure or malfunction of any power source can create a hazard or impair the ability of remaining sources to supply essential loads;
 - (3) The system voltage and frequency (as applicable) at the terminals of essential load equipment can be maintained within the limits for which the equipment is designed, during any probable operating condition;
 - (4) System transients due to switching, fault clearing, or other causes do not make essential loads inoperative, and do not cause a smoke or fire hazard;
 - (5) There are means accessible in flight to appropriate crew members for the individual and collective disconnection of the electrical power sources from the main bus; and
 - (6) There are means to indicate to appropriate crew members the generating system quantities essential for the safe operation of the system, such as the voltage and current supplied by each generator.
- (c) *External power.* If provisions are made for connecting external power to the rotorcraft, and that external power can be electrically connected to equipment other than that used for engine starting, means must be provided to ensure that no external power supply having a reverse polarity, or a reverse phase sequence, can supply power to the rotorcraft's electrical system.
- (d) *Operation with the normal electrical power generating system inoperative.*
- (1) It must be shown by analysis, tests, or both, that the rotorcraft can be operated safely in VFR conditions, for a period of not less than five minutes, with the normal electrical power generating system inoperative, with critical type fuel (from the stand-point of flameout and restart capability), and with the rotorcraft initially at the maximum certificated altitude. Parts of the electrical system may remain on if:
 - (i) A single malfunction, including a wire bundle or junction box fire, cannot result in loss of the part turned off and the part turned on; and
 - (ii) The parts turned on are electrically and mechanically isolated from the parts turned off.
 - (2) Additional requirements for Category A Rotorcraft

- (i) Unless it can be shown that the loss of the normal electrical power generating system is extremely improbable, an emergency electrical power system, independent of the normal electrical power generating system, must be provided with sufficient capacity to power all systems necessary for continued safe flight and landing.
- (ii) Failures, including junction box, control panel or wire bundle fires, which would result in the loss of the normal and emergency systems must be shown to be extremely improbable.
- (iii) Systems necessary for immediate safety must continue to operate following the loss of the normal electrical power generating system, without the need for flight crew action.

CS 29.1353 Electrical equipment and installations

ED Decision 2003/16/RM

- (a) Electrical equipment, controls, and wiring must be installed so that operation of any one unit or system of units will not adversely affect the simultaneous operation of any other electrical unit or system essential to safe operation.
- (b) Cables must be grouped, routed, and spaced so that damage to essential circuits will be minimised if there are faults in heavy current-carrying cables.
- (c) Storage batteries must be designed and installed as follows:
 - (1) Safe cell temperatures and pressures must be maintained during any probable charging and discharging condition. No uncontrolled increase in cell temperature may result when the battery is recharged (after previous complete discharge):
 - (i) At maximum regulated voltage or power;
 - (ii) During a flight of maximum duration; and
 - (iii) Under the most adverse cooling condition likely in service.
 - (2) Compliance with sub-paragraph (c)(1) must be shown by test unless experience with similar batteries and installations has shown that maintaining safe cell temperatures and pressures presents no problem.
 - (3) No explosive or toxic gases emitted by any battery in normal operation, or as the result of any probable malfunction in the charging system or battery installation, may accumulate in hazardous quantities within the rotorcraft.
 - (4) No corrosive fluids or gases that may escape from the battery may damage surrounding structures or adjacent essential equipment.
 - (5) Each nickel cadmium battery installation capable of being used to start an engine or auxiliary power unit must have provisions to prevent any hazardous effect on structure or essential systems that may be caused by the maximum amount of heat the battery can generate during a short circuit of the battery or of its individual cells.
 - (6) Nickel cadmium battery installations capable of being used to start an engine or auxiliary power unit must have:
 - (i) A system to control the charging rate of the battery automatically so as to prevent battery overheating;

- (ii) A battery temperature sensing and over-temperature warning system with a means for disconnecting the battery from its charging source in the event of an over- temperature condition; or
- (iii) A battery failure sensing and warning system with a means for disconnecting the battery from its charging source in the event of battery failure.

CS 29.1355 Distribution system

ED Decision 2003/16/RM

- (a) The distribution system includes the distribution busses, their associated feeders, and each control and protective device.
- (b) If two independent sources of electrical power for particular equipment or systems are required by any applicable CS or operating rule, in the event of the failure of one power source for such equipment or system, another power source (including its separate feeder) must be provided automatically or be manually selectable to maintain equipment or system operation.

CS 29.1357 Circuit protective devices

ED Decision 2003/16/RM

- (a) Automatic protective devices must be used to minimise distress to the electrical system and hazard to the rotorcraft in the event of wiring faults or serious malfunction of the system or connected equipment.
- (b) The protective and control devices in the generating system must be designed to de-energise and disconnect faulty power sources and power transmission equipment from their associated busses with sufficient rapidity to provide protection from hazardous overvoltage and other malfunctioning.
- (c) Each resettable circuit protective device must be designed so that, when an overload or circuit fault exists, it will open the circuit regardless of the position of the operating control.
- (d) If the ability to reset a circuit breaker or replace a fuse is essential to safety in flight, that circuit breaker or fuse must be located and identified so that it can be readily reset or replaced in flight.
- (e) Each essential load must have individual circuit protection. However, individual protection for each circuit in an essential load system (such as each position light circuit in a system) is not required.
- (f) If fuses are used, there must be spare fuses for use in flight equal to at least 50% of the number of fuses of each rating required for complete circuit protection.
- (g) Automatic reset circuit breakers may be used as integral protectors for electrical equipment provided there is circuit protection for the cable supplying power to the equipment.

CS 29.1359 Electrical system fire and smoke protection

ED Decision 2003/16/RM

- (a) Components of the electrical system must meet the applicable fire and smoke protection provisions of [CS 29.831](#) and [29.863](#).
- (b) Electrical cables, terminals, and equipment, in designated fire zones, and that are used in emergency procedures, must be at least fire resistant.

- (c) Insulation on electrical wire and cable installed in the rotorcraft must be self-extinguishing when tested in accordance with CS-25, Appendix F, Part I (a)(3).

CS 29.1363 Electrical system tests

ED Decision 2003/16/RM

- (a) When laboratory tests of the electrical system are conducted:
- (1) The tests must be performed on a mock-up using the same generating equipment used in the rotorcraft;
 - (2) The equipment must simulate the electrical characteristics of the distribution wiring and connected loads to the extent necessary for valid test results; and
 - (3) Laboratory generator drives must simulate the prime movers on the rotorcraft with respect to their reaction to generator loading, including loading due to faults.
- (b) For each flight condition that cannot be simulated adequately in the laboratory or by ground tests on the rotorcraft, flight tests must be made.

LIGHTS

CS 29.1381 Instrument lights

ED Decision 2003/16/RM

The instrument lights must:

- (a) Make each instrument, switch, and other device for which they are provided easily readable; and
- (b) Be installed so that:
 - (1) Their direct rays are shielded from the pilot's eyes; and
 - (2) No objectionable reflections are visible to the pilot.

CS 29.1383 Landing lights

ED Decision 2003/16/RM

- (a) Each required landing or hovering light must be approved.
- (b) Each landing light must be installed so that:
 - (1) No objectionable glare is visible to the pilot;
 - (2) The pilot is not adversely affected by halation; and
 - (3) It provides enough light for night operation, including hovering and landing.
- (c) At least one separate switch must be provided, as applicable:
 - (1) For each separately installed landing light; and
 - (2) For each group of landing lights installed at a common location.

CS 29.1385 Position light system installation

ED Decision 2003/16/RM

- (a) *General.* Each part of each position light system must meet the applicable requirements of this paragraph and each system as a whole must meet the requirements of [CS 29.1387](#) to [29.1397](#).
- (b) *Forward position lights.* Forward position lights must consist of a red and a green light spaced laterally as far apart as practicable and installed forward on the rotorcraft so that, with the rotorcraft in the normal flying position, the red light is on the left side, and the green light is on the right side. Each light must be approved.
- (c) *Rear position light.* The rear position light must be a white light mounted as far aft as practicable, and must be approved.
- (d) *Circuit.* The two forward position lights and the rear position light must make a single circuit.
- (e) *Light covers and colour filters.* Each light cover or colour filter must be at least flame resistant and may not change colour or shape or lose any appreciable light transmission during normal use.

CS 29.1387 Position light system dihedral angles

ED Decision 2003/16/RM

- (a) Except as provided in sub-paragraph (e), each forward and rear position light must, as installed, show unbroken light within the dihedral angles described in this paragraph.
- (b) Dihedral angle L (left) is formed by two intersecting vertical planes, the first parallel to the longitudinal axis of the rotorcraft, and the other at 110° to the left of the first, as viewed when looking forward along the longitudinal axis.
- (c) Dihedral angle R (right) is formed by two intersecting vertical planes, the first parallel to the longitudinal axis of the rotorcraft, and the other at 110° to the right of the first, as viewed when looking forward along the longitudinal axis.
- (d) Dihedral angle A (aft) is formed by two intersecting vertical planes making angles of 70° to the right and to the left, respectively, to a vertical plane passing through the longitudinal axis, as viewed when looking aft along the longitudinal axis.
- (e) If the rear position light, when mounted as far aft as practicable in accordance with [CS 29.1385\(c\)](#), cannot show unbroken light within dihedral angle A (as defined in sub-paragraph (d)), a solid angle or angles of obstructed visibility totalling not more than 0.04 steradians is allowable within that dihedral angle, if such solid angle is within a cone whose apex is at the rear position light and whose elements make an angle of 30° with a vertical line passing through the rear position light.

CS 29.1389 Position light distribution and intensities

ED Decision 2003/16/RM

- (a) *General.* The intensities prescribed in this paragraph must be provided by new equipment with light covers and colour filters in place. Intensities must be determined with the light source operating at a steady value equal to the average luminous output of the source at the normal operating voltage of the rotorcraft. The light distribution and intensity of each position light must meet the requirements of sub-paragraph (b).
- (b) *Forward and rear position lights.* The light distribution and intensities of forward and rear position lights must be expressed in terms of minimum intensities in the horizontal plane, minimum intensities in any vertical plane, and maximum intensities in overlapping beams, within dihedral angles, L, R and A, and must meet the following requirements:
 - (1) *Intensities in the horizontal plane.* Each intensity in the horizontal plane (the plane containing the longitudinal axis of the rotorcraft and perpendicular to the plane of symmetry of the rotorcraft), must equal or exceed the values in [CS 29.1391](#).
 - (2) *Intensities in the vertical plane.* Each intensity in any vertical plane (the plane perpendicular to the horizontal plane) must equal or exceed the appropriate value in [CS 29.1393](#) where I is the minimum intensity prescribed in [CS 29.1391](#) for the corresponding angles in the horizontal plane.
 - (3) *Intensities in overlaps between adjacent signals.* No intensity in any overlap between adjacent signals may exceed the values in [CS 29.1395](#), except that higher intensities in overlaps may be used with the use of main beam intensities substantially greater than the minima specified in [CS 29.1391](#) and [29.1393](#) if the overlap intensities in relation to the main beam intensities do not adversely affect signal clarity.

CS 29.1391 Minimum intensities in the horizontal plane of forward and rear position lights

ED Decision 2003/16/RM

Each position light intensity must equal or exceed the applicable values in the following table:

Dihedral angle (light included)	Angle from right or left of longitudinal axis, measured from dead ahead	Intensity (candelas)
L and R (forward red and green)	0° to 10°	40
	10° to 20°	30
	20° to 110°	5
A (rear white)	110° to 180°	20

CS 29.1393 Minimum intensities in any vertical plane of forward and rear position lights

ED Decision 2003/16/RM

Each position light intensity must equal or exceed the applicable values in the following table:

Angle above or below the horizontal plane	Intensity
0°	1.00 I
0° to 5°	0.90 I
5° to 10°	0.80 I
10° to 15°	0.70 I
15° to 20°	0.50 I
20° to 30°	0.30 I
30° to 40°	0.10 I
40° to 90°	0.05 I

CS 29.1395 Maximum intensities in overlapping beams of forward and rear position lights

ED Decision 2003/16/RM

No position light intensity may exceed the applicable values in the following table, except as provided in [CS 29.1389\(b\)\(3\)](#):

Overlaps	Maximum intensity	
	Area A (candelas)	Area B (candelas)
Green in dihedral angle L	10	1
Red in dihedral angle R	10	1
Green in dihedral angle A	5	1
Red in dihedral angle A	5	1
Rear white in dihedral angle L	5	1
Rear white in dihedral angle R	5	1

Where:

- (a) Area A includes all directions in the adjacent dihedral angle that pass through the light source and intersect the common boundary plane at more than 10° but less than 20°; and
- (b) Area B includes all directions in the adjacent dihedral angle that pass through the light source and intersect the common boundary plane at more than 20°.

CS 29.1397 Colour specifications

ED Decision 2003/16/RM

Each position light colour must have the applicable International Commission on Illumination chromaticity co-ordinates as follows:

- (a) Aviation Red:
 - 'y' is not greater than 0.335; and
 - 'z' is not greater than 0.002.
- (b) Aviation green:
 - 'x' is not greater than 0.440–0.320y;
 - 'x' is not greater than y–0.170; and
 - 'y' is not less than 0.390–0.170x.
- (c) Aviation white:
 - 'x' is not less than 0.300 and not greater than 0.540;
 - 'y' is not less than 'x–0.040' or 'y_o–0.010', whichever is the smaller; and
 - 'y' is not greater than 'x+0.020' nor '0.636–0.400x'.

Where 'y_o' is the 'y' co-ordinate of the Planckian radiator for the value of 'x' considered.

CS 29.1399 Riding light

ED Decision 2003/16/RM

- (a) Each riding light required for water operation must be installed so that it can:
 - (1) Show a white light for at least 4 km (two miles) at night under clear atmospheric conditions; and
 - (2) Show a maximum practicable unbroken light with the rotorcraft on the water.
- (b) Externally hung lights may be used.

CS 29.1401 Anti-collision light system

ED Decision 2003/16/RM

- (a) *General.* If certification for night operation is requested, the rotorcraft must have an anti-collision light system that:
 - (1) Consists of one or more approved anti-collision lights located so that their emitted light will not impair the crew's vision or detract from the conspicuity of the position lights; and
 - (2) Meets the requirements of sub-paragraphs (b) to (f).
- (b) *Field of coverage.* The system must consist of enough lights to illuminate the vital areas around the rotorcraft, considering the physical configuration and flight characteristics of the rotorcraft.

The field of coverage must extend in each direction within at least 30° above and 30° below the horizontal plane of the rotorcraft, except that there may be solid angles of obstructed visibility totalling not more than 0.5 steradians.

- (c) **Flashing characteristics.** The arrangement of the system, that is, the number of light sources, beam width, speed of rotation, and other characteristics, must give an effective flash frequency of not less than 40, nor more than 100, cycles per minute. The effective flash frequency is the frequency at which the rotorcraft's complete anti-collision light system is observed from a distance, and applies to each sector of light including any overlaps that exist when the system consists of more than one light source. In overlaps, flash frequencies may exceed 100, but not 180, cycles per minute.
- (d) **Colour.** Each anti-collision light must be aviation red and must meet the applicable requirements of [CS 29.1397](#).
- (e) **Light intensity.** The minimum light intensities in any vertical plane, measured with the red filter (if used) and expressed in terms of 'effective' intensities, must meet the requirements of subparagraph (f). The following relation must be assumed:

$$I_e = \frac{\int_{t_1}^{t_2} I(t) dt}{0.2 + (t_2 - t_1)}$$

where:

I_e = effective intensity (candelas).

$I(t)$ = instantaneous intensity as a function of time.

$t_2 - t_1$ = flash time interval (seconds).

Normally, the maximum value of effective intensity is obtained when t_2 and t_1 are chosen so that the effective intensity is equal to the instantaneous intensity at t_2 and t_1 .

- (f) **Minimum effective intensities for anti-collision light.** Each anti-collision light effective intensity must equal or exceed the applicable values in the following table:

$I_e = \frac{\int_{t_1}^{t_2} I(t) dt}{0.2 + (t_2 - t_1)}$ Angle above or below the horizontal plane	Effective intensity (candelas)
0° to 5°	150
5° to 10°	90
10° to 20°	30
20° to 30°	15

SAFETY EQUIPMENT

CS 29.1411 General

ED Decision 2003/16/RM

- (a) *Accessibility.* Required safety equipment to be used by the crew in an emergency, such as automatic liferaft releases, must be readily accessible.
- (b) *Stowage provisions.* Stowage provisions for required emergency equipment must be furnished and must:
 - (1) Be arranged so that the equipment is directly accessible and its location is obvious; and
 - (2) Protect the safety equipment from inadvertent damage.
- (c) *Emergency exit descent device.* The stowage provisions for the emergency exit descent device required by [CS 29.809\(f\)](#) must be at the exits for which they are intended.
- (d) *Liferafts.* Liferafts must be stowed near exits through which the rafts can be launched during an unplanned ditching. Rafts automatically or remotely released outside the rotorcraft must be attached to the rotorcraft by the static line prescribed in [CS 29.1415](#).
- (e) *Long-range signalling device.* The stowage provisions for the long-range signalling device required by [CS 29.1415](#) must be near an exit available during an unplanned ditching.
- (f) *Life preservers.* Each life preserver must be within easy reach of each occupant while seated.

CS 29.1413 Safety belts: passenger warning device

ED Decision 2003/16/RM

- (a) If there are means to indicate to the passengers when safety belts should be fastened, they must be installed to be operated from either pilot seat.
- (b) Each safety belt must be equipped with a metal to metal latching device.

CS 29.1415 Ditching equipment

ED Decision 2003/16/RM

- (a) Emergency flotation and signalling equipment required by any applicable operating rule must meet the requirements of this paragraph.
- (b) Each liferaft and each life preserver must be approved. In addition:
 - (1) Provide not less than two rafts, of an approximately equal rated capacity and buoyancy, to accommodate the occupants of the rotorcraft; and
 - (2) Each raft must have a trailing line, and must have a static line designed to hold the raft near the rotorcraft but to release it if the rotorcraft becomes totally submerged.
- (c) Approved survival equipment must be attached to each liferaft.
- (d) There must be an approved survival type emergency locator transmitter for use in one liferaft.

CS 29.1419 Ice protection

ED Decision 2003/16/RM

- (a) To obtain certification for flight into icing conditions, compliance with this paragraph must be shown.
- (b) It must be demonstrated that the rotorcraft can be safely operated in the continuous maximum and intermittent maximum icing conditions determined under [Appendix C](#) within the rotorcraft altitude envelope. An analysis must be performed to establish, on the basis of the rotorcraft's operational needs, the adequacy of the ice protection system for the various components of the rotorcraft.
- (c) In addition to the analysis and physical evaluation prescribed in sub-paragraph (b), the effectiveness of the ice protection system and its components must be shown by flight tests of the rotorcraft or its components in measured natural atmospheric icing conditions and by one or more of the following tests as found necessary to determine the adequacy of the ice protection system:
 - (1) Laboratory dry air or simulated icing tests, or a combination of both, of the components or models of the components.
 - (2) Flight dry air tests of the ice protection system as a whole, or its individual components.
 - (3) Flight tests of the rotorcraft or its components in measured simulated icing conditions.
- (d) The ice protection provisions of this paragraph are considered to be applicable primarily to the airframe. Powerplant installation requirements are contained in Subpart E of this CS-29.
- (e) A means must be identified or provided for determining the formation of ice on critical parts of the rotorcraft. Unless otherwise restricted, the means must be available for night- time as well as daytime operation. The rotorcraft flight manual must describe the means of determining ice formation and must contain information necessary for safe operation of the rotorcraft in icing conditions.

Appendix C – Icing Certification

ED Decision 2003/16/RM

- (a) The maximum continuous intensity of atmospheric icing conditions (continuous maximum icing) is defined by the variables of the cloud liquid water content, the mean effective diameter of the cloud droplets, the ambient air temperature, and the interrelationship of these three variables as shown in figure 1 of this appendix. The limiting icing envelope in terms of altitude and temperature is given in figure 2 of this appendix. The interrelationship of cloud liquid water content with drop diameter and altitude is determined from figures 1 and 2. The cloud liquid water content for continuous maximum icing conditions of a horizontal extent, other than 32.2 km (17.4 nautical miles), is determined by the value of liquid water content of figure 1, multiplied by the appropriate factor from figure 3 of this appendix.
- (b) The intermittent maximum intensity of atmospheric icing conditions (intermittent maximum icing) is defined by the variables of the cloud liquid water content, the mean effective diameter of the cloud droplets, the ambient air temperature, and the interrelationship of these three variables as shown in figure 4 of this appendix. The limiting icing envelope in terms of altitude and temperature is given in figure 5 of this appendix. The interrelationship of cloud liquid water content with drop diameter and altitude is determined from figures 4 and 5. The cloud liquid water content for intermittent maximum icing conditions of a horizontal extent, other than

4.8 km (2.6 nautical miles), is determined by the value of cloud liquid water content of figure 4 multiplied by the appropriate factor in figure 6 of this appendix.

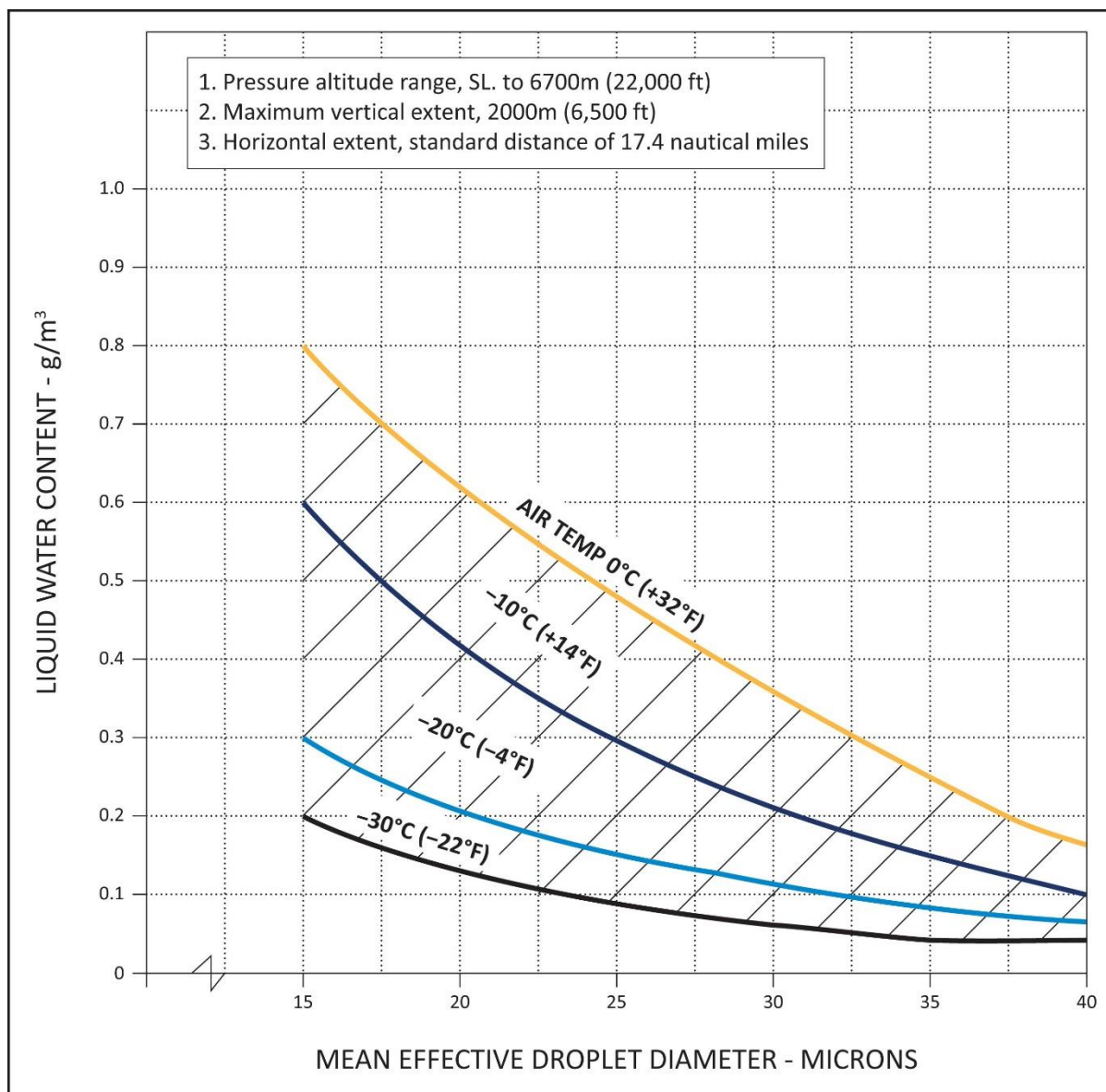


FIGURE 1

CONTINUOUS MAXIMUM (STRATIFORM CLOUDS) ATMOSPHERIC ICING CONDITIONS

LIQUID WATER CONTENT VS MEAN EFFECTIVE DROP DIAMETER

Source of data – NACA TN No. 1855, Class III - M, Continuous Maximum.

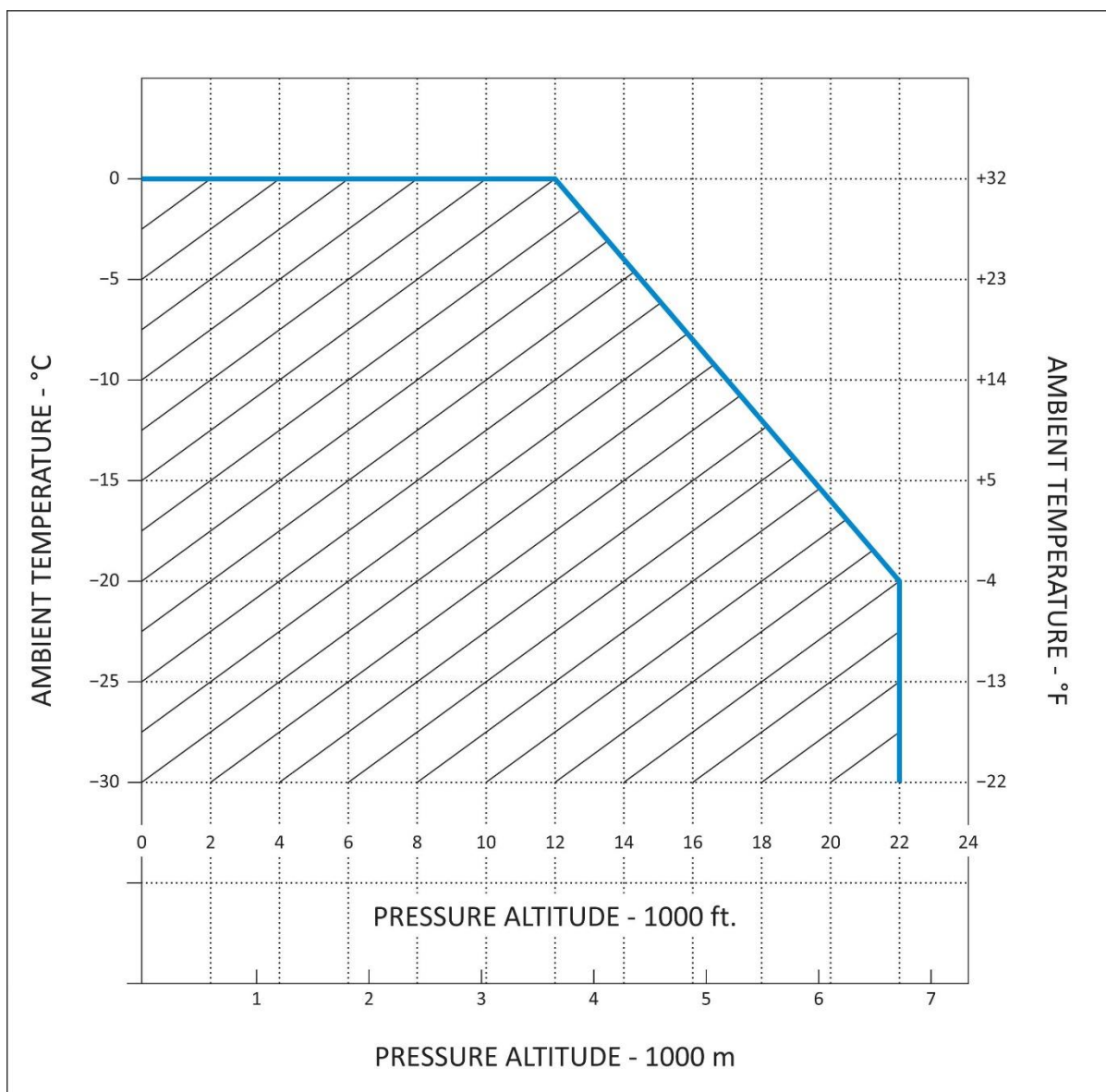
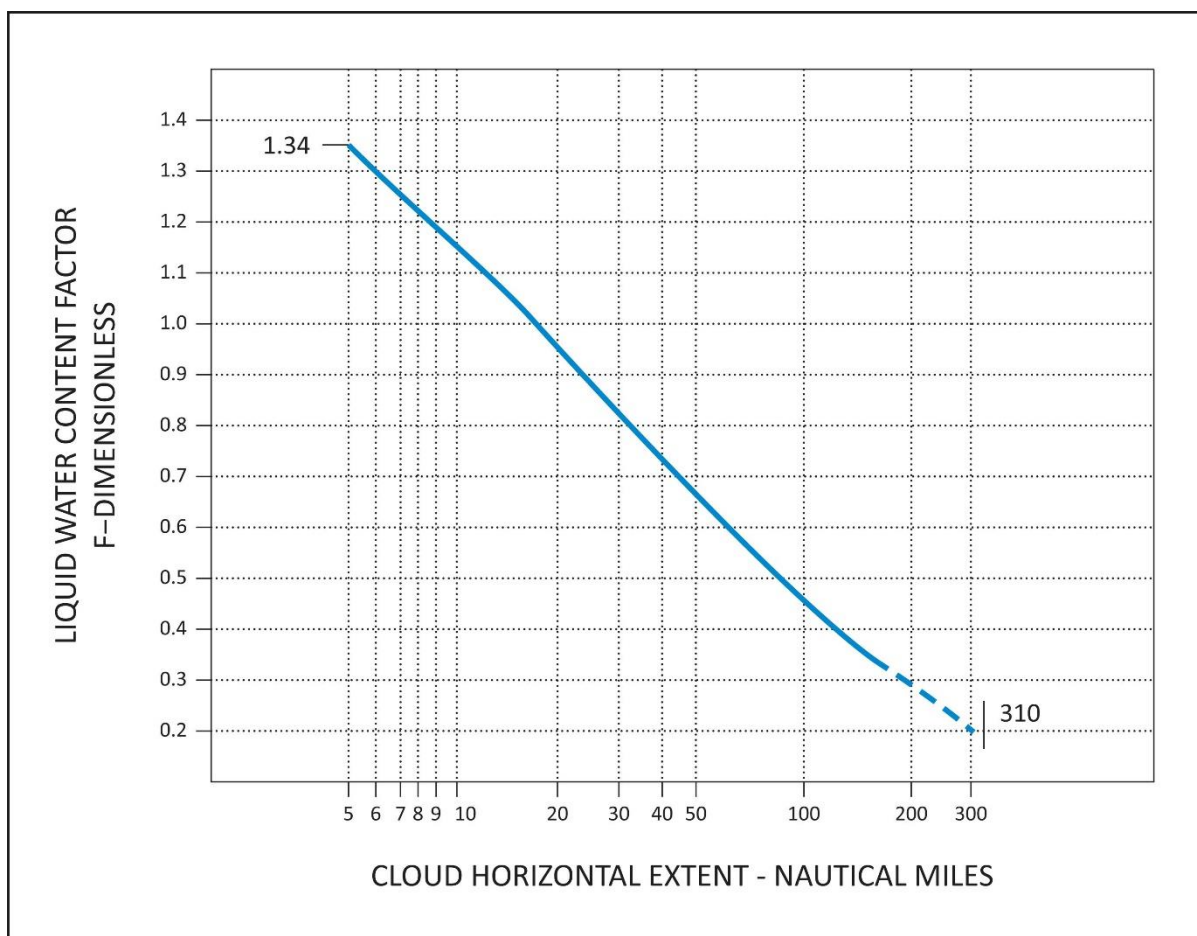


FIGURE 2

CONTINUOUS MAXIMUM (STRATIFORM CLOUDS) ATMOSPHERIC ICING CONDITIONS

AMBIENT TEMPERATURE VS PRESSURE ALTITUDE

Source of data – NACA TN No. 2569.


FIGURE 3
CONTINUOUS MAXIMUM (STRATIFORM CLOUDS) ATMOSPHERIC ICING CONDITIONS
LIQUID WATER CONTENT FACTOR VS CLOUD HORIZONTAL DISTANCE
Source of data – NACA TN No. 2738.

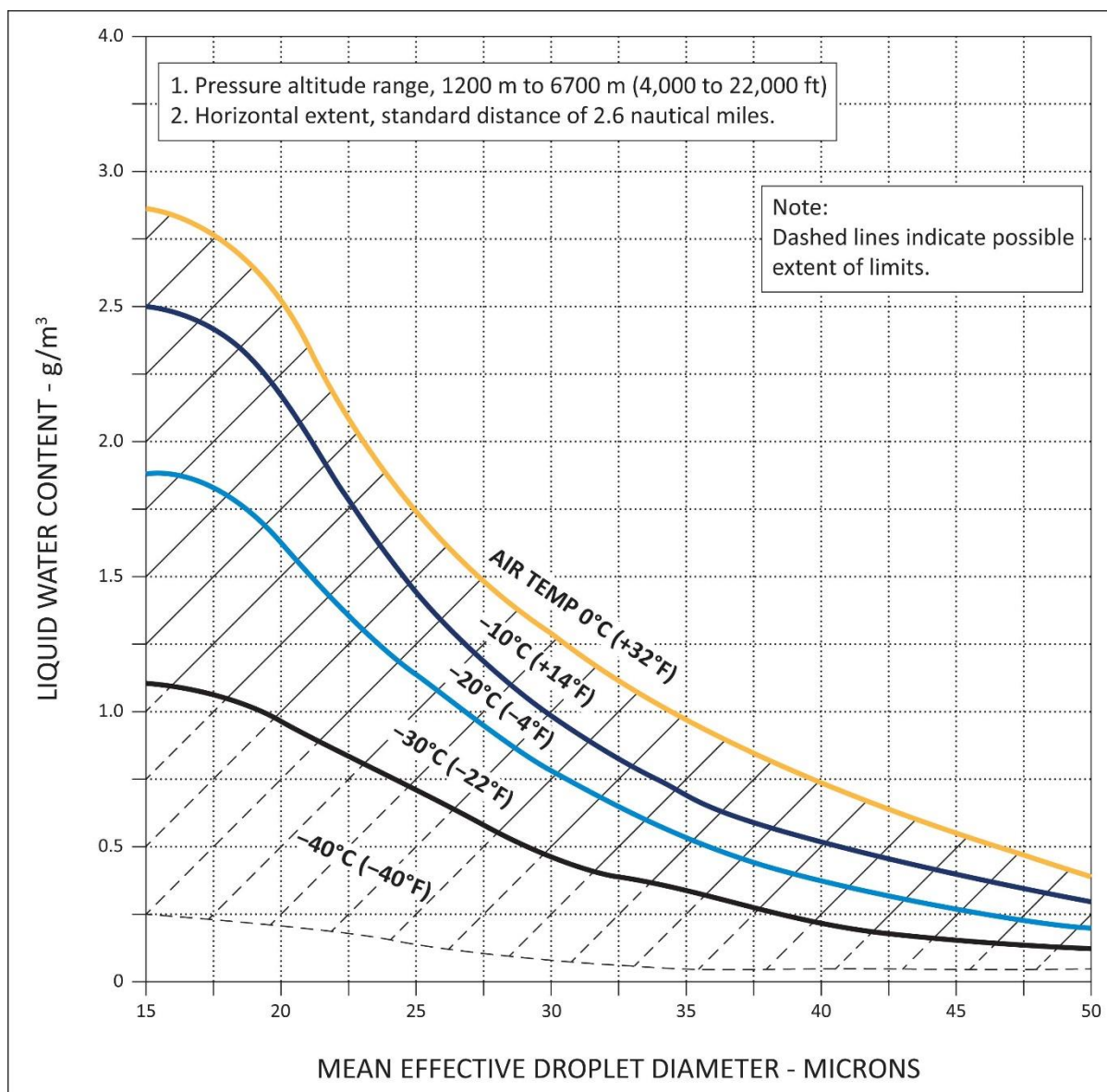


FIGURE 4

INTERMITTENT MAXIMUM (CUMULIFORM CLOUDS) ATMOSPHERIC ICING CONDITIONS

LIQUID WATER CONTENT VS MEAN EFFECTIVE DROP DIAMETER

Source of data – NACA TN No. 1855, Class II - M, Intermittent Maximum.

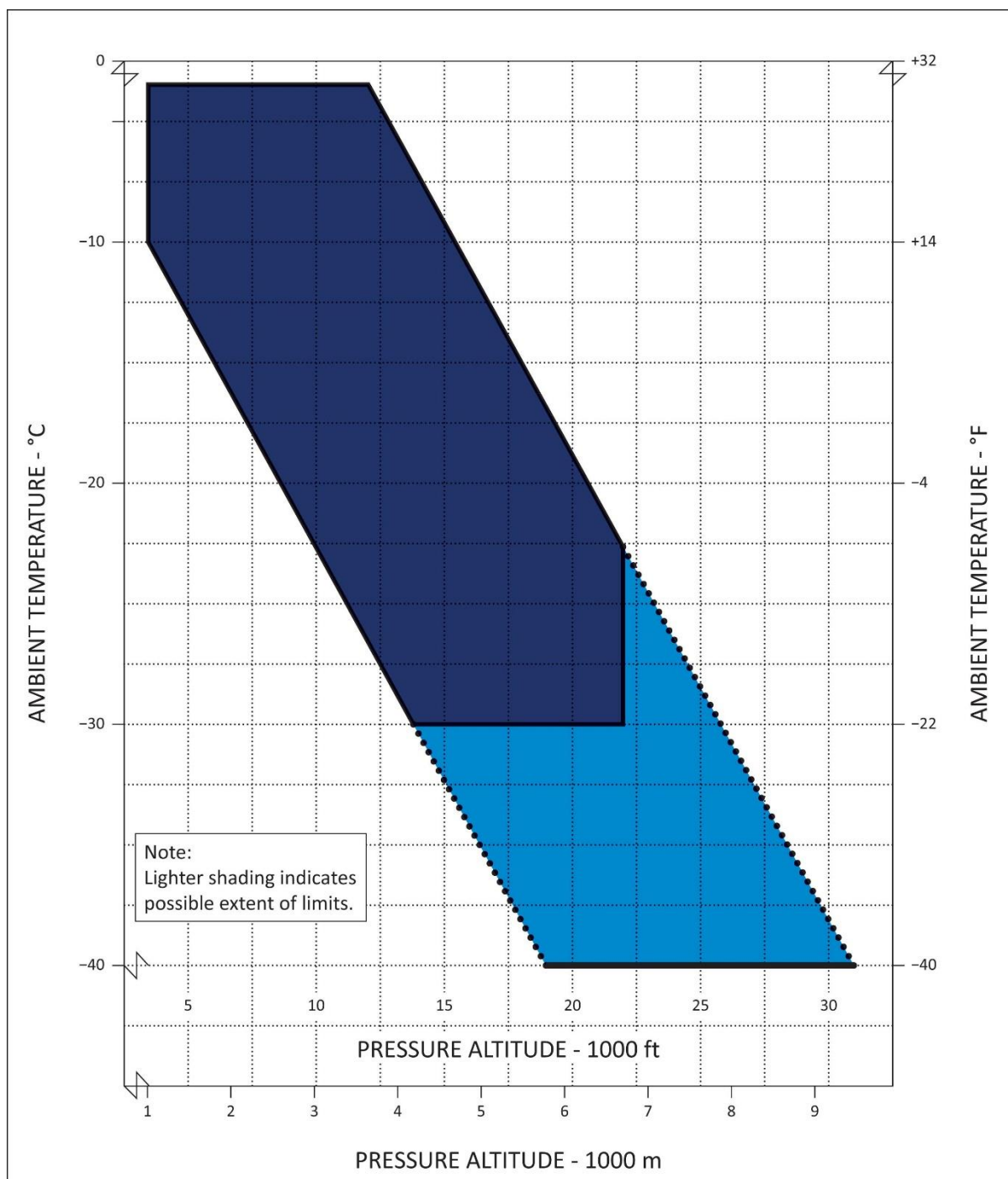


FIGURE 5

INTERMITTENT MAXIMUM (CUMULIFORM CLOUDS) ATMOSPHERIC ICING CONDITIONS

AMBIENT TEMPERATURE VS PRESSURE ALTITUDE

Source of data – NACA TN No. 2569.

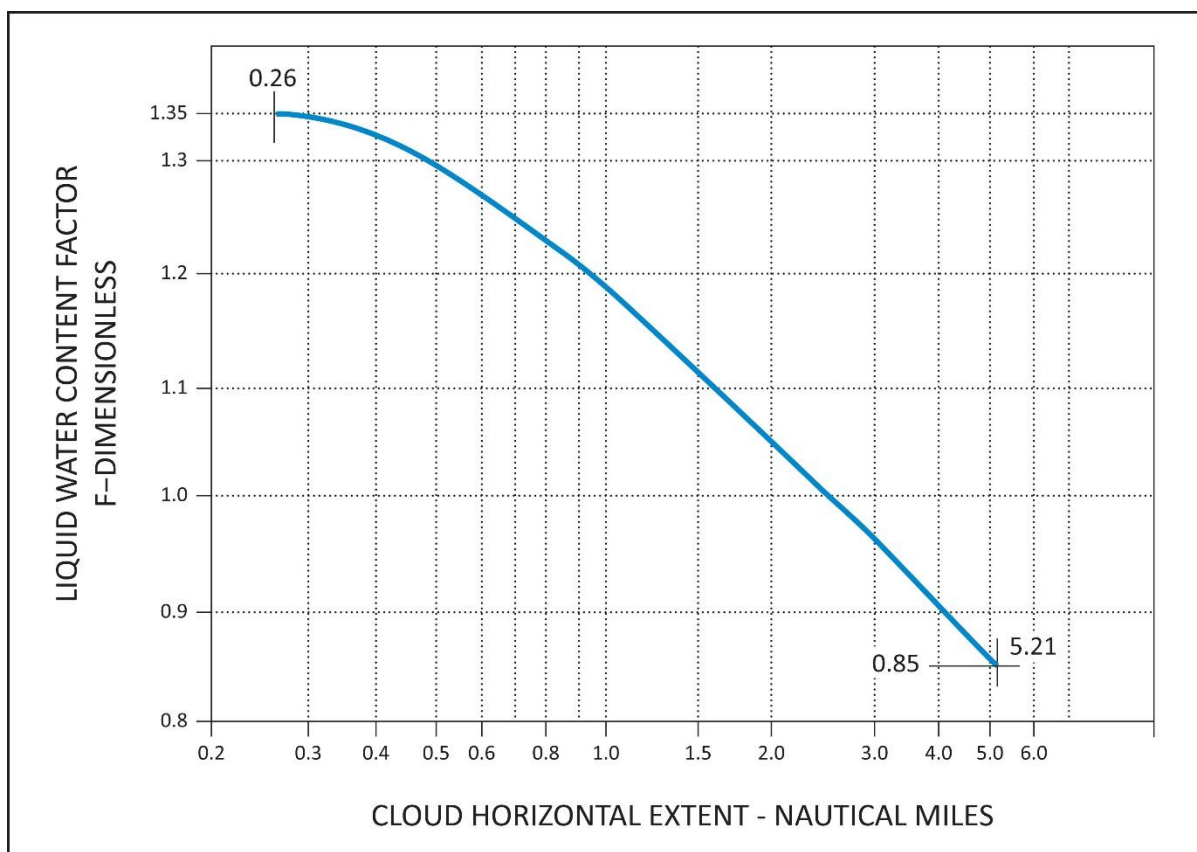


FIGURE 6

INTERMITTENT MAXIMUM (CUMULIFORM CLOUDS) ATMOSPHERIC ICING CONDITIONS

VARIATION OF LIQUID WATER CONTENT FACTOR WITH CLOUD HORIZONTAL EXTENT

Source of data – NACA TN No. 2738.

MISCELLANEOUS EQUIPMENT

CS 29.1431 Electronic equipment

ED Decision 2003/16/RM

- (a) Radio communication and navigation installations must be free from hazards in themselves, in their method of operation, and in their effects on other components, under any critical environmental conditions.
- (b) Radio communication and navigation equipment, controls, and wiring must be installed so that operation of any one unit or system of units will not adversely affect the simultaneous operation of any other radio or electronic unit, or system of units, required by any applicable CS or operating rule.

CS 29.1433 Vacuum systems

ED Decision 2003/16/RM

- (a) There must be means, in addition to the normal pressure relief, to automatically relieve the pressure in the discharge lines from the vacuum air pump when the delivery temperature of the air becomes unsafe.
- (b) Each vacuum air system line and fitting on the discharge side of the pump that might contain flammable vapours or fluids must meet the requirements of [CS 29.1183](#) if they are in a designated fire zone.
- (c) Other vacuum air system components in designated fire zones must be at least fire resistant.

CS 29.1435 Hydraulic systems

ED Decision 2003/16/RM

- (a) *Design.* Each hydraulic system must be designed as follows:
 - (1) Each element of the hydraulic system must be designed to withstand, without detrimental, permanent deformation, any structural loads that may be imposed simultaneously with the maximum operating hydraulic loads.
 - (2) Each element of the hydraulic system must be designed to withstand pressures sufficiently greater than those prescribed in sub-paragraph (b) to show that the system will not rupture under service conditions.
 - (3) There must be means to indicate the pressure in each main hydraulic power system.
 - (4) There must be means to ensure that no pressure in any part of the system will exceed a safe limit above the maximum operating pressure of the system, and to prevent excessive pressures resulting from any fluid volumetric change in lines likely to remain closed long enough for such a change to take place. The possibility of detrimental transient (surge) pressures during operation must be considered.
 - (5) Each hydraulic line, fitting, and component must be installed and supported to prevent excessive vibration and to withstand inertia loads. Each element of the installation must be protected from abrasion, corrosion, and mechanical damage.
 - (6) Means for providing flexibility must be used to connect points, in a hydraulic fluid line, between which relative motion or differential vibration exists.

- (b) *Tests.* Each element of the system must be tested to a proof pressure of 1.5 times the maximum pressure to which that element will be subjected in normal operation, without failure, malfunction, or detrimental deformation of any part of the system.
- (c) *Fire protection.* Each hydraulic system using flammable hydraulic fluid must meet the applicable requirements of [CS 29.861](#), [29.1183](#), [29.1185](#), and [29.1189](#).

CS 29.1439 Protective breathing equipment

ED Decision 2003/16/RM

- (a) If one or more cargo or baggage compartments are to be accessible in flight, protective breathing equipment must be available for an appropriate crew member.
- (b) For protective breathing equipment required by sub-paragraph (a) or by any applicable operating rule:
 - (1) That equipment must be designed to protect the crew from smoke, carbon dioxide, and other harmful gases while on flight deck duty;
 - (2) That equipment must include:
 - (i) Masks covering the eyes, nose, and mouth; or
 - (ii) Masks covering the nose and mouth, plus accessory equipment to protect the eyes; and
 - (3) That equipment must supply protective oxygen of 10 minutes duration per crew member at a pressure altitude of 2438 m (8000 ft) with a respiratory minute volume of 30 litres per minute BTPD.

CS 29.1457 Cockpit voice recorders

ED Decision 2003/16/RM

- (a) Each cockpit voice recorder required by the applicable operating rules must be approved, and must be installed so that it will record the following:
 - (1) Voice communications transmitted from or received in the rotorcraft by radio.
 - (2) Voice communications of flight-crew members on the flight deck.
 - (3) Voice communications of flight-crew members on the flight deck, using the rotorcraft's inter-phone system.
 - (4) Voice or audio signals identifying navigation or approach aids introduced into a headset or speaker.
 - (5) Voice communications of flight-crew members using the passenger loudspeaker system, if there is such a system, and if the fourth channel is available in accordance with the requirements of sub-paragraph (c)(4)(ii).
- (b) The recording requirements of sub-paragraph (a)(2) may be met:
 - (1) By installing a cockpit-mounted area microphone, located in the best position for recording voice communications originating at the first and second pilot stations and voice communications of other crew members on the flight deck when directed to those stations; or
 - (2) By installing a continually energised or voice-actuated lip microphone at the first and second pilot stations.

The microphone specified in this paragraph must be so located and, if necessary, the preamplifiers and filters of the recorder must be so adjusted or supplemented, that the recorded communications are intelligible when recorded under flight cockpit noise conditions and played back. The level of intelligibility must be approved by the Agency. Repeated aural or visual playback of the record may be used in evaluating intelligibility.

- (c) Each cockpit voice recorder must be installed so that the part of the communication or audio signals specified in sub-paragraph (a) obtained from each of the following sources is recorded on a separate channel:
 - (1) For the first channel, from each microphone, headset, or speaker used at the first pilot station.
 - (2) For the second channel, from each microphone, headset, or speaker used at the second pilot station.
 - (3) For the third channel, from the cockpit-mounted area microphone, or the continually energised or voice-actuated lip microphones at the first and second pilot stations.
 - (4) For the fourth channel, from:
 - (i) Each microphone, headset, or speaker used at the stations for the third and fourth crew members; or
 - (ii) If the stations specified in sub-paragraph (c)(4)(i) are not required or if the signal at such a station is picked up by another channel, each microphone on the flight deck that is used with the passenger loudspeaker system if its signals are not picked up by another channel.
 - (iii) Each microphone on the flight deck that is used with the rotorcraft's loudspeaker system if its signals are not picked up by another channel.
- (d) Each cockpit voice recorder must be installed so that:
 - (1) It receives its electric power from the bus that provides the maximum reliability for operation of the cockpit voice recorder without jeopardising service to essential or emergency loads;
 - (2) There is an automatic means to simultaneously stop the recorder and prevent each erasure feature from functioning, within 10 minutes after crash impact; and
 - (3) There is an aural or visual means for pre-flight checking of the recorder for proper operation.
- (e) The record container must be located and mounted to minimise the probability of rupture of the container as a result of crash impact and consequent heat damage to the record from fire.
- (f) If the cockpit voice recorder has a bulk erasure device, the installation must be designed to minimise the probability of inadvertent operation and actuation of the device during crash impact.
- (g) Each recorder container must be either bright orange or bright yellow.

CS 29.1459 Flight recorder

ED Decision 2003/16/RM

- (a) Each flight recorder required by the applicable operating rules must be installed so that:
 - (1) It is supplied with airspeed, altitude, and directional data obtained from sources that meet the accuracy requirements of [CS 29.1323](#), [29.1325](#), and [29.1327](#), as applicable;
 - (2) The vertical acceleration sensor is rigidly attached, and located longitudinally within the approved centre of gravity limits of the rotorcraft;
 - (3) It receives its electrical power from the bus that provides the maximum reliability for operation of the flight recorder without jeopardising service to essential or emergency loads;
 - (4) There is an aural or visual means for pre-flight checking of the recorder for proper recording of data in the storage medium; and
 - (5) Except for recorders powered solely by the engine-driven electrical generator system, there is an automatic means to simultaneously stop a recorder that has a data erasure feature and prevent each erasure feature from functioning, within 10 minutes after any crash impact.
- (b) Each non-ejectable recorder container must be located and mounted so as to minimise the probability of container rupture resulting from crash impact and subsequent damage to the record from fire.
- (c) A correlation must be established between the flight recorder readings of airspeed, altitude, and heading and the corresponding readings (taking into account correction factors) of the first pilot's instruments. This correlation must cover the airspeed range over which the aircraft is to be operated, the range of altitude to which the aircraft is limited, and 360° of heading. Correlation may be established on the ground as appropriate.
- (d) Each recorder container must:
 - (1) Be either bright orange or bright yellow;
 - (2) Have a reflective tape affixed to its external surface to facilitate its location under water; and
 - (3) Have an underwater locating device, when required by the applicable operating rules, on or adjacent to the container which is secured in such a manner that it is not likely to be separated during crash impact.

CS 29.1461 Equipment containing high energy rotors

ED Decision 2003/16/RM

- (a) Equipment containing high energy rotors must meet sub-paragraphs (b), (c), or (d).
- (b) High energy rotors contained in equipment must be able to withstand damage caused by malfunctions, vibration, abnormal speeds, and abnormal temperatures. In addition:
 - (1) Auxiliary rotor cases must be able to contain damage caused by the failure of high energy rotor blades; and
 - (2) Equipment control devices, systems, and instrumentation must reasonably ensure that no operating limitations affecting the integrity of high energy rotors will be exceeded in service.

- (c) It must be shown by test that equipment containing high energy rotors can contain any failure of a high energy rotor that occurs at the highest speed obtainable with the normal speed control devices inoperative.
- (d) Equipment containing high energy rotors must be located where rotor failure will neither endanger the occupants nor adversely affect continued safe flight.

CS 29.1465 Vibration Health Monitoring

ED Decision 2012/022/R

- (a) If certification of a rotorcraft with vibration health monitoring of the rotors and/or rotor drive systems is requested by the applicant, then the design and performance of an installed system must provide a reliable means of early detection for the identified failure modes being monitored.
- (b) If a vibration health monitoring system of the rotors and/or rotor drive systems is required by the applicable operating rules, then the design and performance of the vibration health monitoring system must, in addition, meet the requirements of this paragraph.
 - (1) A safety analysis must be used to identify all component failure modes that could prevent continued safe flight or safe landing, for which vibration health monitoring could provide a reliable means of early detection;
 - (2) All typical VHM indicators and signal processing techniques should be considered in the VHM System design;
 - (3) Vibration health monitoring must be provided as identified in subparagraph (1) and (2), unless other means of health monitoring can be substantiated.

[Amdt 29/3]

AMC 29.1465 Vibration health monitoring

ED Decision 2012/022/R

a. Explanation

- (1) The purpose of this AMC is to provide an Acceptable Means of Compliance and Guidance Material for the design and certification of Vibration Health Monitoring (VHM) applications. VHM is used to increase the likelihood of detection of dynamic component incipient faults in the rotors and rotor drive systems that could prevent continued safe flight or safe landing, by providing timely indications of potential failures to maintenance personnel.
- (2) Designing a VHM system in accordance with this AMC is expected to achieve the required performance together with acceptable levels of system integrity and reliability for compliance with type certification and/or operational regulations that require VHM of rotor and/or rotor drive systems.
- (3) This AMC defines terms, processes, performance and standards that a VHM system should meet and also the support that a VHM approval holder should provide after the system has entered into service.
- (4) VHM systems which satisfy this AMC and that perform functions, the failure of which are categorised as Minor or No Safety Effect (see paragraph p.), can be accepted without the need for additional compliance with AC 29-2C MG15.

Note 1: FAA AC 29-2C Miscellaneous Guidance (MG)15, which addresses the use of HUMS in Maintenance, is complementary to this AMC.

Note 2: If an applicant wishes to install a VHM system that is not compliant with CS 29.1465(a), it may still be accepted for installation on a “No hazard/No credit” basis. However, it cannot replace any existing type-design maintenance instructions or change the established methods of complying with CS-29.

b. Procedures

- (1) [CS 29.1465](#) does not mandate the fitment of VHM systems. However, if a VHM system is installed on the rotorcraft to meet a type-certification or operational rule, then compliance is required. Three typical scenarios are foreseen as to when compliance by the applicant may be requested. The three scenarios in question are:
 - (i) as a means of demonstrating compliance with an operational rule requiring helicopters be fitted with a VHM system and that operators of such helicopters implement procedures covering data collection, analysis and determination of serviceability;
 - (ii) as a selected compensating provision to mitigate the probability of a failure condition, identified from the design assessments of [CS 29.547\(b\)](#) and/or [CS 29.917\(b\)](#), from arising;
 - (iii) on a voluntary basis to meet a customer requirement or company objective.
- (2) [CS 29.1465\(a\)](#) allows non-required and/or partial VHM applications with limited capability to monitor specific failure modes to be approved. Such systems can offer safety benefits and it is not the intention here to discourage their installation and use. However, any installed system must meet [CS 29.1301](#) and be of a kind and design appropriate to its intended function and function properly when installed. The guidance given in this AMC is therefore considered to be applicable to these types of VHM systems.
- (3) Where an operating rule mandates installation of a VHM system, [CS 29.1465\(b\)](#) aims to provide a VHM system capability that maximises the safety benefit. All typical VHM indicators and signal processing techniques should be considered in the VHM design and a system safety assessment undertaken to identify failure modes where VHM could provide early detection of incipient failures. VHM must be provided for all potential failure modes unless other means of health monitoring can be substantiated.
- (4) The safety analysis required by [CS 29.1465\(b\)\(1\)](#) is limited to rotors and rotor drive systems. The existing design assessments of [CS 29.547](#) and [CS 29.917](#) can be used for this purpose. All component failure modes that could prevent continued safe flight or safe landing (Catastrophic and Hazardous failure conditions) and for which vibration health monitoring could provide a reliable means of early detection must be identified. Previous experience together with the guidance in this AMC can be used to determine failure modes that could benefit from VHM and the applicable techniques that can produce reliable indications of incipient failures.
- (5) [CS 29.1465\(b\)\(2\)](#) requires the design and performance of the VHM system to consider indicators and processing techniques used on typical existing VHM installations. A non-exhaustive list is provided in Table 1 of this AMC.
- (6) [CS 29.1465\(b\)\(3\)](#) states that VHM must be provided as identified in subparagraph (b)(1) and (b)(2), unless other means of health monitoring can be substantiated. For many failure modes, there may be other compensating provisions which are capable of

providing protection against the risk of premature failure. In such cases, the added benefit of VHM in increasing the likelihood of early detection should be assessed. It will not be necessary to implement VHM for a given failure mode if no safety benefit can be established.

c. Definitions

- (1) **Alarm:** An Alert that, following additional processing or investigation, has resulted in a maintenance action being required.
- (2) **Alert:** An indication produced by the VHM system that requires further processing or investigation by the operator to determine if corrective maintenance action is required.
- (3) **Commercial Off-the-Shelf (COTS):** This term defines equipment hardware and software that is not qualified to aircraft standards.
- (4) **Controlled Service Introduction (CSI):** A period in-service where capabilities and functions that could not be verified prior to entry into service (including support functions) are evaluated.
- (5) **False Alarm:** An Alert that after further processing or investigation has resulted in unnecessary maintenance action.
- (6) **False Alert:** This is an Alert that after further processing or investigation has been determined to not require any further action.
- (7) **Ground-Based System:** A means of access to VHM data, including Alerts, for immediate post-flight fault diagnosis by the responsible maintenance staff.
- (8) **Prognostic Interval:** The predicted time between an Alarm and the component becoming unairworthy.
- (9) **Vibration Health Monitoring (VHM):** Use of data generated by processing vibration signals to detect incipient failure or degradation of mechanical integrity.
- (10) **VHM Application:** A VHM function implemented for a defined purpose.
- (11) **VHM Indicator:** A VHM Indicator is the result of processing sampled data by applying an algorithm to achieve a single value, which relates to the health of a component with respect to a particular failure mode.
- (12) **VHM System:** Typically comprises vibration sensors and associated wiring, data acquisition and processing hardware, the means of downloading data from the rotorcraft, the Ground-Based System and all associated instructions for operation of the system.

d. Component Monitoring Capability

The scope of the VHM capability is determined by the range of components monitored and their incipient failures which can be detected. For each component to be monitored the range of potential damage being diagnosed should be declared and the principles of the monitoring techniques applied should be described. The health monitoring effectiveness should be demonstrable (see paragraph o).

e. System Design Considerations

- (1) **Sensors:** They are the hardware that measures vibration. They should provide a reliable signal with an appropriate and defined performance. The position and installation of a vibration sensor is as critical as its performance. Sensor selection, positioning and installation should be designed to enable analysis of the processed signals to discriminate the vibration characteristics of the declared monitored component failure modes. Built-

In Test capability is necessary to determine the correct functioning of the sensor. Maintenance instructions should ensure that the correct function, and any calibration, of sensors and their installation are adequately controlled.

- (2) **Signal Acquisition:** It is likely that processed VHM data will be sensitive to the flight regime of the rotorcraft. For this reason it is desirable to focus data acquisition to particular operating conditions or phases of flight. Consideration should be given to the likely operation of rotorcraft that may utilise the VHM system and the practicality of acquiring adequate data from each flight to permit the Alert and Alarm processing to be performed to the required standard. The method of vibration signal acquisition should be designed so that:
- (i) The vibration signal sampling rate is sufficient for the required bandwidth and to avoid aliasing with an adequate dynamic range and sensitivity.
 - (ii) The data acquired from the vibration signal should be automatically gathered in specifically defined regimes at an appropriate rate and quantity for the VHM signal processing to produce robust data for defect detection.
 - (iii) If the mission profile does not allow regular acquisition of complete data sets, then the data acquisition regimes should be capable of reconfiguration appropriate to particular flight operations.
 - (iv) The acquisition cycle should be designed in such a way that all selected components and their defects are monitored with an adequate frequency irrespective of any interruptions in the cycle due to the operational profile.
- (3) **Signal Processing:** The helicopter's rotor and rotor drive systems are a mixture of complex and simple mechanical elements. Therefore, the signal processing or the analysis techniques utilised should reflect the complexity of the mechanical elements being monitored as well as the transmission path of the signal and should be demonstrated as being appropriate to the failure modes to be detected. The objective of processing the sampled data should be to produce VHM Indicators that clearly relate to vibration characteristics of the monitored components, from which the health of these components can be determined. A key part of the success of in-service VHM is the signal-to-noise enhancement techniques such as vibration signal averaging for gears and signal band-pass filtering and enveloping for bearings. These techniques are used to generate enhanced component vibration signatures prior to the calculation of the VHM Indicators. Accordingly, the method of signal enhancement should be shown to be effective. The method of signal processing and the analysis techniques utilised to generate the data used for defect detection should be defined for the claimed defect detection capability (see Table 1 below).

Table 1: Typical Vibration Health Monitoring Indicators & Signal Processing Techniques

Assembly	Component Type	Types of VHM indicators used
Engine to main gearbox input drive shafts	Shafts	Fundamental shaft order and harmonics
Gearboxes	Shafts	Fundamental shaft order and harmonics
	Gears	Gear meshing frequency and harmonics, modulation of meshing waveform, impulse detection and energy measurement, non-mesh-related energy content

	Bearings	High frequency energy content, impulse detection, signal envelope modulation patterns and energies correlated with bearing defect frequencies
Tail rotor drive shaft	Shafts	Fundamental shaft order and harmonics
	Hangar Bearings	As for gearbox bearings, but can utilise simple band-passed signal energy measurements
Oil cooler	Oil Cooler Blower and Drive Shaft	Fundamental shaft order and harmonics, blade pass frequency
Main and Tail rotor	Rotors	Fundamental shaft order and harmonics up to blade pass frequency, plus multiples of this.

Recording and storing of some raw vibration data and the processed vibration signal, from which the Indicators are derived, may also be of significant diagnostic value. Typical signal processing techniques include;

- (i) Asynchronous Power Spectrum where phase information or frequency tracking is not required.
- (ii) Synchronous Spectrum where phase information or frequency tracking is required.
- (iii) Band-pass filtered signal Envelope Power Spectrum Analysis (a recommended technique for gearbox bearings).
- (iv) Synchronous Averaging for time and frequency domain signal analysis (a recommended technique for gearbox gears).
- (v) Band-pass filtering and the measurement of filtered signal statistics, including crest factor (can be used for bearings not within engines or gearboxes).
- (vi) Further signal enhancement techniques are typically required in the calculation of certain VHM indicators targeted at detecting specific defect-related features (e.g. localised signal distortion associated with a gear tooth crack).

Note 1: When showing compliance to [CS 29.1465\(a\)](#), for non-required and/or partial VHM applications with limited capability to monitor specific failure modes, it is not necessary to address the scope of VHM capability stated in Table 1.

Note 2: When showing compliance to [CS 29.1465\(b\)](#), it is not always necessary for the VHM system to cover the complete capability defined in Table 1. However, absence of any of these areas, and/or techniques, should be substantiated. It is acknowledged that the above provides a prescriptive scope for monitoring rotor and rotor drive system components. If alternative methods are proposed, which can be shown to be as effective and reliable as those prescribed and which are to the satisfaction of the Agency, then these can also be accepted.

f. Data Management

The data transfer process from the rotorcraft to the maintenance personnel interface should be sufficient to determine all the VHM Indicators post flight. The upload/download should have minimal impact on flight operations. VHM data should be accessible in order to permit alternative analysis and comparison. The following should be specified:

- (1) Data transfer, processing, networking, data integrity assurance.
- (2) Methods to ensure the reliability of this process.
- (3) The time for upload/download and retrieval of data and/or health report.
- (4) Facilities for the warehousing of all of the data downloaded from the VHM systems and to permit timely access to the data.

g. Alert Management

- (1) VHM Alert Generation: VHM Alert criteria should be applied to every monitored component. VHM Alerts are produced to indicate possible anomalous behaviour or a specific defect.

Note: The fixed or learnt thresholds for each individual health monitoring indicator may have a limited capability to detect incipient failures in a timely manner. This is because the process for threshold setting is sometimes a compromise between increasing sensitivity and incurring a higher risk of false alarms, or reducing sensitivity, which will delay the point at which a rising indicator value will trigger an alert. In-service experience has shown that MGB component fatigue failures can propagate from initiation to failure in a relatively short period of time, thus the use of fixed thresholds alone may not provide a timely indication of impending failure. One characteristic that can often provide an earlier indication of anomalous behaviour is the rate of change of a health monitoring indicator, and automatic trend detection software has been developed and shown to be effective. Another method, commonly referred to as Advanced Anomaly Detection (AAD), combines numerous indicators into multi-dimensional parameters, whereby simultaneous changes of multiple indicators can provide increased confidence of the anomalous behaviour at an earlier point in the failure process. (Further information on AAD can be found in Related documents v.(3)).

- (2) VHM Alert Management: Diagnostic processes are required to determine if VHM driven maintenance of the rotorcraft is necessary.

h. Pilot Interface

Pilot interaction with the VHM system, if any, should be specified and should not adversely impact on pilot workload.

Note: The level of system integrity for VHM provided under this AMC is not sufficient to support the provision of in-flight cockpit VHM alerts.

i. Maintenance Personnel Interface

The person responsible for releasing a rotorcraft into service should be provided with VHM data, maintenance recommendations and VHM system Built-In Test data necessary to release that rotorcraft. This should include the ability to view VHM Indicators, trend data and detection criteria, including thresholds, for relevant VHM parameters from that rotorcraft. These capabilities should be available locally to maintenance personnel for immediate post flight fault diagnosis.

j. Fleet Diagnostic Support Interface

Where an operator has multiple rotorcraft of the same type, facilities should be made available to the operator to support the analysis of all data acquired by the VHM systems in the operator's fleet. The operator and all parties supporting the operator should have remote, multi-user and timely access to the data and the diagnostic processes in order to assist in determining the continued airworthiness of their fleet.

k. VHM system installation

The VHM system installation must comply with CS-29, as applicable to the specific rotorcraft type.

l. Ground-Based System Architecture

Any Ground-Based System Architecture requirements should be specified (see paragraph q. Technical Publications). The Ground-Based System may include COTS hardware, software and services, compatible with the Data Management objectives of paragraph (f) above.

m. Software

(1) For the case where the VHM system is stand alone

All software that makes up the VHM processing, whether airborne or ground-based, is to be produced to the software quality standard required to achieve the necessary level of system integrity.

All COTS software should be identified and should be of a quality standard that does not compromise the overall system's integrity.

All ground-based system software (specifically developed for VHM processing and COTS) should be developed to EUROCAE ED-109A/RTCA DO-278A Assurance Level 5 (AL5). DO 278 Assurance Level 5 (AL5) provides an acceptable method for acceptance of ground-based systems which include COTS.

VHM applications with hazard severity level Major or higher are addressed by MG15 and not AMC 29.1465.

Note: EUROCAE ED-12C/RTCA DO-178C Level D software for airborne systems and EUROCAE ED-109A/RTCA DO-278A Assurance Level 5 for non-airborne systems can be applied where VHM is utilised in addition to traditional helicopter design provisions. This will not require certification to a level any higher than Minor, based on the required reliability for these VHM applications. Should a design be proposed where greater reliance was placed solely on VHM, this would not be in compliance with the "minimise" target of CS 29.917(b) and CS 29.547(b).

(2) For the case where the VHM is integrated into a system with other functions

Software partitioning is addressed in both EUROCAE ED-12C/RTCA DO-178C and EUROCAE ED-109A/RTCA DO-278A.

n. Performance Criteria

(1) Signal Acquisition

The applicant for VHM system certification should specify the rate of acquisition of data sets for defect diagnostics in consistent flight regimes.

As a target, the total data set acquired in a flight should be sufficient for complete and reliable diagnostics to be produced for every flight above a defined duration in stabilised conditions. As a minimum, at least the data set for all components should be automatically obtained on each flight of greater than 30 minutes in stabilised conditions without the need for in-flight pilot action. For operations which do not contain periods of stabilised operation of greater than 30 minutes, alternative procedures need to be incorporated to ensure that the total data set is recorded within a specified number of flying hours related to the minimum adequate frequency of data collection determined under AMC 29.1465(e)(2), and in any case no longer than 25 flying hours.

Where subsystem performance is critical or relied upon to achieve the quoted defect probability of detection or False Alert rate, such as sensor accuracy, dynamic range or bandwidth, then this should be quoted.

(2) Data transfer and Storage Capability

The VHM defect status data should be capable of being downloaded during rotors running turnarounds.

All the data sets acquired should be stored until successfully transferred to the Ground-Based System. The storage capacity should not be less than 25 flying hours.

The applicant should describe the maximum interval between data downloads for which the system memory capacity is not exceeded.

In the event that a complete data set is not recorded, the data transfer process should be capable of downloading a partial data set to the Ground-Based System. In such a case, the ground station should alert maintenance personnel of a missing maintenance log or that the data set provided is incomplete.

(3) VHM Alert generation and fault detection performance

The Alert and Alarm generation processing should be designed to achieve a claimed probability of detection that is acceptable to the Agency for each component defect being monitored. Processing to isolate False Alerts and False Alarms should not result in an unacceptable workload. Also this processing should not compromise the verification and validating evidence of claimed defect detection performance. This workload should be assessed prior to completion of the Controlled Service Introduction (CSI) phase.

o. Performance Validation

The applicant should demonstrate how the VHM system provides an acceptable defect detection performance. Experiences gained during the CSI phase should be reviewed to confirm that this is the case.

(1) Validation methodology

It is not practical to verify predicted component defect detection performance for all failure modes by in-service experience or by trials. Therefore it is necessary that the methodology employed can be clearly substantiated from an understanding of how the failure mechanisms affect vibration and how the diagnostic processing will generate appropriate Alarms. Direct or indirect evidence should be provided as follows:

(i) Direct evidence includes:

- (A) Actual service experience on VHM equipped rotorcraft of the same or of similar type and configuration, including information from module strips, component removals, inspections and other investigations which is relevant to the review of VHM system performance.
- (B) Test rig results.
- (C) Rotorcraft trials, investigating cause and effect (for example, introducing degrees of imbalance or mal-alignment and calibrating the techniques response). This should be supported by flight experience to demonstrate that the False Alert criterion can be met and that all the diagnostic indicators lie within reasonable ranges.

Note: A mechanism should be established for requesting maintenance feedback with respect to component failure/degradation and VHM indication. The cases are as follows:

- to verify component condition following rejection after an Alarm, in order to establish the diagnostic accuracy, probability of detection and the False Alarm rate.
- to inform the TC holder in the event that a failure occurs which is monitored by VHM, where the VHM fails to provide an Alarm. This will provide the missed Alarm rate.

(ii) Indirect evidence includes:

- (A) Evidence as to the provenance of the technology and its suitability for application to rotorcraft.
- (B) Reference to adequate performance in other applications.
- (C) Modelling of the processes

The types of evidence stated in (i) and (ii) above can be used to substantiate:

- (A) That the Alert processing methodology can deliver an adequate False Alarm rate, Prognostic Interval and probability of detection.
- (B) Data acquired in a flight is sufficient for complete and reliable diagnostics to be produced for every flight above a minimum duration in stabilised conditions.
- (C) The sensitivity, dynamic range and bandwidth of the signal acquisition are adequate.
- (D) That the processed vibration signal-to-noise ratio is acceptable and that it is capable of discriminating the features required to identify potential incipient defects for the monitored components.

Typically, the False Alarm Rate and Alert Management performance will be validated during the CSI phase.

p. VHM System Criticality

- (1) It is necessary to understand the criticality of a VHM function in order to determine the appropriate level of integrity required. Criticality describes the severity of the end result of a VHM application failure/malfunction and is determined by an assessment that considers the safety effect that the VHM application can have on the rotorcraft.

Note: The criticality of the VHM function relates only to its contribution to the overall integrity of the component being monitored.

- (2) The criticality categories are defined in FAA AC 29.1309. In order to determine the appropriate level of criticality of the VHM function, it will be necessary to perform a safety assessment or functional hazard analysis on the rotorcraft systems affected. This should be carried out in accordance with standard safety assessment requirements such as CS 29.1309. In performing this assessment it will be necessary to consider the possibility of dormant and common mode failures and the possibility of the VHM system introducing additional risks, e.g. due to the False Alarm rate.
- (3) Different VHM Systems have functions that can have different levels of criticality, such as those described below:

- (i) Many VHM applications provide a method of enhanced health monitoring which adds to traditional techniques that have been used to establish an acceptable level of component integrity. Where a VHM application is not necessary for compliance with [CS 29.547\(b\)](#) and/or [CS 29.917\(b\)](#), the failure effect of these functions is considered to be 'No Safety Effect' when there have been no changes to the traditional techniques.
- (ii) Where a VHM application is identified as a compensating provision in order to comply with [CS 29.547\(b\)](#) and/or [CS 29.917\(b\)](#), then the failure criticality is considered to be 'Minor'. A proposed design that places greater reliance on VHM would not be deemed compliant with the "minimise" target of [CS 29.547\(b\)](#) and [CS 29.917\(b\)](#).
- (iii) When an on-board VHM system is used to replace existing portable test equipment, and is performing an identical function, (though not necessarily utilising the same method of detection), this can be classified as 'No Safety Effect', providing that in such cases there will be no reduction in scheduled component inspection, or extension of overhaul or replacement intervals. A level of system integrity related to Minor criticality supports the reduction or elimination of check flights after standard vibration reduction checks and/or adjustments (rotor track and balance, balancing, absorber tuning, etc.).

As this equipment is airborne equipment, it is considered that a quality standard for the software used is necessary. For this reason software to EUROCAE ED-12C/RTCA DO-178C Level D is necessary.

Note: As there should be no effect on safety of the helicopter as a result of utilising the airborne system, it will not be necessary to carry out recurring independent verification means.

- (iv) When a validated on-board VHM system is used to replace an existing maintenance task, this can be considered to be minor if the validated detection capability and integrity is better than the maintenance task being replaced. For example, VHM system monitoring of grease packed bearings which results in modification to manual inspection intervals.

For use of EUROCAE ED-12C/RTCA DO-178C level D software, it will be necessary to carry out periodic functional verification of the VHM system for dormant hardware or software failure or following a hardware or software change. An alternative approach to periodic functional verification is the retention of the original inspection at an increased interval. These instructions will need to be specified in the ICA.

Note: In cases (iii) and (iv), it is essential that the reliability and accuracy of the VHM must be equal to or better than that of the process it is replacing. This will require direct or indirect verification such as seeded fault testing (bench) or operational experience in accordance with paragraph (o) of this AMC. Compliance with paragraph (o) may require access to the design data and MSG3 analysis (or equivalent) used during substantiation of the original maintenance task.

q. Technical Publications

Appropriate Instructions for Continued Airworthiness (ICA) are required by [CS 29.1529](#) and Appendix A. ICA and other supporting data should be available to operators and maintenance

organisations before entry into service and should be updated whenever necessary during the service life of the system.

ICA should include the following:

- (1) Guidance for the interpretation of the diagnostic information produced by the VHM system for all components monitored, to include Alert and Alarm management, a description of the indicators, and Alert generation methods.
- (2) Maintenance instructions defining the actions to be taken in the event of all Alarms, including the appropriate rotorcraft inspections (or other maintenance) necessary for fault-finding to verify the Alarm.
- (3) Scheduled maintenance to be carried out on the VHM system itself, including inspections to confirm sensor performance and system functionality.
- (4) Instructions for all maintenance of the VHM System, including Illustrated Parts Catalogue/Illustrated Parts Breakdown and wiring diagrams.
- (5) Installation instructions for retrofit VHM systems addressing all aspects of VHM system integration with the rotorcraft.
- (6) A recommendation of the maximum period of unavailability of VHM functions for inclusion in the rotorcraft Master Minimum Equipment List (MMEL) or maintenance instructions, as required.
- (7) Operating Instructions detailing the operation of the VHM system including any ground-based elements or functions.
- (8) Required Flight Manual instructions.

r. Training

Suitable training should be made available with respect to operation and maintenance of the VHM system. This training should be made available prior to initial delivery of the VHM system. Training material and training courses should evolve to include lessons learned from service experience and appropriate diagnostic case studies. Training material and training courses should cover:

- (1) Installation of the VHM system.
- (2) Line maintenance of the VHM system (including VHM system fault-finding, any calibration necessary).
- (3) Use of the VHM System during Line maintenance to monitor the rotorcraft, including the data transfer, interface with data analysis, response to Alerts and Alarm processing, rotorcraft fault-finding and other Line diagnostic actions.
- (4) Necessary system administration functions, covering operational procedures relating to data transfer and storage, recovery from failed down loads and the introduction of hardware and software modifications.
- (5) Any data analysis and reporting functions that are expected to be performed by the operator.

s. **Product Support — System Data and Diagnostic Support**

The necessary support should be provided to operators to ensure that the VHM system remains effective and compliant with any applicable requirements throughout its service life. The

support provided should cover both the VHM system itself (i.e. system support), and the data generated (data and diagnostic support).

The data and diagnostic support provided should ensure that:

- (1) The operator has timely access to approved external data interpretation and diagnostic advice. It is the responsibility of the approval holder to provide this information; however, this may also involve.

t. **Minimum Equipment List (MEL) Recommendation**

The MEL should address the Airborne Element of the VHM system. The maximum period for absence of an assessment of any VHM indicator, to which Alert criteria are applied, should be limited to a suitable period and should not exceed 25 hours.

Note: If the VHM data is subject to close monitoring due to an increased likelihood of a developing mechanical problem, the maximum alleviation of 25 hours provided by the MMEL should be reduced or removed.

It is recommended that the VHM system automatically generates an indication to the operator if no VHM data has been gathered for a particular component for longer than a certain number of hours.

In the absence of any VHM data, reversion to the standard procedures used to ensure component integrity should be made.

u. **Controlled Service Introduction**

- (1) When a VHM system initially enters into service or it is adapted to a new application on a different rotorcraft type, then a Controlled Service Introduction (CSI) phase is usually necessary in order to fully validate the system performance.
- (2) If a CSI phase is considered to be necessary, then this activity should be detailed in a CSI plan to be approved prior to release to service, detailing the VHM applications being developed and the criteria for the successful completion of the CSI. Such criteria should address:
 - (i) The number of rotorcraft, number of operators, calendar time and flying hours.
 - (ii) Validation of specific sensor performance.
 - (iii) If targeted failures or defects occur during the CSI phase, it should be verified that the applicable VHM system applications provide an accurate timely Alarm. (iv) Validate the False Alarm rate.
 - (v) Evolution of Alert criteria.
 - (vi) Validate the timeliness and integrity of the end-to-end data transfer and analysis process.
 - (vii) Demonstration of specific support processes.
 - (viii) System hardware reliability.
 - (ix) System maintainability.
 - (x) System usability (including rotorcraft and ground based man-machine interfaces).
 - (xi) ICA usability.
 - (xii) Effectiveness of training.

- (xiii) Effectiveness and timeliness of diagnostic support.
 - (3) A CSI Plan should be agreed between the applicant for VHM system certification and the Agency prior to initial approval of the VHM system. This plan should then be implemented by the VHM approval holder and the operator(s) and monitored periodically by the Agency. Prior to any VHM function replacing an existing maintenance task, it may be necessary to complete a period of in-service operation. The validation and improvement activities should be detailed in this plan which should also detail the objectives that must be achieved before the CSI can be considered to be completed.
 - (4) Formal CSI meetings should take place in order to review service experience against the CSI criteria. They should involve the VHM system approval holder, the Agency (as applicable), and the operators.
 - (5) Once all parties agree that the intent of the CSI has been satisfied, the CSI phase will be considered closed. The process of review and closure should be recorded.
- v. Related documents
- (1) Federal Aviation Administration (FAA) AC 29-2C MG 15 'Airworthiness Approval of Rotorcraft Health Usage Monitoring Systems (HUMS)'
http://www.faa.gov/regulations_policies/advisory_circulars/
 - (2) CAP 753: Helicopter Vibration Health Monitoring (VHM) — Guidance Material for Operators Utilising VHM in Rotor and Rotor Drive Systems of Helicopters
<http://www.caa.co.uk/docs/33/CAP753.pdf>
 - (3) CAA Paper 2011/01: Intelligent Management of Helicopter Vibration Health Monitoring Data
http://www.caa.co.uk/docs/33/2011_01RFS.pdf

[Amdt 29/3]

SUBPART G — OPERATING LIMITATIONS AND INFORMATION

GENERAL

CS 29.1501 General

ED Decision 2003/16/RM

- (a) Each operating limitation specified in [CS 29.1503](#) to [29.1525](#) and other limitations and information necessary for safe operation must be established.
- (b) The operating limitations and other information necessary for safe operation must be made available to the crew members as prescribed in [CS 29.1541](#) to [29.1589](#).

OPERATING LIMITATIONS

CS 29.1503 Airspeed limitations: general

ED Decision 2003/16/RM

- (a) An operating speed range must be established.
- (b) When airspeed limitations are a function of weight, weight distribution, altitude, rotor speed, power, or other factors, airspeed limitations corresponding with the critical combinations of these factors must be established.

CS 29.1505 Never-exceed speed

ED Decision 2003/16/RM

- (a) The never-exceed speed, V_{NE} , must be established so that it is:
 - (1) Not less than 74 km/h (40 knots) (CAS); and
 - (2) Not more than the lesser of:
 - (i) 0.9 times the maximum forward speeds established under [CS 29.309](#);
 - (ii) 0.9 times the maximum speed shown under [CS 29.251](#) and [29.629](#); or
 - (iii) 0.9 times the maximum speed substantiated for advancing blade tip mach number effects under critical altitude conditions.
- (b) V_{NE} may vary with altitude, rpm, temperature, and weight, if:
 - (1) No more than two of these variables (or no more than two instruments integrating more than one of these variables) are used at one time; and
 - (2) The ranges of these variables (or of the indications on instruments integrating more than one of these variables) are large enough to allow an operationally practical and safe variation of V_{NE} .
- (c) For helicopters, a stabilised power-off V_{NE} denoted as V_{NE} (power-off) may be established at a speed less than V_{NE} established pursuant to sub-paragraph (a), if the following conditions are met:
 - (1) V_{NE} (power-off) is not less than a speed midway between the power-on V_{NE} and the speed used in meeting the requirements of:
 - (i) [CS 29.67\(a\)\(3\)](#) for Category A helicopters;
 - (ii) [CS 29.65\(a\)](#) for Category B helicopters, except multi-engine helicopters meeting the requirements of [CS 29.67\(b\)](#); and
 - (iii) [CS 29.67\(b\)](#) for multi-engine Category B helicopters meeting the requirements of [CS 29.67\(b\)](#).
 - (2) V_{NE} (power-off) is:
 - (i) A constant airspeed;
 - (ii) A constant amount less than power-on V_{NE} ; or

- (iii) A constant airspeed for a portion of the altitude range for which certification is requested, and a constant amount less than power-on V_{NE} for the remainder of the altitude range.

CS 29.1509 Rotor speed

ED Decision 2003/16/RM

- (a) *Maximum power-off (autorotation).* The maximum power-off rotor speed must be established so that it does not exceed 95% of the lesser of:
 - (1) The maximum design rpm determined under [CS 29.309\(b\)](#); and
 - (2) The maximum rpm shown during the type tests,
- (b) *Minimum power-off.* The minimum power-off rotor speed must be established so that it is not less than 105% of the greater of:
 - (1) The minimum shown during the type tests; and
 - (2) The minimum determined by design substantiation.
- (c) *Minimum power-on.* The minimum power-on rotor speed must be established so that it is:
 - (1) Not less than the greater of:
 - (i) The minimum shown during the type tests; and
 - (ii) The minimum determined by design substantiation; and
 - (2) Not more than a value determined under [CS 29.33\(a\)\(1\)](#) and [\(c\)\(1\)](#).

CS 29.1517 Limiting height-speed envelope

ED Decision 2003/16/RM

For Category A rotorcraft, if a range of heights exists at any speed, including zero, within which it is not possible to make a safe landing following power failure, the range of heights and its variation with forward speed must be established, together with any other pertinent information, such as the kind of landing surface.

CS 29.1519 Weight and centre of gravity

ED Decision 2003/16/RM

The weight and centre of gravity limitations determined under [CS 29.25](#) and [29.27](#), respectively, must be established as operating limitations.

CS 29.1521 Powerplant limitations

ED Decision 2003/16/RM

- (a) *General.* The powerplant limitations prescribed in this paragraph must be established so that they do not exceed the corresponding limits for which the engines are type certificated.
- (b) *Take-off operation.* The powerplant take-off operation must be limited by:
 - (1) The maximum rotational speed, which may not be greater than:
 - (i) The maximum value determined by the rotor design; or
 - (ii) The maximum value shown during the type tests;

- (2) The maximum allowable manifold pressure (for reciprocating engines);
 - (3) The maximum allowable turbine inlet or turbine outlet gas temperature (for turbine engines);
 - (4) The maximum allowable power or torque for each engine, considering the power input limitations of the transmission with all engines operating;
 - (5) The maximum allowable power or torque for each engine considering the power input limitations of the transmission with one engine inoperative;
 - (6) The time limit for the use of the power corresponding to the limitations established in sub-paragraphs (b)(1) to (5); and
 - (7) If the time limit established in sub-paragraph (b)(6) exceeds 2 minutes:
 - (i) The maximum allowable cylinder head or coolant outlet temperature (for reciprocating engines); and
 - (ii) The maximum allowable engine and transmission oil temperatures.
- (c) *Continuous operation.* The continuous operation must be limited by:
- (1) The maximum rotational speed, which may not be greater than:
 - (i) The maximum value determined by the rotor design; or
 - (ii) The maximum value shown during the type tests;
 - (2) The minimum rotational speed shown under the rotor speed requirements in [CS 29.1509\(c\)](#);
 - (3) The maximum allowable manifold pressure (for reciprocating engines);
 - (4) The maximum allowable turbine inlet or turbine outlet gas temperature (for turbine engines);
 - (5) The maximum allowable power or torque for each engine, considering the power input limitations of the transmission with all engines operating;
 - (6) The maximum allowable power or torque for each engine, considering the power input limitations of the transmission with one engine inoperative; and
 - (7) The maximum allowable temperatures for –
 - (i) The cylinder head or coolant outlet (for reciprocating engines);
 - (ii) The engine oil; and
 - (iii) The transmission oil.
- (d) *Fuel grade or designation.* The minimum fuel grade (for reciprocating engines) or fuel designation (for turbine engines) must be established so that it is not less than that required for the operation of the engines within the limitations in sub-paragraphs (b) and (c).
- (e) *Ambient temperature.* Ambient temperature limitations (including limitations for winterization installations if applicable) must be established as the maximum ambient atmospheric temperature at which compliance with the cooling provisions of [CS 29.1041](#) to [29.1049](#) is shown.
- (f) *Two and one-half minute OEI power operation.* Unless otherwise authorised, the use of 2½-minute OEI power must be limited to engine failure operation of multi-engine, turbine powered

rotorcraft for not longer than 2½ minutes for any period in which that power is used. The use of 2½-minute OEI power must also be limited by:

- (1) The maximum rotational speed, which may not be greater than:
 - (i) The maximum value determined by the rotor design; or
 - (ii) The maximum value shown during the type tests;
 - (2) The maximum allowable gas temperature;
 - (3) The maximum allowable torque; and
 - (4) The maximum allowable oil temperature.
- (g) *Thirty-minute OEI power operation.* Unless otherwise authorised, the use of 30-minute OEI power must be limited to multi-engine, turbine-powered rotorcraft for not longer than 30 minutes after failure of an engine. The use of 30-minute OEI power must also be limited by:
- (1) The maximum rotational speed, which may not be greater than:
 - (i) The maximum value determined by the rotor design; or
 - (ii) The maximum value shown during the type tests;
 - (2) The maximum allowable gas temperature;
 - (3) The maximum allowable torque; and
 - (4) The maximum allowable oil temperature.
- (h) *Continuous OEI power operation.* Unless otherwise authorised, the use of continuous OEI power must be limited to multi-engine, turbine-powered rotorcraft for continued flight after failure of an engine. The use of continuous OEI power must also be limited by:
- (1) The maximum rotational speed, which may not be greater than:
 - (i) The maximum value determined by the rotor design; or
 - (ii) The maximum value shown during the type tests.
 - (2) The maximum allowable gas temperature;
 - (3) The maximum allowable torque; and
 - (4) The maximum allowable oil temperature.
- (i) *Rated 30-second OEI power operation.* Rated 30-second OEI power is permitted only on multi-engine, turbine-powered rotorcraft also certificated for the use of rated 2-minute OEI power, and can only be used for continued operation of the remaining engine(s) after a failure or precautionary shutdown of an engine. It must be shown that following application of 30-second OEI power, any damage will be readily detectable by the applicable inspections and other related procedures furnished in accordance with paragraph A29.4 of [Appendix A](#) of CS-29. The use of 30-second OEI power must be limited to not more than 30 seconds for any period in which the power is used and by:
- (1) The maximum rotational speed which may not be greater than:
 - (i) The maximum value determined by the rotor design: or
 - (ii) The maximum value demonstrated during the type tests;
 - (2) The maximum allowable gas temperature; and

- (3) The maximum allowable torque.
- (j) **Rated 2-minute OEI power operation.** Rated 2-minute OEI power is permitted only on multi-engine, turbine-powered rotorcraft, also certificated for the use of rated 30-second OEI power, and can only be used for continued operation of the remaining engine(s) after a failure or precautionary shutdown of an engine. It must be shown that following application of 2-minute OEI power, any damage will be readily detectable by the applicable inspections and other related procedures furnished in accordance with paragraph A29.4 of [Appendix A](#) of CS-29. The use of 2-minute OEI power must be limited to not more than 2 minutes for any period in which that power is used, and by:
 - (1) The maximum rotational speed, which may not be greater than:
 - (i) The maximum value determined by the rotor designs; or
 - (ii) The maximum value demonstrated during the type tests;
 - (2) The maximum allowable gas temperature; and
 - (3) The maximum allowable torque.

CS 29.1522 Auxiliary power unit limitations

ED Decision 2003/16/RM

If an auxiliary power unit that meets the requirements of CS-APU is installed in the rotorcraft, the limitations established for that auxiliary power unit including the categories of operation must be specified as operating limitations for the rotorcraft.

CS 29.1523 Minimum flight crew

ED Decision 2003/16/RM

The minimum flight crew must be established so that it is sufficient for safe operation, considering:

- (a) The workload on individual crew members;
- (b) The accessibility and ease of operation of necessary controls by the appropriate crew member; and
- (c) The kinds of operation authorised under [CS 29.1525](#).

CS 29.1525 Kinds of operation

ED Decision 2003/16/RM

The kinds of operations (such as VFR, IFR, day, night, or icing) for which the rotorcraft is approved are established by demonstrated compliance with the applicable certification requirements and by the installed equipment.

CS 29.1527 Maximum operating altitude

ED Decision 2003/16/RM

The maximum altitude up to which operation is allowed, as limited by flight, structural, powerplant, functional, or equipment characteristics, must be established.

CS 29.1529 Instructions for Continued Airworthiness

ED Decision 2003/16/RM

Instructions for continued airworthiness in accordance with [Appendix A](#) to CS-29 must be prepared.

Appendix A – Instructions for Continued Airworthiness

ED Decision 2012/022/R

A29.1 General

- (a) This appendix specifies requirements for the preparation of instructions for continued airworthiness as required by [CS 29.1529](#).
- (b) The instructions for continued airworthiness for each rotorcraft must include the instructions for continued airworthiness for each engine and rotor (hereinafter designated 'products'), for each appliance required by any applicable CS or operating rule, and any required information relating to the interface of those appliances and products with the rotorcraft. If instructions for continued airworthiness are not supplied by the manufacturer of an appliance or product installed in the rotorcraft, the instructions for continued airworthiness for the rotorcraft must include the information essential to the continued airworthiness of the rotorcraft.

A29.2 Format

- (a) The instructions for continued airworthiness must be in the form of a manual or manuals as appropriate for the quantity of data to be provided.
- (b) The format of the manual or manuals must provide for a practical arrangement.

A29.3 Content

The contents of the manual or manuals must be prepared in a language acceptable to the Agency. The instructions for continued airworthiness must contain the following manuals or sections, as appropriate, and information:

- (a) *Rotorcraft maintenance manual or section.*
 - (1) Introduction information that includes an explanation of the rotorcraft's features and data to the extent necessary for maintenance or preventive maintenance.
 - (2) A description of the rotorcraft and its systems and installations including its engines, rotors, and appliances.
 - (3) Basic control and operation information describing how the rotorcraft components and systems are controlled and how they operate, including any special procedures and limitations that apply.
 - (4) Servicing information that covers details regarding servicing points, capacities of tanks, reservoirs, types of fluids to be used, pressures applicable to the various systems, location of access panels for inspection and servicing, locations of lubrication points, the lubricants to be used, equipment required for servicing, tow instructions and limitations, mooring, jacking, and levelling information.
- (b) *Maintenance Instructions.*
 - (1) Scheduling information for each part of the rotorcraft and its engines, auxiliary power units, rotors, accessories, instruments, and equipment that provides the recommended periods at which they should be cleaned, inspected, adjusted, tested, and lubricated, and the degree of inspection, the applicable wear tolerances, and work recommended at

these periods. However, it is allowed to refer to an accessory, instrument, or equipment manufacturer as the source of this information if it is shown that the item has an exceptionally high degree of complexity requiring specialised maintenance techniques, test equipment, or expertise. The recommended overhaul periods and necessary cross references to the airworthiness limitations section of the manual must also be included. In addition, an inspection program that includes the frequency and extent of the inspections necessary to provide for the continued airworthiness of the rotorcraft must be included.

- (2) Trouble-shooting information describing probable malfunctions, how to recognise those malfunctions, and the remedial action for those malfunctions.
 - (3) Information describing the order and method of removing and replacing products and parts with any necessary precautions to be taken.
 - (4) Other general procedural instructions including procedures for system testing during ground running, symmetry checks, weighing and determining the centre of gravity, lifting and shoring, and storage limitations.
- (c) Diagrams of structural access plates and information needed to gain access for inspections when access plates are not provided.
 - (d) Details for the application of special inspection techniques including radiographic and ultrasonic testing where such processes are specified.
 - (e) Information needed to apply protective treatments to the structure after inspection.
 - (f) All data relative to structural fasteners such as identification, discard recommendations, and torque values.
 - (g) A list of special tools needed.

A29.4 Airworthiness Limitations Section

The instructions for continued airworthiness must contain a section titled airworthiness limitations that is segregated and clearly distinguishable from the rest of the document. This section must set forth each mandatory replacement time, structural inspection interval, and related structural inspection required for type-certification. If the instructions for continued airworthiness consist of multiple documents, the section required by this paragraph must be included in the principal manual. This section must contain a legible statement in a prominent location that reads – ‘The airworthiness limitations section is approved and variations must also be approved’.

[Amdt 29/2]

[Amdt 29/3]

MARKINGS AND PLACARDS

CS 29.1541 General

ED Decision 2003/16/RM

- (a) The rotorcraft must contain:
 - (1) The markings and placards specified in [CS 29.1545](#) to [29.1565](#); and
 - (2) Any additional information, instrument markings, and placards required for the safe operation of the rotorcraft if it has unusual design, operating or handling characteristics.
- (b) Each marking and placard prescribed in sub-paragraph (a):
 - (1) Must be displayed in a conspicuous place; and
 - (2) May not be easily erased, disfigured, or obscured.

CS 29.1543 Instrument markings: general

ED Decision 2003/16/RM

For each instrument:

- (a) When markings are on the cover glass of the instrument there must be means to maintain the correct alignment of the glass cover with the face of the dial; and
- (b) Each arc and line must be wide enough, and located to be clearly visible to the pilot.

CS 29.1545 Airspeed indicator

ED Decision 2003/16/RM

- (a) Each airspeed indicator must be marked as specified in sub-paragraph (b), with the marks located at the corresponding indicated airspeeds.
- (b) The following markings must be made:
 - (1) A red radial line:
 - (i) For rotorcraft other than helicopters, at V_{NE} ; and
 - (ii) For helicopters, at V_{NE} (power-on).
 - (2) A red, cross-hatched radial line at V_{NE} (power-off) for helicopters, if V_{NE} (power-off) is less than V_{NE} (power-on).
 - (3) For the caution range, a yellow arc.
 - (4) For the safe operating range, a green arc.

CS 29.1547 Magnetic direction indicator

ED Decision 2003/16/RM

- (a) A placard meeting the requirements of this paragraph must be installed on or near the magnetic direction indicator.
- (b) The placard must show the calibration of the instrument in level flight with the engines operating.
- (c) The placard must state whether the calibration was made with radio receivers on or off.

- (d) Each calibration reading must be in terms of magnetic heading in not more than 45° increments.

CS 29.1549 Powerplant instruments

ED Decision 2003/16/RM

For each required powerplant instrument, as appropriate to the type of instruments –

- (a) Each maximum and, if applicable, minimum safe operating limit must be marked with a red radial or a red line;
- (b) Each normal operating range must be marked with a green arc or green line, not extending beyond the maximum and minimum safe limits;
- (c) Each take-off and precautionary range must be marked with a yellow arc or yellow line;
- (d) Each engine or propeller range that is restricted because of excessive vibration stresses must be marked with red arcs or red lines; and
- (e) Each OEI limit or approved operating range must be marked to be clearly differentiated from the markings of sub-paragraphs (a) to (d) except that no marking is normally required for the 30-second OEI limit.

CS 29.1551 Oil quantity indicator

ED Decision 2003/16/RM

Each oil quantity indicator must be marked with enough increments to indicate readily and accurately the quantity of oil.

CS 29.1553 Fuel quantity indicator

ED Decision 2003/16/RM

If the unusable fuel supply for any tank exceeds 3.8 litres (0.8 Imperial gallon/1 US gallon), or 5% of the tank capacity, whichever is greater, a red arc must be marked on its indicator extending from the calibrated zero reading to the lowest reading obtainable in level flight.

CS 29.1555 Control markings

ED Decision 2003/16/RM

- (a) Each cockpit control, other than primary flight controls or control whose function is obvious, must be plainly marked as to its function and method of operation.
- (b) For powerplant fuel controls:
 - (1) Each fuel tank selector valve control must be marked to indicate the position corresponding to each tank and to each existing cross feed position;
 - (2) If safe operation requires the use of any tanks in a specific sequence, that sequence must be marked on, or adjacent to, the selector for those tanks; and
 - (3) Each valve control for any engine of a multi-engine rotorcraft must be marked to indicate the position corresponding to each engine controlled.
- (c) Usable fuel capacity must be marked as follows:
 - (1) For fuel systems having no selector controls, the usable fuel capacity of the system must be indicated at the fuel quantity indicator.

- (2) For fuel systems having selector controls, the usable fuel capacity available at each selector control position must be indicated near the selector control.
- (d) For accessory, auxiliary, and emergency controls:
 - (1) Each essential visual position indicator, such as those showing rotor pitch or landing gear position, must be marked so that each crew member can determine at any time the position of the unit to which it relates; and
 - (2) Each emergency control must be red and must be marked as to method of operation.
- (e) For rotorcraft incorporating retractable landing gear, the maximum landing gear operating speed must be displayed in clear view of the pilot.

CS 29.1557 Miscellaneous markings and placards

ED Decision 2003/16/RM

- (a) *Baggage and cargo compartments, and ballast location.* Each baggage and cargo compartment, and each ballast location must have a placard stating any limitations on contents, including weight, that are necessary under the loading requirements.
- (b) *Seats.* If the maximum allowable weight to be carried in a seat is less than 77 kg (170 pounds), a placard stating the lesser weight must be permanently attached to the seat structure.
- (c) *Fuel and oil filler openings.* The following apply:
 - (1) Fuel filler openings must be marked at or near the filler cover with:
 - (i) The word 'fuel';
 - (ii) For reciprocating engine powered rotorcraft, the minimum fuel grade;
 - (iii) For turbine-engine-powered rotorcraft, the permissible fuel designations, except that if impractical, this information may be included in the rotorcraft flight manual, and the fuel filler may be marked with an appropriate reference to the flight manual; and
 - (iv) For pressure fueling systems, the maximum permissible fueling supply pressure and the maximum permissible defueling pressure.
 - (2) Oil filler openings must be marked at or near the filler cover with the word 'oil'.
- (d) *Emergency exit placards.* Each placard and operating control for each emergency exit must differ in colour from the surrounding fuselage surface as prescribed in [CS 29.811\(f\)\(2\)](#). A placard must be near each emergency exit control and must clearly indicate the location of that exit and its method of operation.

CS 29.1559 Limitations placard

ED Decision 2003/16/RM

There must be a placard in clear view of the pilot that specifies the kinds of operations (VFR, IFR, day, night or icing) for which the rotorcraft is approved.

CS 29.1561 Safety equipment

ED Decision 2003/16/RM

- (a) Each safety equipment control to be operated by the crew in emergency, such as controls for automatic liferaft releases, must be plainly marked as to its method of operation.
- (b) Each location, such as a locker or compartment, that carries any fire extinguishing, signalling, or other life saving equipment, must be so marked.
- (c) Stowage provisions for required emergency equipment must be conspicuously marked to identify the contents and facilitate removal of the equipment.
- (d) Each liferaft must have obviously marked operating instructions.
- (e) Approved survival equipment must be marked for identification and method of operation.

CS 29.1565 Tail rotor

ED Decision 2003/16/RM

Each tail rotor must be marked so that its disc is conspicuous under normal daylight ground conditions.

ROTORCRAFT FLIGHT MANUAL

CS 29.1581 General

ED Decision 2003/16/RM

- (a) *Furnishing information.* A Rotorcraft Flight Manual must be furnished with each rotorcraft, and it must contain the following:
 - (1) Information required by [CS 29.1583](#) to [29.1589](#).
 - (2) Other information that is necessary for safe operation because of design, operating, or handling characteristics.
- (b) *Approved information.* Each part of the manual listed in [CS 29.1583](#) to [29.1589](#) that is appropriate to the rotorcraft, must be furnished, verified, and approved, and must be segregated, identified, and clearly distinguished from each unapproved part of that manual.
- (c) Reserved.
- (d) *Table of contents.* Each Rotorcraft Flight Manual must include a table of contents if the complexity of the manual indicates a need for it.

CS 29.1583 Operating limitations

ED Decision 2003/16/RM

- (a) *Airspeed and rotor limitations.* Information necessary for the marking of airspeed and rotor limitations on or near their respective indicators must be furnished. The significance of each limitation and of the colour coding must be explained.
- (b) *Powerplant limitations.* The following information must be furnished:
 - (1) Limitations required by [CS 29.1521](#).
 - (2) Explanation of the limitations, when appropriate.
 - (3) Information necessary for marking the instruments required by [CS 29.1549](#) to [29.1553](#).
- (c) *Weight and loading distribution.* The weight and centre of gravity limits required by [CS 29.25](#) and [CS 29.27](#), respectively, must be furnished. If the variety of possible loading conditions warrants, instructions must be included to allow ready observance of the limitations.
- (d) *Flight crew.* When a flight crew of more than one is required, the number and functions of the minimum flight crew determined under [CS 29.1523](#) must be furnished.
- (e) *Kinds of operation.* Each kind of operation for which the rotorcraft and its equipment installations are approved must be listed.
- (f) *Limiting heights.* Enough information must be furnished to allow compliance with [CS 29.1517](#).
- (g) *Maximum allowable wind.* For Category A rotorcraft, the maximum allowable wind for safe operation near the ground must be furnished.
- (h) *Altitude.* The altitude established under [CS 29.1527](#) and an explanation of the limiting factors must be furnished.
- (i) *Ambient temperature.* Maximum and minimum ambient temperature limitations must be furnished.

CS 29.1585 Operating procedures

ED Decision 2003/16/RM

- (a) The parts of the manual containing operating procedures must have information concerning any normal and emergency procedures, and other information necessary for safe operation, including the applicable procedures, such as those involving minimum speeds, to be followed if an engine fails.
- (b) For multi-engine rotorcraft, information identifying each operating condition in which the fuel system independence prescribed in [CS 29.953](#) is necessary for safety must be furnished, together with instructions for placing the fuel system in a configuration used to show compliance with that paragraph.
- (c) For helicopters for which a V_{NE} (power-off) is established under [CS 29.1505\(c\)](#), information must be furnished to explain the V_{NE} (power-off) and the procedures for reducing airspeed to not more than the V_{NE} (power-off) following failure of all engines.
- (d) For each rotorcraft showing compliance with [CS 29.1353\(c\)\(6\)\(ii\) or \(c\)\(6\)\(iii\)](#), the operating procedures for disconnecting the battery from its charging source must be furnished.
- (e) If the unusable fuel supply in any tank exceeds 5% of the tank capacity, or 3.8 litres (0.8 Imperial gallon/1 US gallon), whichever is greater, information must be furnished which indicates that when the fuel quantity indicator reads 'zero' in level flight, any fuel remaining in the fuel tank cannot be used safely in flight.
- (f) Information on the total quantity of usable fuel for each fuel tank must be furnished.
- (g) For Category B rotorcraft, the airspeeds and corresponding rotor speeds for minimum rate of descent and best glide angle as prescribed in [CS 29.71](#) must be provided.

CS 29.1587 Performance information

ED Decision 2008/010/R

Flight manual performance information which exceeds any operating limitation may be shown only to the extent necessary for presentation clarity or to determine the effects of approved optional equipment or procedures. When data beyond operating limits are shown, the limits must be clearly indicated. The following must be provided:

- (a) *Category A.* For each Category A rotorcraft, the rotorcraft flight manual must contain a summary of the performance data, including data necessary for the application of any applicable operating rule, together with descriptions of the conditions, such as airspeeds, under which this data was determined, and must contain –
 - (1) The indicated airspeeds corresponding with those determined for take-off and the procedures to be followed if the critical engine fails during take-off;
 - (2) The airspeed calibrations;
 - (3) The techniques, associated airspeeds, and rates of descent for autorotative landings;
 - (4) The rejected take-off distance determined under [CS 29.62](#) and the take-off distance determined under [CS 29.61](#);
 - (5) The landing data determined under [CS 29.81](#) and [29.85](#);
 - (6) The steady gradient of climb for each weight, altitude, and temperature for which take-off data are to be scheduled, along the take-off path determined in the flight conditions required in [CS 29.67\(a\)\(1\) and \(a\)\(2\)](#);

- (i) In the flight conditions required in [CS 29.67\(a\)\(1\)](#) between the end of the take-off distance and the point at which the rotorcraft is 61 m (200 ft) above the take-off surface (or 61 m (200 ft) above the lowest point of the take-off profile for elevated heliports).
 - (ii) In the flight conditions required in [CS 29.67\(a\)\(2\)](#) between the points at which the rotorcraft is 61 m (200 ft) and 305 m (1000 ft) above the take-off surface (or 61 m (200 ft) and 305 m (1000 ft) above the lowest point of the take-off profile for elevated heliports).
- (7) Hover performance determined under [CS 29.49](#) and the maximum weight for each altitude and temperature condition at which the rotorcraft can safely hover in-ground effect and out-of-ground effect in winds of not less than 31 km/h (17 knots) from all azimuths. This data must be clearly referenced to the appropriate hover charts.
- (b) *Category B.* For each Category B rotorcraft, the Rotorcraft Flight Manual must contain:
 - (1) The take-off distance and the climbout speed together with the pertinent information defining the flight path with respect to autorotative landing if an engine fails, including the calculated effects of altitude and temperature;
 - (2) The steady rates of climb and hovering ceiling, together with the corresponding airspeeds and other pertinent information, including the calculated effects of altitude and temperature;
 - (3) The landing distance, appropriate airspeed and type of landing surface, together with any pertinent information that might affect this distance, including the effects of weight, altitude and temperature;
 - (4) The maximum safe wind for operation near the ground;
 - (5) The airspeed calibrations;
 - (6) The height-speed envelope except for rotorcraft incorporating this as an operating limitation;
 - (7) Glide distance as a function of altitude when autorotating at the speeds and conditions for minimum rate of descent and best glide angle, as determined in [CS 29.71](#);
 - (8) Hover performance determined under [CS 29.49](#) and the maximum safe wind demonstrated under the ambient conditions for data presented. In addition, the maximum weight for each altitude and temperature condition at which the rotorcraft can safely hover in-ground effect and out-of-ground effect in winds of not less than 31 km/h (17 knots) from all azimuths. This data must be clearly referenced to the appropriate hover charts; and
 - (9) Any additional performance data necessary for the application of any applicable operating rule.

[Amdt. No.: 29/1]

[Amdt. No. 29/2]

CS 29.1589 Loading information

ED Decision 2003/16/RM

There must be loading instructions for each possible loading condition between the maximum and minimum weights determined under [CS 29.25](#) that can result in a centre of gravity beyond any extreme prescribed in [CS 29.27](#), assuming any probable occupant weights.

MISCELLANEOUS GUIDANCE

MG4 Full Authority Digital Electronic Controls (FADEC)

ED Decision 2008/010/R

Note: Certification procedures identified in MG4 refer specifically to the FAA regulatory system. For guidance on EASA procedures, reference should be made to Commission Regulation (EC) No 1702/2003 (as amended) (Part-21), AMC-20 (and specifically AMC 20-1 and 20-3) and to EASA internal working procedures, all of which are available on EASA's web site: <http://www.easa.europa.eu/>

[Amdt. No. 29/2]